

Fixed Universal Polynomials

A literal Grill instance, exact degree laws, native witness fibers,
positive-index restoration, an exact negative-index predicate,
nonredundant comparisons, first-index deletion
and the 82- and 83-operation candidates

Research reports 23 (revision 1), 24, 25, 33, 34, 37, 39, 41, 43, 45 and 46 of one AI-assisted research
pipeline,
merged into one report for the ProveIt research-report collection

Sources dated 3 and 4 October 2026; Parts I–IV merged 3 October 2026 (batch 82),
Part V added the same day (batch 83), Parts VI–VIII added the same day (batch 88)

Status. AI-assisted research manuscripts, unrefereed, and not formalized in a proof assistant. The language theorem of Part I is conditional on precisely pinned imports from the Hilbert’s-tenth research tree of this repository; the theorems of Parts III and IV are conditional on, or inherit, pinned native-source theorems as each Part states; Part V inherits the compiler contract and the sign-free bootstrap as premises, and its exact predicate decides neither existence nor emptiness of a negative zero. Parts VI–VIII inherit the same compiler contract; Part VI’s existence theorems rest on two estimates from the literature and on an analytic argument that the programme’s review did not recertify, and the two candidates of Part VIII are refuted, so no Part lowers the research tree’s 84-operation record. The exact-degree theorems of Parts I and II are unconditional algebra on the stated sources. Section 0.6 lists what is and is not claimed.

Merge. Text marked [write] was written when the eleven manuscripts were merged (five in batch 82, Part V in batch 83, Parts VI–VIII in batch 88); all other text is the source’s own. Section 0.2 gives the provenance.

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0.1 [write] What this report contains

This report prints eleven research manuscripts that share one object: the fixed universal Diophantine polynomials of the Hilbert’s-tenth research programme in this repository (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`, below *the research tree*) and the native Pell arithmetic block inside them. They share one spine: first the exact resources and exact degree of one literal instance, then the structure of the witness fibers of the native block. The sixth manuscript, Part V, was added in batch 83; it continues the last appendix of Part IV. Five more, Parts VI–VIII, were added in batch 88: they continue Part V and test cheaper candidates for the research tree’s universal polynomial, whose record is now 84 operations.

- **Part I** (Report 23, revision 1): one literal Grill program with 397,488 run phases and one complete integer polynomial circuit with an ordinary positive input, five program parameters fixed for each represented language, 797,135 positive witnesses and 3,600,546 binary operations. Its language theorem is conditional on pinned imports; its exact total degree 69,339,973 is proved unconditionally.
- **Part II** (Report 24): for every nonempty run table of the same native template, with $N = 2m + g + 5$, the native unit has exact degree $63N - 4$, the largest residual $12N + 10$, and the complete native polynomial $87N + 16$; residual gluing, recoder and loader maxima, and fresh polynomial inputs. Part I’s degree is the instance $N = 797,011$.
- **Part III** (Report 25): the entire positive fiber of the retained twenty-two-coordinate native component at fixed ports: seventeen coordinates forced, five parametrized by two Pell indices, and the two-term height law with a $\zeta(1/2)$ second term.
- **Part IV** (Reports 33 and 34): the complete⁷⁴ signed¹⁹ index-eliminated child has infinitely many positive zeros with negative restored index at every positive input (Report 33), and the height and counting law of one canonical family of these zeros, with a Kesten-type obstruction to a smooth second term (Report 34).
- **Part V** (Report 37): for the two positive interfaces raw²⁹ and positive²¹ that Report 33 leaves open, an exact necessary-and-sufficient arithmetic predicate for a full positive zero with negative restored index, with explicit positive reconstruction of every witness in both interfaces, a finite wrapped odd-input sector, sharper even-input bounds and a certified sub-unit Binet window. Whether the predicate ever passes is open.
- **Part VI** (Report 39): for every valid compiler and input, infinitely many positive tuples satisfying every condition of Part V’s predicate except the main congruence, a fixed-modulus Jacobi filter for that congruence, and a thinned progression on which it fails; hence the raw²⁹ main projection and the positive²¹ main norm are nonredundant comparisons. No full negative zero is constructed; Part V’s question stays open.
- **Part VII** (Report 41): deleting the first-index quotient from the research tree’s 85- and 86-operation universal polynomials gives 81- and 82-operation candidates that accept every positive input on every genuine compiler slice. The tree has since carried the construction to its 84-operation polynomial (an 80-operation candidate, also refuted).
- **Part VIII** (Reports 43, 45 and 46): two reductions of the 84-operation polynomial. For the free-coefficient 83 candidate, structural divisor obstructions and nonnative inner families (Report 43; its open language question has since been answered by the research tree, which refutes the candidate); for the square/product 82 candidate, a full counterfamily at every input (Report 45, a second construction of a refutation the tree also proved); and the exact height and fixed-outer minimality

of Report 45’s certificates, with an even-rank obstruction to extending them to the free-coefficient candidate (Report 46).

0.2 [write] Sources, editions and provenance

0.2.1 The manuscripts

All seven archives of this cluster arrived in commit `db37d18c8` (3 October 2026, “New research reports”) under `docs/incoming/`; they were placed by commit `2f58ab4e9` (batch 82, cluster M3), which retired the archives. The manuscripts are numbered “Research Report NN” by their pipeline; the batch numbers are the manuscript numbers of batch 82. Report 37 arrived later, in commit `3051d1446` (batch 83, manuscript 09), and was placed by commit `a51a439cd` (cluster 83H2) as Part V; its local source number, 15, continues this report’s own sequence after the highest prefix, 14. Table 1 lists them. Each manuscript records its own pins, which are the commits of this repository whose research-tree files it imports or cites; they are kept here as provenance. (*Batch 88.*) Reports 39, 41, 43, 45 and 46 arrived in commit `c5612efa1` (3 October 2026, 21:05 PDT, “New research reports”, ten archives, manuscripts 01–10 of batch 88; they are manuscripts 01, 02, 05, 06 and 08) and were placed by commit `992aaafb3` (cluster 88T2), which retired their archives, as sources 16–20, continuing this report’s own sequence after its highest prefix, 15. The other five archives of that arrival, Reports 40, 42, 44, 47 and 48 on a periodic Langton ant, went to `periodic-turmite-first-revisits` (cluster 88T1).

What each contributed. Source 11 is the base: its `article.tex` and `README.md` were staged as delivered, and its text is Part I. Sources 14, 07, 10 and 05 contributed their manuscripts (Parts II–IV) and their code, data and audit files; their delivered READMEs and PDFs are not shipped. Source 15 (Report 37) contributed its manuscript, printed as Part V, and its code, data and audit files; its delivered README and PDF are not shipped. Sources 16–20 (Reports 39, 41, 43, 45 and 46, batch 88) contributed their manuscripts, printed as Parts VI, VII and VIII, and their code, data and audit files; their delivered READMEs and PDFs are not shipped. Every result, proof, remark, question and limitation of the eleven manuscripts is printed.

The edition of Report 23. Two editions of Report 23 arrived. Archive 11 is revision 1: its `11-grill-poly-REVISION.md` calls it “a versioned extension of Report 23” and anchors the SHA-256 digests of the original PDF, LaTeX source, release inventory and of archive 12 itself (`467e2b47...bf9b`, measured equal). Revision 1 changes no gate, coefficient, witness or comparison; it replaces the original’s propagated upper bound 71,731,007 by the exact degree 69,339,973 (new Section 23.13) and keeps the original’s QA files as `qa/v0-*`. Archive 12 is therefore superseded and nothing of it is staged. Its one open question, “Can the true degree be determined?”, is answered by Section 23.13 and is not printed as open; revision 1 replaces it by “Can a different representation reduce the exact degree?”.

The series citation graph. Report 24 cites Report 23 revision 1 (its optional full regression reads that report’s arithmetic DAG). Report 25 is the sequel of Report 22, “Canonical outer histories and infinite native fibers” (archive `Canonical_Histories_and_Infinite_Fibers_Package.zip` of batch 82, cluster M2, placed by commit `7d2b1b245` as source 23 of the collection report `signal-machine-collision-certificates`, in its new Part VIII); it uses only Report 22’s entire-fiber projection theorem, in the corollary of Section 25.12. Report 34 cites Reports 33 and 25 and carried their LaTeX sources, byte-identical to Parts III and IV, as context (not staged). Reports 23, 24, 25, 33 and 34 all pin research-tree commits. Report 37 continues Report 33’s Appendix 33.B, cites its reduction

Report	Batch, no.	Archive (bytes)	Pins	File prefix	Printed as
23 rev. 1	82, 11	Fixed_Universal_Grill_Polynomial_Package (1).zip (15,508,226)	2d887f0fa	11-grill-poly-	Part I
24	82, 14	Native_Grill_Exact_Degree_Laws_Package.zip (552,226)	2d887f0fa	14-degree-laws-	Part II
25	82, 07	Entire_Native_Witness_Fiber_Package.zip (531,631)	ad634b2d1	07-native-fiber-	Part III
33	82, 10	Failure_of_Positive_Index_Restoration_Package.zip (703,207)	2dde7850f; context 93ad43375	10-index-restore-	Part IV (33.k)
34	82, 05	Counterexample_Height_Expansions_and_Rotation_Discrepancy_Package.zip (591,664)	2dde7850f; transfer 9fa99be93	05-signed19-heights-	Part IV (34.k)
37	83, 09	Exact_Negative_Index_Obstruction_for_Positive_Diophantine_Interfaces_Package.zip (501,623)	d5bd4a67b; refinement 813c1cff4; freshness 0f7d618f4	15-neg-obstruction-	Part V
39	88, 01	Nonredundant_Main_Comparisons_in_Positive_Diophantine_Certificates_Package.zip (622,252)	d5bd4a67b	16-nonredundant-	Part VI
41	88, 02	Failure_of_First_Index_Deletion_Package.zip (547,920)	3f4a974a5; context 93c34e817	17-index-deletion-	Part VII
43	88, 05	Structural_Obstructions_for_the_Free_Coefficient_83_Candidate_Package.zip (552,509)	0d9d1e017; precursor 53c34b012	18-free83-	Part VIII (43.k)
45	88, 06	Full_Positive_Collapse_of_the_Square_Product_82_Candidate_Package.zip (539,173)	6ce2dcaaf	19-square-product82-	Part VIII (45.k)
46	88, 08	Exact_Certificate_Height_and_an_Even_Rank_Obstruction_Package.zip (1,825,258)	6ce2dcaaf, 0d9d1e017, 8cf6239b6	20-cert-height-	Part VIII (46.k)
23 orig.	82, 12	Fixed_Universal_Grill_Polynomial_Package.zip (15,390,453)	2d887f0fa	—	superseded

Table 1: The manuscripts. The file prefix is that of the source’s shipped code, data and audit files in this directory; the README lists them. In the text a source is named by its prefix number (source 11, ..., source 20).

as “prior”, and carried two of Report 33’s evidence files as context (byte copies, shipped once under 10-index-restore-); it does not cite Reports 23–25 or 34. The sibling Report 32 of the same batch, a fixed universal matrix semigroup, is printed as Part V of **group-theoretic-substrates**; it mentions Report 23 only in its delivered audit. (*Batch 88.*) Report 39 continues Report 37 and carried its manuscript as context (a byte copy, not shipped). Report 41 cites Reports 33, 37 and 39. Report 43 cites Report 41; Report 45 cites Reports 41 and 43; Report 46 imports Report 45’s construction and Report 43’s interface and small-norm lemma, and carried Report 45’s complete release and a selection of Report 43’s sources (byte copies, shipped once). All five pin research-tree commits.

Where the merge had to choose.

1. The edition of Report 23 (above).

2. Placement of Report 25. Report 25 continues Report 22, which goes to `signal-machine-collision-certificates`; Report 25 is printed here because its mathematics is purely the native Pell block that Parts I, II and IV also use, and it uses Report 22 in one corollary only.
3. Part IV prints two manuscripts: Report 34 is a direct sequel of Report 33, studies one family inside Report 33's construction, and shipped Report 33 as its context.
4. Numbering. Every source keeps its own numbering, prefixed by its Report number: statement $k.m$ of Report NN is printed as $NN.k.m$, its Section k as Section $NN.k$, its equation (k) as $(NN.k)$, and its Appendix X as Section $NN.X$. Tables are numbered consecutively through the whole report. Front-matter sections are numbered $0.k$.
5. Bibliographies. Each source's bibliography is printed at the end of its own Part, with the source's numbering; entries naming a sibling Report gained a `[write]` pointer to its place.
6. Macros. Only macro names changed, never renderings: Report 24's `\sha` and `\fname` are `\shasmall` and `\fnameB`; Report 33's `\MF` (the native mask M_F^{native}) is `\MFnat` and its `\MFS` (the paid mask M_F^{src}) is `\MFsrc`; Report 34's `\MF`, which denotes the paid mask, is `\MFsrc`. No mathematical symbol was renamed.
7. Labels. Every label carries the prefix `fup:`; the labels of Reports 23, 24, 25, 33 and 34 carry `fup:gp:`, `fup:dl:`, `fup:nf:`, `fup:ir:` and `fup:hc:` before their delivered names, and the front matter uses `fup:fm:`. Report 37's labels carry `fup:nz:` (batch 83).
8. (*Batch 83.*) Placement of Report 37. It continues Appendix 33.B of Report 33, but it is printed as its own Part V rather than as a third source of Part IV, because Part IV is about the signed19 child and Report 37 treats only raw29 and positive21. Its Section 37.3 restates Appendix 33.B and the research tree's refinement note, as the source itself says; it is printed in full (its equations are cited later) with a `[write]` note giving the correspondence, and with no novelty claim. Its re-proof of Lemma 33.3.2 and its use of the tree's auxiliary construction are marked where they occur.
9. (*Batch 83.*) Report 37's macros `\MF` and `\MFnative` are `\MFsrcV` and `\MFnatV` here; their renderings $M_{F,\text{src}}$ and $M_{F,0}$ are kept and denote the masks written M_F^{src} and M_F^{native} in Part IV. Unlike the batch-82 merge, Part V renames some mathematical symbols of its source, to keep one meaning per letter inside the Part; the note at the start of Part V lists every rename.
10. (*Batch 88.*) Placement. The five manuscripts are additions to this report, the only report of the collection that treats these interfaces and the programme's operation record. They form three Parts in report order: Report 39 continues Part V (Part VI); Report 41 is a failure of index elimination like Part IV, for the 85- and 86-operation parents (Part VII); and Reports 43, 45 and 46 form one Part, because Report 46 imports both of the others (Part VIII, after the precedent of Part IV).
11. (*Batch 88.*) Duplicates. Report 45's main theorem was also proved by the research tree, by a different construction, about forty minutes later and without either text citing the other; Report 45 is printed whole as a second route, with the timing stated and no priority asserted. The small-norm lemma is proved in Reports 43 and 46; both proofs are printed, the second marked as such. The odd-quotient, projection and Pell-growth lemmas that several sources re-prove are printed in each, with pointers to Report 33's Lemmas 33.3.2 and 33.3.3 and no novelty claim; Report 39's restatement of Part V's predicate and reconstruction is printed as a recap with pointers; Report 46's recap of Report 45's construction is printed as delivered.
12. (*Batch 88.*) Macros and symbols. The sources' `\MF`, `\MFnative` and `\MFO` are `\MFsrcV` and `\MFnatV`, their `\ya` is `\yaux` and their `\Pos` is `\pos`, all with unchanged renderings. Report 46's

factor names N_a and N_s are printed N_{aux} and N_{strong} , the renderings of Reports 43 and 45 for the same factors; this is the one rendering change. Report 45’s two input files (its pin table and row list) are printed in place. Unlike Part V, Parts VI–VIII rename no mathematical symbol; their opening notes and Table 13 list the letters that collide.

13. (*Batch 88.*) Stale statements. Report 41’s 85-operation record, Report 43’s open language question and Report 45’s repetition of it, and a sentence in Report 39’s first-stage evidence README are kept as delivered, with dated [write] notes; none of them is a retraction of a theorem.

Not shipped. PDFs, checksum ledgers and release seals, the non-base manuscripts and delivery READMEs, byte copies of research-tree files (73 members of the five archives; they match the current files byte for byte and are cited by path), in-archive and cross-archive byte copies (shipped once), Report 34’s context copies of Reports 25 and 33, two heavy regenerable files of Report 23 (the 61,209,290-byte arithmetic DAG and the 1,589,952-byte run table; the README gives the rebuild and the retrieval), and the third-party copies of the Cocke–Minsky (1964) and Neary–Woods (2009) papers that archive 11 contained (PDFs, page scans and text extractions; they are cited in Part I’s bibliography [1, 2] and are not redistributed). All of these survive in the arrival commit. From Report 37’s archive (source 15, batch 83) the manuscript, delivery README and PDF, its two checksum ledgers, three in-archive duplicate receipts, and seven byte copies of repository files (five research-tree files and two of Report 33’s evidence files, cited by path) are not shipped; they survive in its arrival commit 3051d1446. From the five archives of batch 88 (sources 16–20) the manuscripts, delivery READMEs and seven PDFs (including the copies of Reports 43 and 45 inside Report 46’s archive), 20 checksum ledgers, 63 byte copies of research-tree files (cited by path), 40 in-batch byte copies (Report 46’s copy of the complete Report 45 release and of Report 43’s sources, and Report 39’s copy of the Report 37 manuscript) and 33 duplicate receipts are not shipped; they survive in the arrival commit c5612efa1.

0.3 [write] Notation across the Parts

Each Part keeps its source’s letters, and inside a Part a letter always has that Part’s meaning; Part V renames a few symbols of its source to achieve this, as the note at its start lists. Parts VI–VIII rename no symbol: Parts VI and VII keep their sources’ letters (each source flags its own collisions), and each of the three sources of Part VIII keeps its own letters inside its sections, with Table 13 comparing them. Many letters change meaning between Parts. Table 2 lists those that do, with the reading that would be wrong.

Table 2: Letters with different meanings in different Parts.

Letter	Meanings	Tempting false reading
α	I, II: the lane degree $3N_*$ (I) or $3N$ (II). III: the Pell unit $A + \sqrt{\Delta}$, with $\lambda = \log \alpha$. IV, Report 33: a supplied coordinate in Sections 33.1–33.10, and $\alpha_0 = \log \lambda$ in Lemma 33.6.1; Report 34: the growth logarithm $\log \lambda$, with α_{src} for Report 33’s coordinate.	Report 34 swaps Report 25’s letters: its λ is Report 25’s α and its α is Report 25’s λ (Section 34.9 compares the two Parts).
λ	III: $\log(A + \sqrt{\Delta})$. IV: the unit $A + \sqrt{\Delta}$ (Lemma 33.6.1 and Report 34). V: the units $\lambda_A = A + \sqrt{A^2 - 1}$ and $\lambda_P = P + \sqrt{P^2 - 1}$ (the source’s α_*, β_*).	The same letter is a logarithm in Part III and a unit in Parts IV and V.

Letter	Meanings	Tempting false reading
κ	I: $k_* = \eta + \zeta$ is written k_* ; II: $\kappa = \eta + \zeta$ (native register). III: the counting coefficient of $N(H)$. IV: in Report 33 the input coefficient $u + \delta\Delta$; in Report 34 the counting coefficient of $N(B)$, with κ_{in} for Report 33's κ . V: the input coefficient $\kappa = u + \delta\Delta = \psi_A(e)$, as in Report 33.	Part II's register κ and Part III's coefficient κ are unrelated; so are Report 33's and Report 34's.
δ	II: the degree of a fresh polynomial input. IV: a supplied coordinate in Report 33 (δ_{src} in Report 34); $\log(1 + 1/Y)$ in Report 34.	—
N	I: the varying tape integer $N = N_e(x)$, and $N_* = 797,011$ the fixed lane exponent. II: $N = 2m + g + 5$, which is Part I's N_* , not its N ; Part II writes N_{tape} for Part I's N . III: $N(H)$ is a counting function. IV: Report 33's N is a power-of-five exponent; Report 34's $N(B)$ is a count and N_0 that exponent. V: the packing quotient $N = (p + M)/(q^2 - 1)$ of (37.25), and N_s a number of series terms.	Part II's $87N + 16$ is evaluated at Part I's N_* , never at the tape integer.
$K, \mathcal{K}$	I: $K = 2^{2b}$ is the loader radix and $\mathcal{K} = 2^{551890}$ the native radix multiplier. II: K is the native radix multiplier (Part I's $\mathcal{K}$) and K_e the loader radix. IV, V: $K_0 = D_C + BD_R$ is a compiler constant, and K the native support size in the radix bound.	—
$B, \mathcal{B}$	I: $B = 2^{k-1}Q$ in a recoder, $\mathcal{B}$ the native radix. II: $B = KD_0$, Part I's $\mathcal{B}$. III: a generic Pell base, and in Section 25.3 the first Pell base $2XY^2 + 1$. IV: Report 33's $B = 2^d$ is the compiler radix and $P = 2XY^2 + 1$ the first Pell base; Report 34's B is a height bound and B_{comp} the compiler radix. V: $B = 2^{d_{\text{cell}}}$, Report 33's compiler radix, and $P = 2XY^2 + 1$.	Part III's first Pell base B is Parts IV–V's P .
Δ, A	II: $\Delta = 4a + 3$ and $A = a^2 + \Delta$, both computed from the register a . III, IV, V: $A = a + 2$ is a Pell base and $\Delta = A^2 - 1 = a^2 + 4a + 3$; Reports 33 and 37 write $H = 4a + 3$ (Report 34: H_0 ; the programme's review of Report 37: $4A - 5$).	Part II's A is Parts III–IV's Δ , and Part II's Δ is Part IV's H . The registers a are different objects in the two presentations.
$p, \mathfrak{p}$	I: $\mathfrak{p} = 2^{551890}(X + Z_0)V_f J_1$, and p_e a program parameter. II: $p = KLVJ_1$, Part I's $\mathfrak{p}$. III, IV, V: the main Pell index ($p = 2r + 1$ in III).	—
c, a, f, i, j, o, y	I, II: registers and coordinates of the 67-row native kernel of the Grill polynomial. III: coordinates of the twenty-two-coordinate native component pinned at ad634b2d1 . IV: coordinates of the signed19 child. V: coordinates of the raw29 and positive21 children (Part V writes y_{aux} for the auxiliary y).	These are different presentations of the programme's native Pell/AND block; the letters play analogous roles but no coordinate of one Part is identified with a coordinate of another.
R	I: registers R_{15}, R_{16} and residuals R_j . III: $R_m = \Delta\psi_A(m)$ and $R = ic^2$. IV: the restored index $R = k - hXY - 1$; in Report 34 also $R, R(B)$ for a number of progression terms; $R_{\text{rad}} = 2^b$ the inner radix (Section 33.B). V: the restored index, $R = -p$ on the negative branch, and R_{rad} as in IV.	—

Letter	Meanings	Tempting false reading
H	I: a joined kernel port. III: a height bound in $N(H)$. IV, V: $H = 4a + 3$ (Reports 33 and 37).	—
W	II: witness counts $W_{\text{native}}, W_{\text{total}}$. IV: the computed register $W = C - Z$ (Report 33, negative at its zeros), the positive supplied marker $W \equiv 2^v \pmod{H}$ of Section 33.B, $W = 1 - Z$ (Report 34), and the Lambert function W in Section 34.5. V: the positive supplied input marker $W = 2^e \pmod{H}$, $0 < W < q$.	Part V's W is positive by hypothesis; Report 33's main-text W is negative, which is why the signed19 construction cannot be reused.
ζ	I, II, III: a supplied native coordinate. III also uses the Riemann zeta function $\zeta(1/2)$ and warns about the clash.	—
Q	I: a recoder port and the leading form Q_* . II: $Q = \ln(q)$. IV, V: $Q = q^2 - 1$ (Reports 33 and 37); Q_b (IV) and Q_θ (V) the odd quotient polynomials.	—
$M_F^{\text{native}}, M_F^{\text{src}}$	IV: the native and the paid field mask, $M_F^{\text{src}} = M_F^{\text{native}} + B - 1$. V: the same two masks, rendered $M_{F,0} = M_F^{\text{native}}$ and $M_{F,\text{src}} = M_F^{\text{src}}$.	Report 34's own macro for M_F^{src} had the name of Report 33's macro for M_F^{native} ; the renderings always differed and are unchanged. Report 37's macro for $M_{F,\text{src}}$ also had that name.
t	IV: Report 33's binary exponent $t = dN$ of the scale $q = 2^t$ (Section 33.4), and its first-representative quotient t_- with $2n+p-1 = t_-E$ (Section 33.B); Report 34's $t_0 = dN_0$. V: the first-representative quotient t of (37.17), which is Report 33's t_- .	Part V's t is not a binary exponent: on its negative branch q is not known to be a power of two.
e, v	IV: Report 33's $e_A(r) = \chi_A(r) - a\psi_A(r)$, its number e of comparisons (Section 33.1), and the input Pell index v of Section 33.B; Report 34's e_u . V: the input Pell index $e \in \{u, uA\}$ (Report 33's v); Report 33's $e_A(r)$ is written $\mathcal{E}_A(r)$.	—
d, d_{cell}	IV: Report 33's $B = 2^d$, d a power of five; the tree's refinement note writes $d = 2n - p$ for the defect (Section 0.4). V: d_{cell} is Report 33's d , and the defect is $d_{\text{def}} = 2n - p$.	Report 33's d is Part V's d_{cell} , never its defect.
z, z_{quot}	IV: Report 33's transport quotient z in $(K_0 + X)C = F + z(q - 1)$. V: the same quotient, written z_{quot} .	—
U	I: the unit factor of the finalizer $F = U(1 + \sum_j R_j^2) - 1$. IV: the auxiliary root $\chi_{Q_{\text{aux}}}(s_a)/Q_{\text{aux}}$ (Report 33). V: the same auxiliary root, $U = \chi_{T_{\text{aux}}}(\ell)/T_{\text{aux}}$; the source's upper logarithmic bound $U(p)$ is written $\Lambda_+(p)$.	—
L	I: a program parameter. IV: the compiler exponent L_{comp} (Report 33), and $L = E/2$ (Report 34). V: the compiler exponent L in $d_{\text{cell}} = bL$; the source's lower logarithmic bound $L(p)$ is written $\Lambda_-(p)$. VI: $L_0 = q(q - 1)Q$, a lattice modulus. VII: L_* , the factorial base of $t = L_*!$, and $L_0 = 2XY$. VIII: Table 13.	—
u	V, VI: the odd input index $u = 2d_{\text{cell}}x + b$. VII: $u = R/55$, a family parameter; the input index is $I = \ell x + b$. VIII: family parameters (Table 13); the input index is I .	In Parts VII and VIII, u is never the input index.

Letter	Meanings	Tempting false reading
λ_A	V: the unit $A + \sqrt{A^2 - 1}$. VI: its logarithm $\lambda_A(X)$, with $\lambda_P(X)$ likewise. VIII (Report 46): the unit.	Part VI's λ_A is a logarithm.
t (VI–VIII)	VI: $t = 1$, the representative quotient of Part V. VII: $t = L_*$, the binary exponent of $q = 2^t$. VIII: the binary exponent $t = dN$ (Report 43) or $t = 5^r$ (Reports 45, 46); t_{tr} is the transport quotient.	In Parts VII and VIII $q = 2^t$, as in Report 33.
S	VII: the computed auxiliary coefficient $\Delta\psi_A(m_{aux})$. VIII: the supplied free coefficient (Report 43) or $S = i\Delta c^2$ (Reports 45, 46); Table 13.	Only the free-coefficient candidate supplies S .
82, 83	Operation counts of several different candidate polynomials (Table 3).	“The 82-operation candidate” of Part VII (degree 124) is not that of Part VIII (degree 185).

0.4 [write] Relation to the Hilbert’s-tenth research tree

The six manuscripts take the research tree’s own artifacts, at the pinned commits, as their inputs. The tree is maintained separately; this section records what the Parts re-prove or overlap and what is new relative to it at the time of this write (batch 82 for Parts I–IV, batch 83 for Part V, both 3 October 2026).

Byte copies. 73 members of the five batch-82 archives are byte copies of 51 distinct files of `Computability/HilbertTenthProblem/Papers/` (research tree, 1980/ and `verification/`); every one equals the current file. Report 37’s archive holds five more (all in `native-stream-queue/`: the projection receipt, the refinement note, the compiler note, the positive-elimination source and the bootstrap note), equal to the current files, and two byte copies of Report 33’s evidence files. None of these is shipped; the README lists each with its repository path. (*Batch 88.*) The five archives of Parts VI–VIII hold 63 more byte copies, of 33 distinct files of `Computability/HilbertTenthProblem/Papers/` (`research-wip/native-stream-queue/`, 1980/ and `verification/`); every one equals the current file, no pinned file has changed since its pin, and none is shipped. The README lists each with its repository path. The programme’s review counts 48 byte-identical copies of current research-tree files; the difference is the scope of its count, not a disagreement about bytes.

The programme’s reviews. The research tree has reviewed these archives, after their arrival:

- `review_incoming_matrix_grill.md` (commit 32da04296, with a helper and receipt) authenticates both editions of Report 23, Report 24 and Report 32 as data, without running their code. It independently parses every row of the 3,600,546-gate DAG, propagates the syntactic degree 71,731,007, checks liveness of every gate and of all 797,141 supplied coordinates and the 260 finalizer rows, and recomputes $87N + 16 = 69,339,973$ at $N = 797,011$. Its verdict: a “useful full numerical Grill milestone”, not an operation improvement on the programme’s 85-operation universal polynomial (the record at that review; 84 operations since commit 20aafb9a5, below). It did not rerun the exact-degree verifiers.
- `review_incoming_negative_index_fibers.md` (commit 0055e1c4d) reviews Reports 33, 25 and 22. It finds no gap in Report 33’s construction, independently expands the eight residuals, fourteen definitions and the 23-gate finalizer of the signed19 child, and records the refutation of

the signed19 candidate in the programme’s status; it finds Report 25’s classification consistent, notes a domain distinction (Part III, Section 25.4), and does not certify Report 25’s second-term or total-bit-length theorems.

- `review_incoming_counterexample_heights.md` (commit `ceb7222e9`, 3 October 2026, with a helper and receipt) reviews Report 34. (*Correction, batch 83: the batch-82 write, two minutes later, still called Report 34 unreviewed.*) Report 34 “passes this scoped analytic and source-interface review”: the review confirms the eventual height by the auxiliary ordinate y , the Laurent normal-form factorization, the all-orders asymptotic and inverse reasoning and the exact rotation-discrepancy count; it keeps the unproved discrepancy condition $D(R) = o(\log R)$ open, while a two-term law with $O(1)$ remainder is excluded. It confirms the transfer of Appendix 34.A to all eight matching signed19 children as a forward transfer only, with no new gate ledgers, whole-fiber count or parent refutation, and notes that the report improves no universal arithmetic bound.
- `review_exact_negative_obstruction37.md` (commit `308f14072`, with a helper and receipt) reviews Report 37 (Part V): “PASS within the stated fixed-compiler scope”. The predicate is necessary and sufficient, existence and emptiness both remain unresolved, and the programme’s normalized85 polynomial is unaffected because its positive zeros have $F + Z < q$ while every Report 37 negative solution has $F \geq q$. Part V prints its findings after Lemma 37.4.2 and Theorem 37.5.1.
- (*Batch 88.*) `review_incoming_arithmetic_reports_39_46.md` (commit `5d8889017`, 3 October 2026, 21:43 PDT, with a receipt `.json`) reviews Reports 39, 41, 43, 45 and 46 (Parts VI–VIII) within a stated scope: it authenticates the five archives and their 408 members, reads the five delivery READMEs and eleven further proof and source notes in full, executes no archived program and reproduces no package test count; a second reviewer re-verified the hashes and cross-read the note. It is, in its words, “not approval of every manuscript, analytic appendix or saved diagnostic”. It checks Report 39’s Jacobi filter but does not recertify its analytic existence argument; reads Report 41’s full counterfamily and finds its arithmetic consistent; finds Report 43’s descent and divisor classification consistent and its open-language status historical, without reviewing its inner-family and other squarefree addenda in full; reads Report 45’s counterfamily without replaying its receipts; and reads Report 46’s even-rank, height and minimality proofs but not its analytic expansions. Its conclusion: the five reports give “a full obstruction to first-index deletion, exact filters for free-coefficient factor values, and controlled witness-size information for one already-refuted chart. None changes the established84 bound or resolves gamma83.” The same commit adds the tree’s transfer of Report 41 to the 84-operation polynomial, `complete80_first_index_deletion_collapse.md`, with its independent review. Each of Parts VI–VIII prints the findings for its sources.

What the Parts re-prove or overlap.

- The norm cancellation behind the exact degrees, identity (23.10) of Part I and (24.17) of Part II, is identity (4) of `gpcp_shared_selectors774.md` (commit `ee507599f`), also used in three other notes of the tree; neither Report cites it.
- Part II’s law $87N + 16$ is proved by the same method as the tree’s exact law $58N + 28$ for the PCP affine slope-class history (`pcp_affine_slope_class_history.md`, shipped inside archive 11), for a different template.
- Sections 23.6 and 23.10 of Part I re-state and re-certify the tree’s corrected Grill halt bridge and its composed native history; Part I says so and imports them.

- Report 33’s input-index dichotomy (Section 33.B) is proved independently in the tree’s `complete74_negative_index_refinement.md` (commit 813c1cff4), which goes further: it also proves $4^d > X \geq q^3$ for the defect $d = 2n - p$ on the negative branch, and that for fixed q, X, Y and a fixed bounded representative at most one main index p fits the ratio interval; Report 33’s bounded-window reduction imports the tree’s bootstrap (93ad43375) and says so.
- Report 37 (Part V) takes the bounded reduction of Section 33.B and the refinement note’s defect bound and uniqueness as prior results, and re-uses the auxiliary construction of Section 5 of the tree’s bootstrap note `review_complete74_nonlinear_index_bootstrap.md`; it credits all three. Its Section 37.3 is a recap of them, and its identity (37.33) is Lemma 33.3.2 of Report 33.
- (*Batch 88.*) Report 45’s main theorem is the tree’s `complete82_all_input_outer_collapse.md` (commit dc03992d9), proved by a different construction; Report 43’s open language question is answered by the tree’s `free_coefficient83_prime_outer_collapse.md` (commit 8cf6239b6); both postdate the sources’ pins and are not cited by Reports 43 and 45. Report 43’s completion theorem extends the tree’s `free_coefficient83_native_alias.md` (commit 53c34b012), Report 45’s auxiliary lemma extends the tree’s odd- c extension (commit 6ce2dcaaf), and Report 41’s scaled subsystem family is of the kind of the tree’s `first_index_scaled_obstruction.md` (commit 93c34e817); the sources credit all three. The odd-quotient, projection and Pell-growth lemmas that Reports 39, 41, 43, 45 and 46 prove are Report 33’s Lemmas 33.3.2 and 33.3.3 and standard estimates, and Report 39 restates Part V. Conversely, the tree’s `complete80_first_index_deletion_collapse.md` (commit 5d8889017) is a transfer of Report 41’s theorem, which it credits.

What is new relative to the tree. A literal fixed universal Grill program with a paid ordinary-input composition (conditional on the pinned imports), where the tree recorded that “a numerical universal source and its input-language theorem remain open” for the exact-width Grill loader; the exact degree of a Grill polynomial (the tree recorded only upper bounds, with `exact_degree None`); the parametric laws of Part II; the classification and counting of the entire native fiber in Part III; the signed19 counterexample of Report 33, which answers negatively the open obligation of the tree’s nonlinear-index scout for the signed child (raw29 and positive21 stay open); and the canonical-family counting of Report 34 with its Kesten obstruction. From Part V: the deterministic packing recovery, the equivalence of a negative zero in raw29, a negative zero in positive21 and a passing predicate, with complete positive reconstruction, the finite wrapped odd-input sector, the even-input bounds, the $3 \nmid q$ filter and the sub-unit Binet window. Before Report 37 the tree had the bounded reduction and left the question open; it is still open, since raw29 and positive21 are reduced to the predicate, not decided. From Parts VI–VIII: Report 39’s shrinking-target family and its Jacobi nonredundancy theorem; Report 41’s full counterfamily for first-index deletion, which the tree then transferred to its 84-operation polynomial; Report 43’s structural divisor theorems and nonnative inner families; Report 45’s second construction of the square/product 82 refutation; and Report 46’s exact height, fixed-outer minimality and even-rank nonextension.

Records. Part I is not an operation record. At the batch-82 write the tree’s current universal polynomials by other routes were far smaller: 85 operations with 18 positive witnesses and exact degree 175 (commit 109aaca15), and 86 operations with exact degree 131 (commit 55d248dc5); it also has a 557-operation fixed binary-tag universal polynomial and a 97-tile U15 GPCP construction. Part I does not mention these, and disclaims optimality itself. (*Batch 83.*) Since commit 20aafb9a5 (3 October 2026, `complete84_scaled_strong_output.md`) the tree’s record is 84 operations ($84 = 47M + 37A$), with 18 positive witnesses and uniform exact degree 187; the new polynomial is $F_{84} = \Delta \cdot F_{85}$ with

$\Delta > 0$ on every allowed positive tuple, so its positive zeros are those of the 85-operation polynomial. The lower-degree 85/175 and 86/131 constructions remain available, and the comparison-certificate bound remains 74. No Part of this report changes these bounds.

Candidates below the record (batch 88). Parts VII and VIII study candidate polynomials with fewer operations than the record; all those they treat are refuted on their inherited compiler recipe. Several share an operation count, so in this report a candidate is always named by its construction, as in Table 3, which extends the naming table of the programme’s review `review_incoming_arithmetic_reports_39_46.md`. The record is still 84 operations at this write.

Table 3: [write] The research tree’s universal polynomial and the candidates below it, as of this write. W: positive witnesses; deg: exact degree.

Polynomial (tree commit)	Operations	W	deg	Status; where in this report
current84, scaled strong output (20aafb9a5)	47M + 37A	18	187	the record; Section 0.4
normalized85 (109aaca15)	85	18	175	sound; the parent of Part VII’s normalized81
ordinary86 (55d248dc5)	86	18	131	sound; the parent of Part VII’s ordinary82
independent-gamma 83 (d98af7326)	47M + 36A	18	187	language unresolved; not treated here
outer-slack 83 (eaf08931c)	47M + 36A	18	187	refuted by the tree; not treated here
free-coefficient 83 (0d9d1e017)	46M + 37A	18	111	refuted by the tree (8cf6239b6); Part VIII, Reports 43 and 46
auxiliary-product-only 83 (36ed0e524)	46M + 37A	18	185	refuted by the tree after the review; not treated here
square/product 82 (6ce2dcaaf)	45M + 37A	18	185	refuted by the tree (dc03992d9) and by Report 45; Part VIII, Reports 45 and 46
ordinary82, first-index deletion (3f4a974a5)	45M + 37A	17	124	refuted by Report 41; Part VII
normalized81, first-index deletion (3f4a974a5)	46M + 35A	17	168	refuted by Report 41; Part VII
complete80, first-index deletion of current84 (5d8889017)	45M + 35A	17	180	refuted by the tree, transferring Report 41; Part VII’s opening note

Three pairs share both the count and the split, and must be told apart by construction: ordinary82 and square/product 82 (45M + 37A), free-coefficient 83 and auxiliary-product-only 83 (46M + 37A), and independent-gamma 83 and outer-slack 83 (47M + 36A). The programme’s review tabulates six of these rows; the outer-slack and auxiliary-product-only rows and the parents are added from the tree’s README.

0.5 [write] Relation to other reports of the collection

No other report of the collection holds the programme’s universal polynomial, its native block, the Grill route or the complete74 family. Several name the target that Part I approaches, an instantiated audited universal polynomial: `stochastic-and-thermal-exactness`, question `ste:ta:q:instantiate`; `liveness-beyond-halting`, the explicitness boundary of Section `lbh:qd:sec:universal` (“no numerical expanded universal polynomial”); `signal-machine-collision-certificates`, near

`smc:cg:sec:barrier`; and `polynomial-witness-histories`, question `pwh:oc:q:numerical`. Part I bears on them without answering them: it gives a literal circuit, conditional on its imports, with exact degree, not an expanded polynomial, and not their particular source machines. Report 22, the predecessor of Part III, is printed in `signal-machine-collision-certificates`; Report 32 in `group-theoretic-substrates`. No other report of the collection treats the `raw29` and `positive21` interfaces of Parts V and VI, the first-index deletion of Part VII or the candidates of Part VIII.

(Batch 88, reciprocal note, 3 October 2026.) A different, much larger fixed polynomial with positive witnesses is in `periodic-turmite-first-revisits`, Parts III–IV (Research Reports 42, 44, 47 and 48 of the same arrival; written in commit `119a1325d`; its Theorem 44.1.1, `ptr:pc:thm:complete`): through a literal periodic Langton ant it recognizes the fixed U15 sentinel-pair halting language, with 465 positive witnesses, exact degree 2,304,000 and 2,307,457 operations with prescribed coefficients (14,658,934 from the literals 1 and 3; the programme’s later packets reach 14,620,711). It is not an ordinary-input loader for the programme’s input convention, hence not an ordinary-input universal polynomial in the sense of Part I and Table 3; the program-to-tape encoding of U15 is imported, and it does not bear on the 84-operation record. That report’s editorial preface to its four Parts records the same scope (`ptr:it:scope`). No shared theorem.

(Batch 91, reciprocal note, 4 October 2026.) Two further fixed polynomials with positive witnesses and, under imported simulation hypotheses, recursively enumerable complete positive zero sets are in `canonical-diophantine-certificates`, Part XXI (Research Reports 53 and 54 of arrival `0d7f51c44`; written in commit `fb2287290`; its `cdc:rp:cor:recomplete` and `cdc:us:cor:hardness`): 3,865 and 3,262 positive witnesses, exact degree 18, and 17,275 and 14,571 binary gates with integer literals free, on one positive input that codes a periodic sandpile instance (a periodic tile and a finite patch, with a target for Report 53). Their hardness goes through outside computable reductions (Report 53: Cairns’s vertex simulation, with two printed corrections; Report 54: the finite-total loader of Report 35, Part XX of that report, with the imported U15 encoder), not a paid arithmetic front end. So they are not ordinary-input universal polynomials in the sense of Part I and Table 3, and they do not bear on the 84-operation record; the research programme’s review of the four archives of that Part (`review_new_sandpiles_0d7f51c44.md`, commit `bc6e1a62c`) reaches the same conclusion. No shared theorem.

Placement of this report beside the Lean development of `Computability/HilbertTenthProblem` confers no formal status: no statement of this report has been formalized, and no source contains or cites Lean or Rocq code.

0.6 [write] Status, claims and non-claims

The eleven manuscripts are AI-assisted research reports (author lines “Research report 23/24/25”, “Mathematical research report” and “Research Report 43/45/46”), unrefereed, and not formalized. Each Part keeps all its source’s disclaimers; the main ones are collected here.

- Part I: the language theorem is conditional on pinned full native-history, binary-recoder and finite-input U15 universality theorems; only the explicit valid five-parameter slices have the machine interpretation; no new general MRDP theorem, no optimality claim, and no materialized full Pell witness or universal accepting execution. The exact degree is unconditional.
- Part II: source-specific algebraic theorems; no universality theorem, no semantic correctness of arbitrary loader attachments, no minimal degree among equivalent representations.

- Part III: native soundness and positive completeness are inherited from pinned source proofs; no change to the native circuit, no minimality, no finite-fold MRDP result, no numerical Pell witness, no literature-wide novelty claim.
- Part IV: Report 33 refutes only the proposed positive-domain restoration for the signed19 child; no improved operation bound, no machine-specific false acceptance, no materialized full child witness; raw29 and positive21 remain open (Part V reduces both to one exact predicate without deciding it). Report 34 is about one explicit canonical subfamily, not the whole signed19 fiber; its finer second term is equivalent to an unresolved discrepancy estimate.
- Part V: an exact necessary-and-sufficient predicate, not a proof of global restored-index positivity and not a full negative zero; its emptiness and nonemptiness are both unresolved; only q, w are unbounded, and this is not a globally finite decision procedure. The compiler contract and the sign-free bootstrap are inherited premises, not reproved; auxiliary witnesses need not be unique. The finite checks (a 3,536-case relaxed scale grid, auxiliary-only tuples) are not valid-compiler instances, counterexamples or an unbounded search certificate; the Binet window is a necessary test, not an exact acceptance test, with no complexity or separation bound. No operation count, universal polynomial bound, new witness bound or global novelty is claimed.
- Part VI: nonredundancy of two comparisons on positive tuples, in the comparison-deleted systems only; no full negative zero, no main-congruence hit, no unrestricted logical independence, global positivity, complete parametrization, density bounded away from zero, uniformity in a growing compiler or q , or feasible threshold; no large genuine tuple is materialized. The symbol -1 sector and Part V's question remain open. The existence argument rests on two estimates from the literature and was not recertified by the programme's review; finite fixtures are not genuine compiler tuples.
- Part VII: a symbolic existence proof, not a materialized tuple; the counts 81 and 82 and the degrees 168 and 124 describe rejected candidates. No new universal bound, no claim that any count is a lower bound over all circuits, no effect on the parents or on Part V's question. Its "85-operation" record is now 84 (dated note). The smooth radix bounds the radix only.
- Part VIII: Report 43's structural theorems hold on genuine slices and normalize only the stated factors; its inner families are not compiler zeros, and it supplies no zero at a rejected input (the tree has since refuted the candidate by another family). Report 45 refutes the square/product candidate for the inherited compiler, not for a different unprovided compiler, and claims no bound or priority. Report 46's height is exact for its family and minimal only with the outer tuple fixed; its even-rank theorem is not a soundness theorem; its growth is doubly exponential in t and in the numerical input, not in the input's bit length; its analytic inverse is a convergent algorithm, not a canonical inverse transseries, with no interval certification, and was not reviewed by the programme.

The suites delivered with the sources were rerun on copies at placement (README); passing them supports the implementations, not the unbounded theorems.

Part I

A fixed universal Grill polynomial: paid ordinary input and exact width through a literal source program

[write] Source 11 of batch 82: *Research report 23, revision 1* (“Revision 1 with the exact degree theorem”), dated 3 October 2026; archive `Fixed_Universal_Grill_Polynomial_Package (1).zip`; pin `2d887f0fa`; the base of this report. Its sections, statements and equations are numbered *23.k* here.

Abstract of the source. We construct one literal Grill program with 397,488 run phases and one complete integer polynomial circuit attached to it. The circuit accepts an ordinary positive integer input, pays for its conversion into the finite input convention of a fixed universal machine, and binds the initial queue to its exact bit width. Its six external coordinates are the ordinary input and five program parameters fixed for each represented language. It has 797,135 positive existential coordinates and 3,600,546 binary arithmetic operations, comprising 803,517 multiplications and 2,797,029 additions or subtractions. Its exact total degree is 69,339,973, established by an all-tuple norm cancellation and an explicit nonzero leading coefficient on a univariate ray.

The language theorem is conditional on precisely pinned full native-history, binary-recorder, and finite-input U15 universality theorems. The exact-degree theorem is unconditional algebra on the unchanged pinned circuit. The new contribution is the explicit composition: corrected right and left tag macros, a unique accepting-head event, phase-preserving normalization, two distinct complete recoders, signed finite-frame arithmetic, exact unary loading, and a unit-safe finalizer. A complete 61,209,290-byte arithmetic DAG and independent whole-source checks accompany the report. The costs describe this actual emitted circuit in a literal-integer arithmetic model; they are not bit complexity or a minimality claim.

[write] The title states universality before its qualifier: the language theorem, Theorem 23.1.1, is conditional on the pinned imports of Section 23.2, as its name, the abstract and the conclusion say. The exact-degree theorem, Theorem 23.13.1, is unconditional. The programme’s review `review_incoming_matrix_grill.md` (commit `32da04296`) authenticated this source and its original edition as data and checked the DAG independently (Section 0.4); it did not rerun the exact-degree verifiers. The source’s files are shipped with the prefix `11-grill-poly-`.

23.1 Result and its mathematical meaning

The purpose of this report is to close the ordinary-input interface for a concrete Grill route to Diophantine representation. A native queue-history relation by itself is insufficient: it may permit different terminal-zero paddings of the same integer, and a substrate-universality slogan does not identify what language a particular numerical input represents. Here both interfaces are explicit and paid.

Theorem 23.1.1 (Conditional fixed universal polynomial). *Assume the pinned theorem imports in Section 23.2. There exists a single polynomial*

$$F \in \mathbb{Z}[x, a, p, s, C, L, y_1, \dots, y_{797135}]$$

represented by the complete arithmetic DAG in the accompanying archive, such that for every computably enumerable set $S \subseteq \mathbb{Z}_{>0}$ there is a fixed tuple

$$\theta_S = (a_e, p_e, s_e, C_e, L_e) \in \mathbb{Z}_{>0}^5$$

with, for every $x \in \mathbb{Z}_{>0}$,

$$x \in S \iff \exists(y_1, \dots, y_{797135}) \in \mathbb{Z}_{>0}^{797135} : F(x, a_e, p_e, s_e, C_e, L_e, \mathbf{y}) = 0.$$

All five coordinates of θ_S have the finite program-dependent recipes in Section 23.7. The polynomial has 3,600,546 emitted binary arithmetic gates and exact total degree 69,339,973 when all six external coordinates and every witness have degree one. The degree assertion is proved independently of these theorem imports in Section 23.13.

The program parameters are not existentially varied with x . An arbitrary positive tuple (a, p, s, C, L) still gives a polynomial slice, but only the explicitly constructed valid tuples have the stated machine interpretation. The theorem covers positive integer inputs and positive integer witnesses. No statement about $x = 0$, negative inputs, or real zeros is intended.

The construction establishes a concrete representation through a selected computational substrate. It is not a new general MRDP theorem, and the operation or witness counts are not asserted to be optimal. The finite audits establish exact source facts and implementation identities; the unbounded existential equivalence additionally uses the imported complete theorems. In particular, no astronomical full Pell witness tuple or complete universal accepting execution has been materialized.

[write] *Not an operation record.* The research tree’s universal polynomials by other routes are far smaller: 85 operations, 18 positive witnesses and exact degree 175 (commit 109aaca15), and 86 operations with exact degree 131 (commit 55d248dc5), besides a 557-operation fixed binary-tag polynomial. This Part does not mention them. It is the Grill-route instance, and its 3,600,546 operations are about 42,000 times the 85-operation record; the programme’s intake review `review_incoming_matrix_grill.md` (commit 32da04296) calls it a “useful full numerical Grill milestone” and not an operation improvement. (*Batch 83.*) These were the figures at the batch-82 write; since commit 20aafb9a5 the tree’s record is 84 operations at exact degree 187 (Section 0.4).

23.1.1 An overview of the chain

Fix a language S and its finite semidecider. For the varying $x > 0$, the construction follows

$$\begin{aligned} x &\mapsto u = x + 1 \mapsto \text{canonical LSB bits of } u \mapsto N_e(x), \\ \text{U15 finite tape} &\mapsto \text{corrected deletion-2 tag word} \mapsto \text{normalized Genera word} \\ &\mapsto \text{exact Grill E word and width} \mapsto \text{complete native history relation} \mapsto F = 0. \end{aligned}$$

The two recoders serve different jobs. The first selects the canonical bit length of u . The second extracts the unrestricted exponent $N_e(x) + 1$ used to load a unary tag block. Neither exponent is the native history duration. Keeping these three quantities separate is essential.

Object	Exact size or bound
Refined U15 states	30
Nonhalting U15 instructions	29
Tag symbols and total productions	570
Normalized Genera symbols	1,013
Grill run phases	397,488
Positive runs / zero runs	24,300 / 373,188
Distinct run exponents, including zero	2,030
Largest run exponent	275,944
Positive existential coordinates	797,135
Nonunit comparisons / unit condition	86 / 1
Complete arithmetic operations	3,600,546
Exact total degree	69,339,973

Table 4: The literal source and the actual composed circuit. A run exponent describes a fixed block of commands; a run phase is not an individual command.

23.2 Sources and imported complete theorems

All repository theorem imports use the immutable ProveIt commit

2d887f0fa768fd67f3e545d83f8998b5780e530d.

They are under `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`. The reproducibility packet records Git blob identifiers, SHA-256 digests, frozen texts, and the local adaptations needed to replay the new source. A frozen theorem note is an explicit premise, not a claim that its full historical proof dependency graph has been independently formalized here.

The imports have four distinct roles.

1. *Finite-input universality.* The note `neary_woods_explicit_universal_tm.md` supplies the U15 table and effective finite-frame initialization of a selected one-tape semidecider through the bi-tag construction. The primary Neary–Woods paper is frozen and Table 16 was checked visually. This is the finite-tape machine convention, not an infinite periodic background.
2. *Corrected source compilation.* The corrected `grill_tag_halt_bridge.md` supplies the E/L/R compiler, persistent phase alignment, nonhalting simulation, and exact first-empty cleanup. Its hypotheses are proved for the newly literal normalized source in Sections 23.4–23.6.
3. *Complete native history.* The word-closure, weak-cone, phase-sharing, phase-residual206, composed205, and uniform affine-pair unit notes supply the fixed-arity native relation, positive soundness and converse, restoration, and polynomial rewrites. The scalable emitter is proved identical to the pinned complete native polynomial over *all integer tuples*, not just on valid histories.
4. *Complete binary recoding.* The input-dilation130 source and the generic-width bridge, with their complete parent geometry/AND theorem chain, supply all 34 comparisons and 49 positive auxiliaries at every fixed width $k \geq 4$. The positive converse applies at every permitted exponent $h \geq 2$. The duration-extraction argument uses the unrestricted recoder; it does not import the later extra dyadic-duration predicate.

[write] All imported notes are files of the research tree. Twenty-two of them were shipped inside archive 11 as byte copies (24 members, two of them duplicated under `frozen/arithmetic/`); they are not shipped here and are cited by their repository paths, which the README lists, for example `grill_tag_halt_bridge.md`, `neary_woods_explicit_universal_tm.md` (whose copy in the archive differs from the research-tree file in one path-sanitized command line and is therefore shipped, as `11-grill-poly-input-audit-sources-neary_woods_explicit_universal_tm.md`), `native_binary_input_dilation130.json` (the “complete recoder receipt” below) and `pcp_affine_slope_class_history.md`. The pin `2d887f0fa` descends from the halt-bridge commit `8ae0b7c1f` and the exact-width-loader commit `1dd95f928`. Every one of the 24 copies equals the current research-tree file, so the pinned bytes are still the tree’s bytes at this write.

Two small literal data objects anchor native transcription. The complete recoder receipt has SHA-256

175c498990a8e63de7a91d1e3d321ef90a19f129f305074f0c6de10967d0a182,

The 67-row native unit kernel has SHA-256

2d88343037c08fe8b073f67dca531ee73309c0735c1d98a23c20b9ddb1eb08ae.

Four complete upstream native fixtures were authenticated and their frozen rows compared independently. Upstream Python is retained only as inert source evidence; none was executed in this work or is required by the offline replay.

The archive retains earlier semantic and local audit reports with their original historical scope. Statements in those reports that a larger circuit remained to be emitted describe that earlier stage. The complete composition proof, current whole-circuit audit, and this report state the final scope.

23.3 A concrete finite U15 input language

Use $c = 0$, $b = 1$, and blank zero outside a finite tape. Refine a U15 control state u and already-read symbol r to

$$i = 2(u - 1) + r \in \{0, \dots, 29\}.$$

The initial refined state is $0 = (u_1, 0)$; the accepting state is $19 = (u_{10}, 1)$. The unique absent instruction in primary Table 16 is (u_{10}, b) . A primary prose sentence instead names c ; the literal construction follows the table, and the discrepancy is recorded rather than silently reconciled.

For a nonhalting instruction $(u, r) \mapsto (v, \epsilon, D)$, the refined machine writes ϵ , moves in direction D , reads r' , and selects $j_{r'} = 2(v - 1) + r'$. There are 15 left-moving and 14 right-moving nonhalting instructions.

Let M, N be the left and right binary tape integers, with the square nearest the head as the least significant bit. The scanned symbol is omitted because the refined control remembers it. A right move changes

$$(i, M, N) \mapsto (j_{N \bmod 2}, 2M + \epsilon, \lfloor N/2 \rfloor),$$

and a left move changes

$$(i, M, N) \mapsto (j_{M \bmod 2}, \lfloor M/2 \rfloor, 2N + \epsilon).$$

These formulas include zero tape integers and indefinitely blank tails.

23.3.1 The chosen semidecider and physical input blocks

For each $S \subseteq \mathbb{Z}_{>0}$ that is computably enumerable, choose a finite semidecider that reads the canonical LSB-first spelling of $u = x + 1$, decodes it, subtracts one, and recognizes S . This is a fixed program choice. It is not an uncharged external algorithm applied anew to x .

Write the physical binary blocks

$$A_i = (01)^{8i-5}11, \quad B_0 = A_2A_1, \quad B_1 = A_1A_2.$$

Both B_0, B_1 have 32 bits. The U15 head scans the final c of the start marker bc . Its left word is the finite encoded program followed by b ; define M_e by reversing that word and taking its LSB value. Its right word is

$$P_e B_{w_0} \cdots B_{w_{n-1}} S_e, \quad w_i = \text{bit}_i(x+1), \quad n = \text{bitlength}(x+1), \quad (23.1)$$

where

$$P_e = (01)^{8q_B}, \quad S_e = A_{\text{right}}A_{\text{left}}, \quad q_B \geq 2.$$

The alphabet size q_B and the two boundary indices depend only on the selected finite program. All squares beyond the displayed words are blank. Put $N_e(x) = \text{val}_{\text{LSB}}(\text{word in (23.1)})$. The suffix ends in 11, hence $N_e(x) > 0$.

[write] Added 3 October 2026 (batch 82, reciprocal note). Report 32, printed as Part V of **group-theoretic-substrates**, cites this Report's ordinary-input frame (23.1) for the finite $U_{15,2}$ input convention in its delivered dependency audit, for the input hardness of its fixed matrix semigroup only (Theorem **gts:um:thm:semigroup** there); the audit calls this Report's conditional theorem unnecessary for that purpose.

23.4 Corrected tag macros and the accepting head

A deletion-2 tag step reads the first symbol, deletes two symbols, and appends that first symbol's production. The canonical word is

$$C(i, M, N) = A_i \xi_i (a_i \xi_i)^M B_i \xi_i (b_i \xi_i)^N.$$

Internal names are local to the nonhalting refined state. Their numerical identifiers and every production appear in the literal packet.

23.4.1 The right macro

For an instruction writing ϵ and moving right, use the complete rules below. On internal symbols suppress the state index i .

$$\begin{array}{ll} A_i \rightarrow C \xi_i (c \xi_i)^\epsilon, & a_i \rightarrow c \xi_i c \xi_i, \\ B_i \rightarrow S, & b_i \rightarrow s, \\ C \rightarrow D_1 D_0, & c \rightarrow d_1 d_0, \\ S \rightarrow T_1 T_0, & s \rightarrow t_1 t_0, \\ D_1 \rightarrow A_{j_1} \xi_{j_1}, & d_1 \rightarrow a_{j_1} \xi_{j_1}, \\ D_0 \rightarrow \xi_{j_0} A_{j_0} \xi_{j_0}, & d_0 \rightarrow a_{j_0} \xi_{j_0}, \\ T_1 \rightarrow B_{j_1} \xi_{j_1}, & t_1 \rightarrow b_{j_1} \xi_{j_1}, \\ T_0 \rightarrow B_{j_0} \xi_{j_0}, & t_0 \rightarrow b_{j_0} \xi_{j_0}. \end{array}$$

Put $M' = 2M + \epsilon$. Exact intermediate queue cuts are

$$\begin{aligned} B_i \xi_i (b_i \xi_i)^N C \xi_i (c \xi_i)^{M'}, \\ C \xi_i (c \xi_i)^{M'} S s^N, \\ S s^N D_1 D_0 (d_1 d_0)^{M'}. \end{aligned}$$

If $N = 2k + 1$, the shrinking traversal leaves

$$D_1 D_0 (d_1 d_0)^{M'} T_1 T_0 (t_1 t_0)^k,$$

which rotates through the D and T blocks to exactly $\mathcal{C}(j_1, M', k)$. If $N = 2k$, its last deletion consumes D_1 , leaving

$$D_0 (d_1 d_0)^{M'} T_1 T_0 (t_1 t_0)^k.$$

The D_0/d_0 traversal consumes T_1 in its last deletion, including when $M' = 0$, and gives

$$T_0 (t_1 t_0)^k \xi_{j_0} A_{j_0} \xi_{j_0} (a_{j_0} \xi_{j_0})^{M'}.$$

The last T_0/t_0 deletion consumes the extra filler and yields precisely $\mathcal{C}(j_0, M', k)$, including $k = 0$.

The final filler in $t_0 \rightarrow b_{j_0} \xi_{j_0}$ is deliberate. The inspected printed Cocke–Minsky production omits it although the next displayed target retains paired repetitions. With $M = 0, N = 2, \epsilon = 0$, the printed rule produces a five-symbol target after ten steps, missing its sixth filler. The construction here proves its own corrected rule; it does not present an asserted published erratum.

23.4.2 The left macro and explicit rotation

For a left move replace the initial four right rules by

$$\begin{array}{ll} A_i \rightarrow L_i \xi_i, & a_i \rightarrow l_i \xi_i, \\ B_i \rightarrow C \xi_i (c \xi_i)^\epsilon, & b_i \rightarrow c \xi_i c \xi_i, \\ L_i \rightarrow S, & l_i \rightarrow s. \end{array}$$

One traversal gives

$$L_i \xi_i (l_i \xi_i)^M C \xi_i (c \xi_i)^{2N+\epsilon}.$$

Keep the $C/c/S/s$ preparation. Replace target A_j, a_j in the D/d rules by temporary B'_j, b'_j , and replace target B_j, b_j in the T/t rules by actual A_j, a_j . Add

$$B'_j \rightarrow B_j \xi_j, \quad b'_j \rightarrow b_j \xi_j.$$

The right parity calculation with the tape roles exchanged yields

$$B'_j \xi_j (b'_j \xi_j)^{2N+\epsilon} A_j \xi_j (a_j \xi_j)^{\lfloor M/2 \rfloor}, \quad j = j_M \bmod 2.$$

Moving the leading prime block to the tail gives the exact canonical left-move target. This finite rotation is the missing content behind an informal appeal to left/right symmetry.

23.4.3 Safety before acceptance and uniqueness at acceptance

Every canonical word has at least four symbols. All preacceptance productions except the shrinking B_i/b_i or L_i/l_i rules have length at least two. During the shrinking phase an untouched $C\xi_i(c\xi_i)^{M'}$ block of length at least two remains, and an S remains at the finish. Therefore every preacceptance intermediate queue has length at least three. Prime rotation preserves length. No valid rejecting run can halt by an accidental short queue.

The exact cuts create one next canonical head. It is read only after all old-macro symbols preceding it are consumed; at that read the whole word is canonical. In particular the first active A_{19} occurs at a genuine accepting boundary, with no delayed old head lineage elsewhere in the queue.

Set $A_{19} \rightarrow H$, with a fresh halt symbol H . Give the accepting suffix symbols inert productions ZZ , and $Z \rightarrow ZZ$. No other nonhalting rule emits H . Once A_{19} is read, the remaining accepting-state symbols cannot recreate either A_{19} or H . Thus exactly one fresh H is emitted. An occurrence in the second deleted position is never mistaken for an active head event.

23.5 Normalization with a persistent phase

Give each original tag symbol width one. Its unnormalized Genera production is its tag production at phase zero and empty at phase one. Two FIFO reads implement one deletion-2 step. The phase is persistent across generation boundaries, so a generation ending after an odd number of original reads is not reset.

Introduce a width-zero dummy d , with $d \rightarrow dd$. Pad each production to four symbols using trailing d , and replace its two consecutive pairs by fresh width-zero pair symbols. A pair $[u, v]$ expands to uv at either phase. Every normalized production now has exactly two outputs.

Lemma 23.5.1 (Two-layer projection). *Starting in an original-symbol layer, erasing dummies after two normalized generations gives exactly one unnormalized generation, with exactly its final persistent phase.*

Proof. Only original symbols have width one, so only the first pass changes the phase. Its outputs are pairs and width-zero dummies. The second pass expands each pair, preserves order, and leaves the phase unchanged. Dummy expansion erases to the empty word. Applying these facts successively proves both the word and phase identities, also for odd erased word length. $\square$

A first literal H occurs only when its containing pair expands. The preceding prefix of that output contains original symbols and dummies. Their hypothetical next outputs contain only pairs and dummies, never H . This verifies the extra prefix condition in the Genera halt convention, not merely the existence of a halt symbol.

The literal normalized alphabet has 569 nonhalting original symbols, 442 ordered pairs, one dummy, and one halt symbol, for $N_G = 1013$. There are 2,024 nonhalting phase rows, each with two outputs; the two ignored halt rows are also total. Its unique halt-containing pair is (H, d) , numeric ID 694; the halt has ID 1012. Only phase-zero A_{19} produces that pair. The complete literal table has SHA-256

3c8924dbb1b5d6e6b8897e59550b0e38a703654b87442c73487f411e94ae355b.

23.6 The literal corrected Grill program

Set

$$a = 28(N_G + 1) = 28392, \quad b = 7a = 198744, \quad m = 14a = 397488.$$

At one run step a zero is removed without output; a one is removed and appends $g(n_p) = 0(10)^{n_p}$; the phase advances modulo m . An exponent n_p denotes n_p commands 11 followed by command 10. All source times below count run steps.

For a nonhalting source symbol y of width $w_y \in \{0, 1\}$ put $e_y = 7a(1 - w_y)$, and put $e_H = 3a/2$. The corrected blocks are

$$\begin{aligned} E_y &= 0^{14y+7}g(a-3)g(7)g(a-4)0^{3a-14y-10+e_y}, \\ L_y &= 0^{a-3}g(14y+7)g(3)g(3a-14y-10)(10)^{e_y}, \\ R_y &= 0g(14y+7)g(3)g(3a-14y-10+e_y)0^{a-4}. \end{aligned}$$

For $y = H$, the displayed E_H is only a cleanup intermediate E_H^* , not an initial source block. The second long E grill uses $a - 4$, correcting the earlier informal $a - 2$ expression.

23.6.1 Every run exponent is specified

Use zero-based positions relative to the start $7ap$ of each half $p \in \{0, 1\}$ of length $7a$. Place the values $a - 3, 7, a - 4$ in the listed order at

$$28y + \{17, 19, 21\} \quad \text{and} \quad 28y + a + \{13, 15, 17\},$$

respectively, for every symbol y , including H . If nonhalt y outputs (u, v) at source phase p , place

$$14u + 7, \ 3, \ 3a - 14u - 10 + e_u, \ 0, \ 14v + 7, \ 3, \ 3a - 14v - 10 + e_v$$

at the seven positions $14y + 2a + \{3, 5, 7, 9, 11, 13, 15\}$, in that order. All other exponents are zero. There are no E-production positions for H . The source rows therefore determine the entire little-endian uint32 array; its hash is

$$\text{fa658fcdfe1dae2be3a8e2bf97bd613db59558949ea23ca03f549b77201dad01}.$$

Independent reconstruction agrees at every entry and all 1,589,952 bytes.

23.6.2 Phase alignment and exact cleanup

At a normal start phase $7ap$, a nonhalt E block produces the appropriate $L_u R_v$. Its phase advance is $7aw_y \pmod{14a}$. A nonhalt L/R block has length $7a$ or $21a$, both congruent to $7a$ modulo $14a$. A complete two-output LR generation is phase-neutral. These facts preserve the source's persistent phase.

Every active run position modulo $7a$ is odd and below $5a/2$. Translating it by $3a$ lands in the zero-run region. At a start phase shifted by $3a$, every one in an E/L/R block therefore selects exponent zero, producing one zero, while every zero produces nothing.

Suppose the first halting source output is an even word v with its unique H at zero-based index j . The H block has length $10a$ and shifts all later blocks by $3a$. The three remaining generation lengths

from that LR boundary are

$$\begin{aligned} L_1 &= 10a + \sum_{i \neq j} [7a + 14a(1 - w_{v_i})], \\ Z &= \sum_{i > j} [3a + 7a(1 - w_{v_i})], \\ L_2 &= 17a/2 + \sum_{i < j} [7a + 7a(1 - w_{v_i})] + Z, \\ L_3 &= 2a(j + 1). \end{aligned}$$

The last produced word is all zeros. All three lengths are positive, and a FIFO queue cannot empty while an initial-generation suffix remains. Thus $L_1 + L_2 + L_3$ is the exact first-empty time from that boundary. For the actual H-producing pair, the lengths are 880,152, 525,252, and 56,784, totaling 1,462,188 further runs.

Nonhalting E boundaries stay nonempty. Consequently the correctly normalized source halts if and only if the Grill queue from its exact E word empties. This equivalence is at the literal width. Allowing arbitrary high-zero padding of the same numeral would be a different and unjustified interface.

[write] *Re-statement.* The corrected E/L/R compiler, the persistent phase alignment, the $3a$ cleanup shift and the exact first-empty lengths L_1, L_2, L_3 are those of the research tree's `grill_tag_halt_bridge.md` (commit `8ae0b7c1f`), imported in Section 23.2; its $a - 2 \rightarrow a - 4$ correction of the E encoding is the tree's `review_grill_encoding_e.md`. New here are the literal 397,488-entry table and its instantiated totals 880,152, 525,252 and 56,784. The tree's halt bridge itself does not give an ordinary-input universal Grill recognizer; this Part supplies that composition.

23.7 All five valid program parameters

The first three parameters record the finite U15 frame:

$$a_e = \text{val}_{\text{LSB}}(\mathbf{P}_e) = \frac{2(4^{8q_B} - 1)}{3}, \quad p_e = 2^{16q_B}, \quad s_e = \text{val}_{\text{LSB}}(\mathbf{S}_e). \quad (23.2)$$

Each is positive and fixed once the finite program is chosen.

For original symbols define $E_y^\# = \text{val}_{\text{LSB}}(E_y)$. Let

$$\begin{aligned} \gamma(t) &= 2 \sum_{j=0}^{t-1} 4^j = \frac{2(4^t - 1)}{3}, \\ A_* &= 2^7 [\gamma(a - 3) + 2^{2a-5} \gamma(7) + 2^{2a+10} \gamma(a - 4)]. \end{aligned}$$

Then $E_y^\# = A_* 2^{14y}$ and every original block has width b . Put

$$K = 2^{2b}, \quad W_y = E_y^\# + 2^b E_{539}^\#.$$

The literal start IDs are $A_0 = 0$, $a_0 = 270$, $B_0 = 30$, $b_0 = 300$, and $\xi_0 = 539$. The remaining parameters are

$$\begin{aligned} C_e &= W_0 + KW_{270} \sum_{j=0}^{M_e-1} K^j + K^{M_e+1}W_{30}, \\ L_e &= K^{M_e+2}. \end{aligned} \tag{23.3}$$

The empty sum when $M_e = 0$ is zero. These are the value and dyadic width scale of

$$E(A_0\xi_0(a_0\xi_0)^{M_e}B_0\xi_0).$$

Equations (23.2)–(23.3) are finite recipes for fixed program integers. Their powers and finite sums are not runtime operations of the universal polynomial. By contrast, every multiplication involving any of these five *external coordinates* in the universal DAG is charged.

The full initial tag word is the displayed fixed prefix followed by $(b_0\xi_0)^{N_e(x)}$. That repetition count varies with the ordinary input and must be paid inside the polynomial; it is the task of the second recoder.

23.8 Two complete recoders and the signed frame

The complete generic recoder at fixed width $k \geq 4$, on positive input $z < 2^h$ and exponent $h \geq 2$, retains 49 positive auxiliaries and 34 comparison pairs. At a complete zero its ports satisfy

$$\begin{aligned} q &= 2^h, \quad Q = q^k, \quad B = 2^{k-1}Q, \quad P = B^h, \\ J &= \sum_{j < h} B^j, \quad K_{\text{mask}} = \sum_{j < h} (2B)^j, \quad z_{\text{out}} = \sum_{j < h} \text{bit}_j(z)2^{kj}. \end{aligned} \tag{23.4}$$

The full converse supplies every retained positive auxiliary at every such exponent. Selected outer equations alone would not justify this interface.

Typing does not depend circularly on the conclusions. Before the recoder equalities, $B = 2^{k-1}q^k \geq 8q^2$ for positive q and $k \geq 4$, and the conceptual prescribed-scale AND ports $zJ + 1$, $K_{\text{mask}} + 1$, its positive supplied output hat, and qP are positive. Generic width changes only the fixed power chain and coefficient in the complete 130-row source; all remaining comparisons and witnesses remain.

23.8.1 Canonical input with an inclusive upper endpoint

Pay $u = x + 1$ and apply the width-32 recoder with positive spread output R . Its existing positive slack s_0 satisfies $u + s_0 = q_0$. Supply positive β and impose

$$s_0 + \beta = u + 1.$$

Then $u < q_0 \leq 2u$. Since q_0 is dyadic, it is exactly 2^n for $n = \text{bitlength}(u)$. For dyadic u the upper endpoint is necessary and $\beta = 1$. Conversely, $s_0 = 2^n - u > 0$ and $\beta = 2u + 1 - 2^n > 0$ give the required extension. As $x > 0$, $u \geq 2$, so $n \geq 2$ is in the imported domain.

With $K_0 = 2^{32}$ and $m_0 = K_0 - 1$, the literal block values are

$$c_0 = 3941247658, \quad c_1 = 3937053418, \quad c_1 - c_0 = -4194240.$$

Supply positive N and impose the denominator-cleared equation

$$\begin{aligned} m_0 N &= m_0 a_e + p_e c_0 (Q_0 - 1) \\ &\quad + p_e m_0 (c_1 - c_0) R + p_e m_0 s_e Q_0. \end{aligned} \quad (23.5)$$

The recoder gives $Q_0 = K_0^n$ and $R = \sum_{i < n} w_i K_0^i$. Hence (23.5) is exactly the LSB concatenation formula for (23.1). The signed coefficient is retained; it must not be changed to a presumed positive block difference. N is a positive supplied witness, so the off-zero sign of the computed right side does not become an illicit native positivity assumption.

23.8.2 An unrestricted exponent equal to the tape integer plus one

Use a disjoint second recoder at fixed width $k = 2b = 397488$, with input and spread output both equal to one. Its theorem gives $h \geq 2$ and (23.4). It receives no canonical-length guard and no dyadic-duration AND predicate. The two namespaces share no private witness.

Supply positive ℓ, v, g and impose

$$(B - 1)v + \ell = J, \quad \ell + g = B - 1, \quad \ell = N + 1. \quad (23.6)$$

Since $J \equiv h \pmod{B - 1}$, the first equation gives $\ell \equiv h$. The second bounds $1 \leq \ell \leq B - 2$. Also

$$B = 2^{kh+k-1} \geq 2^{4h+3} > h + 1,$$

so $2 \leq h \leq B - 2$. The representatives lie in one modulus interval and are equal: $h = \ell = N + 1$. No modular alias survives.

For the converse choose the full recoder extension at $h = N + 1$ and take

$$v = \sum_{j=1}^{h-1} \sum_{i=0}^{j-1} B^i > 0, \quad g = B - 1 - h > 0, \quad \ell = h.$$

The case $N = 1, h = 2$ is already valid; the inner quotient is positive. $N = 0$ would require the excluded $h = 1$, but the physical suffix has proved $N > 0$. Nondyadic exponents are allowed and are essential.

23.9 Exact unary loading and positive width

The fixed repeated pair has value

$$V = E_{300}^\# + 2^b E_{539}^\# = A_* 2^{14 \cdot 300} + A_* 2^{b+14 \cdot 539}$$

and exact width $k = 397488$. Its value has 319,867 bits. Those two numbers are different because its terminal zeros are part of the word.

The second recoder has $Q_1 = K^{N+1}$. Supply positive T, X and impose

$$KT = Q_1, \quad (23.7)$$

$$(K - 1)X = (K - 1)C_e + L_e V(T - 1),$$

$$P_0 = L_e T. \quad (23.8)$$

The first equation gives $T = K^N$. The second uniquely gives

$$X = C_e + L_e V \sum_{j=0}^{N-1} K^j,$$

precisely the numeral of the full initial E word. Equation (23.8) fixes its exact dyadic width scale.

The native source already has a positive width slack Z_0 and computes $P_0 = X + Z_0$. This coordinate is retained, and (23.8) is an additional comparison. Substituting an arbitrary signed computed difference for a positive input would not be a valid replacement.

Every original E block is nonzero and ends with at least two zero bits. In base 2^b , each block value is below one quarter of that base. A finite geometric sum therefore gives $0 < 3X < P_0$ for the exact word. On a genuine halt the strong native converse applies at that same width, with positive slack $P_0 - 3X$. The established strong-to-weak map yields $Z_0 = P_0 - X > 0$ while preserving the word, width, and history. No optional padding is used.

23.10 A complete scalable native history

The complete native component is much larger than the loader. Its correctness transfer must therefore cover its entire source, not just a few ports or small instances. The accompanying `native_history_proof.md` proves the scalable emitter identically equal to the pinned complete weak native polynomial after a bijective coordinate renaming. The following formulas expose the main identities.

[write] *Re-certification*. The native history polynomial of this section is the research tree’s word-closure, weak-cone, phase-sharing, phase-residual206 and composed205 polynomial for a fixed program; the new content is a scalable emitter proved identical to it over all integer tuples. The tree states for its own small programs that “no ordinary-input decoder or fixed universal Grill recognizer has been supplied” (`grill_tag_native_weak_cone.md`) and that its word closure “supplies no fixed universal Grill program” (`review_grill_native_word_closure.md`).

23.10.1 Coordinates and grouped affine transports

For a fixed program $n_0, \dots, n_{m-1}$, let $s = 2m$ and let g be the number of distinct exponents. The two tiles at phase i are

$$(2, 0, 1, 0), \quad (2, 1, b(n_i), d(n_i)),$$

where $b(n) = 2^{2n+1}$ and $d(n) = 2(4^n - 1)/3$. These are fixed integer coefficients. Distinct exponent classes retain separate selected-history coordinates, while every original chronological selector remains present.

There are s positive selector hats $\hat{S}_i$, g positive history hats $\hat{Z}_j$, seven positive outer coordinates, and sixteen positive native-kernel coordinates. Thus the native count is $s + g + 23$. For the literal program, $s = 794976$ and $g = 2030$, giving 797,029.

Let $I_j = \{2i + 1 : n_i = e_j\}$, with groups ordered by first occurrence, and put

$$\sigma_i = \hat{S}_i - 1, \quad G_j = \sum_{i \in I_j} \sigma_i, \quad z_j = \hat{Z}_j - 1, \quad J = \sum_i \sigma_i.$$

To avoid collision with the loader’s K, B , write the native radix multiplier as $\mathcal{K}$ and radix as $\mathcal{B}$. Its exact fixed value is

$$\mathcal{K} = 2^{\max(3, \lceil \log_2(s+4) \rceil, 2 \max n_i + 2)} = 2^{551890}.$$

The implementation computes the integer ceiling by bit length, without floating point. Define

$$\begin{aligned} P_0 &= X + Z_0, & U_f &= P_0 V_f + X, \\ D &= U_f + q_{\text{init}} + h_{\text{slack}}, & \mathcal{B} &= \mathcal{K}D, & P &= (\mathcal{B} - 1)J + 1. \end{aligned}$$

Grouped transports are

$$\begin{aligned} \text{next } U &= 2H_U + \sum_j G_j, \\ \text{next } V &= H_V + \sum_j (b(e_j) - 1)z_j + \sum_j d(e_j)G_j. \end{aligned}$$

Distributivity gives exactly the ungrouped template. The first three outer comparisons are

$$\begin{aligned} H_U + H_V + \sum_j \hat{Z}_j + g_{\text{bound}} &= P, \\ \mathcal{B} \text{next } U + 1 &= H_U + P U_f, \\ \mathcal{B} \text{next } V + 1 &= H_V + P V_f. \end{aligned}$$

Positive hats in the first equality are intentional; they implement the precise global bound, including its g offset.

23.10.2 The exact phase residual

Let $\pi_i = \hat{S}_{2i} + \hat{S}_{2i+1}$, $S_i = \pi_i - 2$, and

$$T_{\text{ph}} = \sum_{i=1}^{m-1} (m-i)S_i, \quad J = \sum_i S_i.$$

The streamed nested prefix sum gives coefficient $m-i$ on π_i ; subtracting $m(m-1)$ removes its hat offsets. Every prefix and accumulator addition is an emitted gate.

For proof purposes write $\text{Next} = mJ - T_{\text{ph}}$ and $Q = (m+1)J - mS_0 - T_{\text{ph}}$. Substitution yields the polynomial identity

$$\begin{aligned} \mathcal{B} \text{Next} + q_{\text{init}} - Q - mP \\ = m\pi_0 - (\mathcal{B} - 1)T_{\text{ph}} + q_{\text{init}} - (J + 3m). \end{aligned}$$

This is an all-tuple identity over a commutative ring. It assumes neither Boolean selectors nor a valid phase path nor the vanishing of another residual. For $m = 1$ it reduces to $q_{\text{init}} - 1$.

23.10.3 Exact packed ports and the fixed kernel

Use $\text{pack}(v) = \sum_i v_i P^i$ and $\text{rep}(r) = \sum_{i < r} P^i$. Horner packing and the paid identities

$$\text{rep}(2r) = \text{rep}(r)(1 + P^r), \quad \text{rep}(2r+1) = \text{rep}(2r) + P^{2r}$$

are polynomial identities also at $P = 0$ and $P = 1$. All exponents are fixed by the literal program; square-and-multiply emits ordinary paid products.

Define

$$\begin{aligned} S &= \text{pack}(\sigma), \quad M_c = J \text{rep}(s), \quad T_c = H_U + PH_V, \\ G_* &= \text{pack}(G), \quad H_b = H_V \text{rep}(g), \quad M_b = (\mathcal{B} - 1)G_*, \quad Z_b = \text{pack}(z), \\ R_M &= (D - 1)J(1 + P). \end{aligned}$$

With $e_c = g$, $e_r = g + s$, $e_t = g + s + 2$, $e_i = g + s + 4$, put

$$\begin{aligned} W &= P^{e_c} S + P^{e_r} T_c, \\ H &= H_b + W + P^{e_t} \mathcal{B} + P^{e_i} P_0, \\ M &= M_b + P^{e_c} M_c + P^{e_r} R_M + P^{e_t} (\mathcal{B} - 1) + P^{e_i} (P_0 - 1), \\ Z &= Z_b + W, \quad \text{scale} = P^{g+s+5}. \end{aligned}$$

These are the exact controller, range, top/radix, and initial-width lanes of the pinned native adapter. Here the scale exponent is 797,011.

The frozen 67-row kernel receives exactly (H, M, Z, scale) . It retains sixteen positive coordinates, all six native comparison pairs, and a four-factor integer unit

$$U = R_{15} P_{17} \text{first_unit bs}_q.$$

Together with the four outer comparisons this gives ten nonunit native comparisons. In particular the strong norm-coefficient equality is retained. Induction over the frozen rows, combined with the displayed port identities, proves equality of every kernel register, residual, unit factor, and complete native polynomial on all integer tuples.

23.10.4 The semantic import at its positive domain

On supplied positive coordinates and $X > 0$, $P_0 \geq 2$, $D \geq 5$, $\mathcal{B} > D$, and the decoded hats are nonnegative. At a complete zero the global equality forces $P > 1$, $J > 0$ and the history bounds. Native classification gives dyadic scale and width and separates the lanes. The retained transports and phase relation recover a common reverse affine history. Its endpoint yields the word closure, hence an actual halt at or before the proposed closure length.

Conversely a genuine finite halt at the specified admissible P_0 supplies reversed sentinel histories and selector hats. Choose an adequate dyadic height, pack at $\mathcal{B} = \mathcal{K}D$ and $P = \mathcal{B}^t$, and use the positive global-bound estimate from the complete theorem. The prescribed-scale native AND converse supplies its full positive witnesses; the unit projection supplies all sixteen retained coordinates.

A native closure may include a post-halt suffix. Soundness gives a first actual halt no later than the closure length; it does not claim legal physical execution after an empty queue. The separate Grill cleanup theorem identifies the actual first-empty boundary. There is no externally imposed time cutoff.

23.11 One unit-safe finalizer and the full proof

The complete residual list is shown in Table 5. It includes both full recoders, all loader constraints, all native comparisons, and the exact-width equality once.

The emitted polynomial is exactly

$$F = U \left(1 + \sum_{j=0}^{85} R_j^2 \right) - 1. \quad (23.9)$$

Source of nonunit residuals	Number
Complete native history	10
Canonical width-32 recoder	34
Canonical length guard and signed frame	2
Unrestricted width-397488 recoder	34
Exponent extraction and binding	3
Unary scale, unary value, exact native width	3
Total	86

Table 5: All 86 residuals occur inside the same unit product.

For any integer tuple, the second factor is a positive integer. If $F = 0$, it divides one, so it equals one, every $R_j = 0$, and $U = 1$. Conversely those conditions give $F = 0$. This argument does not assume that U or a previously finalized native polynomial is nonnegative away from its zero set. Adding an unweighted external sum of squares to a signed native polynomial would not provide the same implication.

Proof of Theorem 23.1.1. Fix a valid θ_S from Section 23.7. Suppose F has a positive zero at $x > 0$. The finalizer restores every comparison and the native unit. The first recoder fixes the canonical bits of $x + 1$ and the frame equation fixes $N = N_e(x)$. The second recoder and the bounded residue extraction fix $h = N + 1$, hence $T = K^N$. The unary equation and width comparison identify the exact literal E word, with no independent padding choice. Native soundness implies a finite Grill halt. The corrected halt bridge, phase-preserving normalization, and accepting-head invariant imply U15 acceptance. The selected finite program recognizes S on the decoded input x , so $x \in S$.

Conversely let $x \in S$. Its selected U15 computation accepts. The corrected tag macros avoid premature short queues and emit one halt from the accepting head. Normalization satisfies the source halt convention, and the corrected Grill bridge gives a genuine finite first empty time from the exact E word. Choose the full first recoder extension at $n = \text{bitlength}(x + 1)$ and its positive guard. Set the positive genuine N , then choose the full second extension at $h = N + 1$ and the positive ℓ, v, g in Section 23.8. Set $T = K^N$ and the literal positive X . The E-word margin permits the strong native converse at the same width and its positive transfer to the retained weak slack. All private witness sets are disjoint except for the deliberately shared ports. Thus every added coordinate can be positive simultaneously, every residual vanishes, and $U = 1$. Equation (23.9) vanishes. $\square$

23.12 What the arithmetic ledger counts

The source is a straight-line DAG with binary addition, subtraction, and multiplication over integers. Reading an exact literal coefficient is free in this model, but multiplying a runtime register by any nontrivial coefficient is an emitted multiplication. A fixed-exponent runtime power is emitted as binary multiplication gates. There are no runtime division or variable-exponentiation gates.

The two recoders contribute 98 private positive auxiliaries. The first spread output and the seven added loader coordinates $\beta, N, \ell, v, g, T, X$ contribute eight, for 106 new positive witnesses. Added to the 797,029 native witnesses, this gives 797,135. The six external positive coordinates are separate.

The width-32 recoder uses 133 operations. The width-397488 recoder on input/output one uses 150; its one multiplication $1 \cdot J$ is an alias, with all consumers and comparisons retained. The other loader operations total 27. The finalizer consists of 86 differences, 86 squares, 85 sum additions, addition

Stage	Multiplications	Additive gates	Total
Two full recoders and loader	173	137	310
Native history before finalizer	803,257	2,796,719	3,599,976
One finalizer	87	173	260
Complete polynomial	803,517	2,797,029	3,600,546

Table 6: Measured operations in the actual emitted source, not an inherited example.

of one, multiplication by U , and subtraction of one. All 3,600,546 emitted gates and every supplied coordinate reach the output.

23.12.1 Typed constant recipes

The manifest has 8,160 interned constant recipes. The allowed constructors are a decimal integer, a fixed 2^t , a fixed finite geometric sum $(4^t - 1)/3$, and addition, subtraction, or multiplication of earlier constant handles. The exponent t is a fixed nonnegative source integer. A recipe cannot refer to an input or gate. These rules define ordinary finite integer coefficients; they do not smuggle in a witness-dependent power or unproved division.

The largest exact fixed coefficient in the audited pool has 551,891 bits. Constant interning saves repeated storage. It neither removes runtime scalar multiplications nor reduces quantified coordinates. Some interned recipe rows are unused coefficient-construction remnants; they are not arithmetic gates, and do not contradict full gate and input liveness.

23.12.2 Degree and resource scope

Ordinary degree propagation gives $d(A \pm B) \leq \max(d(A), d(B))$ and $d(AB) \leq d(A) + d(B)$, starting every external coordinate and witness at degree one. It produced the valid historical bound 71,731,007. Two initial highest-form probes returned zero and did not certify attainment. The cancellation-aware analysis in Section 23.13 now proves exact total degree 69,339,973 for the very same source. No gate, coefficient, coordinate, or comparison changes.

The DAG occupies 61,209,290 bytes, including its eight-byte header. Its SHA-256 is

a07c1ba41e3a18fafd05c19b8ac39211b0475e30ed8a08ddf43c45a9f5f440f2.

The recorded full emission and backwards liveness audit took 5.47 seconds and 107,515,904 bytes peak RSS. Limits were six million gates, 200,000 constants, 600 seconds, and 1,800 MiB RSS; none was reached. These measurements describe one execution of a streamed construction, not evaluation of a huge positive solution or a performance guarantee.

Earlier 205-operation examples, and the 16,291-operation exact-width illustration with its distinct 16,289-operation factored successor, describe different sources. None supplies the present ledger. The current all-tuple identity proof transfers mathematics between schedules; it does not identify their operation counts.

[write] *Stale bytes in two shipped files.* The frozen reproducibility notes of the delivery keep the original upper-bound wording byte for byte, as `11-grill-poly-REVISION.md` explains: `11-grill-poly-README.md` (delivered `reproducibility/README.md`) says that 71,731,007 “is an upper bound, not a proven exact degree”, `11-grill-poly-arithmetic-README.md` says “no exact-degree claim”, and the manifest field

`degree_upper` keeps 71,731,007. Section 23.13 (revision 1) proves the exact degree 69,339,973; the upper bound remains valid.

23.13 The exact total degree of the unchanged polynomial

The initial edition reported the sound propagated upper bound 71,731,007. A cancellation in the first native norm factor explains why that number was not attained. Resolving this cancellation in the *analysis* of the source gives exact degree 69,339,973. The arithmetic DAG, its coefficients, all comparisons, and all supplied coordinates are unchanged. This is not a new circuit or a replacement polynomial with the same zero set.

Theorem 23.13.1 (Exact degree of the pinned source). *Assign degree one to all 797,141 supplied coordinates of the frozen DAG, including its six external coordinates. The integer polynomial computed by that source has total degree exactly*

$$\deg F = 69,339,973.$$

This assertion is unconditional algebra on the pinned source. It uses neither witness positivity nor a residual equality, and it does not assume any native-history or universality theorem.

The main DAG digest is the one in Section 23.12. The unchanged arithmetic manifest has SHA-256

41e6f754df8f60448e1207ef36e6161729b52004fac719a580d6ec643d039d49.

The kernel digest is the one in Section 23.2. Both new checkers authenticate these three inputs before reading the complete source. Gate and input indices in this section are zero based.

23.13.1 The first norm cancels identically

For local notation in this section, let u, a, c, γ, d denote these actual registers:

Symbol	Native register	Source location
u	<code>and__wn2</code>	gate 3,600,220
a	<code>and__R12</code>	gate 3,600,228
c	<code>and__R10a</code>	gate 3,600,226
γ	<code>and__ga</code>	input 797,137
$d = 4a + 3$	<code>and__a4m5</code>	gate 3,600,232

These local u, a are not the earlier shifted ordinary input or the source alphabet parameter. The first unit factor R_{15} , gate 3,600,240, is

$$R_{15} = (u + ac + \gamma d)^2 - (a^2 + d)c^2.$$

Expanding in four independent indeterminates u, a, c, γ and substituting $d = 4a + 3$ gives the identity

$$R_{15} = u^2 + 2uac + 2u\gamma d + 2ac\gamma d + \gamma^2 d^2 - dc^2. \quad (23.10)$$

The two a^2c^2 terms cancel. There is no appeal to a norm equation, a valid history, or any zero-set restriction. Independent sparse expansion of both sides gives identical coefficient dictionaries with eleven nonzero monomials. The check is in four abstract indeterminates, not an expansion in hundreds of thousands of coordinates.

[write] *Attribution.* Identity (23.10) is identity (4) of the research tree's `gpcp_shared_selectors774.md` (commit `ee507599f`, 1 October 2026), written there as $D^2 - \Delta c^2 = X^2 + 2Xac + 2XG + 2acG + G^2 - (4a+3)c^2$ with $D = X + ac + G$, $G = \gamma(4a+3)$ and $\Delta = a^2 + 4a + 3$; replace X by u and $4a+3$ by d . The tree also uses it in `gpcp_history_computed_fields744.md`, `gpcp_index_linear_units757.md` and `gpcp_coupled_index_units754.md`. This Part does not cite it; the identity is the tree's, and its application to the Grill source with the exact degree that follows is new. Part II proves it again as (24.17).

Before this identity is used, both checkers match all 67 native-kernel rows to the raw DAG, including operation codes, operand handles, supplied coordinates, constants, caller ports and unit factors. Thus (23.10) analyzes the factor actually present in the frozen source.

23.13.2 The outer leading forms

For a nonzero polynomial R , denote its highest homogeneous part by R_* . Put

$$s = 794976, \quad g = 2030, \quad N_* = g + s + 5 = 797011, \quad \alpha = 3N_* = 2391033.$$

Here N_* is a fixed lane exponent, distinct from the varying tape integer N . Let

$$J_1 = \sum_{j=0}^{s-1} \widehat{S}_j, \quad \mathfrak{p} = 2^{551890}(X + Z_0)V_f J_1, \quad Q_* = 16\mathfrak{p}^{N_*}.$$

X is the supplied loader witness, not ordinary input x . From the actual definitions

$$D = (X + Z_0)V_f + X + q_{\text{init}} + h_{\text{slack}}, \quad P = (2^{551890}D - 1)(J_1 - s) + 1,$$

we have $P_* = \mathfrak{p}$ and $\deg P = 3$. Its visibly nonzero leading form has positive coefficients. The independent checker also traverses the raw selector-sum graph and checks that each of the s supplied selector hats occurs exactly once.

The scale port is P^{N_*} , so the kernel's radix $q = 16$ scale has leading form Q_* and exact degree α . In the joined Z port, the uniquely highest summand is $P^{g+s+1}H_V$. Indeed the selector lane has degree $3(g+s) - 2$ and the history lane has degree $3(g+s) + 4$; the low group pack is smaller. Hence

$$Z_* = \mathfrak{p}^{N_*-4}H_V, \quad \deg Z = \alpha - 11.$$

The joined H and M ports have degree at most $\alpha - 2$. These are source-polynomial statements before imposing any comparison.

Write β_* for the native input `and__odd_half`, put $b_* = 2\beta_*$, and put $k_* = \eta + \zeta$. The symbols $w, f, i, \gamma, \tau, \eta, \zeta$ denote their corresponding supplied native auxiliaries; τ is `and__tau_gap`. They are independent coordinates. The relevant kernel leading forms are

$$u_* = wQ_*, \quad (\text{sn}2)_* = b_*Q_*, \quad c_* = k_*b_*Q_*, \quad a_* = wb_*Q_*^2.$$

Their exact degrees are $\alpha + 1$, $\alpha + 1$, $\alpha + 2$, and $2\alpha + 2$, respectively.

23.13.3 The degree of every unit factor

The six terms in (23.10), in the displayed order, have degree bounds

$$2\alpha + 2, \quad 4\alpha + 5, \quad 3\alpha + 4, \quad 5\alpha + 7, \quad 4\alpha + 6, \quad 4\alpha + 6.$$

The fourth is uniquely highest. Since $d_* = 4a_*$, it gives

$$(R_{15})_* = 8\gamma a_*^2 c_* = 8\gamma w^2 b_*^3 k_* Q_*^5 \neq 0.$$

Therefore $\deg R_{15} = 5\alpha + 7 = 11,955,172$. Naive max/sum propagation at the original subtraction gave $6\alpha + 8 = 14,346,206$. The precise reduction is $\alpha + 1 = 2,391,034$.

The other unit-factor calculations use the source's highest homogeneous pieces

$$(R_{16})_* = a_*^2 f^2 = w^2 b_*^2 f^2 Q_*^4, \\ (\text{bs_packed})_* = 16Q_*^3 Z_*, \quad (H_{17})_* = -32Q_*^3 Z_*.$$

Collecting their terms gives Table 7. The linear factor $\tau - k_*$ is nonzero because τ, η, ζ are independent supplied coordinates. Each displayed leading form is consequently nonzero in the unrestricted integer polynomial ring.

Factor	Highest homogeneous part	Exact degree
R_{15}	$8\gamma w^2 b_*^3 k_* Q_*^5$	$5\alpha + 7 = 11,955,172$
P_{17}	$1024w^2 b_*^2 f^2 Q_*^{10} Z_*^2$	$12\alpha - 16 = 28,692,380$
first_unit	$4wb_*^2 k_*(\tau - k_*)Q_*^3$	$3\alpha + 5 = 7,173,104$
bs _q	Q_*	$\alpha = 2,391,033$
U	Product of these four forms	$21\alpha - 4 = 50,211,689$

Table 7: Exact degrees of all four native unit factors and their product.

23.13.4 All residuals and the highest square

The ten native residual bounds, in their actual zero-based order, are

index	0	1	2	3	4	5	6	7	8	9
bound	3	5	4	3	$4\alpha - 11$	$4\alpha - 11$	$4\alpha + 10$	$4\alpha - 11$	$\alpha - 2$	$\alpha - 2$.

The next 75 residuals are the complete loader's comparisons, each of degree at most 397,489. The final exact-width residual has degree at most two. These bounds account for all 86 residuals; they are propagated from the full source, not inferred from satisfying assignments.

Residual 6 is uniquely largest. Its comparison is $\text{ic22} = R_{16}$, where $\text{ic22} = (ic^2)^2$. Therefore

$$(\text{ic22} - R_{16})_* = i^2 c_*^4, \quad \deg(\text{ic22} - R_{16}) = 4\alpha + 10 = 9,564,142.$$

Its square has leading form $i^4 c_*^8$ and degree $8\alpha + 20 = 19,128,284$. Every other residual square has a strictly smaller bound. Thus the leading form of $1 + \sum_{j=0}^{85} R_j^2$ is exactly $i^4 c_*^8$.

Since the polynomial ring over $\mathbb{Z}$ is an integral domain, degrees of products of nonzero polynomials add. Subtracting one changes no positive-degree leading term. Applying this to the unchanged finalizer yields

$$\deg F = (21\alpha - 4) + (8\alpha + 20) = 29\alpha + 16 = 69,339,973,$$

proving Theorem 23.13.1.

[write] Theorem 23.13.1 is the instance $N = N_* = 797,011$ of Part II's law (Theorem 24.1.1): $87 \cdot 797,011 + 16 = 69,339,973$, and also $63N_* - 4 = 21\alpha - 4$ for $\deg U$, $12N_* + 10 = 4\alpha + 10$ for residual 6, and $29N_* - 8$ for the exponent of $\mathbf{p}$. Part II's ray coefficient $-2^{148}(2Ks)^{29N-8}$ with $K = 2^{551890}$ and $s = 794,976$ is (23.12). Both statements are printed: this one is proved on the literal source, Part II's for every table of the template.

23.13.5 The complete leading form and a nonzero ray coefficient

Multiplying the four unit-factor leading forms and the highest residual square, then substituting $b_* = 2\beta_*$ and $Q_* = 16\mathfrak{p}^{N_*}$, gives the entire highest homogeneous component:

$$F_* = 2^{138}\mathfrak{p}^{29N_*-8}\gamma w^5\beta_*^{15}k_*^{10}f^2i^4(\tau - k_*)H_V^2. \quad (23.11)$$

[write] *An observation not made by the source* (checked in this write from (23.11)). The leading form F_* involves only native witnesses and the loader witness X (through $\mathfrak{p}$); none of the six external coordinates x, a, p, s, C, L occurs in it. Every monomial of F of top degree therefore survives any specialization of the external coordinates, so the exact degree 69,339,973 in the remaining coordinates persists on every fixed-program slice, and also with x fixed. Part II's caution that specialization can lower degree (Section 24.7) concerns the loader residuals, which are not the dominant ones here.

The exponent $29N_* - 8$ is 23,113,311. The remaining factors have total degree 40, so (23.11) has degree

$$3(23, 113, 311) + 40 = 69, 339, 973.$$

The independent symbolic checker expands this expression into 23 monomials in the opaque degree-three base $\mathfrak{p}$ and the native input atoms, and checks exact coefficient-dictionary equality with the form propagated independently from raw DAG rows.

There is also an explicit univariate certificate. Set *every supplied coordinate*, including all six externals, equal to a common indeterminate t . Then $\mathfrak{p}$ has leading coefficient $2^{551891}s$, k_* has coefficient two, and $\tau - k_*$ has coefficient minus one. The coefficient of $t^{69339973}$ in this ray specialization is exactly

$$-2^{148}(2^{551891} \cdot 794976)^{23113311} \neq 0. \quad (23.12)$$

This finite negative integer is specified by a compact exact expression; it is not expanded into a giant decimal numeral. Its two independently checked residues are

modulus	coefficient residue
17	3
1, 000, 000, 007	53, 942, 795.

Either nonzero residue alone certifies a nonzero integer coefficient. The certificates are deterministic; they do not invoke random identity testing or require an assumption that a modulus is prime.

23.13.6 Two independent verifiers and their scope

The cancellation-aware verifier reads all 3,600,546 original rows. For each gate it carries a valid degree bound and the coefficient at that bound on the all-coordinate ray, modulo each stated modulus. A zero coefficient at an intermediate bound never promotes that bound to an exact degree. The sole improved rule is at R_{15} , justified by the structurally authenticated identity (23.10). At the output the improved bound and a nonzero coefficient agree, and the coefficient matches (23.12).

The second verifier independently propagates exact sparse highest forms using an opaque degree-three base, authenticates the selector sum and all kernel rows, and proves equality with (23.11). Both execute newly written standard-library code, read the frozen source as data, and retain explicit checks under optimized Python. Their receipts record source pins, kernel and port degrees, all residual bounds, factor degrees, and the two coefficient residues.

The modular verifier uses at most four million source nodes, 800,000 coordinates, 256 monomials for its four-indeterminate identity, 512 MiB address space, and a 180-second ceiling. It keeps compact degree/coefficient arrays, not a degree-sized univariate polynomial or a full multivariate expansion. The independent symbolic verifier has its own explicit limits and receipt. These operational bounds constrain the checks; they are not bounds on witness size or computational complexity of the represented languages.

The historical manifest and original reproducibility subtree remain byte-for-byte unchanged, including their valid syntactic degree bound and earlier inconclusive probes. The separate exact-degree extension supplies the stronger invariant for the *same* source. This preserves an auditable distinction between an improved theorem about a circuit and an alteration of that circuit.

23.14 Independent evidence and reproducibility

The evidence has several layers, with deliberately different purposes.

1. *Semantic audit.* Independent checks cover 29,696 sparse-tape instruction contexts, 1,056 signed input-frame cases, 4,132 canonical-length cases, 192 exponent cases, and 192 exact-width E cases with rejected padding variants. A separate normalization audit checks 819,907 two-generation identities and 237,750 unique-H prefix cases. The accompanying proofs, rather than these finite counts, establish the parametric statements.
2. *Literal-table audit.* All 29 instructions and all numeric tag/Genera productions are reconstructed independently. Dynamics include 18,125 macro comparisons and 2,005,675 tag steps, with all 534 operative nonhalt tag symbols exercised, minimum preacceptance length three, and 300 accepting entries. Original-source code is read as data, never executed.
3. *Grill table audit.* Every one of the 397,488 exponents is derived by an independent position formula and compared byte for byte. Exact local E/L/R, shifted-erasure, and complete cleanup tests check the actual source rows, including the unique halt-producing pair.
4. *Complete arithmetic audits.* Independent readers check the binary DAG's typed references, constant-only recipes, six-external signature, witness count, all comparisons, one-unit finalizer, gate/input liveness, ledger, and degree bound. An exact motif audit consumes every row and compares every runtime coefficient by integer equality; all 67 native rows match authenticated complete fixtures. A separate formula evaluator checks the full source modulo independent primes without importing the producer.

The mathematical text, source emitters, complete DAG, literal table, frozen proof imports, audit code, and results are included in the archive. Revision 1 adds a separate exact-degree proof and two independent verifiers using the existing DAG without duplicating it. The original scientific subtree and its historical degree metadata remain byte-for-byte unchanged. The portable replay uses Python 3.10 or newer on Linux and only its standard library; the resource-limited checkers use the Unix resource module and Linux RSS units. It needs no network or upstream Python execution. It works from a freshly extracted archive and is checked both normally and with `python3 -O`. Explicit exceptions, rather than removable assertions, enforce validation. PDF rebuilding additionally requires a conventional LaTeX installation and the packages named in the preamble.

[write] In this collection the archive's files are shipped under prefixed names (README), except for the arithmetic DAG and the run table, which are rebuilt by the shipped code or extracted from the arrival archive (README, "Reconstructing the excluded data"), the byte copies of research-tree files (cited by

path), the third-party paper copies, and the release seals. The delivered verifiers check sealed inventories, so they run in an extraction of the arrival archive, not in this directory.

The source-provenance record distinguishes original frozen hashes from documented portability transformations, including path adaptation and replacing local validation assertions by explicit exceptions. The inventory authenticates the delivered file set, including the actual arithmetic DAG, and rejects missing, extra, and altered files. Regeneration is performed in a separate work directory; it does not rewrite the supplied scientific evidence. Timing fields may vary, while mathematical metadata and DAG bytes are deterministic.

The binary format is deliberately simple. Its header is CDAGv1 followed by two zero bytes. Each subsequent 17-byte record is little-endian <Bqq: opcode 0 for addition, 1 for subtraction, or 2 for multiplication, followed by two signed 64-bit handles. A negative handle selects a constant recipe; a handle at least 2^{50} selects a supplied coordinate; other nonnegative handles refer to earlier gates. The JSON manifest gives compact coordinate families, exact coefficient recipes, comparisons, and the final output handle. This fully specifies the polynomial without expanding a prohibitively large monomial list.

23.15 Further mathematical questions

Can a different representation reduce the exact degree? The exact degree of this frozen polynomial is now settled, with all program parameters still variable. Lowering degree therefore requires a genuinely different polynomial, rather than a sharper analysis of these same gates. One concrete direction is to isolate the largest native comparison and ask whether a paid auxiliary-variable interface can reduce total degree while controlling witness count, coefficient height and operation count. Such a change needs its own complete positive soundness and converse.

Can ordinary input be loaded without unary expansion? The second recoder pays for an exponent equal to the finite tape integer plus one. This is finite and arithmetically explicit but encodes an enormous unary tag word. A direct binary-input source compiler could remove that conceptual expansion. Such a proposal needs an actual fixed source-language theorem and an exact-width loader, rather than substituting a familiar universality claim.

Which selectors can genuinely be shared? The current emitter preserves all $2m$ chronological selectors. Equal run exponents share justified history classes, not phase identities. Eliminating selectors requires a sound and complete reconstruction of their phase chronology, especially across repeated zero exponents. A smaller numerical table could reduce both selectors and gates, but must retain the complete halt protocol.

Can the five program parameters be tied together? The valid tuple recipes have substantial algebraic dependence on a selected finite program. Encoding those dependences with fewer external parameters is a distinct task: it requires paying for a program decoder and proving that its range is the valid slice. Existentially freeing the present five parameters would change the recognized language and is not an acceptable shortcut.

Can the imported theorem boundary be reduced? A proof-assistant development could first formalize the all-tuple native identities, typed coefficient language, and exact source compiler, leaving

the complete Pell/AND theorem as one explicit interface. A fuller development would certify that interface and the finite U15 initialization. These are separable proof obligations, and neither is supplied by increasing the number of bounded tests.

What should an explicit accepting witness demonstrate? Constructing a full positive tuple for a small genuine accepting source would test the most remote native interfaces, especially prescribed scale and restoration. It would still not prove universality. A feasible experiment should first estimate witness bit lengths and distinguish a full native solution from outer port assignments or modular residues.

What is the right cost comparison? The emitted source offers a reproducible arithmetic-operation baseline. Comparing substrates also requires coefficient height, source-program size, witness arity, maximum witness growth, exact-degree information, and bit cost. A source with fewer arithmetic gates can be worse on the other measures. Reporting those tradeoffs is more informative than collapsing them into a single unqualified complexity figure.

Conclusion

The completed construction has one fixed numerical Grill program, one fully emitted polynomial circuit, explicit ordinary-input and program-parameter semantics, and an exact initial-width interface. The paid composition uses 797,135 positive witnesses and 3,600,546 operations. Its exact total degree is 69,339,973, certified on the unchanged source by the explicit full leading form and nonzero ray coefficient. The central logical safeguards are the accepting-head invariant, phase-preserving normalization, unrestricted second recoder, same-width positive native converse, and one integer-unit finalizer. Under the pinned complete theorem imports, these give the fixed-language equivalence of Theorem 23.1.1 for every positive ordinary input.

References of Report 23 revision 1

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The exact note names, individual blob identifiers, and content hashes used here are listed in the included provenance inventory.
- [6] *Report 23 reproducibility packet, revision 1*, 3 October 2026. Complete composition and component proofs, literal tables, emitted arithmetic DAG, source provenance, independent semantic and circuit audits, the exact-degree proof and certificates, and offline replay. The archive is the accompanying research artifact, not an external theorem citation. [write] Shipped in this directory with the prefix `11-grill-poly-`, except the files the README lists as not shipped.

Part II

Parametric exact degree laws for the native Grill polynomial

[write] Source 14 of batch 82: *Research report 24*, dated 3 October 2026; archive `Native_Grill_Exact_Degree_Laws_Package.zip`; pin 2d887f0fa. It cites Report 23 revision 1, which is Part I. Its sections, statements and equations are numbered 24.*k* here.

Abstract of the source. For the pinned weak-cone native Grill template, a fixed nonempty run table of length m with g distinct exponents determines $N = 2m + g + 5$. We prove that its native unit has exact degree $63N - 4$, its largest retained comparison residual has exact degree $12N + 10$, and its complete native polynomial has exact degree $87N + 16$. The proof covers one-phase and allzero tables, identifies the unrestricted norm cancellation, and gives the full nonzero highest homogeneous part.

Adding arbitrary integer residuals of exact maximum degree D inside the same sum-of-squares finalizer gives degree $63N - 4 + 2 \max(12N + 10, D)$, including ties. The generic k -wide recoder has residual maximum $\max(k + 1, 20)$; the specified two-recoder loader, with canonical width 32, has maximum $\max(34, k + 1)$. A fresh polynomial input of degree $\delta \geq 1$ changes the native degree to $(\delta + 2)(29N - 8) + 40$. Exact witness-count corollaries concern the supplied layouts only.

These are source-specific algebraic theorems. They do not establish a new universality theorem, the semantic correctness of arbitrary loader attachments, or minimal degree among equivalent representations. A portable data-only verification packet supplies small binary fixtures, independent full coefficient expansions, loader checks, and optional regression against the previously studied large circuit.

[write] Theorem 24.1.1 at $N = 797,011$ is Part I's Theorem 23.13.1; both are printed. The programme's review `review_incoming_matrix_grill.md` recomputes the law at that N . Source 14's files are shipped with the prefix `14-degree-laws-`; fourteen of its members are byte copies of source 11's files and are shipped once, under source 11's prefix.

24.1 The theorem and its scope

This report generalizes a degree calculation for one large emitted Grill polynomial to every nonempty fixed run table in the same native source template. The dependence on table length and number of distinct exponents is exact. The values of the exponents change fixed integer coefficients; they do not change the highest-degree lanes.

Throughout, degree means total degree of a polynomial over $\mathbb{Z}$, before any equations on its coordinates are imposed. Every supplied native coordinate and the supplied input X has degree one unless a different substitution is explicitly stated. A fixed numeral, including an integer given by an enormous fixed recipe, has degree zero. Write $\text{in}(A)$ for the highest nonzero homogeneous part of a nonzero polynomial A .

Let $(n_0, \dots, n_{m-1})$ be a fixed table, $m \geq 1$, and let g be its number of distinct exponent values. Set

$$s = 2m, \quad N = s + g + 5, \quad \alpha = 3N. \quad (24.1)$$

In particular, $g \geq 1$, $N \geq 8$, and $\alpha \geq 24$. Let U and $R_1, \dots, R_{10}$ be the unit and comparison differences of the exact source specified in Sections 24.2–24.4. Define

$$F = U \left(1 + \sum_{j=1}^{10} R_j^2 \right) - 1. \quad (24.2)$$

Theorem 24.1.1 (Exact native degree). *For independent degree-one X and native coordinates,*

$$\deg U = 63N - 4, \quad \max_j \deg R_j = 12N + 10, \quad \boxed{\deg F = 87N + 16}.$$

The largest native residual is unique. The statement holds for every table accepted by the pinned emitter: each n_i is an exact integer with $0 \leq n_i \leq 2^{32} - 1$. The same displayed polynomial formulas satisfy the theorem for arbitrary fixed nonnegative integer exponents.

The last sentence is a mathematical extension of the formulas, not an extension of the unchanged implementation’s accepted input type. In particular, it must not be read as permission to pass larger exponents to that emitter. Fixed exponent recipes are not variable-exponent operations.

No positivity, Boolean-selector, history, halting, or universality assumption enters this degree theorem. Such hypotheses matter for the interpretation of zeros; they do not define the ambient polynomial ring. Restricting to a solution set, identifying coordinates, or eliminating a witness can change degree and is a different question.

Theorem 24.1.2 (Arbitrary residual gluing). *Keep the native coordinates independent and of degree one. Let $C_1, \dots, C_r$ be any finite list of integer polynomials in those coordinates and any additional independent coordinates. They may share coordinates with the native part. Let D be the exact maximum of zero and the degrees of the nonzero C_ℓ . Then*

$$\deg \left[U \left(1 + \sum_j R_j^2 + \sum_\ell C_\ell^2 \right) - 1 \right] = 63N - 4 + 2 \max(12N + 10, D). \quad (24.3)$$

No unique largest residual is required in the enlarged family.

The two theorems separate two issues that are easily conflated. A particular loader can be attached algebraically and its degree computed exactly. Showing that its zeros encode the intended input for a particular Grill table requires a separate compiler proof. Section 24.7 gives the exact degree for the supplied loader layout without inferring such a proof for arbitrary pairings.

24.2 The precise source family

The source identity is fixed by `native_history.py`, its accompanying proof note, and `native_unit_kernel.json`. The latter contains 67 binary arithmetic rows, four ports, sixteen native auxiliary coordinates, six comparisons, and the four factors of U . Section 24.11 gives the SHA-256 pins. The source is included as evidence and is never executed by the verification packet.

24.2.1 Coordinates and fixed coefficients

Write $V, h_U, h_V, \phi, \sigma, G_{\text{bd}}, Z_0$ for the seven outer native coordinates, respectively named `Vfinal`, `H_U`, `H_V`, `phase_initial`, `height_slack`, `global_bound`, and `Z0`. Let $S_0, \dots, S_{s-1}$ be the independent supplied

selector hats and let $\widehat{z}_0, \dots, \widehat{z}_{g-1}$ be the independent group hats. They are coordinates, not Boolean quantities in this calculation.

Order the distinct exponent values $e_0, \dots, e_{g-1}$ by first occurrence. Define the nonempty groups

$$I_j = \{2i + 1 : n_i = e_j\}, \quad G_j = \sum_{a \in I_j} (S_a - 1), \quad z_j = \widehat{z}_j - 1.$$

The fixed slope and offset coefficients are

$$b(n) = 2^{2n+1}, \quad d(n) = \frac{2^{2n+1} - 2}{3} = \frac{2(4^n - 1)}{3}.$$

The quotient is an integer. In particular $b(0) - 1 = 1$ and $d(0) = 0$. Set

$$K = 2^{\max\{3, \lceil \log_2(s+4) \rceil, 2 \max_i n_i + 2\}}. \quad (24.4)$$

This is a fixed positive integer, including for an allzero table.

The base and height registers are

$$\begin{aligned} L &= X + Z_0, & U_f &= LV + X, & D_0 &= U_f + \phi + \sigma, \\ B &= KD_0, & J &= \sum_{a=0}^{s-1} (S_a - 1), & P &= (B - 1)J + 1. \end{aligned} \quad (24.5)$$

We use D_0 for the source height register to avoid confusion with the residual-degree parameter D of Theorem 24.1.2. Define the affine transport expressions

$$A_U = 2h_U + \sum_j G_j, \quad A_V = h_V + \sum_j (b(e_j) - 1)z_j + \sum_j d(e_j)G_j.$$

The first three residuals, left minus right, are

$$\begin{aligned} R_1 &= h_U + h_V + \sum_j \widehat{z}_j + G_{\text{bd}} - P, \\ R_2 &= BA_U + 1 - h_U - PU_f, \\ R_3 &= BA_V + 1 - h_V - PV. \end{aligned} \quad (24.6)$$

For $m > 1$ put

$$T_\phi = \sum_{i=1}^{m-1} (m - i)(S_{2i} + S_{2i+1} - 2).$$

The phase residual is exactly

$$R_4 = \begin{cases} \phi - 1, & m = 1, \\ m(S_0 + S_1) - (B - 1)T_\phi + \phi - (J + 3m), & m > 1. \end{cases} \quad (24.7)$$

The one-phase reduction is an unrestricted polynomial identity in this template. Using an unreduced syntactic phase bound would obscure this genuine exceptional degree.

24.2.2 The joined history lanes

For a fixed positive integer r , use polynomial notation

$$\text{rep}(r) = \sum_{a=0}^{r-1} P^a, \quad \text{pack}(v_0, \dots, v_{r-1}) = \sum_{a=0}^{r-1} v_a P^a.$$

These are polynomial sums, including at $P = 0$ and $P = 1$; no division is used. Put

$$\begin{aligned} S &= \text{pack}(S_0 - 1, \dots, S_{s-1} - 1), & M_c &= J \text{rep}(s), \\ T &= h_U + P h_V, & G_p &= \text{pack}(G_0, \dots, G_{g-1}), \\ H_b &= h_V \text{rep}(g), & M_b &= (B - 1)G_p, \\ Z_b &= \text{pack}(z_0, \dots, z_{g-1}), & R_M &= (D_0 - 1)J(1 + P). \end{aligned}$$

The indexed G_j are group selectors; G_{bd} is the separate global-bound coordinate. No supplied coordinate is identified with a selector.

The exact kernel ports are

$$\begin{aligned} H &= H_b + P^g S + P^{g+s} T + P^{g+s+2} B + P^{g+s+4} L, \\ M &= M_b + P^g M_c + P^{g+s} R_M + P^{g+s+2} (B - 1) + P^{g+s+4} (L - 1), \\ Z &= Z_b + P^g S + P^{g+s} T, \quad \text{scale} = P^N. \end{aligned} \tag{24.8}$$

The source uses shared powers, repunits, and a Horner schedule. These evaluation choices represent exactly the displayed polynomials and do not affect their degrees.

24.3 The leading history lanes

Let

$$J_1 = \sum_{a=0}^{s-1} S_a, \quad p = K L V J_1. \tag{24.9}$$

Every factor is nonzero in the free polynomial ring, and $\deg p = 3$. Equation (24.5) gives exact degrees

$$\deg L = 1, \quad \deg U_f = \deg D_0 = \deg B = 2, \quad \deg J = 1, \quad \deg P = 3, \quad \text{in}(P) = p.$$

The nonzero coefficient K may be enormous, but its size has no role in total degree.

Lemma 24.3.1 (Pack degrees). *For $r \geq 1$, $\deg \text{rep}(r) = 3(r - 1)$. A pack of r independent supplied hats minus constants has exact degree $3(r - 1) + 1$. The same degree assertion holds for a pack of r nonempty, disjoint selector-group sums minus constants.*

Proof. The last power of P in the repunit is uniquely largest. In either pack, the last coefficient has a nonzero degree-one homogeneous part. Its contribution has degree $3(r - 1) + 1$, whereas every earlier contribution has degree at most $3(r - 2) + 1$. Thus the last term cannot cancel. Disjointness is more than is needed here; nonvanishing of the last group sum suffices. $\square$

In particular,

$$\text{in}(S) = p^{s-1} S_{s-1}, \quad \deg S = 3s - 2, \quad \text{in}(T) = p h_V, \quad \deg T = 4.$$

The three terms of Z in (24.8) have degrees

$$3g - 2, \quad 3(g + s) - 2, \quad 3(g + s) + 4.$$

The last is strictly largest even when $m = g = 1$. Therefore, with $Z_* = \text{in}(Z)$,

$$Z_* = p^{N-4}h_V, \quad \deg Z = 3N - 11 = \alpha - 11. \quad (24.10)$$

The width lane is strictly largest in both H and M :

$$\text{in}(H) = \text{in}(M) = p^{N-1}L, \quad \deg H = \deg M = \alpha - 2. \quad (24.11)$$

For clarity, the common T lane has degree $\alpha - 11$, the radix lane has degree $\alpha - 7$, and the R_M lane of M has degree $\alpha - 9$. The lower group lanes cannot exceed these width lanes. Thus this step requires neither a positive exponent nor a nonzero appendant.

Finally put

$$q = 16P^N, \quad Q = \text{in}(q) = 16p^N, \quad \deg q = \alpha. \quad (24.12)$$

This notation distinguishes the source register q from its homogeneous part Q .

24.4 The unit and its unrestricted cancellation

The sixteen independent auxiliary coordinates of the kernel are

$$F_0, F_1, F_2, f, h, i, j, o, w, \tau, \eta, \zeta, \gamma, y, \beta, \rho.$$

They correspond, in order, to the source suffixes `F0`, `F1`, `F2`, `f`, `h`, `i`, `j`, `o`, `w`, `tau_gap`, `eta`, `zeta`, `ga`, `y_aux`, `odd_half`, and `bound_beta`, all in the native AND namespace.

The following equations are a compact regrouping of the pinned 67 rows, not a new kernel. Define

$$\begin{aligned} b &= 2\beta, & \kappa &= \eta + \zeta, & v &= (b + 1)q, & u &= wq, \\ c &= \kappa v + \eta, & a &= uv + v, & \Delta &= 4a + 3, & A &= a^2 + \Delta. \end{aligned} \quad (24.13)$$

Its first two norm expressions and packed register are

$$\begin{aligned} R_{15} &= (u + ac + \gamma\Delta)^2 - Ac^2, \\ R_{16} &= A(f^2 - 1), \\ F_3 &= 16Z + 8, & B_p &= F_0 + q(F_1 + q(F_2 + qF_3)). \end{aligned} \quad (24.14)$$

The remaining factors are

$$\begin{aligned} H_{17} &= jc - (2B_p + 1), \\ P_{17} &= R_{16}(H_{17}^2 - y^2) + y^2, \\ U_1 &= \tau^2 + 4(uv)(\kappa v)(\tau - \kappa), \\ q_b &= q - (F_0 + F_1 + F_2 + F_3), \\ U &= R_{15}P_{17}U_1q_b. \end{aligned} \quad (24.15)$$

The six residuals after the four outer residuals are

$$\begin{aligned} R_5 &= B_p + \rho - u, & R_6 &= \kappa - (B_p + 1 + h uv), \\ R_7 &= (ic^2)^2 - R_{16}, & R_8 &= H_{17} - (of - c), \\ R_9 &= F_1 + F_3 - (16H + 12), & R_{10} &= F_2 + F_3 - (16M + 10). \end{aligned} \quad (24.16)$$

The R_{15} , R_{16} , P_{17} names are historical kernel-register names; they are not indices in the ten-residual list.

24.4.1 The cancellation that a syntactic pass misses

From (24.13),

$$\text{in}(u) = wQ, \quad \deg u = \alpha + 1; \quad \text{in}(c) = \kappa bQ, \quad \deg c = \alpha + 2; \quad \text{in}(a) = wbQ^2, \quad \deg a = 2\alpha + 2.$$

The two operands defining R_{15} have the same leading square a^2c^2 . It cancels identically over the entire polynomial ring:

$$R_{15} = u^2 + 2uac + 2u\gamma\Delta + 2ac\gamma\Delta + \gamma^2\Delta^2 - \Delta c^2. \quad (24.17)$$

No equation on witnesses has been used. The six terms in order have degree bounds

$$2\alpha + 2, \quad 4\alpha + 5, \quad 3\alpha + 4, \quad 5\alpha + 7, \quad 4\alpha + 6, \quad 4\alpha + 6.$$

For $\alpha \geq 24$, the fourth is uniquely largest. Since $\text{in}(\Delta) = 4wbQ^2$,

$$\text{in}(R_{15}) = 8\gamma w^2 b^3 \kappa Q^5, \quad \deg R_{15} = 5\alpha + 7. \quad (24.18)$$

A gate-wise rule that replaces the degree of every subtraction by the maximum of operand degrees would overestimate this factor. Its upper bound becomes sharp only after the verified identity (24.17), or after a full expansion that discovers the same cancellation.

[write] *Attribution.* Identity (24.17) is identity (4) of the research tree's `gpcp_shared_selectors774.md` (commit `ee507599f`) and Part I's (23.10); see the note there. This Part does not cite it. The parametric exact law built on it is new for the Grill template; the tree has an exact law of the same shape, by the same method, for its PCP affine slope-class history: factor degrees $5d_0 + 7$, $12d_0 - 4$, $3d_0 + 5$, d_0 with $d_0 = 2N$ and exact degree $58N + 28$ (`pcp_affine_slope_class_history.md`, shipped inside archive 11).

24.4.2 The other three unit factors

Equations (24.10)–(24.15) imply

$$\begin{aligned} \text{in}(R_{16}) &= w^2 b^2 f^2 Q^4, & \deg R_{16} &= 4\alpha + 6, \\ \text{in}(B_p) &= 16Q^3 Z_*, & \deg B_p &= 4\alpha - 11, \\ \text{in}(H_{17}) &= -32Q^3 Z_*, & \deg H_{17} &= 4\alpha - 11, \\ \text{in}(P_{17}) &= 1024w^2 b^2 f^2 Q^{10} Z_*^2, & \deg P_{17} &= 12\alpha - 16, \\ \text{in}(U_1) &= 4wb^2 \kappa(\tau - \kappa)Q^3, & \deg U_1 &= 3\alpha + 5, \\ \text{in}(q_b) &= Q, & \deg q_b &= \alpha. \end{aligned} \quad (24.19)$$

For example, jc has degree $\alpha + 3$, below $4\alpha - 11$, and y^2 cannot compete with H_{17}^2 . The form $\tau - \eta - \zeta$ is nonzero because the supplied coordinates are independent. All four unit factors are therefore nonzero.

Degrees add under multiplication in an integer polynomial ring. Adding the four factor degrees gives

$$\deg U = (5\alpha + 7) + (12\alpha - 16) + (3\alpha + 5) + \alpha = 21\alpha - 4 = 63N - 4. \quad (24.20)$$

The argument explains why $d(0) = 0$ causes no difficulty: it removes only a lower affine term in A_V , not one of these leading factors.

24.5 Residuals and the full leading homogeneous part

The exact residual degrees are as follows. All signs refer to the left-minus-right conventions above.

Residual	Source role	Exact degree
R_1	Global bound	3
R_2	U transport	5
R_3	V transport	4
R_4	Phase	1 if $m = 1$; 3 if $m > 1$
R_5	Packed bound	$4\alpha - 11$
R_6	Packed transport	$4\alpha - 11$
R_7	Strong norm coefficient	$4\alpha + 10$
R_8	Auxiliary root comparison	$4\alpha - 11$
R_9	Padded H	$\alpha - 2$
R_{10}	Padded M	$\alpha - 2$

For the first three, the uniquely largest terms are $-P$, $-PU_f$, and $-PV$. For $m > 1$, the phase residual has degree-three homogeneous part

$$-KLV \sum_{i=1}^{m-1} (m-i)(S_{2i} + S_{2i+1}),$$

which is nonzero. For $m = 1$, equation (24.7) gives exact degree one directly. The top terms of R_5, R_6, R_8 are inherited from B_p or H_{17} ; their alternative operands are strictly smaller. The last two residuals inherit $-16 \ln(H)$ and $-16 \ln(M)$.

The decisive residual is R_7 . Its first term has leading form

$$\ln((ic^2)^2) = i^2 \kappa^4 b^4 Q^4 \quad \text{and degree } 4\alpha + 10,$$

whereas R_{16} has degree $4\alpha + 6$. Thus R_7 is uniquely largest among the ten. Consequently

$$\deg \left(1 + \sum_{j=1}^{10} R_j^2 \right) = 8\alpha + 20.$$

Combining this with (24.20), and observing that subtraction of one cannot remove a positive-degree leading part, proves

$$\deg F = 21\alpha - 4 + 8\alpha + 20 = 29\alpha + 16 = 87N + 16.$$

Proposition 24.5.1 (Exact highest form). *In the notation above,*

$$\boxed{\ln(F) = 2^{138} p^{29N-8} \gamma w^5 \beta^{15} \kappa^{10} f^2 i^4 (\tau - \kappa) h_V^2.} \quad (24.21)$$

In particular this is a nonzero polynomial of total degree $87N + 16$.

Proof. Before substituting $b = 2\beta$, $Q = 16p^N$ and $Z_* = p^{N-4}h_V$, multiplication of the four unit forms and $\ln(R_7)^2$ gives

$$2^{15} \gamma w^5 b^{15} \kappa^{10} f^2 i^4 (\tau - \kappa) Q^{27} Z_*^2.$$

The scalar power becomes $2^{15+15+108} = 2^{138}$, and the power of p becomes $27N + 2(N-4) = 29N - 8$. All factors are nonzero. The factors outside that power of p have total degree 40, so the total is $3(29N - 8) + 40 = 87N + 16$. $\square$

24.5.1 A deterministic ray certificate

Set every supplied coordinate, including X , equal to one indeterminate t . Then $L = 2t$ at highest degree, $J_1 = st$, $p = 2Kst^3$, $\kappa = 2t$, and $\tau - \kappa = -t$. Formula (24.21) gives the exact leading coefficient

$$[t^{87N+16}]F(t, \dots, t) = -2^{148}(2Ks)^{29N-8} \neq 0. \quad (24.22)$$

This single ray witnesses exact degree for every table in the degree-one theorem. A zero residue modulo one chosen prime does not contradict (24.22): the prime may divide its coefficient. For example, a table of length 17 has $s = 34$, so the coefficient vanishes modulo 17. The symbolic integer formula remains nonzero.

24.6 Residual gluing and integer zero sets

The argument needed for arbitrary added residuals is elementary, but the hypothesis on signs matters.

Lemma 24.6.1 (Positive sums of squares). *Let $T_1, \dots, T_v \in \mathbb{Z}[\mathbf{z}]$, and let $e > 0$ be their exact maximum degree, ignoring zero polynomials. Then*

$$\deg \left(1 + \sum_{a=1}^v T_a^2 \right) = 2e.$$

Proof. Let t_a be the degree- e homogeneous part of T_a , possibly zero. The degree- $2e$ part of the sum is $\sum_a t_a^2$, with at least one nonzero t_a . If that sum were identically zero, evaluating at every point of $\mathbb{R}^{|\mathbf{z}|}$ would force each real square, and hence each t_a , to vanish at every point. A real polynomial that vanishes everywhere is the zero polynomial, a contradiction. Adding the constant one does not alter degree $2e$. $\square$

Apply the lemma to the native and added residuals together. Their exact maximum is $\max(4\alpha + 10, D) > 0$. Multiplying by the nonzero U adds $21\alpha - 4$ to the degree, proving Theorem 24.1.2. Tied degrees, shared coordinates, and opposite signs of leading residuals cause no cancellation: the top forms are squared with positive coefficients.

This is a familiar sum-of-squares fact, not a claim of novelty about sums of squares or Diophantine representation generally. It is useful here because it turns an exact residual maximum into an exact degree for this unit-based finalizer.

24.6.1 Why the location of added squares matters

At an integer tuple, let $S = 1 + \sum_j R_j^2 + \sum_\ell C_\ell^2$. Then S is a positive integer. Thus

$$US = 1 \iff U = 1, R_j = 0 \text{ for all } j, C_\ell = 0 \text{ for all } \ell.$$

The forward direction uses both integrality and positivity: $U = 1/S$ is an integer only when $S = 1$. No assumption that U is positive off the zero set is needed. The correct extension is consequently

$$F + U \sum_\ell C_\ell^2,$$

not, in general, $F + \sum_\ell C_\ell^2$. This zero-set observation still does not provide the semantic proof of a loader.

The degree lemma cannot be transferred unchanged to arbitrary coefficient fields or signed differences of squares. Over the complex numbers, $T^2 + (iT)^2 = 0$; over the integers, $T^2 - T^2 = 0$. Nor does a quotient by residual equations retain the ambient polynomial degree. The pinned checks compute the polynomial before any such quotient or substitution.

24.7 Generic recoders and the specified loader

Consider the complete 34-comparison recoder in the pinned 130-row receipt. The generic width replacement is

$$Q_r = q_r^k, \quad B_r = 2^{k-1}Q_r, \quad k \geq 4,$$

with k a fixed integer. The remaining 127 source rows are unchanged. Recoder auxiliaries are independent degree-one coordinates. For the uses considered here, input and output ports are either degree-one affine supplied inputs or the fixed integer one.

Proposition 24.7.1 (Exact recoder residual maximum). *The exact maximum degree of the retained recoder residuals is*

$$D_{\text{recoder}}(k) = \max(k + 1, 20).$$

Proof. Propagation through the complete source gives the following upper bounds for the 34 comparisons in source order:

$$(k + 1, k + 1, 1, k + 1, k, 14, 3, 1, 5, 4, 2, 4, 6, 10, 2, 1, 2, k, \\ 7, 2, 1, 3, 20, 4, 1, 7, 6, 3, 4, 6, 10, 2, 2, 1).$$

When both ports are one, the penultimate bound becomes one; this does not change the maximum. Every bound is at most $\max(k + 1, 20)$.

Two exact source motifs attain the possible maxima. The first residual is

$$(2^{k-1}q_r^k - 1)J_r + 1 - P_r,$$

with leading form $2^{k-1}q_r^k J_r$ and degree $k + 1$. Here J_r, P_r are supplied recoder coordinates, not the computed native registers J, P . The AND norm residual has leading form

$$2^{24}w_r^2\kappa_r^2s_r^4q_r^6P_r^6$$

and exact degree 20. Here w_r, κ_r, s_r are the independent recoder coordinates named `and_w`, `and_k`, and `and_s`; κ_r is not the native sum $\eta + \zeta$. Its other operand has degree two. These two nonzero forms attain the upper envelope, including $k = 19$, where they tie. $\square$

The fixed Pell/AND baseline is essential. For example, at $k = 4$ the maximum is 20, not 5. The supplied data-only checker represents each upper bound as $ak + b$ valid throughout $k \geq 4$, rejects a comparison of crossing degree lines, and authenticates the exact source motifs used for attainment. A list of tested widths alone would not prove the all-width upper bound.

24.7.1 The two-recoder layout

The supplied loader uses a canonical recoder at width 32 and an unrestricted recoder at width k . The canonical input is $x + 1$ and its output is an independent spread coordinate. Both ports of the

unrestricted recoder are one. All six external coordinates and all supplied loader witnesses are assigned degree one.

The canonical recoder maximum is 33. The ordinary-frame residual has the unique degree-34 term

$$-(2^{32} - 1)p_e s_e q_c^{32}, \quad (24.23)$$

where q_c is the canonical recoder's supplied q coordinate. The other terms in that residual have degrees at most 33, 2, and 1. The nonzero fixed coefficient and both independent external factors in (24.23) must be retained when taking degree.

The seven outer loader comparisons, excluding the two groups of 34, have upper bounds in their own source order

$$1, \quad 34, \quad k+1, \quad k, \quad 1, \quad k, \quad 2. \quad (24.24)$$

Their roles are the canonical length guard, ordinary frame, exponent quotient, exponent gap, exponent-to-tape equality, unary scale, and exact unary queue. The exponent-to-tape equation uses a tape coordinate N_{tape} , unrelated to the native lane count N .

For completeness, writing $B_e = 2^{k-1}q_e^k$, their last five source shapes are

$$(B_e - 1)v_e + \ell - J_e, \quad \ell + g_e - (B_e - 1), \quad \ell - (N_{\text{tape}} + 1), \quad K_e T - q_e^k, \\ (K_e - 1)X - ((K_e - 1)C_e + L_e V_e(T - 1)).$$

Here K_e, V_e are fixed integer recipes, $V_e \neq 0$, and the other displayed loader letters are supplied coordinates; K_e is consistently the fixed radix for width k . These formulas explain (24.24) directly. The native-width residual is $X + Z_0 - L_e T$, of exact degree two.

Theorem 24.7.2 (Exact loader maximum and composition). *For this precise two-recoder layout, with the fixed canonical width 32 and consistent fixed width- k radix constants,*

$$D_{\text{loader}}(k) = \max(34, k+1), \quad k \geq 4.$$

Attaching all its residuals and the native-width equality inside the native finalizer gives exact degree

$$63N - 4 + 2 \max(12N + 10, 34, k+1). \quad (24.25)$$

Proof. The first unrestricted recoder residual attains $k+1$, and the canonical frame residual attains 34. Proposition 24.7.1 and (24.24) bound everything else by their maximum. The width equality cannot dominate. Apply Theorem 24.1.2. $\square$

24.7.2 Matching widths and deliberately mismatched tests

If a separately validated compiler has $k = m \geq 4$, then

$$12N + 10 = 24m + 12g + 70 > \max(34, m+1).$$

The complete composition therefore retains exact degree $87N + 16$. The inequality is purely algebraic; the hypothesis that the compiler matches remains separate.

For the frozen width $k = 397488$, the loader maximum is 397489. Its algebraic attachment to a native table is loader-dominated for $N \leq 33123$ and native-dominated for $N \geq 33124$. These integer cutoffs follow from comparing $12N + 10$ with 397489. Arbitrary mismatched attachments are useful degree tests and are not asserted to encode the intended ordinary input.

The degree convention also matters for program slices. The six external coordinates are counted as degree one here even if five will later be fixed to represent a particular language. Specializing those five parameters to integer values is a different polynomial and can lower the loader's degree. The free-coordinate theorem is the stated claim.

24.8 Fresh polynomial inputs and coordinate identifications

Replacing a supplied coordinate by a computed polynomial is not the same operation as retaining that coordinate and adding equations that constrain it. The latter fits Theorem 24.1.2; the former requires a new degree analysis.

Theorem 24.8.1 (A fresh polynomial input). *Substitute $X = \chi$ in the native polynomial, where χ is a nonzero polynomial of exact degree $\delta \geq 1$ in fresh coordinates independent of all native witnesses. Give those fresh coordinates and the native witnesses degree one. Put*

$$d = \delta + 2, \quad \alpha = dN.$$

Then

$$\begin{aligned} \deg U &= 21\alpha + 20 - 8d, \\ \max_j \deg R_j &= 4\alpha + 10, \\ \boxed{\deg F|_{X=\chi} &= d(29N - 8) + 40.} \end{aligned}$$

The highest-form formula (24.21) remains valid with $p = \text{in}(P)$ for the substituted source.

Proof. Let $L_* = \text{in}(\chi + Z_0)$. For $\delta > 1$, $L_* = \text{in}(\chi)$; for $\delta = 1$, $L_* = \text{in}(\chi) + Z_0$. In both cases $L_* \neq 0$ and $\deg L_* = \delta$, because the coordinates of χ are fresh. Thus

$$\deg P = d, \quad p = KL_*VJ_1 \neq 0, \quad Q = 16p^N.$$

The pack argument now uses step size d in place of three. It yields

$$\text{in}(Z) = p^{N-4}h_V, \quad \deg Z = \alpha - 4d + 1, \quad \text{in}(H) = \text{in}(M) = p^{N-1}L_*.$$

The last two forms have degree $d(N-1) + \delta = \alpha - 2$. Their lower radix lanes have degree $\alpha - 2d - 1$, and the R_M lane has degree $\alpha - 3d$, so neither competes with the width lane.

The kernel formulas for u, c, a and the cancellation (24.17) are unchanged, with the new α . Therefore $\deg R_{15} = 5\alpha + 7$. The changed Z degree gives

$$\deg H_{17} = 4\alpha - 4d + 1, \quad \deg P_{17} = 12\alpha + 8 - 8d.$$

The other factors still have degrees $3\alpha + 5$ and α . Adding proves the asserted unit degree. Since $d \geq 3$ and $N \geq 8$, the strong norm residual remains uniquely maximal with degree $4\alpha + 10$. Hence the final degree is

$$(21\alpha + 20 - 8d) + 2(4\alpha + 10) = 29dN - 8d + 40.$$

The same leading-factor multiplication proves (24.21). Every factor is nonzero in the larger polynomial ring, establishing exactness without relying on a particular ray. $\square$

24.8.1 Sharp examples and failed ray certificates

For the one-phase table [0], $N = 8$ and the ordinary degree is 712. Substitution of $X = \xi_1^2$ changes it to

$$4(29 \cdot 8 - 8) + 40 = 936.$$

Thus a claim that arbitrary computed loaders preserve $87N + 16$ would already fail in the smallest case.

For $\chi = \xi_1^3 + \xi_1 \xi_2 + 7$, the degree is 1160. For $\chi = \xi_1^2 - \xi_2^2$, it is 936, witnessed on the ray $\xi_1 = 2t, \xi_2 = t$ with all native coordinates equal to t . On the all-ones ray $\xi_1 = \xi_2 = t$, however, χ vanishes identically and the specialized polynomial has degree 712. The symbolic highest form survives, while that particular test ray does not see it. The portable checks expand the entire specialized polynomial in these examples, with no special norm-cancellation rule.

Freshness is a sufficient hypothesis designed to prevent accidental identities with native factors. It is not a claim to characterize every degree-preserving substitution. For example, setting $\tau = \eta + \zeta$ annihilates the highest form of U_1 , and therefore the displayed highest form of F . Such an identification lies outside both the free-coordinate theorem and the fresh-input theorem.

24.9 Witness counts and concrete instances

The native builder allocates seven named outer coordinates, s selector hats, g group hats, and sixteen kernel auxiliaries. Excluding its supplied X port, its positive-witness ledger is

$$W_{\text{native}} = 7 + s + g + 16 = s + g + 23 = N + 18.$$

Consequently Theorem 24.1.1 can be rewritten as

$$\deg F = 87W_{\text{native}} - 1550. \quad (24.26)$$

The count describes the supplied source variables. Positivity is their semantic type; it is not imposed when computing polynomial degree.

The two recoders contribute $49 + 49$ coordinates. The spread coordinate, canonical guard, tape coordinate, ℓ, v_e, g_e, T, X contribute eight more, for 106 loader witnesses. The loader's X is reused as the supplied native input and is not counted twice. There are six external coordinates, excluded from the existential count. Thus

$$W_{\text{total}} = W_{\text{native}} + 106 = N + 124.$$

Whenever the native residual dominates, including the separately justified matching case $k = m \geq 4$,

$$\deg F_{\text{composed}} = 87W_{\text{total}} - 10772. \quad (24.27)$$

These are exact affine rewritings of this ledger, not lower bounds relating degree to the minimum possible number of variables. Alternative encodings, elimination, or auxiliary introduction can change both quantities.

24.9.1 Small fixtures and the large regression

Run table	m	g	N	Exact native degree
[0]	1	1	8	712
[1]	1	1	8	712
[0, 0]	2	1	10	886
[0, 0, 0]	3	1	12	1060
[0, 1, 1]	3	2	13	1147
[2, 0, 1]	3	3	14	1234
[0, 0, 0, 0, 0]	5	1	16	1408
$[2^{32} - 1]$	1	1	8	712
$[0, 2^{32} - 1]$	2	2	11	973

The four fixtures [0], [1], [0, 1, 1], and [2, 0, 1] are checked against complete frozen binary DAGs and corresponding receipt data. The additional rows are clean-room formula cases, not additional emitted binary fixtures. This distinction is preserved in the certificates.

For [0], the four unit factors have degrees 127, 272, 77, 24, so $\deg U = 500$. The maximum residual degree is 106, and $500 + 2 \cdot 106 = 712$. This explicitly verifies that the phase collapse and the vanishing zero-exponent offset do not invalidate the theorem.

The previously studied full program has

$$m = 397488, \quad g = 2030, \quad N = 797011,$$

with $W_{\text{native}} = 797029$ and $W_{\text{total}} = 797135$. Its exact degree is

$$87 \cdot 797011 + 16 = 69339973.$$

A direct regression reads its 3,600,546-gate frozen DAG, all 86 residual squares, and the actual finalizer. It reproduces highest ray coefficients 3 modulo 17 and 53942795 modulo 1000000007. This is a useful consistency check with Report 23 revision 1; the general theorem is proved above, not inferred from that one large example.

[write] *Consistency with the research tree's recorded bounds.* Every syntactic upper bound that the tree records for a native Grill unit polynomial equals $90N + 17 = (87N + 16) + (3N + 1)$, the exact degree plus the cancellation gap $\alpha + 1$:

Instance	m	g	N	$87N + 16$	$90N + 17$ (recorded)
programs 0, 1 (composed205)	1	1	8	712	737
program 011	3	2	13	1,147	1,187
program 201	3	3	14	1,234	1,277
exact-width loader	1,568	6	3,147	273,805	283,247
Part I, literal program	397,488	2,030	797,011	69,339,973	71,731,007

The loader row is an inference: it assumes that the loader's 3,165 native witnesses follow the ledger $N + 18$ and that its loader residuals stay below $12N + 10$; with that assumption the recorded 283,247 is reproduced exactly. The other rows use the tree's recorded m and g . The programme's intake review recomputes $87N + 16 = 69,339,973$ at Part I's N .

24.10 Verification boundaries and reproducibility

The accompanying packet is self-contained for its default core checks. It includes the four small DAGs and metadata, their receipt fixtures, the unit kernel, recoder receipt, and the inert native and loader source needed for source authentication. Only newly written local verification code is executed. No upstream Python is imported, compiled, or dynamically loaded.

24.10.1 What the separate checks establish

The native leading-term checker structurally matches every one of the 67 kernel rows, the ports, the constants, and the actual finalizer in each frozen fixture. It propagates valid upper bounds and corresponding specialized coefficients modulo two primes. Its only special cancellation is the unrestricted identity (24.17), itself checked by exact sparse expansion. A nonzero coefficient at the upper degree certifies a matching lower degree; a zero modular coefficient is treated as inconclusive.

The direct expansion checker evaluates every gate over $\mathbb{F}_{17}[t]$, retaining all nonzero coefficients. It uses no norm rewrite and no assumed degree truncation. For each small fixture, every coefficient agrees with a separately evaluated receipt. It also checks allzero and maximum-uint32 formula examples and the degree-two input substitution. The fresh-input checker compares complete coefficient dictionaries for the cubic and difference-of-squares inputs discussed in Section 24.8.

The loader checker proves the parametric source-row upper bounds as degree pairs $ak + b$ for all $k \geq 4$, verifies the attaining recoder motifs, and tests representative widths including the transition cases. The seven outer loader formulas are transcribed from authenticated source and proved directly in Section 24.7. The checker does not claim to execute or parse the producer’s Python into a general theorem about that source.

These are finite implementation regressions supporting the source transcription. The symbolic arguments in Sections 24.3–24.8 prove the parameterized statements. Neither an all-gate numerical pass nor a set of modular examples alone is an all-table proof.

24.10.2 Core replay and optional full regression

The archive’s top-level README gives the exact replay command and expected results. Normal Python and Python with optimization enabled are both tested; correctness checks use explicit runtime failures rather than removable assertions. Outputs are written outside the sealed release tree. An exact inventory verifier rejects missing, changed, or unexpected files.

The optional full regression requires an explicit source directory containing the unchanged full-DAG files supplied in Report 23 revision 1. Those large files are not duplicated here. The optional command authenticates the files it consumes and has no automatic workspace search or fallback to a different source directory. A missing optional source does not invalidate the self-contained core; requesting the full regression with missing or mismatched data is an error, never a silent skip.

[write] In this collection Part I’s DAG is not shipped either; the README gives its rebuild, after which the optional regression can be run in an extraction of the two arrival archives.

The packet records original research-file hashes separately from portable-file hashes. Changes needed for relative paths, output discipline, and source selection are documented rather than presented as byte-identical originals. Mutation tests exercise corrupted and missing source files, altered expected results, inventory changes, and invalid optional-full requests. Final fresh-extraction runs verify the packaged bytes, not only an unarchived working directory.

24.11 Source provenance and further questions

The native kernel derives from the composed205 receipt at immutable ProveIt commit

2d887f0fa768fd67f3e545d83f8998b5780e530d.

Its receipt Git blob is 929d89c9f1ac1b727e207d8b518cfab54c8c4125. The source and kernel relationship is described by the included native proof note. The degree claims here use the explicit polynomial identities, not a new import of a semantic universality theorem.

Pinned source	SHA-256
<code>native_history.py</code>	<code>d6f88f09f5ffd48747b60bd53b98814431bcc5515cd16431438e2768a7f67340</code>
<code>native_history_proof.md</code>	<code>714274c3f7a9009f0526bcd4f854396670d6e8f1e2f58fd15cc737779145747</code>
<code>native_unit_kernel.json</code>	<code>2d88343037c08fe8b073f67dca531ee73309c0735c1d98a23c20b9ddb1eb08ae</code>
<code>input_loaders.py</code>	<code>597a028e3a64e8ed97502f0b5e66411f293cdda0c9c1d7ebc470f9c38c897c14</code>
<code>input_recoder130_receipt.json</code>	<code>175c498990a8e63de7a91d1e3d321ef90a19f129f305074f0c6de10967d0a182</code>

The exact source files named in this table are part of the core packet; their inclusion is not a claim that the entire historical proof dependency graph is independently formalized here. The original general theorem text has SHA-256 `4f4281ae58ba6df422f3e56301faa094001e522502362a17aaa302f568de51af`. The release’s provenance record separately pins the frozen research inventory and every adapted artifact.

24.11.1 Questions left open

The degree formulas suggest several concrete next problems.

1. *More general substitutions.* Characterize when coordinate identifications preserve (24.21), and determine exact degrees on exceptional loci such as $\tau = \eta + \zeta$. Nonvanishing of a substituted top form is sufficient, but a vanishing top form requires the next homogeneous layers.
2. *Weighted and multigraded costs.* Assign different weights to input, loader, and native coordinates, derive the resulting piecewise-linear degree function, and locate all leading-lane transitions. The fresh-input theorem is one controlled weighted direction.
3. *Certified source transcription.* Replace manually reviewed outer-formula transcription with a small formally specified arithmetic-source parser or proof-producing checker, while retaining independence from producer execution.
4. *Semantic compiler families.* Determine which width- k loaders and table families are genuinely compatible, then combine those semantic proofs with the exact algebraic gluing law. An arbitrary table-loader pairing is not sufficient.
5. *Alternative representations.* Investigate the cost of replacing high-degree registers by auxiliary variables and quadratic equations. Any tradeoff must recount witnesses, preserve the integer-unit finalizer’s semantics, and distinguish a new representation from a degree computation on this one.

None of these questions is settled by the current degree or arity identities. In particular, this report gives no lower bound on the minimum degree of a Diophantine representation and no improvement to the general MRDP theorem.

References of Report 24

- [1] Native weak-cone source and correctness-transfer note. `native_history.py`, `native_history_proof.md`, and `native_unit_kernel.json`. Exact frozen data included in the accompanying packet. [write] Shipped as `code/11-grill-poly-arithmetic-native_history.py.txt` (inert original) with its executable portable version, `11-grill-poly-arithmetic-native_history_proof.md` and `data/11-grill-poly-arithmetic-native_unit_kernel.json` (source 14’s copies are byte-identical to source 11’s and are shipped once).
- [2] V. Reshetnikov, ProveIt repository, composed205 native receipt, immutable commit `2d887f0fa768fd67f3e545d83f8998b5780e530d`. [Pinned composed205 receipt](#).

- [3] Paid ordinary-input loader and complete recoder receipt. `input_loaders.py` and `input_recoder130_receipt.json`. Exact frozen data included in the accompanying packet. [write] The recoder receipt is the research tree's `native_binary_input_dilation130.json`, not shipped here.
- [4] *A fixed universal Grill polynomial*. Research report 23, revision 1, 3 October 2026. The optional full regression uses its unchanged arithmetic archive. [write] Printed as Part I of this report.

Part III

Entire native fibers: Pell parametrization and height laws

[write] Source 07 of batch 82: *Research report 25*, dated 3 October 2026; archive `Entire_Native_Witness_Fiber_Package.zip`; pin `ad634b2d1`. It is the sequel of Report 22. Its sections, statements and equations are numbered 25.*k* here.

Abstract of the source. For fixed valid padded AND ports and prescribed dyadic scale, we classify every positive witness of the retained twenty-two-coordinate native arithmetic component. Seventeen coordinates are forced. The remaining five are in bijection with two integer indices: $m = M\ell$, $\ell \geq 1$, and $n = p$ or $n = 4mk \pm p$, $k \geq 1$, where $M = pc/\gcd(c, A^2 - 1)$ and $c = \psi_A(p)$. The maximum supplied-coordinate height is exactly the last Pell coordinate, with no native fixed-coordinate plateau. This yields an exact finite floor formula and the unsmoothed two-term law

$$N(H) = \kappa \log H + \frac{\zeta(1/2)}{M\sqrt{\log \alpha}} \sqrt{\log H} + O((\log H)^{1/3}).$$

The coefficient κ includes an exceptional baseline family and an exact convergent series. We prove a corresponding logarithmic ranked-height expansion, allowing ties. Native soundness and positive completeness are explicitly inherited from pinned source proofs; the Pell rank, congruence and analytic arguments are included with their attribution. A companion theorem counts the sum of all coordinate bit lengths. A portable standard-library verification packet accompanies the article.

[write] The classification is conditional on the pinned native soundness and positive-completeness theorems, as the abstract and Section 25.2 say. The programme's review `review_incoming_negative_index_fibers.md` (commit `0055e1c4d`) finds the classification consistent and does not certify the second-term or total-bit-length theorems. In this Part $\alpha = A + \sqrt{\Delta}$ and $\lambda = \log \alpha$; Report 34 in Part IV uses the two letters the other way round (Section 0.3). The source's files are shipped with the prefix `07-native-fiber-`.

25.1 The object being counted

The native component is a fixed integer arithmetic circuit, not an unspecified Diophantine representation. Fix relation arguments

$$P = 2^b \ (b \in \mathbb{Z}_{\geq 0}), \quad h_0 = \widehat{H} - 1, \quad v_0 = \widehat{M} - 1, \quad z_0 = \widehat{Z} - 1, \quad 0 \leq h_0, v_0 < P, \quad z_0 = h_0 \text{ AND } v_0. \quad (25.1)$$

All four relation arguments are positive integers. Its computed padded ports are

$$q = 16P, \quad A_{\text{port}} = 16h_0 + 12, \quad B_{\text{port}} = 16v_0 + 10, \quad F_3 = 16z_0 + 8. \quad (25.2)$$

The twenty-two *supplied* native coordinates, all required to lie in $\mathbb{Z}_{>0}$, are

$$\begin{aligned} &F_0, F_1, F_2, a, c, d, f, h, i, j, k, o, r, s, w, \\ &\tau, \eta, \zeta, ga, y, \text{odd_half}, \text{bound_beta}. \end{aligned} \quad (25.3)$$

Here y denotes the literal coordinate `y_aux`. The symbol ζ in this coordinate list is unrelated to the Riemann zeta function. Computed quantities, including q, F_3, X, Y , are not additional supplied

coordinates. Positivity is imposed on the supplied coordinates; intermediate subtraction registers follow the literal circuit.

We write $\mathcal{F}$ for the full positive fiber at the fixed arguments (25.1). The height and its ordinary counting function are

$$\text{ht}(w) = \max\{\text{the twenty-two coordinates of } w\}, \quad N(H) = \#\{w \in \mathcal{F} : \text{ht}(w) \leq H\}.$$

Different tuples count separately, even if their heights coincide. All logarithms are natural, and all asymptotic constants may depend on the fixed ports and scale.

For an integer $B \geq 2$, define the Pell sequences by

$$\chi_B(t) + \psi_B(t)\sqrt{B^2 - 1} = (B + \sqrt{B^2 - 1})^t \quad (t \in \mathbb{Z}). \quad (25.4)$$

The polynomial and addition identities used below follow either by multiplying these expressions or by the recurrence $u_{t+2} = 2Bu_{t+1} - u_t$. In particular $\chi_B(-t) = \chi_B(t)$ and $\psi_B(-t) = -\psi_B(t)$.

25.1.1 Precisely stated inherited mathematics

All upstream native sources are pinned to the same repository commit

ad634b2d10ad666260f9fdff04ec94b75169ee4b.

The selector soundness theorem [1], its prescribed-scale interface [2], the relaxed-norm proof [3] and fixed-minus proof [4] are supplied as inert text in the accompanying packet. We use these specific conclusions:

1. Every positive native witness has the forced main indices and exponent/rounding values displayed in Section 25.3. These are forward soundness conclusions about *every* witness, established before any preferred converse witness is selected.
2. Every valid tuple (25.1) admits a strictly positive native extension at its prescribed scale. Thus the uniquely forced quotients and differences below are positive integers.
3. The native bootstrap, at the base $A = a + 2$, establishes $c > A(A^2 - 1)^2$ before invoking the relaxed norm. The source obtains the main index $p = 2r + 1$, and $p \equiv 3 \pmod{4}$.

The new classification is conditional on these proved native representation results. It does not purport to rebuild the entire representation theorem from elementary arithmetic. We give the relevant relaxed-norm rank proof and signed congruence argument in full because their exact hypotheses are essential to counting the whole fiber.

The predecessor Report 22 established uniqueness of the canonical outer history and a bijection from its complete positive witness fiber to this native fiber, as well as native infinitude. Section 25.12 records the resulting transport of the present classification. The original Report 22 and its frozen circuits are unchanged.

[write] Report 22 is archive `Canonical_Histories_and_Infinite_Fibers_Package.zip` of batch 82 (cluster M2), placed by commit 7d2b1b245 as source 23 of `signal-machine-collision-certificates`, in its Part VIII. Its theorem “Fixed-scale native infinitude” (three native coordinates vary while nineteen stay fixed) is a corollary of Theorem 25.2.1, which fixes exactly seventeen and parametrizes the other five; Report 22’s elementary argument is a second route and is printed with Report 22.

25.2 The complete result

Using the forced values in Section 25.3, set

$$\begin{aligned} A = a + 2, \quad \Delta = A^2 - 1, \quad p = 2r + 1, \quad c = \psi_A(p), \quad \alpha = A + \sqrt{\Delta}, \quad \lambda = \log \alpha, \\ g = \gcd(c, \Delta) = \gcd(p, \Delta), \quad M = \frac{pc}{g}. \end{aligned} \quad (25.5)$$

The native hypotheses give $A \geq 3$, $p \geq 3$, $p \equiv 3 \pmod{4}$, and $M \geq c > 2p$.

Theorem 25.2.1 (Entire positive native fiber). *Exactly seventeen coordinates of (25.3) are fixed throughout $\mathcal{F}$. The other five, f, i, j, o, y , are parametrized without repetition by*

$$m = M\ell \quad (\ell \geq 1), \quad n = p \quad \text{or} \quad n = 4mk - p, \quad 4mk + p \quad (k \geq 1). \quad (25.6)$$

For each such pair define

$$\begin{aligned} f_m &= \chi_A(m), & R_m &= \Delta\psi_A(m), & i_m &= R_m/c^2, \\ U_{m,n} &= \chi_{R_m}(n)/R_m, & j_{m,n} &= (U_{m,n} + p)/c, \\ o_{m,n} &= (U_{m,n} + c)/f_m, & y_{m,n} &= \psi_{R_m}(n). \end{aligned} \quad (25.7)$$

All five reconstructed coordinates are positive integers. Adjoining the seventeen forced coordinates gives every member of $\mathcal{F}$, and

$$\text{ht}(m, n) = y_{m,n} = \psi_{R_m}(n). \quad (25.8)$$

In fact all seventeen fixed supplied coordinates are strictly smaller than $c^2 \leq R_m < y_{m,n}$.

Thus even the baseline $n = p$ is present for every $m = M\ell$, including $m = M$; the canonical converse choice $m = 2cp$ is not a restriction on the answer.

Put

$$D_m = R_m^2 - 1, \quad \varepsilon_m = R_m + \sqrt{D_m}, \quad \beta_m = \log \varepsilon_m, \quad t_m(H) = \frac{\text{arsinh}(H\sqrt{D_m})}{\beta_m}.$$

For $H \geq 1$, let L_H be the largest $\ell \geq 1$ with $\psi_{R_{M\ell}}(p) \leq H$, or zero if no such ℓ exists.

Theorem 25.2.2 (Finite count and two height laws). *The integer L_H is finite and the exact count is*

$$N(H) = \sum_{\ell=1}^{L_H} \left(\left\lfloor \frac{t_m(H) + 4m - p}{4m} \right\rfloor + \left\lfloor \frac{t_m(H) + p}{4m} \right\rfloor \right), \quad m = M\ell. \quad (25.9)$$

The same formula holds with $t_m(H)$ replaced by $\lfloor t_m(H) \rfloor$. One may also replace the upper summation bound by

$$K_0(H) = \left\lfloor \frac{\log H}{M(p-1)\lambda} \right\rfloor; \quad (25.10)$$

any added summands are zero. For $H < 1$, $N(H) = 0$.

Define the positive constant

$$\kappa = \frac{1}{M(p-1)\lambda} + \sum_{\ell=1}^{\infty} \frac{1}{2M\ell\beta_{M\ell}}, \quad d_*^{\text{count}} = \frac{\zeta(1/2)}{M\sqrt{\lambda}} < 0. \quad (25.11)$$

The series converges absolutely. Pointwise, including exact jump thresholds,

$$N(H) = \kappa \log H + d_*^{\text{count}} \sqrt{\log H} + O((\log H)^{1/3}). \quad (25.12)$$

If H_v is the height of the v -th tuple in nondecreasing height order, counting ties with multiplicity, then

$$\log H_v = \frac{v}{\kappa} - \frac{\zeta(1/2)}{M\sqrt{\lambda}\kappa^{3/2}} \sqrt{v} + O(v^{1/3}). \quad (25.13)$$

This is a logarithmic asymptotic, not a multiplicative equivalent for H_v .

25.3 Seventeen forced coordinates and three varying comparisons

25.3.1 Explicit formulas and universal forcing

The padded ports, checksum and packing comparison first force

$$\begin{aligned} F_1 &= 16(h_0 - z_0) + 4, & F_2 &= 16(v_0 - z_0) + 2, \\ F_0 &= 16(P - h_0 - v_0 + z_0) - 15, & r &= F_0 + qF_1 + q^2F_2 + q^3F_3. \end{aligned} \quad (25.14)$$

Since $h_0 + v_0 - z_0 = h_0$ OR $v_0 < P$, $F_0 \geq 1$; the other fields are positive as well. The checksum gives $F_0 + F_1 + F_2 + F_3 + 1 = q$. Because F_0 is odd and q is even, r is odd, hence $p = 2r + 1 \equiv 3 \pmod{4}$.

The forward selector theorem, Sections 2–3 of [1], gives

$$\begin{aligned} X &= 2^p, & Y &= \left\lfloor \frac{(X+1)^{2r}}{X^r} \right\rfloor = \binom{2r}{r} + \sum_{v=1}^r \binom{2r}{r+v} X^v, \\ w &= X/q, & s &= Y/q, \\ a &= Y(X+1), & c &= \psi_A(p), \quad d = \chi_A(p), \\ E &= XY, & B &= 2XY^2 + 1, \quad k = \psi_B(r+1). \end{aligned} \quad (25.15)$$

The notation B renames the first Pell base to distinguish it from the external scale P . The remaining forced coordinates are

$$\begin{aligned} h &= \frac{k - r - 1}{E}, & \eta &= c - Yk, & \zeta &= (Y+1)k - c, \\ ga &= \frac{d - X - ac}{4a + 3}, & \tau &= \frac{\chi_B(r+1) - 1}{2}, \\ odd_half &= \frac{s - 1}{2}, & bound_beta &= X - r. \end{aligned} \quad (25.16)$$

In particular, the seventeen fixed names are

$$F_0, F_1, F_2, r, w, s, a, c, d, k, h, \eta, \zeta, ga, \tau, odd_half, bound_beta.$$

This enumeration includes no computed register.

Here universal soundness is doing real work: merely writing one canonical witness would not prove these values unique. The source first recovers the indices $p = 2r + 1$ and $r + 1$, and then the exponent X and rounded value Y , for every positive witness. Subsequent divisions have positive denominators. The first norm determines the positive square root

$$(2\tau + 1)^2 - (B^2 - 1)k^2 = 1, \quad 2\tau + 1 = \chi_B(r + 1),$$

so it forces τ as well. Finally the source's positive converse at the same valid ports proves that the unique values in (25.14)–(25.16) are indeed positive integers. This invocation does not impose its auxiliary-index choice on other witnesses.

25.3.2 The complete literal comparison list

Let $U = jc - p$ and $R = ic^2$. The following are all sixteen comparisons of the native block, written as equalities of their actual polynomials. The right column gives incidence in the potentially varying set $\{f, i, j, o, y\}$.

No.	Comparison	Incidence
1	$r = F_0 + qF_1 + q^2F_2 + q^3F_3$	none
2	$F_0 + F_1 + F_2 + F_3 + 1 = q$	none
3	$s = 2\text{odd_half} + 1$	none
4	$r + \text{bound_beta} = X$	none
5	$(E^2 + X)(Yk)^2 = \tau(\tau + 1)$	none
6	$c = Yk + \eta$	none
7	$k = \eta + \zeta$	none
8	$k = r + 1 + hE$	none
9	$a = Y(X + 1)$	none
10	$d = X + ac + ga(4a + 3)$	none
11	$d^2 = 1 + \Delta c^2$	none
12	$(ic^2)^2 = \Delta(f^2 - 1)$	f, i
13	$(ic^2)^2(U^2 - y^2) = 1 - y^2$	i, j, y
14	$U = of - c$	f, j, o
15	$F_1 + F_3 = 16h_0 + 12$	none
16	$F_2 + F_3 = 16v_0 + 10$	none

The parentheses in comparison 12 are essential: its right side is $\Delta(f^2 - 1)$, not $\Delta f^2 - 1$. It is the relaxed main auxiliary norm; comparison 11 is the fixed main Pell norm.

The literal-DAG audit independently expands all sixteen residuals, propagates transitive dependencies, checks the twenty-two supplied names, and checks the acyclic sixty-four-gate source on four retained fixtures. Consequently exactly thirteen comparisons are unaffected when the five coordinates change. All recomputed registers are determined by the unchanged acyclic DAG. Once the forced values are installed, the entire positive fiber is therefore exactly the set of positive solutions of

$$R^2 = \Delta(f^2 - 1), \quad (RU)^2 - (R^2 - 1)y^2 = 1, \quad U = jc - p = of - c, \quad R = ic^2. \quad (25.17)$$

The general valid-port theorem comes from [2]; it is not inferred by extrapolating finite fixture tests.

25.4 The exact main auxiliary progression

The relaxed norm alone can have unwanted rational Pell behavior. The large native bound $c > A\Delta^2$ is what excludes it. The following proof is the rank argument of [3], restated at $A = a + 2$; after its bootstrap the argument only needs the hypotheses displayed here.

Lemma 25.4.1 (Inherited relaxed-norm rank lemma). *Suppose $A \geq 2$, $\Delta = A^2 - 1$, $c = \psi_A(p) > A\Delta^2$, and positive integers R, f satisfy $c^2 \mid R$ and $R^2 = \Delta(f^2 - 1)$. Then for a unique $m \in \mathbb{Z}_{>0}$,*

$$p \mid m, \quad f = \chi_A(m), \quad R = \Delta\psi_A(m).$$

Proof. Write $\Delta = d_0 s_0^2$ with $d_0 > 1$ squarefree. Choose the least positive norm-one unit with integer coefficients in $\mathbb{Z}[\sqrt{d_0}]$, say $F + y_0\sqrt{d_0}$. It exists because $A + s_0\sqrt{d_0}$ is one such unit. Every positive integer-coefficient norm-one unit is its power: division by the appropriate power leaves an integer-coefficient unit in $[1, F + y_0\sqrt{d_0})$, contradicting minimality unless it equals one. Hence for some $e \geq 1$,

$$A = \chi_F(e), \quad S = \psi_F(e), \quad s_0 = y_0 S, \quad \Delta = (F^2 - 1)S^2, \quad \psi_F(ep) = Sc.$$

The relaxed norm gives $s_0 \mid R$. Then $(R/s_0)^2 = d_0(f^2 - 1)$ and squarefreeness gives $d_0 \mid R/s_0$. Thus $f + R/(d_0 s_0)\sqrt{d_0}$ is an integer-coefficient norm-one unit, so for some $v > 0$,

$$f = \chi_F(v), \quad R = T_0 \psi_F(v), \quad T_0 = (F^2 - 1)S = \Delta/S.$$

The sequence ψ_F is strongly divisible:

$$\gcd(\psi_F(u), \psi_F(v)) = \psi_F(\gcd(u, v)).$$

Indeed the norm gives $\gcd(\chi_F(t), \psi_F(t)) = 1$, and the addition identity reduces the gcd by the Euclidean algorithm. Put $b_0 = \gcd(v, ep)$ and $h_1 = \psi_F(b_0)$. Since $c^2 \mid T_0\psi_F(v)$,

$$c \leq \gcd(c^2, T_0\psi_F(ep)) = c \gcd(c, \Delta) \leq T_0 h_1, \quad h_1 \geq c/\Delta.$$

If $b_0 < ep$, then $2b_0 \leq ep$, and doubling yields

$$2h_1^2 < \psi_F(2b_0) \leq \psi_F(ep) = Sc < Ac.$$

Here $S < A$, as follows from $\Delta = (F^2 - 1)S^2 < A^2$ and $F \geq 2$. But $h_1^2 \geq c^2/\Delta^2 > Ac$, a contradiction. Therefore $ep \mid v$. Write $v = em$; Pell composition gives the stated formulas and $p \mid m$. Strict increase of χ_A proves uniqueness. $\square$

Proposition 25.4.2. *For the native fixed parameters, the possible main auxiliary indices are exactly $m = M\ell$, $\ell \geq 1$.*

Proof. The lemma gives $m = pk$ and $R = \Delta\psi_A(pk)$. Expanding $(d + c\sqrt{\Delta})^k$ and dividing its square-root coefficient by c gives

$$\frac{\psi_A(pk)}{c} \equiv kd^{k-1} \pmod{c}.$$

All terms after the linear term contain c^2 . Since $d^2 - \Delta c^2 = 1$, $\gcd(d, c) = 1$. Therefore

$$\begin{aligned} c^2 \mid \Delta\psi_A(pk) &\iff c \mid \Delta kd^{k-1} \iff c \mid \Delta k \\ &\iff (c/g) \mid k \iff M \mid m, \quad g = \gcd(c, \Delta). \end{aligned}$$

This is an equivalence, not only a necessary divisibility condition. Also the binomial expansion modulo Δ gives

$$c \equiv pA^{p-1} \pmod{\Delta}, \quad \gcd(A, \Delta) = 1,$$

so $g = \gcd(p, \Delta)$ and $g \mid p$. Hence M is a multiple of c and $m \geq M \geq c > 2p$. Conversely every $m = M\ell$ has $p \mid m$ and $c^2 \mid R_m$, so $i_m = R_m/c^2$ is a positive integer and the first equation of (25.17) holds. It remains to solve the other two equations; this is done next. $\square$

No evenness condition on m or oddness condition on f_m has been imposed. Those features of some canonical constructions are not requirements of the whole fiber.

[write] *A domain distinction* (from the programme's review `review_incoming_negative_index_fibers.md`, commit 0055e1c4d, which finds the classification consistent). The progression $m = M\ell$, $M = pc/g$, is for the ordinary source, where the coefficient Δ multiplies $\psi_A(m)$. It must not be copied to the tree's normalized strong equation, where $c^2 \mid \psi_A(m)$ forces m to be a multiple of pc . The review's auxiliary-only example $A = 2$, $p = 3$, $c = \psi_2(3) = 15$ has $g = 3$, $M = 15$, against 45 in the normalized case.

25.5 The two normalized Pell progressions

Fix any $m = M\ell$. Then $c \mid m$, $R = R_m$ is a positive multiple of c^2 , and $f = f_m > 2c$. The latter follows from $m > 2p$ and $\chi_A(2p) = 1 + 2\Delta c^2 > 2c$.

25.5.1 Generation and the odd-index congruences

For $B \geq 2$, the least positive integer-coefficient norm-one unit in $\mathbb{Z}[\sqrt{B^2 - 1}]$ is $B + \sqrt{B^2 - 1}$: a positive integer second coordinate is at least one, making the first coordinate at least B . The minimal-unit argument from Lemma 25.4.1 therefore proves that every positive solution is $(\chi_B(n), \psi_B(n))$ for a unique $n \geq 1$. This concerns the integer-coefficient group, not the full ring of integers of the quadratic field; no unjustified fundamental-unit claim is needed.

For a native witness, $U = jc - p \geq c - p > 0$, so the second equation in (25.17) gives

$$RU = \chi_R(n), \quad y = \psi_R(n), \quad n \geq 1. \quad (25.18)$$

Modulo R , the recurrence gives $\chi_R(2h) \equiv (-1)^h$ and $\chi_R(2h + 1) \equiv 0$. Since $R > 1$ and $R \mid \chi_R(n)$, n must be odd.

For $n = 2h + 1$, write $\chi_X(2h + 1) = XQ_h(X^2)$, where

$$Q_0(t) = 1, \quad Q_1(t) = 4t - 3, \quad Q_{h+2}(t) = (4t - 2)Q_{h+1}(t) - Q_h(t).$$

Induction gives

$$Q_h(0) = (-1)^h(2h + 1), \quad Q_h(1 - A^2) = (-1)^h\psi_A(2h + 1).$$

Since $c \mid R$ and $R^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f}$, these identities imply

$$U \equiv (-1)^h n \pmod{c}, \quad U \equiv (-1)^h \psi_A(n) \pmod{f}. \quad (25.19)$$

The reconstruction of j and o requires exactly

$$U \equiv -p \pmod{c}, \quad U \equiv -c \pmod{f}. \quad (25.20)$$

These polynomial identities and the sign analysis below are the fixed-minus argument in [4], with its normalized index renamed n .

25.5.2 Step down and necessity

Lemma 25.5.1 (Plus-sign step down). *If $B \geq 2$, $0 < k_0 \leq m$ and $\chi_B(v) \equiv \chi_B(k_0) \pmod{\chi_B(m)}$, then $v \equiv \pm k_0 \pmod{4m}$.*

Proof. The identities give $\chi_B(v + 2m) \equiv -\chi_B(v)$ modulo $\chi_B(m)$ and $\chi_B(-v) = \chi_B(v)$. Thus reduce to a representative $\pm\chi_B(l)$, $0 \leq l \leq m$. For $k_0 < m$ and $l < m$,

$$\chi_B(l) + \chi_B(k_0) \leq 2\chi_B(m - 1) < \chi_B(m).$$

Strict monotonicity therefore excludes the negative representative and forces $l = k_0$ for the positive one. The case $l = m$ is zero modulo the modulus and cannot occur when $k_0 < m$. When $k_0 = m$, only $l = m$ is zero. Translating these representatives gives exactly the two residues asserted. $\square$

Squaring the second congruence in (25.19)–(25.20) and using $\chi_A(2t) = 1 + 2\Delta\psi_A(t)^2$ yields

$$\chi_A(2n) \equiv \chi_A(2p) \pmod{f}.$$

Since $0 < 2p < m$, Lemma 25.5.1 gives

$$n = ep + 2mt, \quad e \in \{1, -1\}, \quad t \in \mathbb{Z}. \quad (25.21)$$

This division by two is a division of an integer equality after obtaining the residue modulo $4m$; it is not cancellation of a possibly nonunit in a residue ring.

The addition identities give

$$\psi_A(n) \equiv e(-1)^t c \pmod{f}, \quad n \equiv ep \pmod{c}.$$

With $p = 2r + 1$ and $h = (n - 1)/2$, both signs satisfy $(-1)^h e = (-1)^{r+mt}$. The two original, unsquared minus congruences become

$$(-1)^{r+mt} p \equiv -p \pmod{c}, \quad (-1)^{r+(m+1)t} c \equiv -c \pmod{f}.$$

Because $c > 2p$ and $f > 2c$, the respective exponents must both be odd. Their difference is t , which is therefore even. Consequently

$$n \equiv p \text{ or } -p \pmod{4m}. \quad (25.22)$$

The same argument recovers the necessary oddness of r , consistent with the padded fields.

25.5.3 Converse and full-fiber bijection

Conversely let $n > 0$ have either residue in (25.22). It is odd, so $U = \chi_R(n)/R$ is a positive integer. Write $n = ep + 4mk$. The same identities give

$$\psi_A(n) \equiv ec \pmod{f}, \quad n \equiv ep \pmod{c}, \quad (-1)^{(n-1)/2} e = (-1)^r = -1.$$

Thus both congruences (25.20) hold. Hence $j = (U + p)/c$ and $o = (U + c)/f$ are positive integers, and (25.18) supplies the norm. All three varying comparisons hold, while the other thirteen keep their zero values. This proves positive completeness for every allowed pair in (25.6).

The representatives p and $4m - p$ are distinct and lie in $(0, 4m)$ because $0 < p < 2m$. Their increasing merge is

$$p, 4m - p, 4m + p, 8m - p, 8m + p, \dots$$

The zero-index Pell solution is excluded by $y > 0$; negative indices have negative y , and a negative first coordinate contradicts $U > 0$. Strict increase of $\chi_A(m)$ recovers m from f , and strict increase of $\psi_{R_m}(n)$ then recovers n from y . There is no double counting. Moreover f, i vary as m increases, and j, o, y vary within each fixed- m progression. This proves the exact seventeen/five partition and the bijection of Theorem 25.2.1.

25.6 The native maximum is always the last Pell coordinate

First control the fixed coordinates. The native bounds give

$$q < r < X < a < A < c.$$

The formulas for the fields, $w, s, a, \text{odd_half}, \text{bound_beta}$ therefore make each less than c . The relations give $k < c/Y < c$, $h < k$, $\eta < c$, and $\zeta < k$. Also

$$\frac{d}{c} = \sqrt{\Delta + c^{-2}} < A, \quad d < Ac < c^2, \quad ga < \frac{d}{4a + 3} < c,$$

because $4a + 3 > A$. The coordinate c itself is below c^2 .

For the remaining coordinate τ , note $E = XY < a$ and $X < 2E + 1$. Thus

$$\tau^2 < \tau(\tau + 1) = (E^2 + X)(Yk)^2 < (E + 1)^2(Yk)^2,$$

whence $\tau < (E + 1)Yk < (E + 1)c < Ac < c^2$. This accounts for all seventeen fixed supplied coordinates.

For the five varying ones, the relaxed norm and $\Delta \geq 3, f \geq 2$ give

$$R^2 = \Delta(f^2 - 1) > f^2, \quad i = R/c^2 \leq R.$$

Since $n \geq p \geq 3$,

$$y = \psi_R(n) \geq \psi_R(3) = 4R^2 - 1 > R \geq c^2.$$

The normalized norm yields

$$0 < U \leq y, \quad U^2 = y^2 - \frac{y^2 - 1}{R^2}.$$

Using $p < c$ and $c < f$,

$$U + p \leq y + c - 1 \leq cy, \quad U + c \leq y + f - 1 \leq fy.$$

Therefore $j, o \leq y$. This proves (25.8) for the actual full native tuple.

An expression $\max(C, y)$ remains correct if one fixes other coordinates and calls their maximum C . Within the native twenty-two-coordinate fiber, however, the inequalities just proved always give $C < y$: there is no native initial plateau. A genuinely independent external fixed-coordinate maximum can create such a cutoff; Section 25.12 keeps that case separate.

The smallest native height is attained uniquely at $(m, n) = (M, p)$. This follows from strict increase of R_m in m and of $\psi_R(n)$ in either argument in the present range. No assertion about componentwise minimality of the entire five-coordinate tuple is needed for this height conclusion.

25.7 Exact finite counts and the fixed-slice phase

For fixed m ,

$$\psi_{R_m}(n) = \frac{\sinh(n\beta_m)}{\sqrt{D_m}},$$

so $y_{m,n} \leq H$ if and only if $n \leq t_m(H)$. Counting the two distinct residues yields the summand in (25.9). Since the smallest admissible index is p , only $\ell \leq L_H$ contribute. The baseline height $\psi_{R_{M\ell}}(p)$ increases strictly to infinity, proving finiteness.

For $n \geq 1$ the geometric-sum identity is

$$\psi_{R_m}(n) = \varepsilon_m^{n-1} \frac{1 - \varepsilon_m^{-2n}}{1 - \varepsilon_m^{-2}} \geq \varepsilon_m^{n-1}. \quad (25.23)$$

Also $R_m > f_m = \cosh(m\lambda)$, so $\beta_m > m\lambda$. Thus a baseline point under height H satisfies $\log H \geq (p - 1)m\lambda$, proving (25.10). The outer floors are unchanged on replacing t_m by its integer part because their shifts and denominators are integers.

25.7.1 The exact staircase inside one main-index slice

For fixed m , put

$$\mathcal{P}_m(x) = 1 - \left\{ \frac{x-p}{4m} \right\} - \left\{ \frac{x+p}{4m} \right\},$$

where braces denote fractional part. The function is bounded and $4m$ -periodic. For every $H \geq 1$, the fixed- m count is exactly

$$N_m(H) = \frac{t_m(H)}{2m} + \mathcal{P}_m(t_m(H)). \quad (25.24)$$

In particular $N_m(H) = \log H / (2m\beta_m) + O(1)$. The admissible density is $1/(2m)$ among all positive indices, or $1/m$ among odd indices.

The smooth inverse phase has the convergent expansion, for $H\sqrt{D_m} > 1$,

$$t_m(H) = \frac{1}{\beta_m} \left[\log(2\sqrt{D_m}H) + \sum_{k \geq 1} \frac{(-1)^{k-1} \binom{2k}{k}}{2k 4^k D_m^k H^{2k}} \right]. \quad (25.25)$$

To verify it, expand the derivative of $\operatorname{arsinh} x - \log(2x)$ by the convergent binomial series for $x > 1$, and integrate toward infinity. The first terms are $1/(4x^2) - 3/(32x^4) + 5/(96x^6)$. This smooth expansion must not be substituted approximately into a discontinuous floor with a claim of uniform correctness at jumps. The exact phase or exact integer comparisons are needed there.

The ranks within this slice have explicit indices

$$n_{2k+1} = 4mk + p \quad (k \geq 0), \quad n_{2k} = 4mk - p \quad (k \geq 1),$$

and exact heights $\psi_{R_m}(n_v)$. There are no initial repeated heights caused by fixed native coordinates. The least slice triple (j, o, y) occurs at $n = p$; selecting it mathematically is not a new arithmetic gate or a finite-fold implementation.

25.8 The leading coefficient and its exceptional baseline

Before obtaining the second term, it is useful to prove the leading law with a simpler uniform estimate. Write $T = \log H$ and separate

$$N(e^T) = B(T) + Q(T),$$

where B counts $n = p$ over varying m , and Q counts $n = 4mk \pm p$, $k \geq 1$.

Since

$$R_m = \frac{\sqrt{\Delta}}{2} \alpha^m (1 - \alpha^{-2m}),$$

we have $\log R_m = m\lambda + O(1)$. For fixed p , $\psi_R(p)$ is a polynomial of degree $p-1$ with positive leading coefficient 2^{p-1} ; moreover $R^{p-1} \leq \psi_R(p) \leq (2R)^{p-1}$ for $R \geq 2$. Hence

$$\log \psi_{R_m}(p) = (p-1)m\lambda + O(1), \quad B(T) = \frac{T}{M(p-1)\lambda} + O(1). \quad (25.26)$$

The $p-1$ is essential: the varying Pell denominator must not be discarded.

The smallest nonbaseline index is $4m-p$. Since $m > 2p$ and $p \geq 3$, $4m-p-1 \geq 3m$. Equation (25.23) gives

$$\log \psi_{R_m}(4m-p) \geq 3\lambda m^2.$$

Thus nonbaseline slices are supported only at $m \leq J(T) = \sqrt{T/(3\lambda)}$, whereas the baseline occupies $O(T)$ slices.

Uniformly in m and $H \geq 1$,

$$0 \leq t_m(H) - \frac{\log H}{\beta_m} \leq 1. \quad (25.27)$$

The lower bound follows from $\operatorname{arsinh}(H\sqrt{D_m}) \geq \log H$. For the upper bound, $\operatorname{arsinh}(H\sqrt{D_m}) - \log H$ decreases with $H \geq 1$ and equals $\operatorname{arsinh}\sqrt{D_m} = \beta_m$ at $H = 1$. The two floors now give $N_m(e^T) = T/(2m\beta_m) + O(1)$ with an absolute uniform error. Subtracting the baseline indicator preserves this bound. Summing only over $m \leq J(T)$ therefore gives

$$Q(T) = T \sum_{M\ell \leq J(T)} \frac{1}{2M\ell\beta_{M\ell}} + O(\sqrt{T}).$$

Because $\beta_{M\ell} > M\ell\lambda$, the series tail after $K \geq 1$ terms satisfies

$$0 < \sum_{\ell > K} \frac{1}{2M\ell\beta_{M\ell}} \leq \frac{1}{2\lambda M^2 K}. \quad (25.28)$$

Extending the sum costs $O(\sqrt{T})$ and proves the weaker law $N(e^T) = \kappa T + O(\sqrt{T})$. Also

$$0 < \kappa - \frac{1}{M(p-1)\lambda} \leq \frac{\pi^2}{12\lambda M^2}.$$

One cannot obtain the correct coefficient by simply summing fixed-slice leading terms: that misses the exceptional baseline. Nor can one sum a fixed-slice $O(1)$ error over the $O(T)$ baseline range.

25.9 A pointwise second term without smoothing

We now strengthen the remainder and identify its square-root term. The method is the classical hyperbola decomposition and Euler summation, applied to exact Pell heights. The exact sign-specific boundary below avoids any averaging of floor phases.

25.9.1 Exact energy and uniform row and column estimates

For this section set

$$q_0 = M\lambda, \quad b_0 = \frac{1}{2} \log \Delta, \quad a_0 = 4\lambda M^2, \quad \beta_\ell = \beta_{M\ell}, \quad g_\ell = 4M\ell\beta_\ell.$$

The elementary bounds are

$$q_0\ell \leq \beta_\ell \leq q_0\ell + b_0. \quad (25.29)$$

The lower bound was proved above. For the upper bound use $R_{M\ell} = \sqrt{\Delta} \sinh(q_0\ell)$ and $\cosh(q_0\ell + b_0) > \sqrt{\Delta} \sinh(q_0\ell)$.

For $\sigma \in \{-1, 1\}$ define the nonbaseline energy

$$E_\sigma(\ell, k) = \log \psi_{R_{M\ell}}(4M\ell k + \sigma p), \quad \ell, k \geq 1.$$

Taking logarithms in (25.23) gives the exact identity

$$E_\sigma(\ell, k) = g_\ell k + (\sigma p - 1)\beta_\ell + \theta_{\ell, k, \sigma}, \quad 0 \leq \theta_{\ell, k, \sigma} \leq d_0, \quad (25.30)$$

where $d_0 = -\log(1 - e^{-2q_0})$. The bound holds because every normalized index here is at least p . In particular

$$|E_\sigma(\ell, k) - a_0 k \ell^2| \leq C k \ell, \quad C = 4M b_0 + (p+1)(q_0 + b_0) + d_0. \quad (25.31)$$

Also $g_\ell = a_0 \ell^2 + O(\ell)$, so

$$S = \sum_{\ell \geq 1} \frac{1}{g_\ell} < \infty, \quad \sum_{\ell > L} \frac{1}{g_\ell} = \frac{1}{a_0 L} + O(L^{-2}). \quad (25.32)$$

The exact series is retained here. Replacing every β_ℓ by $q_0 \ell$ in its definition would generally change the linear coefficient.

Let $q_\sigma(T, \ell)$ be the number of $k \geq 1$ with $E_\sigma(\ell, k) \leq T$. Exact inversion gives

$$q_\sigma(T, \ell) = \max \left(0, \left\lfloor \frac{t_{M\ell}(e^T) - \sigma p}{4M\ell} \right\rfloor \right).$$

By (25.27), uniformly for $T \geq 0$ and $\ell \geq 1$,

$$q_\sigma(T, \ell) = T/g_\ell + O(1). \quad (25.33)$$

The positive-part clipping and floor each cost a bounded amount, since $T/g_\ell \geq 0$ and $p/(4M\ell)$ is uniformly bounded.

Let $r_\sigma(T, k)$ be the number of $\ell \geq 1$ with $E_\sigma(\ell, k) \leq T$, and put $x = \sqrt{T/(a_0 k)}$ and $D_0 = C/a_0$. From (25.31),

$$\ell^2 - D_0 \ell \leq E_\sigma(\ell, k)/(a_0 k) \leq \ell^2 + D_0 \ell.$$

Every integer $\ell \leq x - D_0$ is admitted, and every admitted ℓ satisfies $\ell \leq x + D_0$. Including endpoint rounding and the possible empty range proves the uniform estimate

$$r_\sigma(T, k) = \sqrt{T/(a_0 k)} + O(1). \quad (25.34)$$

This estimates the count directly. It never assumes that a floor remains unchanged after replacing its argument by an asymptotic approximation.

The energy is strictly increasing in each positive integer variable. For fixed k , both $R_{M\ell}$ and $4M\ell k + \sigma p > 1$ increase with ℓ ; $\psi_R(n)$ increases with R and n in this range. Finiteness follows, for example, from

$$E_\sigma(\ell, k) \geq (4M\ell k - p - 1)M\ell \lambda \geq 3\lambda M^2 k \ell^2.$$

25.9.2 An exact boundary rectangle

For sufficiently large T set

$$L = \lfloor T^{1/3} \rfloor, \quad K_\sigma = q_\sigma(T, L).$$

Equations (25.33) and (25.29) give

$$K_\sigma = \frac{T}{a_0 L^2} + O(T/L^3 + 1) = \frac{T}{a_0 L^2} + O(1) \asymp T^{1/3}. \quad (25.35)$$

Let $Q_\sigma(T)$ count all points of sign σ . Monotonicity gives the exact hyperbola identity

$$Q_\sigma(T) = \sum_{\ell \leq L} q_\sigma(T, \ell) + \sum_{k \leq K_\sigma} r_\sigma(T, k) - L K_\sigma. \quad (25.36)$$

Every point in $1 \leq \ell \leq L$, $1 \leq k \leq K_\sigma$ is admitted because $E_\sigma(L, K_\sigma) \leq T$. No point with both $\ell > L$ and $k > K_\sigma$ is admitted, since already $E_\sigma(L, K_\sigma + 1) > T$. The two strips therefore cover the region and overlap in exactly that rectangle. This argument remains true when T is exactly an energy, and uses a separate exact cutoff for each sign.

Put $u = \sqrt{T/a_0}$. Substituting (25.32), (25.33), and (25.34) into (25.36) gives

$$Q_\sigma(T) = ST - \frac{T}{a_0 L} + u \sum_{k \leq K_\sigma} k^{-1/2} - LK_\sigma + O(L + T/L^2 + K_\sigma). \quad (25.37)$$

The classical Euler-summation formula is

$$\sum_{k=1}^K k^{-1/2} = 2\sqrt{K} + \zeta(1/2) + O(K^{-1/2}). \quad (25.38)$$

For completeness, the identity recorded in DLMF 25.2.8 [6] at $s = 1/2$ is

$$\zeta(1/2) = \sum_{k=1}^K k^{-1/2} - 2\sqrt{K} - \frac{1}{2} \int_K^\infty \{x\} x^{-3/2} dx.$$

The integral contribution has absolute value at most $K^{-1/2}$, proving (25.38). The alternating-series identity DLMF 25.2.3 gives $\zeta(1/2) < 0$ because its numerator is positive and its factor $1 - \sqrt{2}$ is negative.

The apparent boundary mismatch cancels quadratically:

$$-\frac{T}{a_0 L} + 2u\sqrt{K_\sigma} - LK_\sigma = -L \left(\sqrt{K_\sigma} - u/L \right)^2 = O(1). \quad (25.39)$$

Indeed (25.35) gives $K_\sigma - u^2/L^2 = O(1)$, and hence the squared difference is $O(T^{-1/3})$. The Euler-sum error is $O(u/\sqrt{K_\sigma}) = O(T^{1/3})$, as are all other errors in (25.37). Consequently

$$Q_\sigma(T) = ST + \frac{\zeta(1/2)}{2M\sqrt{\lambda}} \sqrt{T} + O(T^{1/3}). \quad (25.40)$$

The signs are disjoint, and disjoint from the exceptional baseline. Adding both copies of (25.40) to (25.26) proves (25.12), since $2S$ is the series term in (25.11).

This proof is unsmoothed and pointwise. The fixed-slice staircases remain present, but only $O(T^{1/3})$ short rows and columns contribute independent bounded discrepancies. No generic-position assumption, equidistribution claim, or unproved cancellation of floor phases is used.

25.10 Ranked heights and binary length

Let $F(T) = N(e^T)$, and write $d = d_*^{\text{count}}$. The fiber has finitely many tuples below any finite height and is infinite, so the nondecreasing ranked heights H_v exist and tend to infinity. Let $T_v = \log H_v$. Irrespective of any ties,

$$F(T_v - 1) < v \leq F(T_v). \quad (25.41)$$

The lower count is strictly below the height of the v -th tuple; no bound on jump multiplicity is needed. Applying (25.12) at both arguments gives

$$v = \kappa T_v + d\sqrt{T_v} + O(T_v^{1/3}).$$

First $T_v \sim v/\kappa$, then $T_v - v/\kappa = O(\sqrt{v})$, and therefore $\sqrt{T_v} = \sqrt{v/\kappa} + O(1)$. Substituting gives (25.13).

Equivalently, for a suitable fixed constant C and all sufficiently large v ,

$$\exp\left(\frac{v}{\kappa} - \frac{d}{\kappa^{3/2}}\sqrt{v} - Cv^{1/3}\right) \leq H_v \leq \exp\left(\frac{v}{\kappa} - \frac{d}{\kappa^{3/2}}\sqrt{v} + Cv^{1/3}\right).$$

The square-root correction is positive because $d < 0$. The unbounded $O(v^{1/3})$ uncertainty in the exponent is why a relative $1 + o(1)$ height equivalent has not been proved.

For maximum binary coordinate length at most the integer $b \geq 1$, substitute $H = 2^b - 1$ to obtain

$$N_{\text{bits}}(b) = \kappa(\log 2)b + \frac{\zeta(1/2)\sqrt{\log 2}}{M\sqrt{\lambda}}\sqrt{b} + O(b^{1/3}).$$

This is maximum bit length, not the sum of coordinate lengths; those are different size conventions.

25.11 The sum of all coordinate bit lengths

There is a different two-term law when the budget is the sum of all twenty-two coordinate bit lengths. For $x \in \mathbb{Z}_{>0}$ let $b(x) = \lfloor \log_2 x \rfloor + 1$, and set

$$S(w) = \sum_{x \text{ a supplied coordinate of } w} b(x), \quad N_{\text{sum}}(B) = \#\{w \in \mathcal{F} : S(w) \leq B\}.$$

This is a raw unsigned coordinate-bit budget: it includes all twenty-two supplied coordinates and excludes delimiters, signs, metadata and computed registers.

Theorem 25.11.1 (Total coordinate-bit budget). *For fixed valid native ports and real $B \rightarrow \infty$,*

$$N_{\text{sum}}(B) = \kappa_s(\log 2)B + \frac{\zeta(1/2)\sqrt{\log 2}}{M\sqrt{3\lambda}}\sqrt{B} + O(B^{1/3}), \quad (25.42)$$

where

$$\kappa_s = \frac{1}{M(3p-2)\lambda} + \sum_{\ell \geq 1} \frac{1}{6M\ell\beta_{M\ell}} = \frac{\kappa}{3} - \frac{1}{M\lambda(3p-2)(3p-3)}. \quad (25.43)$$

All exact integer cutoffs and tied tuples are included.

Proof. The reconstruction formulas have the exact product cancellation

$$fijoy = \frac{R_m y (U + p)(U + c)}{c^3}. \quad (25.44)$$

Put $\beta = \beta_m$ and $q_0 = M\lambda$. The hyperbolic expressions give

$$\log R_m = \beta - \log 2 + r_m, \quad \log y = (n-1)\beta + \theta_{m,n}, \quad \log U = (n-1)\beta + u_{m,n},$$

with, uniformly on every admissible pair,

$$0 \leq r_m = \log(1 + e^{-2\beta}) \leq r_* := \log(1 + e^{-2q_0}), \\ 0 \leq \theta_{m,n} \leq d_0 := -\log(1 - e^{-2q_0}), \quad -r_* \leq u_{m,n} = \log(1 + e^{-2n\beta}) - r_m \leq 0.$$

Also $U \geq 1$. Taking logarithms in (25.44) therefore proves

$$\log(fijoy) = (3n-2)\beta_m + O(1) \quad (25.45)$$

uniformly over the entire pair family. For example the absolute error is bounded by $\log 2 + 3 \log c + 3r_* + d_0 + \log(1+p) + \log(1+c)$.

Define the comparison energy $E_0(m, n) = (3n - 2)\beta_m$, and let $G(T)$ count admissible pairs with $E_0 \leq T$. For $n = p$,

$$E_0(M\ell, p) = (3p - 2)M\lambda\ell + O(1),$$

so its count is $T/[M(3p - 2)\lambda] + O(1)$. For each sign, the exact nonbaseline energy is

$$E_{0,\sigma}(\ell, k) = 12M\ell\beta_{M\ell}k + (3\sigma p - 2)\beta_{M\ell}.$$

Use the proof of Section 25.9 with g_ℓ replaced by $12M\ell\beta_{M\ell}$ and a_0 by $12\lambda M^2$. The row estimate is still $T/g_\ell + O(1)$ because the offset divided by g_ℓ is $(3\sigma p - 2)/(12M\ell)$. The reciprocal tail is still $1/(a_0L) + O(L^{-2})$, and $|E_{0,\sigma} - a_0k\ell^2| \leq Ck\ell$ follows from (25.29). The energy is strictly increasing in both variables, so the exact sign-specific rectangle (25.36) applies unchanged. The same Euler sum and quadratic cancellation yield

$$G(T) = \kappa_s T + \frac{\zeta(1/2)}{M\sqrt{3\lambda}}\sqrt{T} + O(T^{1/3}).$$

In particular, the exceptional baseline has coefficient $3p - 2$, not $3(p - 1)$, which proves the correction in (25.43).

Let C_{bits} be the fixed sum of the seventeen forced bit lengths. For the five varying coordinates,

$$\log_2(fijoy) < b(f) + b(i) + b(j) + b(o) + b(y) \leq \log_2(fijoy) + 5.$$

Together with (25.45), this gives a fixed constant C for which

$$|(\log 2)S(w) - E_0(m, n)| \leq C.$$

Hence, for every sufficiently large real B ,

$$G((\log 2)B - C) \leq N_{\text{sum}}(B) \leq G((\log 2)B + C).$$

Apply the asymptotic for G to both sides. A fixed shift changes its linear term by $O(1)$ and its square-root term by $O(B^{-1/2})$, proving (25.42). This global sandwich includes powers of two and every rounding boundary; individual floors need not be stable. $\square$

If B_v is the v -th total bit length, in its own nondecreasing order with multiplicity, put

$$a_s = \kappa_s \log 2, \quad e_s = \frac{\zeta(1/2)\sqrt{\log 2}}{M\sqrt{3\lambda}}.$$

The inequalities $N_{\text{sum}}(B_v - 1) < v \leq N_{\text{sum}}(B_v)$ and the same inversion as in Section 25.10 give

$$B_v = \frac{v}{a_s} - \frac{e_s}{a_s^{3/2}}\sqrt{v} + O(v^{1/3}).$$

The ordering by total bits need not agree with the ordering by maximum height. Both formulas are for their stated encoding convention.

25.12 Transport to the canonical outer history

This section is a consequence of the original Report 22’s entire-fiber projection theorem [7], rather than a new audit of its compiler or physical-clock construction.

[write] Report 22’s entire-fiber projection theorem is printed with Report 22 (source 23, Part VIII of `signal-machine-collision-certificates`). The corollary below is conditional on it and was not re-audited in this write.

[write] Added 3 October 2026 (batch 82, reciprocal note). Part VIII of `signal-machine-collision-certificates` is now written: there, Report 22’s entire-fiber projection is Theorem `smc:ch:thm:projection` (154.1) and its fixed-scale native infinitude Theorem `smc:ch:thm:infinite` (155.1), with a note crediting Theorem 25.2.1.

Fix an accepted nonempty input and one of its retained clock models. Determinism and first-halt semantics fix the whole trace, final payload and physical clock. The selected outer height is the unique power of two in $(S, 2S]$, where S is the sum of the two encoded endpoint values and the physical clock. The physical duration bounds already dominate every true quotient, so the packed trace fits at this exact height. The retained no-wrap argument recovers the exact clock from its congruence. All outer supplied coordinates and all native boundary ports are thereby fixed.

Write w_{out} for this unique outer tuple. Report 22 checked that the five nonnative comparisons have no native witness dependency and that the full native block is retained literally. Projection to the twenty-two native coordinates is therefore a bijection: two complete witnesses with the same native projection have the same fixed outer tuple, and adjoining w_{out} to any native witness makes all native and outer comparisons vanish.

Corollary 25.12.1 (Conditional transport through the established outer bijection). *For each such accepted nonempty input, all complete witnesses are*

$$(w_{\text{out}}, w_{m,n}), \quad (m, n) \text{ as in (25.6).}$$

If C_{out} is the maximum supplied outer coordinate, their complete height is

$$\max(C_{\text{out}}, \psi_{R_m}(n)).$$

Their count is zero below C_{out} and equals $N(H)$ for $H \geq C_{\text{out}}$. Thus the two-term asymptotic and logarithmic ranked-height law have the same coefficients. At finite ranks the complete height is $\max(C_{\text{out}}, H_v)$.

This is the only plateau possibility discussed here: it comes from additional fixed outer coordinates. It is distinct from the native height law (25.8). The predecessor’s separate no-witness initial-halt circuits are outside this nonempty-fiber corollary.

25.13 Reproduction and scope

The packet distinguishes a mathematical proof, finite auxiliary evidence, and a sealed reproducibility artifact. Its README gives the precise commands and inventory. The core checker independently audits every literal native comparison with sparse integer-polynomial arithmetic, verifies the required pinned inputs, and uses explicit failures that remain active under optimized Python. It imports no

upstream author program. Exact integer tests check the main-index divisibility equivalence, positive reconstruction, congruence residues, height domination and count values immediately below, at and above Pell-height jumps.

The second-term verification supplements the analytic proof with exact boundary tests and numerical auxiliary models. Small test values are not examples of a complete padded native tuple unless the native hypotheses have separately been checked; the astronomical native tuple has not been materialized. Floating-point agreement is never used to prove the classification, floors at a boundary, or the asymptotic theorem.

The release includes source provenance, independently reviewed mathematical statements, portable checker adaptations, deterministic receipts and a strict manifest. Replays run the substantive checks in both normal and optimized Python, test rejection of altered or missing required material, and compare the entire release inventory and bytes before and after. A fresh extracted ZIP is the final replay target. The PDF build requires a local LaTeX installation; the verification code itself uses only the Python standard library. Source hashes and any portable text/code adaptation hashes are kept distinct in the provenance record.

25.13.1 Attribution and limits

Pell generation, strong divisibility, step-down congruences, fixed-base logarithmic counting and Euler–Maclaurin summation are classical. The native repository already contains the crucial relaxed-norm integrality recovery, fixed-minus necessity and parity analysis, and a positive converse. The present compiler-specific deduction is the exact seventeen/five split, the full two-index parameter set with its converse, native maximum-coordinate domination, the baseline-corrected whole-fiber coefficient, and the pointwise two-term count for that exact fiber.

These statements do not change the native circuit, add an uncharged operation, prove circuit minimality, establish a general finite-fold MRDP result, or produce a full numerical Pell witness. Nor is a literature-wide open-problem or novelty claim made. The asymptotic parameters are fixed: neither the error constants nor its onset are asserted uniform across different external inputs.

Binary powering gives effective evaluation and exact comparison in the two quadratic rings. One can locate integer cutoffs by monotone searches rather than floating-point inversion. This is computability, not a small-witness claim: a coordinate at normalized index n requires $\Theta(n \log \varepsilon_m)$ output bits, and the fixed native coordinates can already be enormous. The least-height pair and exact counts are mathematical descriptions; they do not implement a canonical choice by the retained polynomial.

25.13.2 Further mathematical questions

Several bounded questions remain. Can the $O((\log H)^{1/3})$ error be improved, or can further oscillatory terms be identified without smoothing? Can the onset and constants be controlled uniformly while the external ports vary? What is the cost of enforcing a preferred auxiliary-index choice in the retained arithmetic model, with every operation charged? How do other explicit encodings change the coefficient and secondary term? None of these improvements is claimed here.

References of Report 25

- [1] V. Reshetnikov, *Native controller binary selector 56*, pinned repository proof, Sections 2–3 and 5. [Pinned source](#). Local snapshot: `native_controller_binary_selector56.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.

- [2] V. Reshetnikov, *Native binary masked selection 63*, prescribed-scale AND interface, Section 2. [Pinned source](#). Local snapshot: `native_binary_masked_selection63.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
- [3] V. Reshetnikov, *Recovering integrality after relaxing an auxiliary Pell norm*. [Pinned source](#). Local snapshot: `PELL_RELAXED_AUXILIARY_PROOF.md`. [write] Shipped as `07-native-fiber-source-PELL_RELAXED_AUXILIARY_PROOF.md`; it differs from the research-tree file only by one added final newline.
- [4] V. Reshetnikov, *Fixed-minus index parity*, Sections 2–3. [Pinned source](#). Local snapshot: `EXPLORATION_FIXED_MINUS_INDEX_PARITY.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
- [5] V. Reshetnikov, *Odd-index Pell signs*. [Pinned source](#). Local snapshot: `EXPLORATION_ODD_INDEX_PELL_SIGNS.md`. [write] The local snapshot is a byte copy of the research-tree file and is not shipped.
- [6] NIST Digital Library of Mathematical Functions, Section 25.2, [equation 25.2.8](#) and [equation 25.2.3](#). Accessed 3 October 2026.
- [7] *Canonical outer histories and infinite native fibers*, Research report 22, original release, 3 October 2026. Entire-fiber projection theorem and its dependency audit. This prior result is used only for Section 25.12; its original release remains unchanged. [write] Batch 82, cluster M2; source 23 of `signal-machine-collision-certificates`, in its Part VIII (placed by commit 7d2b1b245).

Part IV

Failure of positive index restoration, and the height and counting of one counterexample family

[write] This Part prints two manuscripts. Report 33 (Sections 33.*k*) proves that the signed₁₉ child of the complete₇₄ parent has positive zeros with negative restored index at every positive input; Report 34 (Sections 34.*k*) counts one canonical family of those zeros. Report 34 is a direct sequel of Report 33 and shipped Report 33’s source, byte-identical to the text printed here, as context.

Report 33. Failure of Positive Index Restoration: the complete₇₄ signed₁₉ eliminated child

[write] Source 10 of batch 82: *Research Report 33*, author line “Mathematical research report”, dated 3 October 2026; archive `Failure_of_Positive_Index_Restoration_Package.zip`; pin 2dde7850f, context 93ad43375.

Abstract of the source. Deleting a positive Diophantine witness by a polynomial graph substitution need not preserve the positive existential projection. We prove that the complete₇₄ signed₁₉ index-eliminated child has infinitely many strictly positive supplied-witness zeros at every strictly positive ordinary input, for every actual fixed compiler export in the pinned source. At every constructed zero, the uniquely restored index is negative. The proof incorporates the full transport, shifted packing, first and main Pell norms, strong auxiliary block, and signed input block. It uses one prime in a primitive progression, compatible linear congruences, an irrational rotation, and exact Pell identities. All nineteen supplied child coordinates are positive; a negative computed intermediate is permitted. The established parent theorem and the previously audited graph identity remain intact. No improved universal operation bound, machine-specific false acceptance, or fully materialized full child witness is claimed. A portable independently authored finite-check packet accompanies the proof and is explicitly separated from the unbounded existence argument.

[write] The programme’s review `review_incoming_negative_index_fibers.md` (commit 0055e1c4d) finds no gap in the construction of Sections 33.4–33.9 and also passes the generalized supplement of Section 33.A. Thirty-eight members of the archive are byte copies of research-tree files (README). In this file Report 33’s macro for the native mask is `\MFnat` and for the paid mask `\MFsrc` (Section 0.2). The source’s files are shipped with the prefix `10-index-restore-`.

33.1 Result and exact scope

The relevant source is pinned to the ProveIt repository at commit

2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff.

The nonlinear-index scout removes one positive supplied coordinate from each of three complete₇₄ parent presentations. It correctly proves an all-value polynomial graph identity, but leaves positivity of the restored coordinate open. This report resolves that open obligation negatively for its signed₁₉ child. The names *signed₂₀* and *signed₁₉* concern allowed signed intermediate registers. Their supplied existential coordinates are still required to be strictly positive integers.

[write] The research tree now records this: its review `review_incoming_negative_index_fibers.md` (commit 0055e1c4d) finds no gap in the construction and states that Report 33 “settles the signed19 case”; the programme’s README calls the signed19 candidate refuted, and raw29 and positive21 open. The tree’s later refinement `complete74_negative_index_refinement.md` (commit 813c1cff4) had excluded the signed child from its own scope (“Domain limitation”).

Theorem 33.1.1 (Full positive child zeros with negative restoration). *Fix any actual half-binomial compiler export used by the complete74 signed parent. In particular let $B = 2^d \geq 16$, with d a power of five, let b be positive and odd, and let $K_0 = D_C + BD_R \geq 0$. Retain the actual fixed masks M_C, M_F^{native} unchanged and put $M_F^{\text{src}} = M_F^{\text{native}} + B - 1$. For every $x \in \mathbb{Z}_{>0}$, the complete signed19 child has infinitely many zeros in its nineteen strictly positive supplied coordinates such that*

$$R = k - hXY - 1 < 0.$$

The computed input root μ is positive at these zeros.

The source additionally specifies $b < B$, $0 < M_C, M_F^{\text{native}} < B - 1$, and the appropriate fixed mask congruences. The construction below uses no alteration of these numerals and works for every such export. In fact it only needs the displayed structural conditions on B, d, b, K_0 , with the remaining mask numerals fixed integers. The stronger compiler facts are retained when interpreting the result as a theorem about that particular source.

Corollary 33.1.2. *The positive existential projection of this signed19 child onto its positive ordinary input is all of $\mathbb{Z}_{>0}$. The proposed restoration map from all positive child zeros to positive parent zeros fails.*

This is a failure of the proposed positive-domain equivalence. The original source audit did not assert that equivalence. The positive parent retains $R > 0$ as a hypothesis; no constructed assignment here belongs to that parent domain. The report makes no claim about the acceptance status of a particular input for a chosen machine. It also makes no corresponding existence or nonexistence claim for the separate raw29 and positive21 children.

33.1.1 Operation counts are context only

The pinned scout and its source review give the following costs for the emitted children, using the original parent labels. These are source accounting statements, not universal bounds newly established here.

Parent label	Child witnesses	Equations	Full SOS cost	Degree
raw30	29	18	127	52
positive22	21	10	103	84
signed20	19	8	97	84

Each comparison schedule is $40M + 34A = 74$, and the full sum-of-squares cost is $74 + 3e - 1$ for e comparisons. Thus the signed candidate has $48M + 49A = 97$. These figures do not certify a universal representation on the intended positive domain: Theorem 33.1.1 invalidates this signed candidate’s proposed restoration route. The established universal74 and universal86 parent bounds are not changed by this report.

[write] At the pin the programme's established bounds were 74 operations for a complete certificate and 86 for one universal polynomial. At the batch-82 write they were 74 and 85 (commit 109aaca15, exact degree 175), with an 86-operation alternative of exact degree 131 (commit 55d248dc5). (*Batch 83.*) Since commit 20aafb9a5 the universal polynomial has 84 operations (exact degree 187, with the same positive zeros as the 85-operation one) and the certificate bound is still 74. Report 33 changes none of them.

33.2 The exact polynomial interface

We state the entire child interface so that neither its scope nor a hidden sign convention can be imported from a different presentation. The ordinary input x and precisely the coordinates

$$J, F, \alpha, z, f, h, i, j, o, s, w, \tau, \eta, \zeta, y, Z, \delta, \rho, \sigma \text{ belong to } \mathbb{Z}_{>0}. \quad (33.1)$$

All other quantities below are computed expressions; they are not additional existential coordinates. Define

$$q = (B - 1)J + 1, \quad X = wq^3, \quad Y = sq^3, \quad E = XY. \quad (33.2)$$

$$k = \eta + \zeta, \quad a = Y(X + 1), \quad c = kY + \eta, \quad (33.3)$$

$$A = a + 2, \quad H = 4a + 3, \quad \Delta = a^2 + H = A^2 - 1, \quad (33.4)$$

$$\gamma = \rho + \sigma, \quad D = X + ac + \gamma H, \quad R = k - hE - 1, \quad (33.5)$$

$$u = 2dx + b, \quad \kappa = u + \delta\Delta, \quad C = q - \alpha - 2dx, \quad (33.5)$$

$$W = C - Z, \quad \mu = W + a\kappa + \rho H.$$

For clarity, the source's register named **A** is our Δ , whereas our $A = a + 2$ is a mathematical abbreviation. The source's **Jrep**, **zquot**, and **y_aux** are J, z, y ; its **R10b**, **R10a**, **R12**, and **R14** are k, c, a, D . Its paid field-mask port is M_F^{src} , not the native mask M_F^{native} .

The eight retained equations are exactly

$$(K_0 + X)C = F + z(q - 1), \quad (33.6)$$

$$R = (q^2 - Z - qF)(q^2 - 1) + (M_C + qM_F^{\text{src}})J, \quad (33.7)$$

$$\tau^2 - XY^2(XY^2 + 1)k^2 = 1, \quad (33.8)$$

$$D^2 - \Delta c^2 = 1, \quad (33.9)$$

$$(ic^2)^2 = \Delta(f^2 - 1), \quad (33.10)$$

$$(ic^2)^2((jc - R)^2 - y^2) = 1 - y^2, \quad (33.11)$$

$$jc - R = of - c, \quad (33.12)$$

$$\mu^2 - \Delta\kappa^2 = 1. \quad (33.13)$$

The child polynomial F_{19} is the sum of the squares of their residuals. Over the integers, it vanishes if and only if all eight equations hold. The first factored paid comparison is

$$(EkY)(EkY + k) = \tau^2 - 1.$$

Because $E = XY$, its left side equals $k^2XY^2(XY^2 + 1)$, exactly as in (33.8). No weaker auxiliary square has been substituted in (33.10) or (33.11).

33.2.1 Why the graph identity does not supply positivity

In the signed20 parent, R is itself a positive supplied coordinate, and there is a ninth equation $k = R + 1 + hE$. Its index-eliminated descendant substitutes $R = k - hE - 1$ in both packing and $jc - R$, then removes only that now-identical ninth residual. Consequently

$$F_{19}(v) = F_{20}(v, R = k(v) - hE(v) - 1)$$

as a polynomial identity over every commutative ring. On integer zero sets it gives inverse deletion and restoration maps without sign restrictions. On positive zero sets, the inverse is valid exactly on the slice $k > hE + 1$. Theorem 33.1.1 supplies positive child zeros outside this slice.

The source proof for the parent uses positive R before deriving a packing bound. More explicitly, its shifted-mask rewrite is

$$\begin{aligned} S' &= Z + qF - 1, \\ T' &= M_C J + 1 + q(M_F^{\text{native}} J - 1), \\ R &= (q^2 - S')(q^2 - 1) + T', \quad 0 < T' < q^2 - 1. \end{aligned}$$

Positive R rules out $S' > q^2$. Without that hypothesis the same formula permits negative representatives. Importing the parent's consequent bounds into the eliminated child would be circular.

33.3 Elementary Pell identities used in the proof

For an integer $A \geq 2$ and an integer $r \geq 0$, define $\chi_A(r), \psi_A(r)$ by

$$\chi_A(r) + \psi_A(r)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^r. \quad (33.14)$$

These are integer polynomials in A . Their initial values are $(\chi_A(0), \psi_A(0)) = (1, 0)$ and $(\chi_A(1), \psi_A(1)) = (A, 1)$; both obey $v_{r+1} = 2Av_r - v_{r-1}$. The positive coefficient sequence $\psi_A(r)$ is strictly increasing for $r \geq 1$.

Lemma 33.3.1 (Norms and congruences). *Write $\Delta = A^2 - 1$. For nonnegative indices,*

$$\chi_A(r)^2 - \Delta\psi_A(r)^2 = 1, \quad (33.15)$$

$$\psi_A(r+s) = \psi_A(r)\chi_A(s) + \chi_A(r)\psi_A(s), \quad (33.16)$$

$$\psi_A(r) \equiv rA^{r-1} \pmod{\Delta} \quad (r \geq 1). \quad (33.17)$$

If A is even, $\psi_A(r) \equiv r \pmod{2}$. If $A \equiv 1 \pmod{m}$, then $\psi_A(r) \equiv r \pmod{m}$. If $r \geq 3$ is odd, then $\psi_A(r) > r$ and $\psi_A(r) \equiv r \pmod{\Delta}$.

Proof. Multiplication of (33.14) proves the addition law; conjugation proves the norm. In the binomial expansion the only term surviving in the square-root coefficient modulo Δ is rA^{r-1} . For odd r , $A^{r-1} \equiv 1 \pmod{\Delta}$. The stated parity and modulus- m formulas follow from the recurrence and initial values. Finally $\psi_A(2) = 2A > 2$, and induction in the recurrence gives $\psi_A(r+1) > \psi_A(r) + 1$ for $r \geq 2$. $\square$

Lemma 33.3.2 (Projection congruence and growth). *Let $a \geq 1$, $A = a + 2$, $H = 4a + 3$, and set*

$$e_A(r) = \chi_A(r) - a\psi_A(r).$$

Then $e_A(0) = 1$, $e_A(1) = 2$, and

$$e_A(r) \equiv 2^r \pmod{H}, \quad e_A(r) = 2\psi_A(r) - \psi_A(r-1) > \psi_A(r) \quad (r \geq 1). \quad (33.18)$$

In particular $e_A(r)$ grows exponentially with r for fixed A .

Proof. The sequence e_A has recurrence coefficient $2A$. The candidate 2^r satisfies that recurrence modulo H because $2A \cdot 2 - 1 = 4a + 7 \equiv 4 \pmod{H}$, and its initial values agree. The standard recurrence identity $\chi_A(r) = A\psi_A(r) - \psi_A(r-1)$ gives the second formula; strict growth of ψ_A gives the inequality. Formula (33.14), or its conjugate, gives exponential growth. $\square$

Lemma 33.3.3 (Odd quotient polynomial). *There is an integer polynomial $\mathcal{Q}_b(z)$ for each $b \geq 0$ such that*

$$\chi_Q(2b+1)/Q = \mathcal{Q}_b(Q^2), \quad (33.19)$$

$$\mathcal{Q}_b(0) = (-1)^b(2b+1), \quad (33.20)$$

$$\mathcal{Q}_b(1-A^2) = (-1)^b\psi_A(2b+1). \quad (33.21)$$

The first identity is a polynomial identity, with the quotient interpreted polynomially even at $Q = 0$.

Proof. The odd polynomial $\chi_Q(2b+1)$ is divisible by Q . Its quotient is a polynomial in Q^2 . More explicitly,

$$\mathcal{Q}_0(z) = 1, \quad \mathcal{Q}_1(z) = 4z - 3, \quad \mathcal{Q}_{b+1}(z) = (4z - 2)\mathcal{Q}_b(z) - \mathcal{Q}_{b-1}(z).$$

This follows from the two-step Pell recurrence. At $z = 0$ the initial values and recurrence give (33.20). At $z = 1 - A^2$, its recurrence coefficient is $2 - 4A^2$. The sequence $(-1)^b\psi_A(2b+1)$ has this coefficient and the same initial values $1, 1 - 4A^2$, proving (33.21). $\square$

Lemma 33.3.4 (A divisibility lift). *If $c = \psi_A(p)$ and $m = cp$ with $p, c \geq 1$, then $c^2 \mid \psi_A(m)$.*

Proof. Write $D = \chi_A(p)$. The square-root coefficient in $(D + c\sqrt{\Delta})^c$ is $\psi_A(cp)$. Its term with one square root is $c^2 D^{c-1}$. Every higher odd term contains at least c^3 . Thus each term is divisible by c^2 . $\square$

33.4 Fixing the input and a compatible scale

Fix an arbitrary $x \in \mathbb{Z}_{>0}$ and an actual fixed compiler export. Choose a power of five N large enough that

$$q = B^N > 2dx + 1.$$

Put

$$\begin{aligned} t &= dN, & J &= \frac{q-1}{B-1}, & p_0 &= 12t + 7, \\ X &= 2^{p_0}, & w &= 2^{p_0-3t}, & Q &= q^2 - 1. \end{aligned} \quad (33.22)$$

Here t is a power of five, $q = 2^t$, $p_0 \equiv 3 \pmod{4}$, and $p_0 \equiv 1 \pmod{6}$. Thus J, w are strictly positive integers and $X = wq^3$. Set

$$C = 1, \quad \alpha = q - 1 - 2dx, \quad z = 1, \quad F = K_0 + X - q + 1, \quad M = (M_C + qM_F^{\text{src}})J. \quad (33.23)$$

The coordinates α, z, F are positive, since $X \geq q^3 > q$ and $K_0 \geq 0$. The definition of C and the transport equation (33.6) are now exact. The fixed input index

$$u = 2dx + b$$

is odd and at least three.

33.4.1 A prime progression removes the remaining congruence obstruction

Define

$$D_0 = \frac{4q^3(X+1)}{3}. \quad (33.24)$$

This is an integer because p_0 is odd.

Lemma 33.4.1. *One has $\gcd(D_0, Q) = 1$.*

Proof. Since t is a power of five, $\gcd(p_0, 2t) = \gcd(12t + 7, 2t) = 1$. We show $\gcd(2^{p_0} + 1, 2^{2t} - 1) = 3$. Any common divisor g is odd. Choose integers r, s with $rp_0 + 2st = 1$. The integer r is odd, so, using inverses modulo g if necessary, $2 = (2^{p_0})^r (2^{2t})^s \equiv (-1)^r = -1 \pmod{g}$. Thus $g \mid 3$, and both expressions are divisible by three. Moreover $p_0 \equiv 1 \pmod{6}$ gives $X + 1 \equiv 3 \pmod{9}$. Removing its sole factor of three makes $(X + 1)/3$ coprime to Q . Since Q is odd and coprime to q , (33.24) proves the assertion. $\square$

For each prime divisor r of the odd integer Q , avoid the two distinct residues 0 and $-D_0^{-1}$ modulo r . This is possible because $r \geq 3$. The Chinese remainder theorem, followed by any lift to the prime powers if needed, provides s_0 with

$$\gcd(s_0, Q) = \gcd(1 + D_0 s_0, Q) = 1. \quad (33.25)$$

The progression

$$\ell \equiv 1 + D_0 s_0 \pmod{D_0 Q} \quad (33.26)$$

is primitive: its residue is one modulo D_0 and a unit modulo Q . Dirichlet's theorem therefore supplies arbitrarily large primes in it [1]. Fix one with $\ell > 3$ and put

$$\begin{aligned} s &= (\ell - 1)/D_0, & Y &= sq^3, & E &= XY, \\ a &= Y(X + 1), & A &= a + 2, & H &= 4a + 3 = 3\ell, \\ \Delta &= A^2 - 1, & P &= 2XY^2 + 1. \end{aligned} \quad (33.27)$$

The coordinate s is a positive integer congruent to s_0 modulo Q . In particular it is coprime to Q and not divisible by three. The latter follows because $3 \mid Q$. As D_0 is also a unit modulo three and ℓ is a unit modulo three, $\ell - 1 = D_0 s$ is nonzero modulo three; consequently $\ell \equiv 2 \pmod{3}$.

Let $O = \text{ord}_H(2)$, which exists because H is odd. Since $H = 3\ell$,

$$O \mid \text{lcm}(2, \ell - 1) = \ell - 1 = D_0 s.$$

The number $D_0 s$ is coprime to Q , to three, and to ℓ , so $\gcd(QH, O) = 1$. Separately $\gcd(QH, 2E) = 1$: the factors of Q do not divide s or the powers of two X, q ; neither three nor ℓ divides s , and both are odd. Therefore, with

$$T = \text{lcm}(4, 2E, O), \quad L = E/2, \quad (33.28)$$

we have the crucial compatibility

$$\gcd(QH, T) = 1. \quad (33.29)$$

The scale, including this one chosen prime, is now held fixed. Dirichlet's theorem is used only here; the infinitely many full child zeros below come from varying the main Pell index.

33.5 The entire outer and input block in one free progression

Before choosing the main index, set

$$\kappa = \psi_A(u), \quad \mu_0 = \chi_A(u), \quad e_u = e_A(u), \quad \delta = \frac{\kappa - u}{\Delta}. \quad (33.30)$$

Lemma 33.3.1 makes δ a strictly positive integer. Impose the two congruences

$$p \equiv p_0 \pmod{T}, \quad (33.31)$$

$$p \equiv -M - Q(q^2 - qF + e_u - 1) \pmod{QH}. \quad (33.32)$$

By (33.29), their solutions form one nonempty arithmetic progression

$$p = p_* + Vr, \quad V = TQH, \quad r \in \mathbb{Z}. \quad (33.33)$$

In particular p remains a freely varying linear index. It has not been restricted to values of a Pell sequence or to another nonlinear sparse set. Every solution has

$$p \equiv 3 \pmod{4}, \quad 2^p \equiv X \pmod{H}, \quad p \equiv p_0 \pmod{2E}. \quad (33.34)$$

Choose an integer n_0 representing $(1 - p_0)/2$ modulo $L = E/2$. For every

$$n = n_0 + Lv \quad (33.35)$$

and every p satisfying (33.31),

$$2n \equiv 1 - p \pmod{E}. \quad (33.36)$$

For sufficiently large p in (33.33), define

$$Z = q^2 - qF + \frac{p + M}{Q}, \quad \rho = \frac{Z + e_u - 1}{H}, \quad W = 1 - Z. \quad (33.37)$$

Reduction of (33.32) modulo Q makes Z integral; division by Q in the same congruence makes ρ integral. Both are eventually strictly positive: as functions of p , their slopes are respectively $1/Q$ and $1/(QH)$. The computed root is

$$W + a\kappa + \rho H = e_u + a\kappa = \chi_A(u) = \mu_0 > 0. \quad (33.38)$$

Hence the complete input definition, the definition $\kappa = u + \delta\Delta$, and the input norm (33.13) all hold. Also

$$(q^2 - Z - qF)Q + M = -p. \quad (33.39)$$

Thus the packing equation is exact with $R = -p$. Notice that $W = 1 - Z$ is eventually negative. It is a computed intermediate, not one of the nineteen supplied coordinates in (33.1). No positivity condition on W may be added to this child.

33.6 An irrational rotation supplies the strict Pell ratio

The only missing coupling between the main and first Pell indices is the strict interval

$$Y < \frac{\psi_A(p)}{2\psi_P(n)} < Y + 1. \quad (33.40)$$

It gives the two positive decomposition witnesses η, ζ . We must obtain it on the already imposed progressions, without discarding any outer equation.

Lemma 33.6.1 (Independent quadratic growth rates). *For the fixed parameters above, put*

$$\lambda = A + \sqrt{\Delta}, \quad \nu = P + \sqrt{P^2 - 1}, \quad \alpha_0 = \log \lambda, \quad \beta_0 = \log \nu.$$

Then α_0/β_0 is irrational.

Proof. The integer A is even, hence Δ is odd. On the other hand

$$P^2 - 1 = 4XY^2(XY^2 + 1), \quad v_2(P^2 - 1) = 2 + p_0 + 2v_2(Y)$$

has odd 2-adic valuation. Thus the squarefree part of Δ is odd and that of $P^2 - 1$ is even. Both numbers are nonsquares: for any integer $z \geq 2$, $z^2 - 1$ lies strictly between consecutive squares. Their real quadratic fields are therefore distinct and have intersection $\mathbb{Q}$.

If α_0/β_0 were rational, there would be positive integers r, s with $\lambda^r = \nu^s$. This common positive number would belong to the intersection $\mathbb{Q}$. Its norm in either quadratic field is one; the norm of a rational number q is q^2 , so positivity would force it to be one. This contradicts $\lambda, \nu > 1$. $\square$

Lemma 33.6.2 (Density on the imposed progressions). *There are infinitely many pairs of positive integers (p, n) satisfying (33.33), (33.35), and (33.40), with both p and n tending to infinity.*

Proof. For a positive irrational θ , every forward tail of the rotation $r\theta$ modulo one is dense. One elementary proof is to use the pigeonhole principle to obtain nonzero integer multiples arbitrarily close to zero modulo one; successive steps by such a multiple approximate any point. The closure of the integer orbit is thus the entire circle. The closure of a forward tail is the same, because positive multiples approach zero and hence approximate the negative multiples as well. In particular every nonempty open arc is visited infinitely often.

Apply this fact to

$$\theta = \frac{V\alpha_0}{L\beta_0},$$

which is irrational by Lemma 33.6.1. Write

$$c_0 = \log \frac{\sqrt{P^2 - 1}}{2\sqrt{\Delta}}.$$

Choose a nondegenerate closed interval $I = [a_0, b_0]$ strictly inside $(\log Y, \log(Y + 1))$, small enough that its length is less than $L\beta_0$. The affine expression

$$(p_* + Vr)\alpha_0 - n_0\beta_0 + c_0$$

is dense modulo $L\beta_0$ as positive integers r vary. Infinitely many such r admit an integer v for which

$$(p_* + Vr)\alpha_0 - (n_0 + Lv)\beta_0 + c_0 \in I. \quad (33.41)$$

To see that the closed interval causes no endpoint issue, apply density to its nonempty interior. Since the left side is bounded in (33.41),

$$v = \frac{V\alpha_0}{L\beta_0}r + O(1),$$

so v , and then $n = n_0 + Lv$, tend to positive infinity along the selected hits. So does $p = p_* + Vr$.

The exact Binet formula gives

$$\begin{aligned} \log \frac{\psi_A(p)}{2\psi_P(n)} &= p\alpha_0 - n\beta_0 + c_0 + \varepsilon_{p,n}, \\ \varepsilon_{p,n} &= \log(1 - e^{-2p\alpha_0}) - \log(1 - e^{-2n\beta_0}). \end{aligned}$$

The error tends to zero. The positive distance between I and the complement of $(\log Y, \log(Y+1))$ absorbs this error at every sufficiently large selected hit. Exponentiating proves (33.40). $\square$

No density assertion about prime indices or Pell-valued indices is being used. The prime fixes the scale once; the simultaneous congruences leave a full ordinary arithmetic progression for p .

33.7 All main and first witnesses are positive

For each sufficiently large selected pair from Lemma 33.6.2, define

$$c = \psi_A(p), \quad D = \chi_A(p), \quad k = 2\psi_P(n), \quad \tau = \chi_P(n), \quad (33.42)$$

$$\eta = c - kY, \quad \zeta = k - \eta, \quad \gamma = \frac{D - ac - X}{H}, \quad h = \frac{k + p - 1}{E}, \quad (33.43)$$

$$\sigma = \gamma - \rho, \quad R = -p.$$

The strict ratio (33.40) says exactly $0 < c - kY < k$. Thus η, ζ are positive integers, $k = \eta + \zeta$, and $c = kY + \eta$. The two Pell norms imply

$$D^2 - \Delta c^2 = 1, \quad \tau^2 - 1 = (P^2 - 1)\psi_P(n)^2 = XY^2(XY^2 + 1)k^2.$$

This establishes the two norms (33.8) and (33.9).

By Lemma 33.3.2 and (33.34),

$$D - ac = e_A(p) \equiv 2^p \equiv X \pmod{H}.$$

Hence γ is integral. Moreover $e_A(p) > c$ grows exponentially with p , so γ is eventually positive. The already fixed expression ρ is affine in p . Therefore $\sigma = \gamma - \rho$ is also a strictly positive integer for all sufficiently large selected pairs. In particular the actual stronger source definition $\gamma = \rho + \sigma$, with both summands positive, holds exactly.

Because $P \equiv 1 \pmod{E}$, Lemma 33.3.1 gives $\psi_P(n) \equiv n \pmod{E}$. Consequently

$$k \equiv 2n \equiv 1 - p \pmod{E},$$

so h is a positive integer and

$$k - hE - 1 = -p = R. \quad (33.44)$$

Thus R is exactly the eliminated source's restored expression, not an independently imposed variable. Equation (33.39) is therefore its retained packing comparison.

33.8 The complete strong auxiliary block

We now choose the four remaining auxiliary supplied coordinates f, i, j, o and the Pell coordinate y , verifying their integrality, positivity, and all three retained auxiliary equations.

Since A is even and $p \equiv 3 \pmod{4}$, Lemma 33.3.1 makes $c = \psi_A(p)$ odd. Define

$$m = cp, \quad f = \chi_A(m), \quad Q_{\text{aux}} = \Delta\psi_A(m), \quad i = Q_{\text{aux}}/c^2. \quad (33.45)$$

By Lemma 33.3.4, i is a strictly positive integer. The norm at index m gives

$$Q_{\text{aux}} = ic^2, \quad Q_{\text{aux}}^2 = \Delta(f^2 - 1). \quad (33.46)$$

This is precisely the actual strong square required by (33.10).

Let

$$s_a = p + 2m, \quad b_a = (s_a - 1)/2, \quad U = \frac{\chi_{Q_{\text{aux}}}(s_a)}{Q_{\text{aux}}}, \quad y = \psi_{Q_{\text{aux}}}(s_a). \quad (33.47)$$

Both m and p are odd, and $p \equiv 3 \pmod{4}$, so $s_a \equiv 1 \pmod{4}$ and b_a is even. Lemma 33.3.3 proves that U is a positive integer. We need two exact congruences.

First, $c^2 \mid Q_{\text{aux}}$, so (33.20) gives

$$U = Q_{b_a}(Q_{\text{aux}}^2) \equiv s_a \equiv p \pmod{c}. \quad (33.48)$$

Second, (33.46) gives $Q_{\text{aux}}^2 \equiv -\Delta = 1 - A^2 \pmod{f}$. Therefore (33.21), with b_a even, gives

$$U \equiv \psi_A(p + 2m) \pmod{f}.$$

The doubling formulas are

$$\chi_A(2m) = 2f^2 - 1, \quad \psi_A(2m) = 2f\psi_A(m).$$

Using the addition law (33.16), we obtain

$$U \equiv -\psi_A(p) = -c \pmod{f}. \quad (33.49)$$

Define

$$j = \frac{U - p}{c}, \quad o = \frac{U + c}{f}. \quad (33.50)$$

Equations (33.48)–(33.49) prove their integrality. For all sufficiently large selected p , one has $c > p$, $Q_{\text{aux}} \geq c^2$, and $s_a \geq 3$. Monotonicity of the positive Pell root then gives

$$U \geq \frac{\chi_{Q_{\text{aux}}}(3)}{Q_{\text{aux}}} = 4Q_{\text{aux}}^2 - 3 > c > p.$$

Thus j, o are strictly positive. Since the restored index is $R = -p$,

$$U = jc + p = jc - R = of - c,$$

which proves (33.12). Finally the norm at parameter Q_{aux} is

$$(Q_{\text{aux}}U)^2 - (Q_{\text{aux}}^2 - 1)y^2 = 1.$$

Rearranging and using $Q_{\text{aux}} = ic^2$ yields

$$(ic^2)^2(U^2 - y^2) = 1 - y^2,$$

exactly (33.11). This completes the strong auxiliary block, including the sign that is essential for a negative restored index.

33.9 Completion and domain audit

Every definition in Section 33.2 has now been checked. To collect the strict supplied-coordinate signs without suppressing a domain condition:

- J, w are positive integer powers or repunit expressions; s is positive by the prime choice

- α, z, F are positive by (33.23) and $q > 2dx + 1$
- $\delta > 0$ by the odd input Pell congruence and strict growth
- $Z, \rho > 0$ eventually by their positive affine slopes in p
- $\tau > 0$ and $\eta, \zeta > 0$ by the Pell norm and strict ratio
- $h > 0$ by its positive integral quotient in (33.43)
- $\sigma > 0$ eventually because exponential γ dominates affine ρ
- $f, i, j, o, y > 0$ by the full auxiliary construction

These are precisely the nineteen coordinates in (33.1). Finitely many eventual lower bounds on p can be met simultaneously because Lemma 33.6.2 gives unbounded selected pairs. Equations (33.6)–(33.13) all hold, and therefore $F_{19} = 0$. Yet (33.44) gives $R = -p < 0$. Different selected values of p yield different Z , whose affine slope is $1/Q > 0$. Thus there are infinitely many distinct witness tuples at the same arbitrary fixed x . This proves Theorem 33.1.1 and Corollary 33.1.2.

The computed $\mu = \chi_A(u)$ is positive, while $W = 1 - Z$ is eventually negative. All supplied variables remain positive. Neither a signed input root nor a nonpositive supplied auxiliary coordinate is used to evade a hypothesis.

33.9.1 What remains outside this result

The positive21 child retains a positive supplied W and a stronger positive input-gap structure; raw29 has its own positive supplied interface. The signed19 construction cannot be transferred to those domains because its W is negative. The general kernel family alone also does not settle their additional outer conditions. The signed restoration failure established here should therefore not be relabelled as a failure of every complete74 presentation.

The proof does not derive $X = 2^R$, an accepting typed word, or a no-wrap exponent estimate. Those are conclusions of a positive-index theorem whose sign hypothesis is unavailable here. It also does not produce a new universal Diophantine operation bound. The invalidated signed shortcut and the established positive parent occupy different mathematical domains.

33.10 Proof provenance and reproducibility

The mathematical theorem is established by Sections 33.2–33.9. Finite arithmetic tests do not prove Dirichlet’s theorem, the density step, eventual positivity, or the quantified conclusion. In particular they are not a numerical construction of one full giant witness tuple.

33.10.1 Pinned source evidence

The source definitions are taken from the signed-projection note [2]; the exact substitution, literal domains, and candidate costs are taken from the nonlinear-index scout and its independent source review [3, 4]. The complete74 factored parent receipt supplies the literal arithmetic interface [5]. The half-binomial compiler note explicitly preserves powers-of-five b, L, d and $B = 2^d$ [6]. The unchanged nonnegative sparse definitions of D_C, D_R also support $K_0 \geq 0$; the packet includes the relevant compiler source files as inert text.

A later repository contribution at commit 93ad43375a4a5610de0cfb25d64f84ec018d02f7 records the sign-free bootstrap $p \geq 13$, positivity of the auxiliary argument, raw/positive exact-sign reductions,

and a positive auxiliary realization of $R = -p$ at arbitrarily large indices [7]. We inspected that note and credit the overlapping auxiliary construction. It expressly leaves the full child/compiler system unresolved. Theorem 33.1.1 goes beyond that auxiliary-only scope by satisfying the transport, literal packing, first norm, and complete signed input block simultaneously. This later contextual note does not change the earlier equation pin; its separate cached bytes and manifest are included under `context/`.

All sixteen cached upstream files have their repository paths, exact pinned URLs, SHA256 hashes, and Git blob SHA1 identities recorded in the packet's source manifest. The Git blob hash is computed as SHA1 of the canonical bytes `blob <length>` followed by a zero byte and the file bytes. Reading these files as text or JSON is distinct from executing the upstream programs. No upstream compiler, historical checker, or emitted arithmetic schedule was executed for this report, and no upstream repository file was modified.

33.10.2 Independent proof review

The full all-input proof was independently audited against the exact source interface, including all eight literal residuals, all nineteen signs, the shifted native mask, the CRT coprimality, the unrestricted linear main-index progression, and the strong auxiliary sign. The reviewed proof and audit are byte-bound in the accompanying evidence:

Proof SHA256

b109e2fd1142cdad84a5b519acc5dc5055160f1d8e7617d648a561538fd956cd

Independent audit SHA256

cd62529a6efcdcfbab43f1426c6b3fe5930575fda2cc865f12c40a5ec6c3bda6

The audit verdict is PASS for the full signed19 theorem stated here. The article supplies the supporting identities directly, rather than requiring the reader to combine chronological exploratory reports.

33.10.3 Finite checks and limits

The portable replay uses newly authored Python, the standard library, explicit checks that remain active with Python optimization, and exact-type receipt comparison. Upstream source bytes are only authenticated and read as data. Its identity gate rejects a changed packet before running the authored check program. Normal and optimized replay run from an unrelated working directory with isolated imports and temporary outputs.

The finite evidence comprises an exact recursive Lucas primality certificate for a 98-bit prime, scale and CRT coprimality checks, and 32 complete outer/input progression fixtures at toy fixed numerals. These toy masks are not represented as actual compiler exports. Additional bounded exact identities corroborate the elementary Pell and auxiliary congruence algebra: 451 main Pell/projection cases, 160 first-norm factoring cases, 25 coefficientwise odd-quotient identities and 200 evaluations, and six strong-auxiliary congruence cases. One small auxiliary-only block is materialized; it is not a full child zero. The replay does not numerically locate a density hit or materialize the full main and auxiliary coordinates. The unbounded proof applies to the actual arbitrary fixed exports; the toy fixtures test algebraic components only.

The packet README provides the exact replay commands and scope of each receipt. A failed identity or receipt check is an error rather than a silently regenerated expected result. The article's LaTeX source and a local build script are included; PDF compilation is separate from arithmetic replay and requires a standard LaTeX installation. The archive contains no third-party PDF cache and requires no network access to replay its arithmetic evidence.

33.A A broader algebraic parameter theorem

The powers-of-five convention is sufficient for the actual-compiler result, but is not necessary for failure of restoration in the same algebraic interface. The following extension is separately independently reviewed. It does not assert that arbitrary numeral ports implement a compiler.

Theorem 33.A.1. *Consider exactly the definitions and eight equations in Section 33.2, but assume only $d \in \mathbb{Z}_{>0}$, $B = 2^d$, b a positive odd integer, $K_0 \in \mathbb{Z}_{\geq 0}$, and arbitrary fixed integers M_C, M_F^{src} . At every $x \in \mathbb{Z}_{>0}$, there are infinitely many strictly positive supplied-witness zeros with restored $R < 0$ and computed $\mu > 0$.*

Proof. Choose any positive integer N large enough that $q = B^N > 2dx + 1$, put $t = dN$, and retain $J = (q - 1)/(B - 1)$. Write

$$t = 2^{a_2} 3^{a_3} t_0, \quad \gcd(t_0, 6) = 1.$$

Because $\gcd(12, t_0) = 1$, CRT provides p_0 with

$$p_0 \equiv 7 \pmod{12}, \quad p_0 \equiv 1 \pmod{t_0}.$$

When $t_0 = 1$ the second condition is vacuous. Increase p_0 by a positive multiple of $12t_0$ until $p_0 > 3t$. This replaces only the specific choice $p_0 = 12t + 7$ in the main proof. Every prime factor of t either is two or three, or divides t_0 . Therefore

$$\gcd(p_0, 2t) = 1, \quad p_0 \equiv 3 \pmod{4}, \quad p_0 \equiv 1 \pmod{6}.$$

Set $X = 2^{p_0}$ and $w = 2^{p_0-3t}$ as before. The proof of Lemma 33.4.1 uses exactly these congruences, so $D_0 = 4q^3(X + 1)/3$ is integral and coprime to $Q = q^2 - 1$. The primitive prime progression construction works without change.

There is one parity detail worth making explicit. Here $D_0 \equiv (-1)^t \pmod{3}$, which need not be two. No particular nonzero residue is needed: both D_0 and s are units modulo Q , and $3 \mid Q$, so $\ell - 1 = D_0 s$ is nonzero modulo three. Since ℓ is also a unit modulo three, $\ell \equiv 2 \pmod{3}$ follows. Thus all coprimality culminating in $\gcd(QH, T) = 1$ remain valid.

The input $u = 2dx + b$ is odd and at least three, so the same $\delta > 0$ works without an upper bound on b . Also $F = K_0 + X - q + 1 > 0$. Arbitrary signs of the fixed masks affect only the constant $M = (M_C + qM_F^{\text{src}})J$. CRT still accepts any resulting residue in (33.32); the positive slopes of Z, ρ in (33.37) are unchanged.

Finally q, Y are even for every $t \geq 1$, whereas p_0 is odd. Hence the quadratic-field separation in Lemma 33.6.1 remains true even when t is even. The free progression, density interval, all Pell definitions, the exponential dominance over ρ , and the auxiliary construction use only the properties just established. Every subsequent step of Sections 33.5–33.9 therefore applies and proves the claim. $\square$

This extension changes the algebraic parameter range, not the main theorem's source attribution. For actual exports the paid mask remains $M_F^{\text{src}} = M_F^{\text{native}} + B - 1$ and all compiler conditions remain in force. The independent supplement review is bundled separately from the originally approved main proof and audit.

33.B The unresolved raw29 and positive21 domains

This appendix records a separate audited necessary-condition reduction. It does not prove positivity of the restored index and does not construct a full negative zero in either domain. The source receipts

call the parent modes `raw30` and `positive22`; after deletion the children have 29 and 21 supplied coordinates. Unlike `signed19`, both retain a positive supplied W and a positive input-gap witness. The theorem of the main text has $W < 0$ and therefore cannot settle these cases.

33.B.1 Literal assumptions and imported bootstrap

In addition to the common Pell and scale equations, a positive zero in either domain has

$$\begin{aligned} q &= (B - 1)J + 1, & C + \alpha + 2dx &= q, \\ C &= Z + W, & (K_0 + X)C &= F + z(q - 1), \\ R &= (q^2 - Z - qF)(q^2 - 1) + (M_C + qM_F^{\text{src}})J, \\ \kappa &= u + \delta\Delta, & c &= \kappa + \phi, \\ \mu &= W + a\kappa + \rho H, & \mu^2 - \Delta\kappa^2 &= 1, \end{aligned}$$

where W, Z, α, ϕ and the supplied coordinates are strictly positive. In `raw29`, the appropriate raw quantities are supplied and the equations are retained comparisons; in `positive21` the triangular quantities are computed positive expressions. Thus

$$q \geq B \geq 16, \quad 0 < Z < C < q, \quad 0 < W < C < q, \quad F < (K_0 + X)C, \quad 0 < u < 2q. \quad (33.51)$$

The last bound uses the actual compiler's $b < B \leq q$.

The sign-free bootstrap is an imported, independently audited prerequisite here, as also recorded in the later source note [7]; it is not needed for the constructive main theorem. Its precise conclusions are

$$\begin{aligned} c &= \psi_A(p), & p &\geq 13 \text{ odd}, & R &\equiv \pm p \pmod{c}, \\ k &= 2\psi_P(n), & \frac{p+1}{2} &\leq n \leq p-1, & 2n &\equiv R+1 \pmod{E}, \\ c &> 2p, & c &> A^{12}, & P &= 2XY^2 + 1. \end{aligned} \quad (33.52)$$

This bootstrap derives positivity of the auxiliary argument and the rank hypotheses without $R > 0$. It invokes the ordinary strong norm, not a weakened replacement. We cite its rank/step-down argument rather than claiming to reprove it in this appendix; its reviewed snapshot and audit are included with the packet.

33.B.2 Why the actual compiler gives the required coefficient bound

The reduction needs $K_0 < B^2$ for actual compiler exports; it would not be legitimate to impose that inequality on arbitrary evaluator ports. Here is the source provenance. Write $R_{\text{rad}} = 2^b$ for the inner radix, and $B = R_{\text{rad}}^{L_{\text{comp}}}$. If S is the sum of the positive center coefficients, the `complete76` sparse D_C consists of four unit monomials, two copies of those coefficients, and at most one optional unit monomial. Every coefficient is therefore at most $2S + 5$. All exponents, including the optional high monomial, are below L_{comp} . The half-binomial wrapper raises the radix to at least

$$R_{\text{rad}} \geq 4(K + 2)(2S + 6) + 8 > 2S + 5,$$

where K is the number of native positions. Hence there are no radix carries and $0 < D_C < B$. The inherited $D_R = R_{\text{rad}}^{H_{\text{comp}}}$ has $0 < H_{\text{comp}} < L_{\text{comp}}$, so $0 < D_R < B$ too.

The materializers are inherited from `complete78` through `complete77` and `complete76`. `Complete77` changes only the masks/export wrapper. The sparse construction does not access the cached properties

B, D_C, D_R before the half-binomial wrapper raises the radix; hence the export contains no stale smaller-radix cached value. Its `new_constants` function changes only M_C, M_F^{native} after materializing the other numerals. The literal alias chain identifies `Kconstant` with $D_C + BD_R$ and the paid MF with $M_F^{\text{native}} + B - 1$. These facts are checked against the pinned source files in the separate raw/positive audit. Consequently

$$0 < K_0 = D_C + BD_R \leq (B - 1) + B(B - 1) = B^2 - 1 \leq q^2 - 1. \quad (33.53)$$

33.B.3 Only two exact index representatives survive

The shifted-mask identity from Section 33.2 holds with $0 < T' < q^2 - 1$ using only the native mask ranges and the repunit equation; it does not require $q = B^N$. It implies $R \neq 0$ and $R < q^4$. If $R < 0$, dropping the positive mask contribution in the original packing formula and using (33.51) gives

$$\begin{aligned} -R &< (Z + qF)q^2 \\ &< Cq^3(X + K_0 + 1/q) \\ &< CXq^3(1 + 1/q) \\ &\leq (q - 1)Xq^3(1 + 1/q) < Xq^4. \end{aligned}$$

The third step follows from $K_0 + 1/q < q^2 \leq X/q$, by (33.53) and $X \geq q^3$. The positive branch already has $0 < R < q^4 < Xq^4$. Thus $|R| < Xq^4$ in either case. Since $A > XY \geq Xq^3$, one has $A^2 > 2Xq^4$, and (33.52) gives $c > 2Xq^4$. Both $|R|$ and p are therefore less than $c/2$. The congruence $R \equiv \pm p \pmod{c}$ can have only its zero multiple of c , so

$$R = p \quad \text{or} \quad R = -p. \quad (33.54)$$

Both alternatives are odd. Were q odd, the repunit equation with even B would make J even; both terms in the literal packing formula would then be even. Hence q is even, so X, Y, A are even and Δ is odd. This is a parity conclusion, not a proof that q is a power of two.

33.B.4 A negative zero must lie in a bounded window

If $R = -p$, the first-index congruence in (33.52) becomes

$$2n + p - 1 = t_- E$$

for a positive integer t_- . This quotient is not the supplied witness h and is unrelated to the main proof's binary exponent $t = dN$. The interval for n gives

$$2p \leq t_- E \leq 3p - 3.$$

Combining with $p < Xq^4$ yields the necessary restrictions

$$E/3 + 1 \leq p < Xq^4, \quad s < 3q, \quad 1 \leq t_- < 3q/s. \quad (33.55)$$

Packing further requires

$$p = (Z + qF - 1 - q^2)(q^2 - 1) - T', \quad F \geq q.$$

At $F = q$ it requires $Z \geq 2$. Thus $F < q$, a conclusion of the old positive-index proof, cannot be silently imposed here. The window is finite for fixed q, X, Y and compiler numerals, but is not uniformly finite as the witnesses vary.

33.B.5 The retained positive input gives an exact dichotomy

Since the input root and coefficient are positive, the Pell norm supplies an index $v \geq 1$ with $\kappa = \psi_A(v)$ and $\mu = \chi_A(v)$. For $A \geq 2$ the unit $A + \sqrt{A^2 - 1}$ is fundamental: a smaller positive solution would have square-root coefficient at least one and root at least A . Thus the standard positive Pell parametrization applies. The retained $c > \kappa$ gives $v < p$. Also

$$p = |R| < Xq^4 < \Delta,$$

because $\Delta > (XY)^2 \geq X^2q^6$. From $0 < u < 2q < A - 1$, both u and uA belong to $(0, \Delta)$. The discriminant congruence of Lemma 33.3.1 now gives

$$\psi_A(v) \equiv \begin{cases} v & v \text{ odd,} \\ vA & v \text{ even} \end{cases} \pmod{\Delta}.$$

As $\kappa \equiv u \pmod{\Delta}$, uniqueness of the representatives in $(0, \Delta)$ proves

$$v = u \quad \text{or} \quad v = uA. \tag{33.56}$$

For the even case, multiply $vA \equiv u$ by A and use $A^2 \equiv 1 \pmod{\Delta}$ before selecting representatives. Since u is odd and A even, uA is indeed even. On the negative branch this alternative forces $uA < p < Xq^4$, hence $uY < q^4$ and $us < q$.

[write] *The same dichotomy in the research tree.* The tree's `complete74_negative_index_refinement.md`, Section 2 (commit `813c1cff4`, 3 October 2026), proves for `raw30` and `positive22` that the input Pell index is exactly $e = u$ or $e = uA$, by the same discriminant-representative argument. It is later than this Report's pins and not cited by it, and this Report is not cited there: the two proofs are independent, and no priority is claimed either way. The tree's note goes further: on $r = -p$ it proves $4^d > X \geq q^3$ for the defect $d = 2n - p$ (so $d \geq 7$), and that for fixed q, X, Y and a fixed bounded representative at most one main index p satisfies the ratio interval. The programme's review of this Report notes the same coincidence.

The positive branch actually excludes $v = uA$: it has $p = R < q^4 < A$, whereas $v = uA \geq A$. On $R = -p$, no such exclusion has been proved. The remaining projections are only

$$W \equiv 2^v \pmod{H}, \quad 0 < W < q, \quad X \equiv 2^p \pmod{H}.$$

Neither $p < \Delta$ nor $v < p$ rules out exponent wrapping. The unresolved task is to exclude or realize the negative branch while simultaneously meeting (33.55), (33.56), the exact Pell ratio, and every outer and input equation. This appendix establishes neither outcome.

[write] *Re-scoped, 3 October 2026 (batch 83).* Report 37, printed as Part V, converts this task into an exact predicate: Theorem 37.5.1 proves, at each fixed valid compiler slice and input x , that `raw29` has a full positive zero with $R < 0$ if and only if `positive21` has one, if and only if the arithmetic predicate of Section 37.4 passes for some q, w (the only unbounded parameters) and finitely many choices of s, t, p, e ; every passing predicate reconstructs full positive witnesses in both interfaces. The task stays open in its yes-or-no form: neither outcome is established there either. Part V also sharpens the bound $us < q$ above to (37.39), isolates a finite wrapped odd-input sector (Section 37.6), and gives a certified sub-unit Binet window (Theorem 37.7.1).

References of Report 33

- [1] P. G. L. Dirichlet, *There are infinitely many prime numbers in all arithmetic progressions with first term and difference coprime*, original proof 1837, English translation by R. Stephan, arXiv:0808.1408v2 (2014). <https://arxiv.org/abs/0808.1408v2>. The prime-progression theorem is used only in Section 33.4.
- [2] ProveIt repository (VladimirReshetnikov), *A 101-operation polynomial from a signed projection*, `complete75_signed_projection_elimination101.md`, pinned commit specified in Section 1, especially Sections 1–3. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_signed_projection_elimination101.md.
- [3] ProveIt repository (VladimirReshetnikov), *Complete74 nonlinear index projection scout*, `complete74_nonlinear_index_projection_scout.md` and its JSON receipt, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_nonlinear_index_projection_scout.md.
- [4] ProveIt, *Independent review of the complete74 nonlinear index scout*, `review_complete74_nonlinear_index_projection.md`, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_complete74_nonlinear_index_projection.md.
- [5] ProveIt, `complete74_factored_first_norm.json`, literal parent source receipt, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_factored_first_norm.json.
- [6] ProveIt repository (VladimirReshetnikov), *Modified compiler masks for the half-binomial75*, `complete75_half_binomial_compiler.md`, same pinned commit, Section 1. https://github.com/VladimirReshetnikov/ProveIt/blob/2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.md.
- [7] ProveIt, *Signed first-index bootstrap for the complete74 nonlinear projection*, `review_complete74_nonlinear_index_bootstrap.md`, later commit 93ad43375a4a5610de0cfb25d64f84ec018d02f7. Its theorem is auxiliary-only as to existence of negative solutions. https://github.com/VladimirReshetnikov/ProveIt/blob/93ad43375a4a5610de0cfb25d64f84ec018d02f7/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_complete74_nonlinear_index_bootstrap.md.
- [8] ProveIt, `pell_kernel_half_binomial42.md` and the fixed-minus parity note, `EXPLORATION_FIXED_MINUS_INDEX_PARITY.md`, same pinned commit. These motivate the Pell auxiliary identities; Section 33.3 proves the identities used here independently. Exact repository locations and hashes are in the bundled source manifest.

Report 34. Height and Counting of a Signed19 Family: canonical witnesses with negative restored index

[write] Source 05 of batch 82: *Research Report 34*, author line “Mathematical research report”, dated 3 October 2026; archive `Counterexample_Height_Expansions_and_Rotation_Discrepancy_Package.zip`; pins 2dde7850f (construction) and 9fa99be93 (transfer appendix). Its sections, statements and equations are numbered 34.*k* here.

Abstract of the source. We fix the input, compiler export, repunit and one prime-scale choice in the signed19 construction of Report 33, and count only its prescribed canonical auxiliary witnesses. Each main index in one Chinese-remainder progression admits at most one first-Pell index in the required strict ratio window. The maximum of the nineteen supplied coordinates is eventually the last auxiliary Pell coordinate, whose double logarithm has an explicit exponentially corrected expansion and a Lambert W inverse. The height count satisfies $N(B) = \kappa \log \log B + O((\log \log B)^{1-\varepsilon})$ for some effective fixed-parameter $\varepsilon > 0$. A logarithmic-form bound makes the exact selection eventually identical to a fixed irrational-rotation test. The resulting exact discrepancy term is unbounded by Kesten’s theorem, ruling out a smooth two-term count with bounded remainder. The finer two-term law with $o(\log \log \log B)$ remainder is equivalent to a specific unresolved $o(\log R)$ discrepancy estimate. This is a theorem about one explicit subfamily, not the whole signed19 fiber.

[write] Report 34 has no executable mathematical checker by design. (*Corrected in batch 83: the batch-82 write said it had not been reviewed.*) The programme reviewed it in `review_incoming_counterexample_heights.md` (commit `ceb7222e9`) and passes it within its scope (Section 0.4). Its letters differ from Part III’s: here $\lambda = A + \sqrt{\Delta}$ and $\alpha = \log \lambda$; κ is a counting coefficient and κ_{in} is Report 33’s κ ; $\alpha_{\text{src}}, \delta_{\text{src}}$ are Report 33’s coordinates α, δ ; and its paid mask is written M_F^{src} as in Report 33. The source’s files are shipped with the prefix 05-signed19-heights-.

34.1 The result and the object being counted

Report 33 proves that, for every positive ordinary input and every actual fixed compiler export in its pinned source, the complete74 signed19 index-eliminated child has infinitely many positive supplied-witness zeros with negative restored index [1]. We study the size and number of one canonical family within that construction. Report 33 and its source are unchanged.

The term *negative restoration* means that the omitted, algebraically computed index is $R = -p < 0$. It does not mean that any of the nineteen supplied coordinates is negative. A negative computed intermediate is allowed by the signed presentation. Nothing here identifies a machine input’s acceptance status or changes a positive-parent theorem.

Fix all construction data in Section 34.2. In particular the prime used to choose the scale is fixed once and for all. Write

$$\begin{aligned} \Delta &= A^2 - 1, & D_P &= P^2 - 1, & \lambda &= A + \sqrt{\Delta}, & \nu &= P + \sqrt{D_P}, \\ \alpha &= \log \lambda, & \beta &= \log \nu, & L &= E/2, & V &= TQH_0, \\ \delta &= \log(1 + 1/Y), & d_{\text{rot}} &= \frac{\delta}{L\beta}, & \kappa &= \frac{\delta}{2\alpha V L \beta}, & C_0 &= \frac{\alpha}{2\Delta}. \end{aligned} \tag{34.1}$$

All logarithms are natural. The growth logarithm α is distinct from the source coordinate α_{src} ; δ and κ likewise differ from δ_{src} and κ_{in} . The letter B below is a height bound, whereas B_{comp} denotes the fixed compiler radix.

For each eligible pair (p, n) , let $\mathbf{w}(p, n)$ be the full tuple in Section 34.2, and put

$$\text{ht}(\mathbf{w}) = \max\{\text{its nineteen supplied coordinates}\}.$$

We count a sufficiently late tail $\mathcal{C}$ of these tuples, retaining all pairs in that tail which satisfy the exact congruence and strict ratio test. Define

$$N(B) = \#\{\mathbf{w} \in \mathcal{C} : \text{ht}(\mathbf{w}) \leq B\}.$$

The tail begins at one fixed main-index progression representative p_b . Its onset may be enlarged to ensure positivity, height dominance and the exact rotation reduction. Finite changes to this convention affect N by $O(1)$ only.

Theorem 34.1.1 (Canonical family height and count). *For the fixed data just specified, the following statements hold.*

- (i) *Each $p = p_b + Vr$, $r \in \mathbb{Z}_{\geq 0}$, has at most one positive $n \equiv n_0 \pmod{L}$ satisfying*

$$Y < \frac{\psi_A(p)}{2\psi_P(n)} < Y + 1. \quad (34.2)$$

The proportion of progression terms admitting such an n is $d_{\text{rot}} > 0$.

- (ii) *On $\mathcal{C}$, the height is exactly $y = \psi_{Q_{\text{aux}}}(S)$ and strictly increases with p . With $b_0 = (\log \Delta)/\alpha - 1$,*

$$\log y = C_0 p^2 e^{2\alpha p} (1 + O(e^{-\alpha p})), \quad (34.3)$$

$$\log \log y = 2\alpha p + 2 \log p + \log C_0 + \sqrt{\Delta} \left(1 + \frac{b_0}{p}\right) e^{-\alpha p} + O(e^{-2\alpha p}). \quad (34.4)$$

- (iii) *If B_j is the j th height in increasing order, then*

$$N(B) \sim \kappa \log \log B, \quad \log \log B_j \sim j/\kappa. \quad (34.5)$$

For some effectively computable $\varepsilon > 0$ depending on the fixed data,

$$N(B) = \kappa \log \log B + O((\log \log B)^{1-\varepsilon}), \quad \log \log B_j = \frac{j}{\kappa} + O(j^{1-\varepsilon}). \quad (34.6)$$

- (iv) *There is an exact fixed-rotation discrepancy $D(R)$, defined in Section 34.7, such that, for $t = \log \log B$,*

$$N(B) = \kappa t - 2\kappa \log t + D(R(B)) + O(1), \quad R(B) \sim \frac{t}{2\alpha V}. \quad (34.7)$$

The smooth all-height formula $N(B) = \kappa t - 2\kappa \log t + O(1)$ is false. The weaker formula with $o(\log t)$ in place of $O(1)$ holds if and only if $D(R) = o(\log R)$; that condition is not established here.

The height inverse is given more precisely in Section 34.5. All constants and thresholds may depend on the fixed input, compiler, repunit and prime-scale choice. No parameter-uniform exponent or useful numerical onset is asserted. Equations (34.5) and (34.6) are statements on the double-logarithmic scale, not multiplicative equivalents for the heights.

34.2 The canonical family and its source interface

This section defines every tuple being counted. We inherit from the audited construction in Report 33 its integrality, positivity on a sufficiently late tail, and satisfaction of the full child equations. We recall the exact formulas and the relevant hypotheses rather than reproducing its whole existence proof.

34.2.1 Fixed compiler and scale choices

For an integer $D \geq 2$ define the Pell sequences by

$$\chi_D(v) + \psi_D(v)\sqrt{D^2 - 1} = (D + \sqrt{D^2 - 1})^v \quad (v \in \mathbb{Z}_{\geq 0}). \quad (34.8)$$

Their norm is $\chi_D(v)^2 - (D^2 - 1)\psi_D(v)^2 = 1$.

Fix $x \in \mathbb{Z}_{>0}$ and an actual compiler export from Report 33: $B_{\text{comp}} = 2^d \geq 16$, with d a power of five, b positive and odd, $K_0 \geq 0$, and the fixed masks M_C, M_F^{native} . Put $M_F^{\text{src}} = M_F^{\text{native}} + B_{\text{comp}} - 1$. All further compiler requirements remain those of the inherited source; no toy mask replaces these numerals.

Choose and fix a power of five N_0 such that $q = B_{\text{comp}}^{N_0} > 2dx + 1$. Define

$$\begin{aligned} t_0 &= dN_0, & p_0 &= 12t_0 + 7, & X &= 2^{p_0}, & Q &= q^2 - 1, \\ J &= \frac{q - 1}{B_{\text{comp}} - 1}, & w &= 2^{p_0 - 3t_0}, & \alpha_{\text{src}} &= q - 1 - 2dx, & z &= 1, \\ F &= K_0 + X - q + 1, & M &= (M_C + qM_F^{\text{src}})J, & C &= 1. \end{aligned} \quad (34.9)$$

Thus $q = 2^{t_0}$ and $X = wq^3$. Set $D_0 = 4q^3(X + 1)/3$. The construction proves $D_0 \in \mathbb{Z}_{>0}$ and $\gcd(D_0, Q) = 1$. Choose s_0 with

$$\gcd(s_0, Q) = \gcd(1 + D_0 s_0, Q) = 1,$$

and fix one prime $\ell > 3$ in the primitive progression $1 + D_0 s_0 \pmod{D_0 Q}$, large enough that $s = (\ell - 1)/D_0 > 0$. Dirichlet's theorem supplies such primes; this report neither varies nor counts them. Define

$$\begin{aligned} Y &= sq^3, & E &= XY, & a &= Y(X + 1), & A &= a + 2, & H_0 &= 4a + 3 = 3\ell, \\ \Delta &= A^2 - 1 = a^2 + H_0, & P &= 2XY^2 + 1, \\ O &= \text{ord}_{H_0}(2), & T &= \text{lcm}(4, 2E, O), & L &= E/2. \end{aligned} \quad (34.10)$$

The inherited scale argument gives

$$\gcd(QH_0, T) = 1. \quad (34.11)$$

This fact is what allows a free main-index progression after imposing all outer constraints.

34.2.2 Indices and all nineteen coordinates

The positive odd input index $u = 2dx + b \geq 3$ is fixed. Set

$$\kappa_{\text{in}} = \psi_A(u), \quad \mu_0 = \chi_A(u), \quad e_u = \mu_0 - a\kappa_{\text{in}}, \quad \delta_{\text{src}} = \frac{\kappa_{\text{in}} - u}{\Delta}. \quad (34.12)$$

Choose p in the single simultaneous progression

$$p \equiv p_0 \pmod{T}, \quad p \equiv -M - Q(q^2 - qF + e_u - 1) \pmod{QH_0}. \quad (34.13)$$

By (34.11), its step is $V = TQH_0$. In particular $p \equiv 3 \pmod{4}$. Fix $n_0 \equiv (1 - p_0)/2 \pmod{L}$. Retain the pairs (p, n) with $n \in \mathbb{Z}_{>0}$, $n \equiv n_0 \pmod{L}$, and the exact test (34.2).

For each retained pair define

$$\begin{aligned}
c &= \psi_A(p), & D &= \chi_A(p), & k &= 2\psi_P(n), & \tau &= \chi_P(n), \\
\eta &= c - kY, & \zeta &= k - \eta, & h &= \frac{k + p - 1}{E}, & R &= -p, \\
Z &= q^2 - qF + \frac{p + M}{Q}, & \rho &= \frac{Z + e_u - 1}{H_0}, \\
\gamma &= \frac{D - ac - X}{H_0}, & \sigma &= \gamma - \rho, & W &= 1 - Z.
\end{aligned} \tag{34.14}$$

The *canonical auxiliary choice* in this report is exactly

$$\begin{aligned}
m &= pc, & f &= \chi_A(m), & Q_{\text{aux}} &= \Delta\psi_A(m), & i &= \frac{Q_{\text{aux}}}{c^2}, \\
S &= p + 2m, & U &= \frac{\chi_{Q_{\text{aux}}}(S)}{Q_{\text{aux}}}, & y &= \psi_{Q_{\text{aux}}}(S), \\
j &= \frac{U - p}{c}, & o &= \frac{U + c}{f}.
\end{aligned} \tag{34.15}$$

The supplied tuple, in source order, is

$$\mathbf{w}(p, n) = (J, F, \alpha_{\text{src}}, z, f, h, i, j, o, s, w, \tau, \eta, \zeta, y, Z, \delta_{\text{src}}, \rho, \sigma). \tag{34.16}$$

All its coordinates are positive integers for sufficiently large eligible p . This is the construction theorem inherited from Report 33, Sections 4–9, and its independent audit [2, 3]. In particular, Q_{aux}/c^2 , U , j , and o are integers by the divisibility and odd-quotient Pell identities proved there. Their integrality is not inferred from the asymptotic analysis below.

[write] Report 33's Sections 4–9 are Sections 33.4–33.9 of this Part. The construction packet and audit cited here are shipped once, as source 10's `10-index-restore-repro-evidence-FULL-SIGNED-COUNT EREXAMPLE.md` and `10-index-restore-repro-evidence-INDEPENDENT-AUDIT.md`; source 05's copies, `evidence/FULL-SIGNED-COUNTEREXAMPLE.md` and `evidence/CONSTRUCTION-INDEPENDENT-AUDIT.md`, are byte-identical and not shipped.

There is no choice of a second auxiliary index, a different multiple of pc , another prime, or a new scale hidden in $N(B)$. Once (p, n) is selected, every entry of (34.16) is prescribed. Distinct p already yield distinct tuples because Z is affine in p with positive slope.

34.2.3 Link to the complete child equations

For clarity, the child consists of the sum of squares of the following eight residuals, with the computed definitions in (34.9)–(34.15) and $\mu = W + a\kappa_{\text{in}} + \rho H_0$:

$$\begin{aligned}
(K_0 + X)C &= F + z(q - 1), \\
R &= (q^2 - Z - qF)(q^2 - 1) + (M_C + qM_F^{\text{src}})J, \\
\tau^2 - XY^2(XY^2 + 1)k^2 &= 1, & D^2 - \Delta c^2 &= 1, \\
(ic^2)^2 &= \Delta(f^2 - 1), \\
(ic^2)^2((jc - R)^2 - y^2) &= 1 - y^2, \\
jc - R &= of - c, & \mu^2 - \Delta\kappa_{\text{in}}^2 &= 1.
\end{aligned} \tag{34.17}$$

The restored expression is $R = k - hE - 1 = -p$. Also $c = kY + \eta$, $\gamma = \rho + \sigma$, $D = X + ac + \gamma H_0$, $\kappa_{\text{in}} = u + \delta_{\text{src}}\Delta$, and $\mu = \mu_0 > 0$. Thus the family satisfies the whole signed19 interface, rather than just an auxiliary subsystem. The validity and positivity proof of these statements is the inherited theorem. Only the size and count of this prescribed family are newly analyzed here.

34.3 The exact first index window

The two quadratic fields are distinct. Indeed A is even, so Δ is odd, while

$$D_P = 4XY^2(XY^2 + 1)$$

has odd 2-adic valuation $2 + p_0 + 2v_2(Y)$. The squarefree part of D_P is therefore even and that of Δ is odd. Both define real quadratic fields, and their intersection is $\mathbb{Q}$. Consequently α/β is irrational: a rational relation would put equal positive powers of λ and ν in that intersection, and the norm-one property would force such a positive rational power to be 1, which is impossible.

Put $c_p = \psi_A(p)$ and

$$g = \log \frac{\sqrt{D_P}}{2\sqrt{\Delta}}.$$

The hyperbolic Pell formulas make (34.2) exactly equivalent to

$$a_p < n\beta < b_p, \quad a_p = \operatorname{arsinh} \frac{c_p \sqrt{D_P}}{2(Y+1)}, \quad b_p = \operatorname{arsinh} \frac{c_p \sqrt{D_P}}{2Y}. \quad (34.18)$$

The derivative of $t \mapsto \operatorname{arsinh}(e^t)$ lies strictly between 0 and 1. Integrating across a logarithmic interval of width δ gives

$$0 < b_p - a_p < \delta < L\beta. \quad (34.19)$$

Here $\delta \leq \log 2$, $L \geq 1$, and $\beta = \operatorname{arcosh} P > \log 2$. Distinct allowed n have spacing L , so at most one can satisfy (34.18). This is an exact uniqueness result, before any limiting approximation. Both endpoints tend to infinity, and any large-index solution has

$$n = (\alpha/\beta)p + O(1) > 0. \quad (34.20)$$

For $v \in \{Y, Y+1\}$ define

$$\epsilon_v(p) = \operatorname{arsinh} \frac{c_p \sqrt{D_P}}{2v} - (\alpha p + g - \log v).$$

Binet's formula and $\operatorname{arsinh} z = \log(2z) + (4z^2)^{-1} + O(z^{-4})$ give

$$\epsilon_v(p) = \left(\frac{4v^2\Delta}{D_P} - 1 \right) e^{-2\alpha p} + O(e^{-4\alpha p}). \quad (34.21)$$

The -1 term is the Binet correction. It is essential to the first endpoint coefficient.

Write the main progression as $p = p_b + Vr$, $r \geq 0$, and set

$$\theta = \frac{V\alpha}{L\beta}, \quad \xi = \frac{\alpha p_b + g - \log Y - n_0\beta}{L\beta}, \quad I = (0, d_{\text{rot}}) \pmod{1}. \quad (34.22)$$

The limiting interval contains a point of $n_0\beta + L\beta\mathbb{Z}$ precisely when

$$\{\xi + r\theta\} \in I. \quad (34.23)$$

To check the phase convention, put $z = (\alpha p + g - \log Y - n_0 \beta)/(L\beta)$. The condition is that some integer lies strictly between $z - d_{\text{rot}}$ and z , equivalent to $0 < \{z\} < d_{\text{rot}}$.

Because θ is irrational, the proportion of fixed-interval hits is d_{rot} . This transfers to the exact moving window without a logarithmic-form theorem: for any $\epsilon > 0$, after finitely many indices the exact and limiting indicators can differ only within ϵ -neighborhoods of the two endpoints of I . Equidistribution bounds the upper density of these discrepancies by 4ϵ . Letting ϵ tend to zero proves assertion (i) and positive density. This argument alone does not imply finitely many discrepancies; Section 34.7 supplies that stronger fact separately.

34.4 The maximum supplied coordinate and its growth

34.4.1 Eventual exact height and no ties

The strict ratio gives $k < c/Y$, hence $\eta, \zeta = O(c)$ and $\tau = O(c)$ by the first norm. Also $h = O(c + p)$; Z and ρ are affine in p ; and $\sigma = O(c + p)$ by the formula for γ . The remaining nonauxiliary coordinates in (34.16) are fixed. All these estimates have fixed-data implied constants.

For the auxiliary coordinates, $Q_{\text{aux}} = ic^2$ with $i \geq 1$, and

$$Q_{\text{aux}}^2 = \Delta(f^2 - 1), \quad y \geq \psi_{Q_{\text{aux}}}(3) = 4Q_{\text{aux}}^2 - 1 > Q_{\text{aux}} \geq c^2.$$

Since $\Delta \geq 3$ and $f \geq 2$, we have $f < Q_{\text{aux}}$ and $i \leq Q_{\text{aux}}$. The auxiliary norm is exactly

$$U^2 = (1 - Q_{\text{aux}}^{-2})y^2 + Q_{\text{aux}}^{-2} < y^2. \quad (34.24)$$

Therefore $j = (U - p)/c < y$ and

$$o = \frac{U + c}{f} < \frac{y + c}{2} < y.$$

Finally $y \geq 4c^4 - 1$ dominates every $O(c + p + 1)$ coordinate because c grows exponentially and $p = o(c)$. This checks all nineteen entries of (34.16). Thus $\text{ht}(\mathbf{w}) = y$ with strict dominance on a sufficiently late tail.

Extend the formulas to real $p \geq 1$ by

$$\begin{aligned} c &= \frac{\sinh(\alpha p)}{\sqrt{\Delta}}, \quad m = pc, \quad Q_{\text{aux}} = \sqrt{\Delta} \sinh(\alpha m), \quad S = p + 2m, \\ b_{\text{aux}} &= \text{arcosh } Q_{\text{aux}}, \quad y(p) = \frac{\sinh(Sb_{\text{aux}})}{\sinh b_{\text{aux}}}. \end{aligned} \quad (34.25)$$

Here $c(1) = m(1) = 1$, $Q_{\text{aux}}(1) = \Delta \geq 3$, and $S(1) = 3$. The functions m , Q_{aux} , and S increase strictly. For $S > 1$ and $b > 0$, $\sinh(Sb)/\sinh b$ increases in both arguments: the derivative of its logarithm in b is $S \coth(Sb) - \coth b > 0$, because $z \coth z$ increases on $(0, \infty)$. Thus $y(p)$ is strictly increasing, and the canonical family has no height ties after the fixed tail onset.

34.4.2 An exponentially corrected expansion

Put $d_0 = \sqrt{\Delta}$ and $\ell_0 = \log d_0$. From (34.25),

$$b_{\text{aux}} = \alpha m + \ell_0 + O(e^{-2\alpha m}), \quad \log(2 \sinh b_{\text{aux}}) = \alpha m + \ell_0 + O(e^{-2\alpha m}).$$

The logarithm of the numerator in y contributes the remaining term $\log(1 - e^{-2Sb_{\text{aux}}})$, which is smaller. Consequently

$$\log y = (2m + p - 1)(\alpha m + \ell_0) + O((m + p)e^{-2\alpha m}). \quad (34.26)$$

The error is superexponentially small in p . Substituting

$$m = \frac{p}{2d_0} e^{\alpha p} (1 - e^{-2\alpha p})$$

gives the first two scales explicitly:

$$\log y = C_0 p^2 e^{2\alpha p} + \frac{p}{2d_0} [\alpha(p - 1) + \log \Delta] e^{\alpha p} + O(p^2). \quad (34.27)$$

Divide by $C_0 p^2 e^{2\alpha p}$ and take the logarithm. The resulting first correction is

$$d_0 \left(1 + \frac{(\log \Delta)/\alpha - 1}{p} \right) e^{-\alpha p},$$

and the rest is $O(e^{-2\alpha p})$. This proves (34.3)–(34.4), including the coefficient $C_0 = \alpha/(2\Delta)$.

34.5 The real height cutoff and Lambert inversion

Strict monotonicity gives, for every sufficiently large B , a unique real p_B with $y(p_B) = B$. The exact height condition on the tail is $p \leq p_B$.

Let W be the positive real branch of the Lambert function, satisfying $W(z)e^{W(z)} = z$, and set

$$t_B = \frac{1}{\alpha} W(\sqrt{2\alpha\Delta \log B}). \quad (34.28)$$

It solves the leading model exactly:

$$C_0 t_B^2 e^{2\alpha t_B} = \log B, \quad 2\alpha t_B + 2 \log t_B + \log C_0 = \log \log B.$$

Write $b_0 = (\log \Delta)/\alpha - 1$. Equation (34.4), with the positive derivative $2\alpha + 2/p$ of the leading logarithmic model, first brackets $p_B - t_B$ by $O(e^{-\alpha t_B})$. Taylor expansion of the displayed smooth correction then gives

$$p_B = t_B - \frac{\sqrt{\Delta}(1 + b_0/t_B)e^{-\alpha t_B}}{2\alpha + 2/t_B} + O(e^{-2\alpha t_B}). \quad (34.29)$$

The derivative of the first correction is $O(e^{-\alpha t_B})$ in this neighborhood; multiplying by the first bracket error produces the stated $O(e^{-2\alpha t_B})$ residual. In particular $p_B < t_B$ eventually.

A coarser expression is

$$\begin{aligned} p_B &= \frac{1}{\alpha} W(\sqrt{2\alpha\Delta \log B}) + O\left(e^{-W(\sqrt{2\alpha\Delta \log B})}\right) \\ &= \frac{\log \log B - 2 \log \log \log B + \log(8\alpha\Delta)}{2\alpha} + O\left(\frac{\log \log \log B}{\log \log B}\right). \end{aligned} \quad (34.30)$$

The latter follows either by expanding W or by substitution into the strictly increasing leading model. It is an inverse-height expansion, not a counting second term.

The difference $|p_B - t_B|$ is eventually smaller than V , so using t_B instead of p_B in a progression cutoff changes the count by at most one. It need not give equality at every jump. We keep p_B in the exact floor formula below.

The density in Section 34.3 already proves

$$N(B) \sim \frac{d_{\text{rot}}}{V} p_B \sim \kappa \log \log B.$$

If p_j is the j th selected main index, then $p_j \sim Vj/d_{\text{rot}}$. Inserting this in (34.4) proves $\log \log B_j \sim j/\kappa$. Only the double logarithm is substituted; an $o(p_j)$ index error must not be silently exponentiated into a multiplicative height equivalent.

34.6 Height and cutoff expansions to every order

The first correction above admits an all-orders refinement. It affects the continuous height and its inverse, without improving the rotation discrepancy. It is useful to separate a convergent analytic *normal form* from the smaller remainder of the exact quantity.

Put $d_0 = \sqrt{\Delta}$, $\epsilon = e^{-\alpha p}$, and define

$$A_p = d_0(1 - 1/p), \quad B_p = \frac{d_0 \log \Delta}{\alpha p}, \quad G_0(p) = 2\alpha p + 2 \log p + \log C_0. \quad (34.31)$$

The symbols A_p, B_p here are coefficient functions, distinct from the fixed Pell base A and the height bound B .

Proposition 34.6.1 (All orders at fixed parameters). *For each fixed integer $N \geq 1$,*

$$\log \log y(p) = G_0(p) + \sum_{n=1}^N a_n(p) e^{-n\alpha p} + O(e^{-(N+1)\alpha p}), \quad (34.32)$$

where $a_n(p)$ is a polynomial in $1/p$ of degree at most n . Define

$$S_0(C) = 2, \quad S_1(C) = -C, \quad S_n(C) = -CS_{n-1}(C) + S_{n-2}(C), \quad a_n(p) = -\frac{S_n(A_p) + S_n(B_p)}{n}. \quad (34.33)$$

For $t = t_B$ from (34.28), the exact cutoff has, for every fixed N ,

$$p_B = t + \sum_{n=1}^N c_n(t) e^{-n\alpha t} + O(e^{-(N+1)\alpha t}). \quad (34.34)$$

Each $c_n(t)$ is rational in t and the fixed constants, bounded as $t \rightarrow \infty$, and analytic as a function of $1/t$ near zero. A triangular recurrence is given in (34.38). Constants may depend on N and all the fixed data.

Proof. The leading expression $F(p) = (2m + p - 1)(\alpha m + \frac{1}{2} \log \Delta)$ in (34.26) has the exact factorization

$$F(p) = C_0 p^2 \epsilon^{-2} (1 + A_p \epsilon - \epsilon^2) (1 + B_p \epsilon - \epsilon^2). \quad (34.35)$$

Indeed, $m = p(\epsilon^{-1} - \epsilon)/(2d_0)$, so the two original factors respectively equal $(p/d_0)\epsilon^{-1}(1 + A_p \epsilon - \epsilon^2)$ and $(\alpha p/(2d_0))\epsilon^{-1}(1 + B_p \epsilon - \epsilon^2)$. Since $F(p) \sim 2\alpha m^2$ and $m \geq p$, dividing the error in (34.26) by $F(p)$ and taking a logarithm gives

$$\begin{aligned} \log \log y(p) &= G_0(p) + \log(1 + A_p \epsilon - \epsilon^2) + \log(1 + B_p \epsilon - \epsilon^2) + \mathcal{R}(p), \\ \mathcal{R}(p) &= O(e^{-2\alpha m}/m) = o(e^{-K\alpha p}) \quad \text{for every fixed } K > 0. \end{aligned} \quad (34.36)$$

The roots $r_{\pm}(C) = (-C \pm \sqrt{C^2 + 4})/2$ satisfy

$$1 + C\epsilon - \epsilon^2 = (1 - r_+(C)\epsilon)(1 - r_-(C)\epsilon).$$

Their power sums are $S_n(C)$ in (34.33), and hence

$$\log(1 + C\epsilon - \epsilon^2) = - \sum_{n \geq 1} \frac{S_n(C)}{n} \epsilon^n$$

whenever $|\epsilon| \max(|r_+|, |r_-|) < 1$, taking branches that vanish at zero. For $p \geq 1$, both A_p and B_p lie in a fixed bounded nonnegative interval. A constant $K'_0 \geq 1$ therefore bounds all four root moduli uniformly. For $|\epsilon| \leq 1/(2K'_0)$, the sum of the two tails after order N has absolute value at most

$$4 \sum_{n > N} \frac{(K'_0 |\epsilon|)^n}{n} \leq \frac{8(K'_0)^{N+1}}{N+1} |\epsilon|^{N+1}.$$

This proves uniform convergence of the two-logarithm normal form. The recurrence shows that S_n is a polynomial of degree at most n , proving the coefficient assertion and (34.32).

For inversion put $p = t + u$ and $w_0 = e^{-\alpha t}$. Define

$$\begin{aligned} H_t(u, w_0) &= 2\alpha u + 2 \log(1 + u/t) \\ &\quad + \log(1 + A_{t+u} w_0 e^{-\alpha u} - w_0^2 e^{-2\alpha u}) \\ &\quad + \log(1 + B_{t+u} w_0 e^{-\alpha u} - w_0^2 e^{-2\alpha u}). \end{aligned} \tag{34.37}$$

Introduce $z_0 = 1/t$, replacing $1/(t + u)$ by $z_0/(1 + z_0 u)$. The resulting function $H(u, z_0, w_0)$ is jointly analytic near $(0, 0, 0)$, using logarithms near 1. It satisfies

$$H(0, z_0, 0) = 0, \quad \partial_u H(0, z_0, 0) = 2\alpha + 2z_0,$$

which is nonzero near zero. The analytic implicit-function theorem supplies a unique analytic small root $\mathcal{U}(z_0, w_0)$ on a fixed neighborhood, with $\mathcal{U}(z_0, 0) = 0$. Its series in w_0 converges uniformly on a smaller polydisk and has uniformly bounded fixed-order remainders.

Explicitly, write $u_{<n} = \sum_{j=1}^{n-1} c_j(t) w_0^j$. Then

$$c_n(t) = - \frac{[w_0^n] H_t(u_{<n}, w_0)}{2\alpha + 2/t}. \tag{34.38}$$

Adding $c_n(t) w_0^n$ affects the coefficient of order n only through the displayed linear derivative. Expansion of the rational functions, exponentials and logarithms proves inductively that these coefficients are rational in t and the fixed constants; joint analyticity proves regularity and boundedness at $1/t = 0$.

It remains to justify inversion for the exact height, rather than just its normal form. The already proved first-order inverse gives $u_{\text{ex}} = p_B - t = O(w_0)$. Equation (34.36) gives

$$H_t(u_{\text{ex}}, w_0) = -\mathcal{R}(t + u_{\text{ex}}).$$

In this real neighborhood $\partial_u H_t \geq \alpha > 0$ for large t . The mean value theorem bounds the distance from u_{ex} to $\mathcal{U}(1/t, w_0)$ by $\alpha^{-1} |\mathcal{R}(t + u_{\text{ex}})|$. Since $m(t + u_{\text{ex}})/m(t) \rightarrow 1$, this is

$$O(e^{-\alpha m(t)}/m(t)) = o(w_0^K) \quad \text{for every fixed } K > 0.$$

The uniformly convergent normal-form series therefore gives the actual fixed-order expansion (34.34). $\square$

For reference, $a_1 = A_p + B_p$ and $a_2 = -2 - (A_p^2 + B_p^2)/2$, while

$$c_1(t) = -\frac{d_0[1 + ((\log \Delta)/\alpha - 1)/t]}{2\alpha + 2/t}.$$

These reproduce the earlier first corrections. The convergent series sum to the *normal forms*; the exact height and cutoff still differ by beyond-all-orders terms. Neither series is asserted to equal the exact quantity. Smooth expansions of arbitrarily high fixed order do not remove the counting discrepancy analyzed next.

34.7 An eventual exact rotation count

We now strengthen the density argument using a standard effective theorem on linear forms in logarithms. For fixed positive algebraic numbers $z_1, \dots, z_s$, there are effective constants $c, C > 0$, depending on those numbers, such that every nonzero real form

$$\Lambda = \sum_{i=1}^s b_i \log z_i, \quad b_i \in \mathbb{Z},$$

satisfies $|\Lambda| \geq c\mathcal{B}^{-C}$, where $\mathcal{B} = \max(3, |b_1|, \dots, |b_s|)$. The sufficient statement is Theorem 1.1, equation (1.2), and its fixed-number consequence in Bugeaud [5]; Baker–Wüstholz [6] is a classical additional reference. No multiplicative-independence assumption is needed beyond nonvanishing of the particular form.

For $v = Y, Y + 1$ and an integer n near (34.20), consider

$$\Lambda_v(p, n) = p \log \lambda - n \log \nu + \log \frac{\sqrt{D_P}}{2v\sqrt{\Delta}}. \quad (34.39)$$

This form is nonzero. If it vanished, squaring the associated positive multiplicative equality would give

$$\lambda^{2p} = \frac{4v^2 \Delta}{D_P} \nu^{2n}.$$

Its left side lies in $\mathbb{Q}(\sqrt{\Delta})$ and its right side in $\mathbb{Q}(\sqrt{D_P})$, hence both are rational. The norm-one property forces the positive rational λ^{2p} to equal 1, contrary to $p > 0$.

The coefficients in (34.39) are $O(p)$ by (34.20). Thus, uniformly for either endpoint and all relevant large p ,

$$|\Lambda_v(p, n)| \geq c_1 p^{-C_1} \quad (34.40)$$

for effective fixed-data constants. This eventually exceeds the $O(e^{-2\alpha p})$ endpoint shifts in (34.21). A disagreement between an exact and limiting indicator would require an allowed lattice point within that exponentially small distance of a limiting endpoint, contradicting (34.40). Therefore only finitely many disagreements occur.

Enlarge p_b within its progression beyond these exceptions and the positivity and height thresholds. Use the phase ξ corresponding to this new p_b in (34.22). For every $r \geq 0$, the exact selection is now precisely (34.23). Define

$$D(R) = \sum_{r=0}^{R-1} \mathbf{1}_I(\{\xi + r\theta\}) - d_{\text{rot}} R \quad (R \in \mathbb{Z}_{\geq 0}), \quad (34.41)$$

with the empty sum zero. In $D(R)$ and $R(B)$, R denotes a nonnegative number of progression terms, not the restored coordinate $R = -p$ used in the construction. For all sufficiently large B ,

$$R(B) = \max \left(0, 1 + \left\lfloor \frac{p_B - p_b}{V} \right\rfloor \right). \quad (34.42)$$

The exact identity for this tail is

$$N(B) = d_{\text{rot}} R(B) + D(R(B)). \quad (34.43)$$

This retains the fixed phase, the open strict-ratio interval and the true real cutoff. No asymptotic expression has been passed through a floor as if exact.

Combining (34.30) and (34.42) gives, for $t = \log \log B$,

$$d_{\text{rot}} R(B) = \kappa t - 2\kappa \log t + O(1), \quad R(B) \sim t/(2\alpha V),$$

which proves (34.7). Changing to another finite initial convention merely adds a fixed count offset and does not remove the discrepancy.

34.8 Effective errors and the second term obstruction

34.8.1 A fixed parameter power saving

For $h \geq 1$, choose an integer k_h nearest to $h\theta$. The nonzero form

$$hV\alpha - k_h L\beta$$

has integer coefficients $O(h)$, and its nonvanishing follows from irrationality of θ . The same logarithmic-form bound therefore supplies effective $c_2 > 0$ and $\mu \geq 1$ such that

$$\|h\theta\| \geq c_2 h^{-\mu}. \quad (34.44)$$

The exponent can be weakened to ensure $\mu \geq 1$. The finite geometric sum gives

$$\left| \sum_{r=0}^{R-1} e^{2\pi i h(\xi + r\theta)} \right| \leq \min \left(R, \frac{1}{2\|h\theta\|} \right) \ll h^\mu.$$

The unnormalized Erdős–Turán inequality [9, 10], with an absolute implied constant, yields for each integer $K \geq 1$

$$\begin{aligned} |D(R)| &\ll \frac{R}{K+1} + \sum_{h=1}^K \frac{1}{h} \left| \sum_{r=0}^{R-1} e^{2\pi i h(\xi + r\theta)} \right| \\ &\ll \frac{R}{K} + K^\mu. \end{aligned} \quad (34.45)$$

Taking K of order $R^{1/(\mu+1)}$ gives

$$D(R) = O(R^{1-\varepsilon}), \quad \varepsilon = \frac{1}{\mu+1} > 0. \quad (34.46)$$

Since $R(B)$ is of order t and $\varepsilon \leq 1/2$, the analytic $\log t$ correction is absorbed by this error. Equations (34.43) and (34.7) prove the counting part of (34.6).

There are no height ties, so $N(B_j) = j$. The leading law gives $t_j = \log \log B_j \sim j/\kappa$; substituting this into $j = \kappa t_j + O(t_j^{1-\varepsilon})$ proves the ranked error in (34.6). This effective error may be very weak, and remains much larger than the scale $\log \log B$ of the putative second term.

34.8.2 Why a bounded smooth remainder is impossible

Kesten's interval bounded-remainder theorem states that an interval for an irrational rotation has bounded discrepancy if and only if its length belongs to $\mathbb{Z} + \theta\mathbb{Z}$ [7, 8]. It applies to a fixed phase: translating the interval absorbs that phase. Open versus half-open endpoint conventions change at most two orbit hits.

In our case $d_{\text{rot}} \notin \mathbb{Z} + \theta\mathbb{Z}$. Otherwise integers a_1, b_1 would give

$$\delta = a_1 L\beta + b_1 V\alpha, \quad \frac{Y+1}{Y} = \nu^{a_1 L} \lambda^{b_1 V}.$$

The numbers λ and ν and their inverses are algebraic integers, since their monic integer polynomials have constant term 1. The right side is an algebraic unit. A rational algebraic unit must be 1 or -1 , whereas $(Y+1)/Y > 1$. This contradiction proves the claim. Kesten's theorem now implies that $D(R)$ is unbounded.

As B ranges through large real values, $R(B)$ attains every sufficiently large integer: take $B = y(p_b + V(R-1))$. The bounded residual in (34.7) cannot cancel an unbounded discrepancy. Hence

$$N(B) = \kappa \log \log B - 2\kappa \log \log \log B + O(1)$$

is false as an all-height assertion. The conclusion is about the claimed smooth remainder, not a failure of the leading law or of the precise height inverse.

34.8.3 The exact unresolved criterion

For this fixed rotation, the following statements are equivalent:

$$\begin{aligned} N(B) &= \kappa t - 2\kappa \log t + o(\log t) && \text{as all real } B \rightarrow \infty, \quad t = \log \log B; \\ D(R) &= o(\log R) && \text{as integers } R \rightarrow \infty. \end{aligned} \tag{34.47}$$

Sufficiency follows from (34.7) and $R(B) \sim t/(2\alpha V)$. For necessity take $B = y(p_b + V(R-1))$. Then $R(B) = R$ exactly, and $\log t = \log R + O(1)$ by (34.4). Subtracting the smooth terms in (34.7) gives the required $o(\log R)$ estimate.

Unboundedness of $D(R)$ does not disprove this finer condition. An $O(\log R)$ bound alone would also be insufficient to identify the second coefficient, since its error has exactly that scale. Equidistribution supplies only $o(R)$, and the effective power saving (34.46) is still too large. No $o(\log R)$ estimate is claimed.

34.9 Comparison and further research

34.9.1 The scope differs from Report 25

Report 25 classifies an entire fixed positive native fiber: seventeen supplied coordinates are forced, and all remaining tuples are parametrized by two Pell indices [4]. Its height law follows from exact finite floor counts, exceptional baseline terms, hyperbola decomposition and Euler summation. In its notation the leading count is logarithmic in height, with a proved unsmoothed square-root second term and a smaller power error.

Here the input, compiler, repunit and prime-scale choice are fixed, but only the prescription $m = p\psi_A(p)$ and $S = p + 2m$ is counted. The free main index runs through one CRT progression,

selected by an irrational-rotation window. No classification of all admissible auxiliary indices or all signed19 witnesses has been proved. The count is double-logarithmic in height, and its second-term question is governed by the explicit discrepancy (34.41). Report 25's two-index floor-sum error cannot be imported into this problem.

Pell growth, Binet expansion and careful height inversion are standard shared tools. The specific issue resolved here is the persistence of the exact rotation discrepancy through exponentially small strict-ratio endpoint perturbations, and the consequent bounded-remainder obstruction. This comparison is a statement of mathematical scope, not a literature-wide novelty claim.

34.9.2 Concrete questions left open

1. **The finer discrepancy scale.** Determine whether $D(R) = o(\log R)$ for the explicit fixed θ, ξ, I in (34.22). By (34.47), this is exactly the missing estimate for the stated triple-logarithmic second coefficient. A result only on a subsequence or after averaging would not establish the all-height claim.
2. **Usable effective constants.** Specialize explicit logarithmic-form constants to one fixed compiler, input and scale, and bound the last possible endpoint disagreement. The present proof gives effectivity in principle, without a practical numerical exponent or onset. It does not materialize the enormous supplied tuples.
3. **Larger families.** Characterize other permitted auxiliary choices at a fixed main index, then count their union without duplication. The current canonical prescription offers a lower-bound subfamily inside the full fiber, not an upper bound or an exhaustive parametrization.
4. **Dependence on the scale.** Study how density and height constants vary over prime-scale choices, with genuinely uniform error bounds if possible. Summing fixed-scale asymptotics without such control is not justified by this report.

The valid positive-index parent retains its positivity hypothesis. The separate raw29 and positive21 restoration questions remain outside the present theorem. No new universal operation count, degree bound, full numerical witness or machine-specific acceptance claim follows.

34.A Transfer to asymmetric signed19 orientations

A separate pinned source comparison [11] extends the applicability of this fixed family. At ProveIt commit

[write] The pinned sources of this appendix are research-tree files shipped inside archive 05 as byte copies and not shipped here: `complete74_asymmetric_scale_transfer.md` and `.json`, `complete74_equation_orientation_census.md` and `.json` (the latter 1,508,808 bytes), and `review_complete74_equation_orientation_census.md`, all in the research tree's `native-stream-queue/` directory and equal to the current files. The transfer proof itself is shipped as `05-signed19-heights-evidence-TRANSFER-PROOF.md`. (*Corrected in batch 83: the batch-82 write said Report 34 had not been reviewed.*) The programme's review `review_incoming_counterexample_heights.md` (commit `ceb7222e9`) inspects all eight signed20 orientation sources and confirms this transfer to the eight matching signed19 children as a forward transfer only.

9fa99be93fb479294d5d5a2458e59e5fba8444b6,

the asymmetric scale replaces $X = wq^3$ by $X = wq$, retaining $Y = sq^3$. The supplied w has no other consumer. Hence the coordinate map

$$\Phi : \quad w_{\text{new}} = q^2 w_{\text{old}} \quad (34.48)$$

preserves X and every subsequent computed quantity, including R , while leaving all other supplied coordinates unchanged.

There are four relevant equation orientations at each scale. Define

$$T_2 = (ic^2)^2, \quad K = \Delta(f^2 - 1), \quad U_0 = jc - R, \quad U_1 = of - c.$$

Each orientation retains *both* protecting comparisons $T_2 = K$ and $U_0 = U_1$, and selects its auxiliary norm from

$$C_e(U_v^2 - y^2) = 1 - y^2, \quad C_0 = T_2, \quad C_1 = K, \quad (e, v) \in \{0, 1\}^2.$$

For each parent define the corresponding child by substituting $r = R = k - hXY - 1$ everywhere and deleting only the resulting identical index comparison. These are mathematically defined descendants, not a claim that every eliminated packet has been published upstream.

For each *matching* orientation, the scale map (34.48) preserves every residual, so

$$G_{(1,3)}^{(e,v)}(\Phi(\mathbf{w})) = G_{(3,3)}^{(e,v)}(\mathbf{w})$$

is an all-value polynomial identity over any commutative ring. Across *different* orientations, the retained protecting comparisons give equality of comparison-zero loci over every commutative ring, and hence equality of sum-of-squares zero loci over the reals. This is not an assertion that the differently oriented sum-of-squares polynomials agree away from their zero sets.

The Report 33 tuples therefore give positive supplied-witness zeros with $R = -p$ in all four symmetric orientations and, after Φ , all four asymmetric orientations. In the family of this report, q and w are fixed. The map changes just one fixed coordinate to another fixed value, while leaving $y(p)$ and every index-selection test unchanged. After increasing the finite height threshold if necessary, the maximum is still exactly y . All the canonical-family height and counting conclusions of Report 34 consequently transfer with the same constants and discrepancy, up to the finite-tail convention.

This uses only the forward map. It asserts neither an integral inverse on every asymmetric-child zero nor a classification of that child's full fiber. The valid twenty-coordinate parents still supply a positive r and retain their nine comparisons. No conclusion for raw29, positive21, normalized87 or coupled88 is imported. No parent degree or operation ledger is assigned to these children.

34.B Evidence and the portable release

The construction source is pinned to ProveIt commit

2dde7850ffeb9c923a7bd92e3f2ee2b5af0e36ff.

The report was prepared from the frozen construction, its independent audit, and the separately audited analytic proof packet. The original reviewed analytic proof, before path normalization in the packaged copy, has SHA256 identity

ff9d116c90b058be082c2c9d6ee051cc
f537d80821c14d10b28aec9a96d5a533.

The two lines concatenate to the complete digest. The whole-packet audit reports a mathematical pass for that exact version, including the effective error and the fixed-phase bounded-remainder application. `PROVENANCE.json` maps this original identity to the packaged digest and records every path-only portability normalization of the accompanying analytic notes.

The evidence has three distinct roles:

1. The inherited construction and its audit establish full child membership, positive supplied coordinates, and the fixed arithmetic progression.
2. The analytic proof and independent height, rotation and whole-packet checks establish the asymptotic formulas and external-theorem applications.
3. The local release verifier checks file identities; the offline build script rebuilds the PDF from the supplied LaTeX. These are packaging checks, not finite tests of the unbounded mathematical assertions.

No executable mathematical checker is included or needed for this pure analytic follow-on. No upstream code, source arithmetic schedule or historical finite checker was run to establish its claims. The source component checks are analytic notes. No third-party paper PDF is redistributed in the archive; the bibliography links to the research sources. The report is a new release and does not alter Report 33.

References of Report 34

- [1] *Failure of Positive Index Restoration: The complete⁷⁴ signed¹⁹ eliminated child*, Research Report 33, 3 October 2026. Frozen delivered report; its LaTeX source is included as context, unchanged. The construction used here is in Sections 4–9. [write] Printed in Part IV of this report, as Sections 33.1–33.B; its Sections 4–9 are Sections 33.4–33.9.
- [2] *Full negative-restoration family for the signed¹⁹ child*, frozen analytic construction packet, 3 October 2026. Included as `evidence/FULL-SIGNED-COUNTEREXAMPLE.md`; SHA256 recorded in the release provenance. Based on the ProveIt source at the commit identified in Appendix 34.B. [write] Shipped once, as `10-index-restore-repro-evidence-FULL-SIGNED-COUNTEREXAMPLE.md`.
- [3] *Independent audit of the full signed¹⁹ negative-restoration family*, 3 October 2026. Included as `evidence/CONSTRUCTION-INDEPENDENT-AUDIT.md`. This audit supports the inherited construction; it is distinct from the whole-packet growth audit. [write] Shipped once, as `10-index-restore-repro-evidence-INDEPENDENT-AUDIT.md`.
- [4] *Entire native fibers: Pell parametrization and height laws*, Research Report 25, 3 October 2026. Frozen delivered report, Theorems 2.1–2.2; its LaTeX source is included unchanged as context. The comparison here is not a new audit of that entire article. [write] Printed as Part III of this report; its Theorems 2.1–2.2 are Theorems 25.2.1–25.2.2.
- [5] Y. Bugeaud, B' , arXiv:2209.00275v1 (2022), Theorem 1.1, equation (1.2), printed pp. 2–3, and the ensuing fixed-number consequence. <https://arxiv.org/abs/2209.00275v1>. The theorem collects estimates attributed there to Waldschmidt and Matveev; its fixed-number consequence is the sufficient bound used in this report.
- [6] A. Baker and G. Wüstholz, *Logarithmic forms and group varieties*, Journal für die reine und angewandte Mathematik **442** (1993), 19–62. <https://doi.org/10.1515/crll.1993.442.19>. Classical background reference; the explicit sufficient statement inspected for this report is [5].

- [7] H. Kesten, *On a conjecture of Erdős and Szűsz related to uniform distribution mod 1*, Acta Arithmetica **12** (1966), 193–212. <https://doi.org/10.4064/aa-12-2-193-212>.
- [8] M. Kelly and L. Sadun, *Pattern equivariant cohomology and theorems of Kesten and Oren*, arXiv:1404.0455v2 (2014), Theorem 1. <https://arxiv.org/abs/1404.0455v2>. A new proof of Kesten’s criterion, with a fixed starting point in its setup.
- [9] P. Erdős and P. Turán, *On a problem in the theory of uniform distribution I*, Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen **51** (1948), 1146–1154, Theorem III, printed p. 1150. https://www.renyi.hu/~p_erdos/1948-02.pdf.
- [10] P. Erdős and P. Turán, *On a problem in the theory of uniform distribution II*, Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen **51** (1948), 1262–1269. https://www.renyi.hu/~p_erdos/1948-03.pdf.
- [11] *Report 33 transfer to the asymmetric signed19 orientation children*, source-pinned transfer proof, 3 October 2026. Included as `evidence/TRANSFER-PROOF.md`; source identities are recorded in `PROVENANCE.json`. The pinned asymmetric scale transfer and equation orientation census descriptions are identified in Appendix 34.A. [Pinned asymmetric transfer source](#). [write] Shipped as `05-signed19-heights-evidence-TRANSFER-PROOF.md`.

Part V

Exact full negative zero obstruction for positive Diophantine interfaces

[write] Source 15 of this report, manuscript 09 of batch 83: *Research Report 37*, subtitled “Deterministic packing recovery and complete positive witness reconstruction”, author line “Mathematical research report”, dated 3 October 2026; archive `Exact_Negative_Index_Obstruction_for_Positive_Diophantine_Interfaces_Package.zip` (arrival commit 3051d1446, placed by a51a439cd); pin d5bd4a67b, targeted refinement 813c1cff4, freshness check 0f7d618f4. Its sections, statements and equations are numbered 37.*k* here.

Abstract of the source. For the fixed-compiler raw29 and positive21 nonlinear-index interfaces, we give a necessary-and-sufficient predicate for a full strictly positive supplied tuple whose restored index is negative. The predicate completes the previously established reduction to two unbounded scale parameters. At each scale, a bounded representative determines at most one possible main Pell index. Packing then forces the marker and field word, and either of two input indices determines the remaining input marker. Every passing predicate admits explicit positive witnesses for all outer, first, main, input, strong and auxiliary equations. A necessary Binet window, intersected with the admissible index range, has length less than $q^{-3} + q^{-5} < 1$ and admits outward rational evaluation. The odd-input wrapped sector is finite for each fixed compiler and input; the even-input sector obeys stronger bounds involving the actual outer marker. These results neither prove global restored-index positivity nor construct a full negative zero. Exact finite tests and separate audits corroborate the algebra and certified interval exclusions; they are not valid-compiler counterexamples or an unbounded search certificate.

[write] *What the title means.* The “obstruction” of the title is an exact necessary-and-sufficient predicate (Theorem 37.5.1), not a proof that no negative zero exists. Whether the predicate ever passes is open: a full negative zero of raw29 or positive21 is neither constructed nor excluded.

Place in this report. Report 37 continues Appendix 33.B of Report 33, whose closing task (“to exclude or realize the negative branch while simultaneously meeting” the window, the input dichotomy, the exact Pell ratio and every outer and input equation) it converts into this predicate with a complete positive inverse witness map. That task is therefore re-scoped, not closed (dated note at the end of Appendix 33.B). Its evidence packet carries two of Report 33’s own evidence files as context, and its Section 37.3 restates Appendix 33.B and the research tree’s refinement note; that section is printed as a recap, with pointers. Report 37 is printed as its own Part rather than as a third source of Part IV because Part IV is about the signed19 child, whereas Report 37 treats only raw29 and positive21.

The programme’s review. `review_exact_negative_obstruction37.md` in the research tree’s `native-stream-queue/` directory (commit 308f14072, with the helper `review_exact_negative_obstruction37.py` and its receipt `.json`) gives “PASS within the stated fixed-compiler scope”: the predicate is necessary and sufficient and closes the conditional witness-reconstruction obligations, but proves neither that it has a solution nor that it has none (“Those two positive inverse problems remain unresolved”). Its scope precisions follow Theorem 37.5.1; its separation from the programme’s universal polynomial follows Lemma 37.4.2.

Notation in this Part. Report 37 uses Report 33’s letters (Section 33.2) with these differences: d_{cell} is Report 33’s d in $B = 2^d$; z_{quot} is Report 33’s transport quotient z ; the paid and native masks are rendered $M_{F,\text{src}}$ and $M_{F,0}$, the objects written M_F^{src} and M_F^{native} in Part IV; T_{aux} and ℓ are Report 33’s Q_{aux} and s_a ; the first-representative quotient t is Report 33’s t_- (Appendix 33.B), not the binary exponent $t = dN$ of Section 33.4; the input Pell index e is Report 33’s v in Appendix 33.B; and W is the positive supplied input marker. To keep one meaning per letter inside this Part, the merge renamed these symbols of the source (renderings otherwise unchanged, no normalization changed): $E_r^* \rightarrow \mathcal{E}_A(r)$, which is Report 33’s $e_A(r)$ (E

stays XY); $Q_{h_0}, h_0 \rightarrow Q_\theta, \theta$, which are Report 33's Q_b, b_a (h stays the supplied quotient and Q stays $q^2 - 1$); $a_\pm(V), a_\pm, b_\pm \rightarrow \omega_\pm(V), \omega_\pm^A, \omega_\pm^P$ (a stays the register $Y(X + 1)$ and b the odd compiler constant); $r_\pm \rightarrow \xi_\pm$; $\alpha_*, \beta_* \rightarrow \lambda_A, \lambda_P$ (α stays the raw slack; λ_A is Part IV's unit λ); $L(p), U(p) \rightarrow \Lambda_-(p), \Lambda_+(p)$ (L stays the compiler exponent, U the auxiliary root); $J_{\log} \rightarrow I_{\text{Bin}}$ (J stays the repunit coordinate); and in the logarithm series of Section 37.7 $r, k, t, z, j, N \rightarrow \vartheta, \nu, \vartheta_0, z_0, l, N_s$ and the endpoints $l, h \rightarrow p_{\text{lo}}, p_{\text{hi}}$, since k, t, j, N, h are witnesses or quotients elsewhere in this Part. The inner radix $R_{\text{rad}} = 2^b$ and the defect $d_{\text{def}} = 2n - p$ keep the source's subscripts; R alone is the restored index. Section 0.3 lists the letters that differ between Parts.

The source's files are shipped with the prefix `15-neg-obstruction-`.

37.1 Conclusion and exact scope

The unresolved question is whether deleting an explicit positive-index coordinate from the two positive interfaces can introduce a full positive zero with a negative restored index. Earlier work reduced a putative negative zero to $R = -p$, where p is the positive main Pell index, and bounded all index choices after two scale parameters are fixed. Necessary conditions alone did not settle whether omitted input or auxiliary sign constraints could prevent a reconstruction.

This report closes that reconstruction gap. A passing arithmetic predicate gives a complete strictly positive witness in *both* interfaces, and every negative zero gives a passing predicate. The remaining problem is exactly the simultaneous feasibility of that predicate. Its emptiness and nonemptiness are both unresolved.

The source receipt uses the inherited mode names `raw30` and `positive22`. After the index elimination their supplied lists have respectively 29 and 21 positive coordinates; we call those children *raw29* and *positive21*. Their saved finalizers are sums of squared integer residuals. The signed19 child has different input-sign freedoms and is excluded throughout. A signed-interface negative construction does not supply the bounded positive markers required here.

What is new and what is inherited. The two-unbounded-parameter reduction, exact alternatives $R = \pm p$, evenness of q , the two input indices, defect exclusion, and at-most-one main index are prior results. The new conclusions are deterministic packing recovery, a full-zero equivalence with every sign obligation discharged, the finite wrapped-odd-input sector, the stronger even-input bounds, and a quantitative sub-unit necessary real window. The all-odd-index auxiliary construction is explicitly credited to the prior bootstrap. No operation count, universal polynomial bound, or new witness bound is claimed.

[write] In this report the prior results are Appendix 33.B of Report 33 and the research tree's `complete74_negative_index_refinement.md` (commit 813c1cff4), Sections 2–4; the bootstrap is the tree's `review_complete74_nonlinear_index_bootstrap.md` (commit 93ad43375). Section 37.3 gives the correspondence.

Source boundary. The mathematical target is the pinned ProveIt projection [1] at commit

d5bd4a67b41b89688a079997574a94bcd855bb83.

The five source files and their immutable URLs, byte counts, SHA256 digests and Git blob identities are supplied in `evidence/source_manifest.json`. The targeted refinement was last changed at commit

813c1cff42f9238a0158228f5e003a6a5405444a. A separate freshness receipt records that all five relevant blobs were unchanged at the later inspected tip 0f7d618f43bcb6361cee432273ba644e7defd951. This is a time-stamped source check, not a promise about future upstream changes. No upstream Python module or saved arithmetic schedule was executed for this report. Their role is read-only source evidence.

[write] The source manifest and the freshness receipt are shipped as `data/15-neg-obstruction-evidence-source_manifest.json` and `data/15-neg-obstruction-evidence-final_source_freshness.json`. The five source files themselves are byte copies of research-tree files and are not shipped; the bibliography of this Part gives their repository paths. At this write all five are byte-identical to the current files, and d5bd4a67b is an ancestor of the commit at which this Part was written.

37.2 Fixed compiler contract and literal equations

37.2.1 A genuine compiler slice rather than arbitrary numeral ports

Fix a valid export of the complete⁷⁵ modified half-binomial compiler [4] and an ordinary positive integer input x . Its fixed integers include

$$B = 2^{d_{\text{cell}}}, \quad K_0 = D_C + BD_R, \quad M_C, M_{F,0}, \quad M_{F,\text{src}} = M_{F,0} + B - 1.$$

The valid compiler contract, not a free choice of mask numerals, gives

$$B \geq 16, \quad 1 \leq b < B, \quad b \text{ odd}, \quad 0 < D_C, D_R < B, \quad K_0 \leq B^2 - 1, \quad 0 < M_C, M_{F,0} < B - 1. \quad (37.1)$$

Put

$$u = 2d_{\text{cell}}x + b. \quad (37.2)$$

Thus u is odd and $u \geq 3$.

For clarity, the fixed source recipe retains $b, L, d_{\text{cell}} = bL$ as powers of five, an inner radix $R_{\text{rad}} = 2^b$, and $B = R_{\text{rad}}^L$. It retains the original separated native layout, selectors, duplicated center-clause bands, four anchors, and the optional high monomial ensuring $2D_C - D_R \not\equiv 0 \pmod{5}$. If e_* is the specified ignored dummy exponent, the modified masks are

$$M_C = M_C^{\text{old}} - 2R_{\text{rad}}^{e_*}, \quad M_{F,0} = M_F^{\text{old}} + 4.$$

The fixed radix is enlarged until

$$R_{\text{rad}} \geq \max \left(4(K+2) \left(2 \sum_e c_e + 6 \right) + 8, 2\mu_{\text{mask}} + 4, 16 \right).$$

Here $K, c_e, \mu_{\text{mask}}$ are the native compiler's support size, center coefficients and clause mask, not varying witnesses. The export is the modified `new_constants(compile_windows(...))` interface. Using only an inherited baseline `constants()` object would select the wrong masks. In particular the native bound $M_{F,0} < B - 1$ must not be applied to the shifted source coefficient $M_{F,\text{src}}$.

The source contract and sign-free bootstrap are inherited mathematical premises. This report is self-contained for the new obstruction and reconstruction under those premises; it does not reprove the fixed machine's complete compiler theorem or the generic ordinary auxiliary rank theorem. The supplied prior reduction explains the genuine coefficient bound: the sparse D_C coefficients remain below the enlarged radix, its support and the D_R exponent remain below L , and the modified export changes only the masks. Thus $D_C, D_R < B$ is provenance from the actual compiler, not a numerical-port assumption silently substituted for it.

[write] The coefficient bound is the one derived in Section 33.B (“Why the actual compiler gives the required coefficient bound”, equation (33.53)), where L is written L_{comp} ; in the radix bound the summation index e and the coefficients c_e are the compiler’s, unrelated to the input index e and the coordinate c below. The programme’s review stresses the same point: $K_0 < B^2$ is a property of the actual numerals, not an assumption on arbitrarily chosen numeral ports.

37.2.2 Scale notation and Pell sequences

For positive scale coordinates J, q, w, s , set

$$q = (B - 1)J + 1, \quad X = wq^3, \quad Y = sq^3, \quad E = XY, \quad (37.3)$$

$$a = Y(X + 1), \quad A = a + 2, \quad \Delta = A^2 - 1, \quad H = 4a + 3, \quad (37.4)$$

$$P = 2XY^2 + 1, \quad Q = q^2 - 1, \quad M = (M_C + qM_{F,\text{src}})J.$$

For every integer base $V \geq 2$, define the integer Pell sequences by

$$\chi_V(r) + \psi_V(r)\sqrt{V^2 - 1} = \left(V + \sqrt{V^2 - 1}\right)^r \quad (r \geq 0). \quad (37.5)$$

They have initial values $\chi_V(0) = 1$, $\chi_V(1) = V$, $\psi_V(0) = 0$, $\psi_V(1) = 1$ and recurrence $f(r + 1) = 2Vf(r) - f(r - 1)$. Thus every comparison below is an effective comparison of integers, however large those integers may be.

The mathematical base $A = a + 2$ must not be confused with the saved source register named **A**, which represents Δ . The saved source name **d** represents the positive main root D , and **Jrep** represents J .

37.2.3 The complete retained system

The restored index is the expression $R = k - hE - 1$, not a separately supplied positive coordinate. The outer comparisons are

$$q = (B - 1)J + 1, \quad C + \alpha + u - b = q, \quad (37.6)$$

$$(K_0 + X)C = F + z_{\text{quot}}(q - 1), \quad C = Z + W, \quad (37.7)$$

$$R = (q^2 - Z - qF)Q + M.$$

The first and main blocks impose

$$\tau^2 - XY^2(XY^2 + 1)k^2 = 1, \quad c = kY + \eta, \quad k = \eta + \zeta, \quad (37.8)$$

$$a = Y(X + 1), \quad D = X + ac + g_a H, \quad D^2 - \Delta c^2 = 1.$$

Writing $T = ic^2$ and $U = jc - R$, the ordinary strong and auxiliary comparisons are

$$T^2 = \Delta(f^2 - 1), \quad T^2(U^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2, \quad U = of - c. \quad (37.9)$$

Finally the entire input block is

$$\kappa = u + \delta\Delta, \quad c = \kappa + \phi, \quad \mu^2 = 1 + \Delta\kappa^2, \quad \mu = W + a\kappa + \rho H. \quad (37.10)$$

All 29 supplied coordinates in the raw mode are strictly positive integers. In positive21, eight coordinates are eliminated by triangular source expressions; the retained comparisons and supplied positive coordinates force their same positive values. In particular the main quotient g_a is independent of ρ . The split $g_a = \rho + \sigma$ belongs to the signed source and is not an equation of these two interfaces.

[write] Equation (37.6) writes the raw bound as $C + \alpha + u - b = q$; Section 33.B writes the same equation as $C + \alpha + 2dx = q$, since $u - b = 2d_{\text{cell}}x$. The signed19 system of Section 33.2 instead has $\gamma = \rho + \sigma$ and a computed, possibly negative, $W = C - Z$.

37.3 Inherited reduction and bounded indices

[write] *A recap, not new.* This section restates, with the source’s own credit, results already printed in this report or proved in the research tree. Equations (37.11)–(37.12) are the imported sign-free bootstrap (33.52) of Section 33.B, from the tree’s `review_complete74_nonlinear_index_bootstrap.md` (commit 93ad43375). Equation (37.13) is the shifted-mask identity of Section 33.2. Equation (37.14), the exact alternatives $R = \pm p$ and the parity paragraph are the chain before (33.54), equation (33.54) itself and the paragraph after it. Equations (37.15)–(37.16) are the input dichotomy (33.56), where the index is called v ; it is also proved, independently, in Section 2 of the tree’s `complete74_negative_index_refinement.md` (commit 813c1cff4; see the note after (33.56)). Equation (37.17) and the bounds after it are the window (33.55), where t is written t_- . Equation (37.18) and its argument are Section 3 of the refinement note, and the uniqueness of Lemma 37.4.1 is its Section 4. The “prior raw/positive reduction” and “prior independent review” that the source cites [6] are Report 33’s own evidence files (bibliography). The section is printed in full because the later sections cite its equations.

37.3.1 The bootstrap input

The prior sign-free bootstrap and its independent review [2, 6] classify every positive child zero as follows:

$$c = \psi_A(p), \quad D = \chi_A(p), \quad p \geq 13 \text{ odd}, \quad R \equiv p \text{ or } -p \pmod{c}, \quad (37.11)$$

with $c > 2p$ and $c > A^{12}$. It classifies the first norm by

$$k = 2\psi_P(n), \quad \tau = \chi_P(n), \quad \frac{p+1}{2} \leq n \leq p-1. \quad (37.12)$$

The ratio $Y < c/k < Y + 1$ supplies the index range. The ordinary strong norm and the mixed-sign auxiliary analysis, with its positive root checked before classification, supply the rank congruence. These are conclusions of the cited bootstrap, not consequences asserted from a finite sample.

37.3.2 Exact signs and input representatives

The shifted-mask identity is

$$R = (q^2 - S')Q + T', \quad S' = Z + qF - 1, \quad T' = M_C J + 1 + q(M_{F,0} J - 1), \quad 0 < T' < Q. \quad (37.13)$$

It gives $R \neq 0$ and $R < q^4$. On the negative branch the actual positive outer markers, transport, and (37.1) sharpen the other direction to

$$\begin{aligned} -R &< (Z + qF)q^2 < Cq^3(X + K_0 + 1/q) \\ &< CXq^3(1 + 1/q) < Xq^4. \end{aligned} \quad (37.14)$$

The third comparison uses $K_0 + 1/q < q^2 \leq X/q$. Since $A > XY \geq Xq^3$, the Pell growth gives $c > 2Xq^4$. Hence $|R| < c/2$ and $p < c/2$ turn (37.11) into the exact alternatives $R = p$ or $R = -p$.

If q were odd, its repunit equation with even B would force J even. Both terms in the original packing expression would then be even, contradicting odd $R = \pm p$. Therefore q is even; so are X, Y, A , while Δ is odd. This does not prove that q is a power of two.

The positive input norm gives $\kappa = \psi_A(e)$ and $\mu = \chi_A(e)$ for an integer $0 < e < p$, using the positive gap $c > \kappa$. The recurrence modulo Δ gives

$$\psi_A(r) \equiv \begin{cases} r & r \text{ odd,} \\ rA & r \text{ even} \end{cases} \pmod{\Delta}. \quad (37.15)$$

The established size bounds give $p < Xq^4 < \Delta$ and $0 < u, uA < \Delta$. Reducing $\kappa = u + \delta\Delta$ and, in the even case, multiplying by A proves

$$e = u \quad \text{or} \quad e = uA. \quad (37.16)$$

Here u is odd and uA is even. Both alternatives must be retained.

37.3.3 The first representative and prior defect exclusion

On $R = -p$, the first-index congruence is

$$2n + p - 1 = tE, \quad t \geq 1. \quad (37.17)$$

It initially gives $2p \leq tE \leq 3p - 3$. Combining $p < Xq^4$ with $tE < 3Xq^4$ gives $ts < 3q$, and in particular $s < 3q$.

[write] (*Batch 88.*) On the unwrapped odd-input sector Report 39 sharpens $ts < 3q$ to $ts < 3C$ (equation (39.72) in Section 39.12).

The prior refinement [3] proves that the positive odd defect $d_{\text{def}} = 2n - p$ satisfies

$$4^{d_{\text{def}}} > X \geq q^3, \quad d_{\text{def}} \geq 7, \quad 2p + 6 \leq tE \leq 3p - 3. \quad (37.18)$$

Its argument uses exact Pell growth, not a small-index approximation. From the ratio it obtains

$$a > \frac{X^{(p+d_{\text{def}})/(2d_{\text{def}})}}{3}.$$

If $X \geq 4^{d_{\text{def}}}$, this would force $a > 2^p$. Since $0 < X < a$ and the main projection is $X \equiv 2^p \pmod{H}$, both representatives would lie in $(0, H)$, forcing $X = 2^p$. But then $E \leq tE \leq 3p - 3 < 2^p = X \leq E$, a contradiction. This conditional no-wrap argument excludes the small-defect sector only; it is not permission to replace every exponential congruence by equality.

37.4 An exact arithmetic predicate

37.4.1 Finite choices after two scale parameters

Enumerate positive integers q, w with

$$q \geq B, \quad q \text{ even}, \quad q \equiv 1 \pmod{B-1}, \quad q > u - b + 2. \quad (37.19)$$

Define $J = (q - 1)/(B - 1)$ and the quantities in (37.3). Choose only

$$1 \leq s \leq 3q - 1, \quad 1 \leq t \leq \left\lfloor \frac{3q - 1}{s} \right\rfloor. \quad (37.20)$$

The permissible main indices are odd integers in

$$I = \left[\max \left(13, \left\lceil \frac{tE}{3} \right\rceil + 1 \right), \min \left(Xq^4 - 1, \left\lfloor \frac{tE - 6}{2} \right\rfloor \right) \right]. \quad (37.21)$$

An empty interval is rejection. For each candidate define

$$n = \frac{tE + 1 - p}{2}. \quad (37.22)$$

The +1 is essential. Since E is even and p odd, n is integral; (37.21) implies $n \leq p - 1$ and $2n - p \geq 7$.

Require the exact Pell ratio

$$2Y\psi_P(n) < \psi_A(p) < 2(Y + 1)\psi_P(n), \quad (37.23)$$

and the filters

$$4^{2n-p} > X, \quad 2^p \equiv X \pmod{H}. \quad (37.24)$$

The defect test is necessary and useful, but its separate inclusion is not needed to make the later reconstruction sufficient.

Lemma 37.4.1 (Prior uniqueness with an exact search). *For fixed q, w, s, t , at most one odd $p \in I$ satisfies (37.23). There is a terminating exact procedure to find or reject that candidate.*

Proof. As p increases by two, n decreases by one. Thus the ratio $\psi_A(p)/(2\psi_P(n))$ strictly increases. More strongly, its numerator grows by more than $(2A - 1)^2$ and its denominator decreases. Since $(2A - 1)^2 Y > Y + 1$, two distinct odd indices cannot both lie in the target ratio interval. Binary search finds the first odd index, if any, that exceeds the lower threshold; only that index needs the upper-threshold test. Integer recurrence or powering gives all exact comparisons. This is an effectiveness statement, not a feasible running-time bound. $\square$

[write] The uniqueness is the tree's (Section 4 of `complete74_negative_index_refinement.md`, "at most one main index per representative"); the factor- $(2A - 1)^2$ argument and the exact-search formulation are the source's.

37.4.2 Packing recovery and the two input tests

Require the divisibility $Q \mid (p + M)$, and define

$$N = \frac{p + M}{q^2 - 1}, \quad Z = N \bmod q \quad (0 \leq Z < q), \quad F = q + \frac{N - Z}{q}. \quad (37.25)$$

Reject $Z = 0$. For each $e \in \{u, uA\}$, define the least nonnegative residue

$$W = 2^e \bmod H, \quad C = Z + W. \quad (37.26)$$

Accept that branch exactly when

$$e < p, \quad 0 < W < q, \quad C + u - b < q, \quad (37.27)$$

and

$$(K_0 + X)C - F \text{ is a strictly positive multiple of } q - 1. \quad (37.28)$$

Together (37.19)–(37.28) constitute the predicate. The Pell ratio is the sole exact Pell acceptance step in candidate extraction; modular powers can be evaluated without constructing their full exponential integers.

Lemma 37.4.2 (Forced marker and field word). *For fixed q, w, s, t, p , the values in (37.25) are the only possible Z, F satisfying negative packing and $0 < Z < q$. They satisfy $F \geq q$. If $F = q$, the shifted-mask remainder forces $Z \geq 2$.*

Proof. The equation $-p = (q^2 - Z - qF)Q + M$ is equivalent to $N = Z + qF - q^2$. Reduction modulo q fixes Z and then fixes F . Conversely these choices give the equation. Since $p, M > 0$, $N > 0$ and its nonnegative quotient by q gives $F \geq q$. If $F = q$ and $Z = 1$, then $S' = q^2$ in (37.13), giving $R = T' > 0$, impossible. $\square$

[write] Section 33.B already records the necessary conditions $F \geq q$ and, at $F = q$, $Z \geq 2$ (after (33.55)); the unique recovery (37.25) is new.

Separation from the programme's universal polynomial (from the programme's review, commit 308f14072). Every negative solution of this Part has $F \geq q$. The programme's normalized85 universal polynomial instead computes $C_{85} = q - F - Z - \alpha - 2d_{\text{cell}}x$ and $N_t = (K_{\text{const}} + w)C_{85} + q - F - t_{\text{quot}}(q - 1)$, with K_{const} its `Kconstant` and t_{quot} its `tquot`. At a positive zero of that polynomial N_t is an integer unit, its second part $q - F - t_{\text{quot}}(q - 1)$ is at most zero and $K_{\text{const}} + w \geq 2$, so $C_{85} \geq 0$ and $F + Z < q$, which is incompatible with $F \geq q$. This is a pretyping argument that comes before any native rank or index-sign recovery, and the review concludes that the normalized85 theorem is unaffected. The review names only normalized85; it predates by about five minutes the programme's 84-operation polynomial (commit 20aafb9a5, `complete84_scaled_strong_output.md`), which satisfies $F_{84} = \Delta \cdot F_{85}$ identically with $\Delta = (a + 1)(a + 3) > 0$ on every allowed positive tuple. The two polynomials therefore have the same positive zeros, and the separation holds for the 84-operation polynomial too. (Here α, w and t_{quot} are that source's coordinates.)

37.5 Full negative zero equivalence

Theorem 37.5.1 (Exact full-zero obstruction). *Fix a valid compiler slice and $x > 0$ as in Section 37.2. The following are equivalent:*

1. *The raw29 child has a full strictly positive integer supplied tuple with $R < 0$*
2. *The positive21 child has a full strictly positive integer supplied tuple with $R < 0$*
3. *Some q, w and finite choices s, t, p, e pass the predicate of Section 37.4*

Every passing predicate reconstructs a full tuple in both modes. Only q, w remain independently unbounded; this is not a globally finite decision procedure. Auxiliary witnesses need not be unique.

[write] *Scope, as the programme's review records it* (`review_exact_negative_obstruction37.md`, commit 308f14072; PASS within the stated fixed-compiler scope). The theorem closes the conditional witness-reconstruction obligations; it proves neither that the predicate has a solution nor that it has none. The review adds these precisions.

- The mode labels remain `raw30` and `positive22`, while the child interfaces have 29 and 21 positive witnesses, 18 and 10 comparisons and 127 and 103 full sum-of-squares gates (the table of Section 33.1).
- The omitted coordinate is restored as $R = k - hXY - 1$ with the actual raw k supplied.
- Neither the inherited reduction nor Report 37 assumes that q is a power of two on the negative branch.
- $K_0 < B^2$ is a property of the actual numerals, not an assumption on arbitrary ports.
- The signed19 density theorem of Part IV (Section 33.6) cannot be substituted for this question: it allows $W < 0$ and lacks the retained positive gap.
- Report 37 uses the symmetric scales $X = wq^3$, the ordinary strong equation and an independent g_a ; it proves no asymmetric-scale transfer (contrast Section 34.A for signed19).
- The odd-input split of Section 37.6 does not eliminate the unbounded unwrapped sector, and the even branch remains open.
- The Binet window of Section 37.7 is necessary, not an exact ratio test.
- The 3,536-case relaxed grid of Section 37.8 was not rerun or independently certified by the review and is not a set of genuine compiler instances.

The review writes $H = 4A - 5$, which is $H = 4a + 3$ since $A = a + 2$. Its helper authenticates the archive (SHA-256 018b960e...69f9), four decisive evidence members and the five upstream sources, matches both literal witness and comparison inventories and both finalizer suffixes, checks the normalized85 separation (note after Lemma 37.4.2), and adds 8 auxiliary divisibility and parity cases with $p \geq 13$, 24 input-projection cases, 36 odd-quotient specializations and 16 deterministic packing recoveries. These are supplementary algebra cases, not valid compiler exports or full zeros, and the review claims no new universal bound.

37.5.1 Necessity

The prior results in Section 37.3 give $R = -p$, even q , the range (37.21), the ratio, the defect filter and the two possible input indices. The source main-root equation gives $2^p \equiv X \pmod{H}$. The strict raw bound and $C = Z + W \geq 2$ give $q \geq u - b + 3$, which is exactly the preliminary bound in (37.19). The inequalities $s < 3q$ and $ts < 3q$ become (37.20) without losing an endpoint.

Lemma 37.4.2 gives (37.25). The input-root equation gives $W \equiv 2^e \pmod{H}$; since $0 < W < q < H$, W is the unique least residue. The raw bound is precisely $C + \alpha + u - b = q$ with $\alpha > 0$, and transport is precisely (37.28) with positive quotient z_{quot} . Thus all tests are necessary in both modes.

37.5.2 First and main witnesses and every outer sign

For a passing predicate put

$$\alpha = q - C - u + b, \quad z_{\text{quot}} = \frac{(K_0 + X)C - F}{q - 1}, \quad (37.29)$$

$$c = \psi_A(p), \quad D = \chi_A(p), \quad (37.30)$$

$$k = 2\psi_P(n), \quad \tau = \chi_P(n), \quad (37.31)$$

$$\eta = c - kY, \quad \zeta = k - \eta, \quad (37.32)$$

$$h = \frac{k + p - 1}{E}, \quad g_a = \frac{D - ac - X}{H}.$$

The predicate makes $\alpha, z_{\text{quot}}, \eta, \zeta$ strictly positive integers. To check h , note $P \equiv 1 \pmod{E}$ and hence $\psi_P(n) \equiv n \pmod{E}$ by recurrence. Consequently $k + p - 1 \equiv 2n + p - 1 = tE \equiv 0 \pmod{E}$. Its numerator is positive, so $h > 0$ and $k - hE - 1 = -p$.

The Pell identities give the main norm and the first norm exactly:

$$D^2 - \Delta c^2 = 1, \quad \tau^2 - 1 = (P^2 - 1)\psi_P(n)^2 = XY^2(XY^2 + 1)k^2.$$

The factor of four is absorbed by $k = 2\psi_P(n)$; it cannot be omitted. Packing, scale and transport now hold by definition.

To prove the sign of g_a , define $\mathcal{E}_A(r) = \chi_A(r) - a\psi_A(r)$. The exact recurrence gives

$$\mathcal{E}_A(0) = 1, \quad \mathcal{E}_A(1) = 2, \quad \mathcal{E}_A(r) \equiv 2^r \pmod{H}, \quad \mathcal{E}_A(r) = 2\psi_A(r) - \psi_A(r-1) > \psi_A(r) \quad (r \geq 1). \quad (37.33)$$

The congruence holds because 2^r satisfies the same recurrence modulo H : its characteristic residual is $4 - 4A + 1 = -H$. For $r \geq 3$,

$$\psi_A(r) \geq 4A^2 - 1 > H + X \quad \text{and} \quad \psi_A(r) > H + q,$$

since $a > X, q$. The main projection therefore makes $(D - ac - X)/H$ an integer greater than zero. The positive main root is explicitly $D = \chi_A(p)$, not an arbitrary sign of a square root.

[write] Equation (37.33) is Lemma 33.3.2 of Report 33, where $\mathcal{E}_A(r)$ is written $e_A(r)$; the source writes E_r^* . The proof here is the source's own.

37.5.3 The entire positive input block

Define

$$\kappa = \psi_A(e), \quad \mu = \chi_A(e), \quad \delta = \frac{\kappa - u}{\Delta}, \quad \phi = c - \kappa, \quad \rho = \frac{\mu - a\kappa - W}{H}. \quad (37.34)$$

If $e = u$ is odd, (37.15) gives $\kappa \equiv u \pmod{\Delta}$. If $e = uA$ is even, it gives $\kappa \equiv uA^2 \equiv u \pmod{\Delta}$. Hence δ is integral in both branches. Since $u \geq 3$ and $e \geq u$, strict Pell growth gives $\kappa > u$, so $\delta > 0$. The condition $e < p$ gives $\phi > 0$.

The modulus H is odd, so $W = 2^e \pmod{H}$ is nonzero. Equation (37.33) gives $\mu - a\kappa > H + q$, while the predicate gives $W < q$. Thus ρ is an integer and is strictly positive. The norm, discriminant increment, positive gap and exponent equation in (37.10) all hold, including the positive sign of the supplied input root μ .

37.5.4 Ordinary strong and mixed-sign auxiliary completion

We use the already verified bootstrap construction for *every odd* p . Define

$$m = \begin{cases} cp, & p \equiv 3 \pmod{4}, \\ 2cp, & p \equiv 1 \pmod{4}, \end{cases} \quad f = \chi_A(m), \quad T_{\text{aux}} = \Delta\psi_A(m), \quad i = \frac{T_{\text{aux}}}{c^2}. \quad (37.35)$$

Next put

$$\ell = p + 2m, \quad U = \frac{\chi_{T_{\text{aux}}}(\ell)}{T_{\text{aux}}}, \quad y_{\text{aux}} = \psi_{T_{\text{aux}}}(\ell), \quad j = \frac{U - p}{c}, \quad o = \frac{U + c}{f}. \quad (37.36)$$

We verify integrality and positivity rather than simply invoking a solvable auxiliary norm.

Write $m = pv$, with $v = c$ or $2c$. The square-root coefficient of $(D + c\sqrt{\Delta})^v$ is $\psi_A(m)$. Its first odd term has a factor vc , divisible by c^2 , and every higher odd term has a factor c^3 . Hence $c^2 \mid \psi_A(m)$ and i is a positive integer. The Pell identity gives

$$T_{\text{aux}}^2 = \Delta(f^2 - 1).$$

Since A is even and p odd, c is odd. The choice (37.35) makes $\ell \equiv 1 \pmod{4}$; in the second case the extra factor two is essential.

For odd $\ell = 2\theta + 1$, the quotient $\chi_T(\ell)/T$ is an integer polynomial $\mathcal{Q}_\theta(T^2)$. Its exact identities are

$$\mathcal{Q}_\theta(0) = (-1)^\theta(2\theta + 1), \quad \mathcal{Q}_\theta(1 - A^2) = (-1)^\theta\psi_A(2\theta + 1).$$

Here $\theta = (\ell - 1)/2$ is even. Because $c \mid T_{\text{aux}}$ and $c \mid m$, these identities give $U \equiv \ell \equiv p \pmod{c}$. Also $T_{\text{aux}}^2 \equiv -\Delta = 1 - A^2 \pmod{f}$, so $U \equiv \psi_A(\ell) \pmod{f}$. The Pell addition formulas imply

$$\chi_A(2m) \equiv -1 \pmod{f}, \quad \psi_A(2m) \equiv 0 \pmod{f},$$

and hence $\psi_A(p + 2m) \equiv -c \pmod{f}$. This proves the integrality of j and o .

Finally $T_{\text{aux}} \geq c^2$, $\ell \geq 3$, and Pell growth give

$$U \geq 4T_{\text{aux}}^2 - 3 > p.$$

Thus j, o are positive. The Pell norm at base T_{aux} reads

$$T_{\text{aux}}^2(U^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2,$$

and $U = jc + p = jc - R = of - c$. These are all comparisons in (37.9). The construction supplies positive $f, i, j, o, y_{\text{aux}}$ and discharges the auxiliary sign obligations. No uniqueness is asserted for this completion.

[write] *Credit*. The construction is Section 5 of the tree's bootstrap note `review_complete74_nonlinear_index_bootstrap.md` (commit 93ad43375), "A parametric positive auxiliary realization of $r = -p$ ", as the source says. Report 33 used its case $p \equiv 3 \pmod{4}$ with the same tools: Lemmas 33.3.3 and 33.3.4 and Section 33.8, where T_{aux} , ℓ , θ and y_{aux} are written Q_{aux} , s_a , b_a and y . Report 37 re-verifies the integrality and positivity of i, j, o for every odd p , including $p \equiv 1 \pmod{4}$ with $m = 2cp$, inside the full `raw29` and `positive21` systems.

37.5.5 The two literal supplied lists

The `raw29` tuple, in saved source order after the stated aliases, is

$$(C, F, J, W, Z, a, \alpha, c, D, \delta, \eta, f, g_a, h, i, j, k, \kappa, \mu, o, \phi, \\ q, \rho, s, \tau, w, y_{\text{aux}}, \zeta, z_{\text{quot}}).$$

The `positive21` tuple is

$$(J, F, \alpha, z_{\text{quot}}, f, h, i, j, o, s, w, \tau, \eta, \zeta, g_a, y_{\text{aux}}, Z, W, \delta, \phi, \rho).$$

The omitted coordinates are exactly the positive triangular expressions established above. The `raw` mode has 18 retained comparisons; `positive21` retains 10 of them. Table 9 accounts for each `raw`

comparison. The positive21 comparisons correspond to raw indices 1, 2, 3, 5, 10, 11, 12, 13, 15, 16; the other eight become source definitions.

Table 9: Complete retained residual map

Index	Literal comparison	Reason for equality
0	repunit = qm1	$q = (B - 1)J + 1$
1	raw_bound = q	Positive $\alpha = q - C - u + b$
2	innerC = local_rhs_sum	Positive transport quotient z_{quot}
3	restored_r = r_lhs	$k - hE - 1 = -p$ and recovered packing
4	C = marked_rhs	$C = Z + W$
5	L9 = R9	First Pell norm and $k = 2\psi_P(n)$
6	c = R10a	$\eta = c - kY$
7	k = R10b	$\zeta = k - \eta$
8	a = R12	$a = XY + Y$
9	d = R14	Positive integral $g_a = (D - ac - X)/H$
10	L15 = R15	Positive main Pell norm
11	ic22 = R16	Ordinary strong norm
12	L17 = P17	Auxiliary Pell norm at T_{aux}
13	H17 = aux_u_rhs	$U = jc + p = of - c$
14	kappa = index_rhs	Positive integral $\delta = (\kappa - u)/\Delta$
15	c = pell_gap	Positive $\phi = c - \kappa$
16	mu2 = norm_rhs	Positive input Pell norm
17	mu = exponent_rhs	Positive integral $\rho = (\mu - a\kappa - W)/H$

The static inventory also checks the complete saved sum-of-squares suffix: one squared residual for every listed comparison and its full sum chain. Hence vanishing of this complete list gives vanishing of the saved integer polynomial in either mode. This finishes the constructive direction and Theorem 37.5.1.

37.6 Input and packing consequences

37.6.1 The finite wrapped odd sector

On the odd branch $e = u$, the integer 2^u is fixed once the compiler and input are fixed. If $H > 2^u$, then $W = 2^u$ exactly. Since $Z, \alpha \geq 1$, the raw bound gives

$$q \geq 2^u + u - b + 2. \quad (37.37)$$

If instead $H \leq 2^u$, parity improves it to $H \leq 2^u - 1$. Substituting the exact H gives

$$4swq^6 + 4sq^3 + 4 \leq 2^u, \quad w \leq \left\lfloor \frac{2^u - 4sq^3 - 4}{4sq^6} \right\rfloor, \quad 4q^6 < 2^u. \quad (37.38)$$

There are consequently only finitely many q, w, s, t, p in this wrapped odd sector at each fixed compiler and input. Enumerating those exact bounds and applying the predicate decides that sector. This does not require factoring H or finding a multiplicative order.

Corollary 37.6.1 (Odd-branch forbidden interval). *There is no odd-input negative zero satisfying both*

$$4q^6 \geq 2^u \quad \text{and} \quad q < 2^u + u - b + 2.$$

Moreover $u \leq 6d_{\text{cell}} + 2$ excludes the wrapped odd sector. In the actual compiler, inputs $x = 1, 2$ have no wrapped odd branch.

Proof. The first claim separates the exhaustive wrapped and unwrapped cases. Since $q \geq B = 2^{d_{\text{cell}}}$, wrapping requires $2^{6d_{\text{cell}}+2} < 2^u$. Finally $d_{\text{cell}} = bL$ with $L \geq 1$ gives $b \leq d_{\text{cell}}$, so for $x = 1, 2$ one has $u \leq 5d_{\text{cell}} < 6d_{\text{cell}} + 2$. $\square$

No finite enumeration at the actual, potentially huge compiler constants has been performed. The even branch is unaffected by this corollary.

37.6.2 The even branch uses the same outer marker

For $e = uA$ we have $uA < p$. Combining this with (37.14) and $A > sXq^3$ gives

$$us < C(1 + 1/q) < C + 1,$$

because $C < q$. Integrality and the strict raw bound therefore imply

$$us \leq C \leq q - u + b - 1, \quad u(s + 1) \leq q + b - 1, \quad s \leq \left\lfloor \frac{q - u + b - 1}{u} \right\rfloor. \quad (37.39)$$

Also $uA < p < tE/2$ and $A > E$ imply

$$t > 2u, \quad \text{hence } t \geq 2u + 1. \quad (37.40)$$

The improvement over $us < q$ is that the same $C = Z + W$ must simultaneously pass packing, transport and the ordinary input's raw bound.

[write] The bound $us < q$ improved here is the one recorded at the end of Section 33.B, after (33.56).

37.6.3 A local exclusion independent of the sign

No full child zero of either sign has $3 \mid q$. Indeed this would make $3 \mid X, Y, H$, while the main projection $2^p \equiv X \pmod{H}$ would make a power of two divisible by three. This elementary filter can be applied before candidate extraction. It is not the full obstruction theorem.

37.7 A certified sub unit real window

Exact Pell powering is definitive but may involve very large integers. A necessary interval can reject parameter choices without materializing those integers. The interval is stronger than the prior uniqueness statement in its quantitative width, but is not an exact acceptance test.

37.7.1 Rational bounds for the Binet formula

For an integer $V \geq 2$ put

$$\omega_-(V) = 2V - 1/V, \quad \omega_+(V) = 2V - 1/(2V).$$

Squaring shows

$$V - 1/V < \sqrt{V^2 - 1} < V - 1/(2V), \quad \omega_-(V) < V + \sqrt{V^2 - 1} < \omega_+(V).$$

Abbreviate $\omega_{\pm}^A = \omega_{\pm}(A)$ and $\omega_{\pm}^P = \omega_{\pm}(P)$, and define positive rational prefactors

$$\xi_- = \frac{P - 1/P}{2A} \left(1 - (2A - 1)^{-26}\right), \quad \xi_+ = \frac{P}{2(A - 1/A)} \frac{1}{1 - (2P - 1)^{-14}}. \quad (37.41)$$

With $\lambda_A = A + \sqrt{A^2 - 1}$ and $\lambda_P = P + \sqrt{P^2 - 1}$, Binet's identity is exactly

$$\frac{\psi_A(p)}{2\psi_P(n)} = \frac{\sqrt{P^2 - 1}}{2\sqrt{A^2 - 1}} \frac{\lambda_A^p}{\lambda_P^n} \frac{1 - \lambda_A^{-2p}}{1 - \lambda_P^{-2n}}.$$

For $p \geq 13$ and $n \geq 7$, the exponents 26 and 14 in (37.41) uniformly bound the two tails. Thus

$$\begin{aligned} \Lambda_-(p) &= \log \xi_- + p \log \omega_-^A - n \log \omega_+^P < \log \frac{\psi_A(p)}{2\psi_P(n)} \\ &< \log \xi_+ + p \log \omega_+^A - n \log \omega_-^P = \Lambda_+(p). \end{aligned} \quad (37.42)$$

These bounds make no assumption that p/A is small.

Substitute $n = (tE + 1 - p)/2$. The strict ratio requires

$$\frac{\log Y - \log \xi_+ + (tE + 1) \log \omega_-^P/2}{\log \omega_+^A + \log \omega_-^P/2} < p < \frac{\log(Y + 1) - \log \xi_- + (tE + 1) \log \omega_+^P/2}{\log \omega_-^A + \log \omega_+^P/2}. \quad (37.43)$$

Both denominators are positive. Let I_{Bin} denote this open real interval.

Theorem 37.7.1 (Sub-unit necessary window). *The intersection $I_{\text{Bin}} \cap I$, with I as in (37.21), has length less than $q^{-3} + q^{-5} < 1$. It contains at most one integer, before odd parity is imposed. Every ratio-admissible index lies in that intersection.*

Proof. The exponential-base errors satisfy

$$\log(\omega_+^A/\omega_-^A) = \log\left(1 + \frac{1}{4A^2 - 2}\right) < \frac{1}{3A^2}, \quad \log(\omega_+^P/\omega_-^P) < \frac{1}{3P^2}.$$

Here $P > A$. The square-root-prefactor log error is less than

$$\frac{1}{A^2 - 1} + \frac{1}{P^2 - 1} < \frac{3}{A^2},$$

and the two logarithmic tail errors sum to less than $1/A^2$. For instance $-\log(1 - z) < 2z$ at each present $0 < z < 1/2$, and each resulting tail is below $1/(2A^2)$. On I , where $n < p$ and $p \geq 13$, this yields

$$\Lambda_+(p) - \Lambda_-(p) < \frac{p + n}{3A^2} + \frac{4}{A^2} < \frac{p}{A^2} < \frac{1}{Xq^2} \leq \frac{1}{q^5}. \quad (37.44)$$

The penultimate comparison uses the actual outer bound $p < Xq^4$ and $A > XY \geq Xq^3$.

Take any two real points $p_1 < p_2$ in the intersection $I_{\text{Bin}} \cap I$. At p_1 , $\Lambda_+(p_1) > \log Y$, so $\Lambda_-(p_1) > \log Y - q^{-5}$. At p_2 , $\Lambda_-(p_2) < \log(Y + 1)$. Hence

$$\Lambda_-(p_2) - \Lambda_-(p_1) < \log(1 + 1/Y) + q^{-5} < q^{-3} + q^{-5}.$$

The affine slope of Λ_- is $\log \omega_-^A + \frac{1}{2} \log \omega_+^P > 1$. Dividing proves the claimed width. Restriction to I is essential because the uniform error estimate was proved there. $\square$

37.7.2 Outward rational evaluation and exact fallback

For a positive rational ϑ , range-reduce $\vartheta = 2^\nu \vartheta_0$ with $1 \leq \vartheta_0 < 2$. Set $z_0 = (\vartheta_0 - 1)/(\vartheta_0 + 1)$. The elementary series gives

$$\log \vartheta_0 = 2 \sum_{l=0}^{N_s-1} \frac{z_0^{2l+1}}{2l+1} + \varepsilon_{N_s}, \quad 0 \leq \varepsilon_{N_s} \leq \frac{2z_0^{2N_s+1}}{(2N_s+1)(1-z_0^2)}. \quad (37.45)$$

The same formula at $\vartheta_0 = 2$ encloses $\log 2$. All endpoints and errors are rational. Correct signed interval arithmetic for $\nu \log 2$, and division only after the denominator's positive lower bound has been certified, give outward enclosures of (37.43).

An outward lower endpoint p_{lo} and upper endpoint p_{hi} safely restrict strict integer membership to $\lfloor p_{lo} \rfloor + 1 \leq p \leq \lceil p_{hi} \rceil - 1$, followed by the index-range and odd-parity restrictions. If that set is empty the parameter choice is rejected exactly. A surviving integer, or an endpoint that cannot be separated by the present enclosure, still needs exact Pell comparison. Refinement isolates or rejects; it does not by itself certify strict ratio acceptance.

The logarithmic precision depends on the parameters. Exact final acceptance can still require extremely large Pell integers, and no uniform polynomial-time complexity or nonzero separation bound is proved. Binary search from Lemma 37.4.1 remains a terminating exact fallback regardless of an unresolved real endpoint.

37.8 Checked evidence and reproducibility

[write] *Shipped names*. The packet's files are shipped with the prefix `15-neg-obstruction-` and the delivery directories flattened into the name (README): `evidence/check_exact_obstruction.py` is `code/15-neg-obstruction-evidence-check_exact_obstruction.py`, `evidence/independent/audit_exact_obstruction.py` and `audit_log_windows.py` are `code/15-neg-obstruction-evidence-independent-audit_exact_obstruction.py` and `code/15-neg-obstruction-evidence-independent-audit_log_windows.py`, and their frozen receipts are the three `data/15-neg-obstruction-evidence-*expected*.json`. The packet's `evidence/sources/` and `evidence/context/` hold byte copies of repository files and are not shipped; the checkers read them, so the README gives a rerun route that restores them from the repository. The delivered release verifier refuses to run on NTFS, whose file modes it cannot match; at placement the three mathematical programs, run directly on a copy, passed in normal and optimized Python, with receipts equal to the frozen ones (up to CRLF line ends of the author receipt on Windows). The programme's review did not rerun the 3,536-case grid below.

37.8.1 What the finite checks establish

The new mathematical checker authenticates the five cached upstream files by SHA256 and Git blob SHA1, reads the source JSON as data, and verifies both exact supplied witness lists and comparison inventories. The independent checker additionally verifies both complete sum-of-squares finalizer suffixes. Neither evaluates a saved source schedule.

Table 10 records the author's exact test families. All integer and interval decisions use exact arithmetic. The floating maximum-window-width value in the receipt is a display statistic only; the asserted rational bound is $1/100000$. The full grid takes $q \in \{16, 20, 32\}$, $w \in \{1, 2, 3, 4\}$, every $1 \leq s \leq 3q - 1$ and every $1 \leq t \leq \lfloor (3q - 1)/s \rfloor$. It uses 36-term outward rational series bounds. Every one of the 3,536 necessary windows has no odd integer; the maximum outward width is strictly less than $1/100000$. This is a

Check family	Count
Negative outer-packing arithmetic reconstructions	18,360
Input discriminant and residue fixtures	288 each
Fully materialized auxiliary-only tuples	4
Large-index auxiliary divisibility and parity fixtures	16
Direct rational Binet envelope cases	450
Relaxed scale grid choices with empty odd-index windows	3,536

Table 10: Author checker results, with subsystem and scale scopes retained

relaxed arithmetic scale grid, *not* 3,536 valid compiler instances. No surviving main-index candidate or full negative compiler zero was materialized.

The four complete auxiliary-only fixtures include $p = 1$ and $p = 3$, covering both odd residue classes modulo four. They are outside the full main-index condition $p \geq 13$. The larger-index modular fixtures and the general construction address that condition without claiming to have materialized a full tuple.

37.8.2 Independent corroboration

The independent review [8] concludes that the full-negative-zero equivalence, finite-sector consequences and sub-unit necessary interval are valid under the stated compiler and prior-bootstrap premises. Its exact checker records:

- Five source identity checks and both full witness, comparison and finalizer inventories
- 61 odd-quotient polynomial identities; 1,620 main/input projection, discriminant and growth instances; 360 positive input-delta cases
- 240 auxiliary divisibility/parity cases, six auxiliary modular completions with exact f , and three materialized auxiliary-only tuples
- 9,030 index-range boundary cases and 448 packing reconstructions, including $F = q$
- 162 rational Binet/error cases, 126 representative uniform-error cases, and 1,728 wrapped-sector threshold cases

The three materialized independent auxiliary tuples all have $p = 3$; their U bit lengths are 7,939, 56,928 and 214,545. They are explicitly subsystem evidence.

A separate independent interval implementation uses 48-term rational logarithm bounds on the *full rational prefactors*, rather than the author’s split-tail implementation. It checks 16 selected scale points covering all three q values, interior choices and extreme representatives. All 16 independently exclude odd integers. This corroborates selected points; it does not replace or inflate the author’s 3,536-case full-grid count.

37.8.3 Portable execution contract

The package preserves the complete pinned evidence packet, including the proof, prior-context snapshots, independent review, static upstream sources, three new mathematical executables and frozen expected receipts. The relevant executable paths are

```
evidence/check_exact_obstruction.py
evidence/independent/audit_exact_obstruction.py
evidence/independent/audit_log_windows.py.
```

Each defaults to canonical JSON on stdout only. The `-expect` option checks a separately frozen receipt; `-output` permits an external destination and rejects an in-packet write. Normal Python and `python3` are both supported because all checks use explicit exceptions rather than removable assertions. The cached upstream Python file is inert source evidence, not an executable of this package; its historical assertions are not altered or used as a release gate.

The release verifier checks pinned payload and executable identities, exact recursive JSON types and values, and copied-source preservation. Its normal/optimized replays run from an external copy, with outputs outside the packet. Tamper and malformed-input regressions test failure paths. The archive tools enforce exact inventory, unique safe paths, and payload identities. The PDF build disables shell escape, runs outside the release tree and checks reproducible bytes. The README gives the executable commands; the integrity record separates these reproducibility properties from the mathematical scope. Reproducible checks corroborate a proof under stated premises; they do not prove the unresolved existential conclusion.

37.9 What remains open and useful next directions

Theorem 37.5.1 reduces the unresolved full-zero question to a precisely specified intersection. Its two unbounded sectors are:

1. The odd branch with the exact fixed marker $W = 2^u$ and $q \geq 2^u + u - b + 2$
2. The even branch with $us \leq C \leq q - u + b - 1$ and $t \geq 2u + 1$

The finite wrapped odd sector is an additional, decidable component at each fixed compiler and input. Finite decidability here need not mean practical enumeration.

A useful next proof would exploit the simultaneous constraints on the uniquely recovered Z, F , the two modular exponent tests, and the exact ratio. A contradiction in one of the listed sectors would eliminate that sector; a candidate would need to pass *every* test and then use the explicit reconstruction. The full witness formulas show that an auxiliary sign obstacle will not independently eliminate a passing candidate.

The certified interval suggests an exact rejection-first implementation: apply local filters, isolate the possible main index with rational bounds, then pay for exact Pell acceptance only if a candidate survives. A meaningful complexity theorem would need explicit precision/separation estimates or a different exact comparison algorithm. None is presently supplied.

Some tempting shortcuts fail. Replacing $W \equiv 2^u \pmod{H}$ by $W = 2^u$ discards the finite wrapped sector, and is unjustified on the even branch. Density at a fixed scale does not force an index into the bounded representative range. The signed19 progression lets a marker grow freely, whereas $0 < Z, W < C < q$ and deterministic recovery constrain it here. Arbitrary small mask ports are not genuine compiler exports, even if they satisfy a few numerical inequalities.

The decisive remaining question is therefore unchanged in its yes-or-no form: does any valid fixed-compiler choice admit a passing full predicate, or can all such choices be excluded? The contribution is an exact obstruction with a complete inverse witness map, explicit sector boundaries and certifiable finite rejection machinery. It is neither a global positivity theorem nor a counterexample.

[write] *Re-scoped (batch 88)*. Part VI (Report 39) works in the first unbounded sector above. At $q = B^{2x+2}$, with $W = 2^u$, $Z = 1$, $C = W + 1$ and $s = t = 1$, it proves that for every valid compiler and input infinitely many scales satisfy every condition of the predicate except the main congruence $2^p \equiv X \pmod{H}$ (Theorem 39.7.1); that an odd-index hit of that congruence requires the Jacobi symbol $\mathcal{J}(w, 4q^3 + 3)$ to be

-1 (Lemma 39.8.1), a fixed-modulus test of the kind suggested above; and that on a thinned progression with symbol $+1$ infinitely many scales pass every other condition and fail the congruence (Theorem 39.9.1). No sector is eliminated: the scales of that family with symbol -1 , the rest of the odd sector and the whole even sector remain, and the yes-or-no question above is still open. The programme’s review of Report 39 checks the Jacobi filter but does not recertify its analytic existence argument (note at the start of Part VI).

37.A Source provenance and dependency map

The package’s proof note `evidence/EXACT-OBSTRUCTION.md` is frozen at SHA256

918b82b7666b4f5cbab5fd7a0b8298275ac62fe84f83261ef88ea863765d455f.

The authoritative mathematical projection JSON has 74,001 bytes and SHA256

ea982f9585eb4e16d756dfc33c73041e73e5e18a38ea37c2302d57c1fe104c92.

The source manifest provides the remaining full digests and byte counts without shortening identities. The unchanged projection bytes also match the prior reviewed receipt. The original source recipe’s powers-of-five conditions, fixed high-monomial correction, native/shifted mask distinction, and positive-interface domain are retained throughout.

The logical dependencies are deliberately separated:

1. Valid fixed compiler export and ordinary rank bootstrap supply the inherited source-specific premises
2. The prior raw-positive reduction sharpens the absolute index bound and restores exact signs and input representatives
3. The upstream refinement supplies the defect lower bound and one-index uniqueness
4. This report supplies deterministic packing, complete positive reconstruction, sector consequences and a sub-unit necessary real interval
5. Newly written exact checkers and independent audits corroborate specified identities and finite cases; static source inventories connect the witness map to every retained comparison

The presence of cached sources does not amount to independently reproving every inherited theorem. The full theorem has exactly the source and bootstrap conditions stated in its hypotheses. No historical checker is rerun as a substitute for those logical dependencies.

[write] The proof note is shipped as `15-neg-obstruction-evidence-EXACT-OBSTRUCTION.md` (its SHA-256 measured equal to the digest above); the projection JSON is the research-tree file named in [1], measured equal (74,001 bytes). Item 2 is Section 33.B of this report, item 3 the tree’s refinement note, and item 1 the tree’s compiler note and bootstrap note.

References of Report 37

- [1] ProveIt, *Complete⁷⁴ nonlinear-index projection scout*, saved JSON at pinned commit d5bd4a67.
<https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855>

- `bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_nonlinear_index_projection_scout.json`. [write] A byte copy is in the archive (`evidence/sources/`), not shipped; it equals the current `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_nonlinear_index_projection_scout.json`.
- [2] ProveIt, *Signed first-index bootstrap for the complete74 nonlinear projection*, pinned source note. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_complete74_nonlinear_index_bootstrap.md. [write] Also cited by Report 33 as [7]; byte copy not shipped, equal to the current file of that name in `native-stream-queue/`.
 - [3] ProveIt, *Three further restrictions on the negative first-index branch*, pinned source note. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_negative_index_refinement.md. [write] Introduced by commit 813c1cff4; byte copy not shipped, equal to the current file.
 - [4] ProveIt, *Modified compiler masks for the half-binomial75*, pinned source note. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.md. [write] Also cited by Report 33 as [6]; byte copy not shipped, equal to the current file.
 - [5] ProveIt, *Complete75 positive elimination*, source read as text only. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_positive_elimination.py. [write] Byte copy not shipped, equal to the current file; never executed.
 - [6] *Raw/positive restored-index reduction* and *Prior independent review*. Preserved contextual proof snapshots in `evidence/context/`. These establish the sharper $p < Xq^4$ bound used here. [write] Both are byte copies of Report 33's own evidence, shipped once as `10-index-restore-repro-evidence-RAW-POSITIVE-REDUCTION.md` and `10-index-restore-supplements-evidence-INDEPENDENT-REVIEW.md`; the reduction is printed as Section 33.B.
 - [7] *Exact full-zero obstruction for raw29 and positive21*. Frozen proof note and author receipts in `evidence/`. [write] Shipped as `15-neg-obstruction-evidence-EXACT-OBSTRUCTION.md` and `data/15-neg-obstruction-evidence-expected_check_results.json`.
 - [8] *Independent audit of the exact negative-zero obstruction*. Complete review and exact receipts in `evidence/independent/`. [write] Shipped as `15-neg-obstruction-evidence-independent-INDEPENDENT-REVIEW.md` and `data/15-neg-obstruction-evidence-independent-*.json`.

Part VI

Nonredundant main comparisons in positive Diophantine certificates

[write] Source 16 of this report, manuscript 01 of batch 88: *Research Report 39*, subtitled “Infinite genuine compiler counterfamilies after deleting one comparison”, author line “Mathematical research report”, dated 3 October 2026; archive `Nonredundant_Main_Comparisons_in_Positive_Diophantine_Certificates_Package.zip` (arrival commit `c5612efa1`, placed by `992aaafb3`); pin `d5bd4a67b`, the pin of Report 37. Its sections, statements and equations are numbered $39.k$ here.

Abstract of the source. For every valid fixed complete75 compiler export and ordinary input $x \geq 1$, we construct infinitely many positive tuples violating exactly one retained comparison: the main projection in `raw29`, or the main norm in `positive21`. First, exact affine packing and a quantitative shrinking target argument construct a broader family satisfying every condition of Report 37’s negative-zero predicate except $2^p \equiv X \pmod{4Y(X+1)+3}$. The compiler numerals stay genuine and fixed, with $q = B^{2x+2}$, $W = 2^{2d_{\text{cell}}x+b}$, $Z = 1$, $C = W + 1$ and $s = t = 1$. A later Jacobi-symbol argument proves the fixed-modulus necessary filter $(w/(4q^3+3)) = -1$ for any odd-index congruence hit. Restricting $w = 1 + (q-1)(4q^3+3)j$ instead gives symbol $+1$ and certifies misses. For every sufficiently large N , at least $\delta_0 N / (32L_0 \log N)$ such scales with $N \leq j < 2N$ satisfy every other condition. Their exact raw sum of squares is $r_{\text{main}}^2 > 0$; the `positive21` sum is $[r_{\text{main}}(2D - r_{\text{main}})]^2 > 0$. This proves nonredundancy of the indicated comparison in each interface. It supplies no full negative zero and no global restored-index positivity theorem. No enormous genuine tuple is numerically materialized, and no practical threshold is claimed.

[write] *What the title means.* The comparisons are *nonredundant* in a precise and limited sense: deleting the `raw29` main projection, or the `positive21` main norm, from the respective interface admits infinitely many strictly positive tuples that violate the deleted comparison (Theorem 39.9.1). These tuples are zeros of the comparison-deleted systems, not of `raw29` or `positive21`. The yes-or-no question of Part V, whether a full positive zero with negative restored index exists (Section 37.9), stays open; the source says so in its first section and in Section 39.13.

Place in this report. Report 39 continues Part V. It works inside the first of the two unbounded sectors that Section 37.9 lists, the odd branch with the fixed marker $W = 2^u$, at the explicit scale $q = B^{2x+2}$ with $Z = 1$, $C = W + 1$ and $s = t = 1$, and it settles what that section calls “a useful next proof” for one infinite subfamily: every condition of Part V’s predicate except the main congruence holds infinitely often (Theorem 39.7.1), and a fixed-modulus Jacobi symbol, one of the “modular exponent tests” that section asks to exploit, excludes the main congruence on a thinned progression (Lemma 39.8.1, Theorem 39.9.1). Part V’s note at the end of Section 37.9 records this re-scoping. Section 39.2 restates Part V’s compiler contract and predicate (Sections 37.2 and 37.4); Section 39.10 specializes Part V’s positive reconstruction. Both are printed in full, because later equations cite them, with pointers and no novelty claim. The source carried Report 37’s manuscript as context (a byte copy of the text printed as Part V; not shipped).

The programme’s review. The Hilbert’s-tenth programme reviewed this archive with Reports 41, 43, 45 and 46 in `review_incoming_arithmetic_reports_39_46.md` (research tree, commit `5d8889017`, with the receipt `.json`). It checks the Jacobi identity (39.46) and finds it valid, calls the thinning to multiples of h_0 or $4h_0$ “a concrete obstruction, not an inference from a search”, and finds that the transformed second derivative and discrepancy estimates of Section 39.9 have the expected scaling. *It did not read the parent analytic proof or its long independent audit in full, and does not recertify the complete analytic existence argument:* the shrinking-target construction of Sections 39.3–39.7, on which Theorems 39.7.1 and 39.9.1 rest, is certified only by the source’s own independent review (shipped as `16-nonredundant-evidence-independent-INDEPENDENT-REVIEW.md`). The review’s verdict on scope: the result “supplies neither a full zero of a

smaller polynomial nor an improved universal operation count”. A second reviewer’s provenance cross-read does not recertify these analytic dependencies either. The two analytic estimates of Section 39.5 are cited from the literature, as the source states.

A historical statement in the shipped evidence. `16-nonredundant-evidence-README.md` (line 9) says that no genuine candidate has been proved either to satisfy or to fail the omitted congruence. That sentence belongs to the first frozen stage and was accurate at its freeze; the later Jacobi addendum, integrated here as Sections 39.8–39.9, proves infinitely many certified failures. The source says this itself (“Historical evidence stages”, Section 39.1). The delivered file is kept byte for byte. No main-congruence *hit* is proved anywhere.

Notation in this Part. Report 39 uses Part V’s letters (and Part V’s macros $M_{F,\text{src}}$, $M_{F,0}$ for the masks), with no symbol renamed by the merge. Its own letters that differ from Part V: $\lambda_A(X)$, $\lambda_P(X)$ are *logarithms* of the Pell units (Part V’s λ_A, λ_P are the units themselves); $L_0 = q(q-1)Q$ is a lattice modulus (Part V’s L is the compiler exponent in $d_{\text{cell}} = bL$); N bounds the progression index and N_{pack} is Part V’s packing quotient N ; $h_0 = 4q^3 + 3$ is a fixed Jacobi modulus, while the odd-quotient index that Part V writes θ is h_{aux} here; E_j^* is Part V’s $\mathcal{E}_A(j)$; $K(X)$, $S(X)$, $G(z, v)$, $g(r)$, $a_0 = \log(2Y)$, κ_0 , δ_0 , K_* belong to the analytic argument and are unrelated to the compiler constant K_0 , the free coefficient S of Part VIII, the input coefficient κ and the register a ; $\mathcal{F}$ is the analytic centre (not the field word F); T_{thin} is a thinning factor (not T_{aux}); f_{aux} is Part V’s auxiliary root f . The letter j is the thinned progression index in Sections 39.8–39.9 and the supplied auxiliary coordinate in Section 39.10; the source warns about this. Section 0.3 lists the letters that differ between Parts.

The source’s files are shipped with the prefix `16-nonredundant-`.

39.1 Conclusion and boundaries of the result

The raw main projection and the positive21 main norm are genuinely nonredundant comparisons. For each actual fixed compiler and input slice, deleting the respective comparison admits infinitely many strictly positive tuples that violate it, with restored index $R = -p$. These are counterfamilies for the *comparison-deleted* systems. They are not zeros of the original full interfaces and do not by themselves resolve whether deleting the explicit positive-index coordinate can introduce a full negative zero.

The proof has two stages. The broad family theorem realizes the exact affine packing lattice, positive input loading, exact Pell ratio, bounded first-index conditions and all remaining sign obligations. A separate arithmetic argument then filters by a fixed-modulus Jacobi symbol and thins the progression to a certified infinite miss subfamily. Both stages retain genuine compiler numerals throughout.

This report proves a precise nonimplication of one comparison by the others on positive integer tuples. It does not assert unrestricted logical independence of axioms, a full zero, global positivity, a complete parametrization, or a feasible search threshold. Abstract finite fixtures with small numeral ports are not actual compiler exports and are not the genuine counterfamilies proved here.

Historical evidence stages. The frozen first-stage proof stated that it had established neither a hit nor a miss of the omitted congruence. That statement remains accurate for its freeze. The separately frozen Jacobi addendum supplies the later exclusion argument, and this report integrates both without changing either source packet. No main-congruence hit is proved. The symbol -1 sector and full negative-zero existence remain unresolved.

Relationship to Report 37. Report 37 gives an exact necessary-and-sufficient arithmetic predicate for a full strictly positive supplied tuple with restored index $R = -p$ in the raw29 and positive21 interfaces. The present report proves a new sufficiency result for its one-condition deletion. The source/compiler contract, sign-free bootstrap and prior full-zero equivalence remain inherited premises.

The analytic proof and the later Jacobi extension are given in full, and every positive witness required for the reduced system is reconstructed below. Neither the complete universal compiler theorem nor the generic ordinary auxiliary rank theorem is reproved here.

Literal mode names. The preserved projection names the modes `raw30` and `positive22`. After eliminating the explicit index coordinate, their supplied lists contain 29 and 21 positive coordinates; we call them `raw29` and `positive21`. Both finalizers are sums of squared integer residuals. The `signed19` interface is excluded. A signed-interface construction does not discharge the positive markers required here.

Fixed parameters before asymptotics. The compiler, input, q , and hence Y, δ_0, L_0 are fixed before r or N tends to infinity. All implicit constants and the eventual threshold may depend on those fixed data. There is no uniform assertion for a growing compiler or growing q .

39.2 The inherited compiler and predicate

39.2.1 The genuine source contract

Fix an actual valid export of the `complete75` modified half-binomial compiler and $x \geq 1$. Retain its fixed integers

$$B = 2^{d_{\text{cell}}}, \quad K_0 = D_C + BD_R, \quad M_C, M_{F,0}, \quad M_{F,\text{src}} = M_{F,0} + B - 1.$$

The inherited contract supplies

$$\begin{aligned} B \geq 16, \quad d_{\text{cell}} = bL \geq 4, \quad 1 \leq b \leq d_{\text{cell}}, \quad b \text{ odd}, \\ 0 < D_C, D_R < B, \quad K_0 \leq B^2 - 1, \quad 0 < M_C, M_{F,0} < B - 1. \end{aligned} \tag{39.1}$$

In the genuine recipe b, L, d_{cell} are powers of five, the inner radix is $R_{\text{rad}} = 2^b$, and $B = R_{\text{rad}}^L$. The separated native layout, selectors, duplicated center-clause bands, four anchors and optional high monomial are retained. The last ensures $2D_C - D_R \not\equiv 0 \pmod{5}$. For the specified ignored dummy exponent e_* , the modified masks are

$$M_C = M_C^{\text{old}} - 2R_{\text{rad}}^{e_*}, \quad M_{F,0} = M_F^{\text{old}} + 4.$$

The radix is enlarged until

$$R_{\text{rad}} \geq \max \left(4(K+2) \left(2 \sum_e c_e + 6 \right) + 8, 2\mu_{\text{mask}} + 4, 16 \right).$$

Here $K, c_e, \mu_{\text{mask}}$ belong to the fixed native compiler, not to varying witnesses. This is the modified `new_constants(compile_windows(...))` export, not the baseline `constants()` masks. In particular, $M_{F,0} < B - 1$ must not be misapplied to the shifted coefficient $M_{F,\text{src}}$.

These conditions are provenance from a genuine export, not a license to choose mask numerals freely. The source snapshots and Report 37 theorem are immutable evidence in the accompanying packet; no upstream module or saved arithmetic schedule is executed.

39.2.2 Source specific evenness of the main mask

The present construction additionally uses the source-specific fact that M_C is even. This is not deduced merely from the abbreviated numerical bounds in (39.1). The modified compiler note explicitly states it in equations (2)–(3), lines 34–46 [3]. The retained materializer formula, lines 73–76 [4], is

$$M_C^0 = B - 1 - \sum_{\substack{e \in \mathcal{P} \\ e \neq 1}} R_{\text{rad}}^e, \quad M_C = M_C^0 - 2R_{\text{rad}}^{e*}. \quad (39.2)$$

The position construction retains the Start selector 0 exactly once and every other retained exponent is positive. Since R_{rad} is even, the sum in (39.2) has exactly one odd term, $R_{\text{rad}}^0 = 1$. Both it and $B - 1$ are odd, so M_C^0 is even, and the modification preserves parity. No stronger congruence for M_C is assumed. Both source files are cached as inert text with immutable commit and digest references in Appendix 39.A.

39.2.3 Notation and Pell sequences

Put

$$u = 2d_{\text{cell}}x + b, \quad (39.3)$$

so u is odd and at least 3. For positive scale coordinates let

$$\begin{aligned} q &= (B - 1)J + 1, & X &= wq^3, & Y &= sq^3, & E &= XY, \\ a &= Y(X + 1), & A &= a + 2, & \Delta &= A^2 - 1, & H &= 4a + 3, \\ P &= 2XY^2 + 1, & Q &= q^2 - 1, & M &= (M_C + qM_{F,\text{src}})J. \end{aligned} \quad (39.4)$$

For each integer $V \geq 2$, define the integer Pell sequences by

$$\chi_V(j) + \psi_V(j)\sqrt{V^2 - 1} = (V + \sqrt{V^2 - 1})^j \quad (j \geq 0). \quad (39.5)$$

They satisfy the recurrence $f(j + 1) = 2Vf(j) - f(j - 1)$ with

$$\chi_V(0) = 1, \quad \chi_V(1) = V, \quad \psi_V(0) = 0, \quad \psi_V(1) = 1.$$

The mathematical base A differs from the source register **A**, which holds Δ . We write D for the main positive Pell root, k_{first} for the first coefficient, α_{src} for the raw loading slack, f_{aux} for the auxiliary root, and $\mathcal{F}$ for the analytic center. Thus neither an analytic logarithm nor a lattice quotient is confused with a supplied coordinate.

39.2.4 The exact predicate being reduced

Report 37 enumerates $q, w > 0$ with

$$q \geq B, \quad q \text{ even}, \quad q \equiv 1 \pmod{B - 1}, \quad q > u - b + 2,$$

and finite choices

$$1 \leq s \leq 3q - 1, \quad 1 \leq t \leq \left\lfloor \frac{3q - 1}{s} \right\rfloor.$$

Its candidate main indices are the odd integers in

$$I = \left[\max\left(13, \left\lceil \frac{tE}{3} \right\rceil + 1\right), \min\left(Xq^4 - 1, \left\lfloor \frac{tE - 6}{2} \right\rfloor\right) \right], \quad n = \frac{tE + 1 - p}{2}. \quad (39.6)$$

The tests are

$$2Y\psi_P(n) < \psi_A(p) < 2(Y+1)\psi_P(n), \quad (39.7)$$

$$4^{2n-p} > X, \quad 2^p \equiv X \pmod{H}. \quad (39.8)$$

Require $Q \mid p+M$ and recover

$$N_{\text{pack}} = \frac{p+M}{Q}, \quad Z = N_{\text{pack}} \bmod q \quad (0 \leq Z < q), \quad F = q + \frac{N_{\text{pack}} - Z}{q}. \quad (39.9)$$

Reject $Z = 0$. For an input branch $e \in \{u, uA\}$, put

$$W = 2^e \bmod H \quad (0 \leq W < H), \quad C = Z + W.$$

Finally require

$$e < p, \quad 0 < W < q, \quad C + u - b < q, \quad (39.10)$$

and that $(K_0 + X)C - F$ be a strictly positive multiple of $q - 1$.

The reduced predicate keeps every one of these conditions except $2^p \equiv X \pmod{H}$. In particular it keeps the defect test even though Report 37's necessity proof for that filter uses the main congruence. Here the filter will be proved directly from the constructed indices. The construction uses the odd-input branch $e = u$ throughout.

[write] This section restates Part V. The contract (39.1) is (37.1), with $d_{\text{cell}} = bL \geq 4$ and $1 \leq b \leq d_{\text{cell}}$ stated in place of $1 \leq b < B$; the scales (39.4) are (37.3); and the predicate of Section 37.4 appears as (39.6) (Part V's (37.21)), (39.7)–(39.8) (Part V's (37.23) and (37.24)), (39.9) (Part V's (37.25), whose quotient N is written N_{pack} here) and (39.10) (Part V's (37.27)). The one new ingredient is the evenness of M_C in Section 39.2.2; the two research-tree passages it cites were checked at this write (the compiler note's lines 34–46 and the materializer's lines 73–76 say what the source quotes) and are unchanged since the pin.

39.3 The fixed scale and exact affine lattice

39.3.1 An explicit admissible scale

Set $W = 2^u = 2^b B^{2x}$. Choose any even q such that

$$q \equiv 1 \pmod{B-1}, \quad 3 \nmid q, \quad q \geq W + u - b + 2. \quad (39.11)$$

The convenient choice for every genuine compiler and input is

$$q = B^{2x+2}. \quad (39.12)$$

Indeed $b \leq d_{\text{cell}}$ gives $W \leq B^{2x+1}$, while

$$q - W \geq (B-1)B^{2x+1} > 2d_{\text{cell}}x + 1 = u - b + 1.$$

For the final strict inequality, $2^{d_{\text{cell}}(2x+1)} > d_{\text{cell}}(2x+1) > 2d_{\text{cell}}x + 1$ since $d_{\text{cell}} \geq 4$. The desired integer bound follows. The power of two in (39.12) is even, is not divisible by 3, and is 1 modulo $B-1$.

Fix from now on

$$J = \frac{q-1}{B-1}, \quad Q = q^2 - 1, \quad M = (M_C + qM_{F,\text{src}})J, \quad (39.13)$$

$$C = W + 1, \quad Z = 1, \quad \alpha_{\text{src}} = q - C - u + b, \quad s = t = 1, \quad Y = q^3, \quad L_0 = q(q-1)Q.$$

Then $\alpha_{\text{src}} \geq 1$, $0 < W < q$, $C + u - b < q$, C is odd, and M is even. These are exact facts before any asymptotic argument. All compiler constants remain unchanged.

39.3.2 A common lattice for every progression index

For integer $r \geq 1$ define

$$w_r = 1 + (q - 1)r, \quad X_r = q^3 w_r, \quad E_r = X_r Y, \quad (39.14)$$

and the fixed integers

$$\begin{aligned} P_0 &= Q((1 + q(K_0 + q^3))C - q^2 - W) - M, \\ p_0 &= P_0 \bmod L_0, \quad 0 \leq p_0 < L_0, \quad p = p_0 + L_0 k. \end{aligned} \quad (39.15)$$

Since Q, C are odd and q, W, M are even, P_0 is odd. Since L_0 is even, p_0 is odd and nonzero. Every index in this lattice is odd; no coprimality or extra favorable mask residue is needed.

Lemma 39.3.1 (Exact packing and transport lattice). *For every $r \geq 1$, the lattice in (39.15) is equivalent to the combined packing and transport congruence at the fixed markers in (39.13). With*

$$\ell_{\text{lat}} = \frac{p - P_0}{L_0} \in \mathbb{Z},$$

the recovered coordinates are exactly

$$F = (K_0 + q^3)C + (q - 1)\ell_{\text{lat}}, \quad z_{\text{quot}} = q^3 C r - \ell_{\text{lat}}. \quad (39.16)$$

Their remaining sign conditions are strict linear inequalities in r, ℓ_{lat} .

Proof. Use $\mathcal{D}_r = 1 + q(K_0 + X_r)$, not the Pell root D . Direct expansion gives

$$Q(\mathcal{D}_r C - q^2 - W) - M = P_0 + L_0 q^3 C r. \quad (39.17)$$

Therefore $p \equiv P_0 \pmod{L_0}$ is precisely

$$p + M \equiv Q(\mathcal{D}_r C - q^2 - W) \pmod{L_0}. \quad (39.18)$$

It first makes $N_{\text{pack}} = (p + M)/Q$ integral. Dividing the congruence by Q and reducing modulo q gives

$$N_{\text{pack}} \equiv C - W = 1 \pmod{q}.$$

The recovered least marker is exactly $Z = 1$. Thus

$$F = \frac{N_{\text{pack}} - 1 + q^2}{q}, \quad z_{\text{quot}} = \frac{(K_0 + X_r)C - F}{q - 1}.$$

Substituting $p = P_0 + L_0 \ell_{\text{lat}}$ gives (39.16), proving every division integral. Conversely those formulas give exactly

$$-p = (q^2 - 1 - qF)Q + M, \quad (K_0 + X_r)C = F + z_{\text{quot}}(q - 1). \quad (39.19)$$

The two forms of the lattice coordinate differ only by the fixed integer $(P_0 - p_0)/L_0$. The analytic argument below chooses an index in this already-compatible lattice; it does not introduce another congruence problem. $\square$

39.4 The exact ratio center and its curvature

39.4.1 The center is defined from the exact Binet expression

With $Y = q^3$ fixed, let X be a sufficiently large positive real variable and define

$$\begin{aligned} A(X) &= Y(X+1) + 2, & P(X) &= 2XY^2 + 1, \\ \lambda_A(X) &= \log(A + \sqrt{A^2 - 1}), & \lambda_P(X) &= \log(P + \sqrt{P^2 - 1}), \\ K(X) &= \frac{\sqrt{P^2 - 1}}{2\sqrt{A^2 - 1}}, & \delta_0 &= \frac{1}{2} \log(1 + 1/Y) > 0, & S(X) &= \lambda_A(X) + \frac{1}{2}\lambda_P(X). \end{aligned} \quad (39.20)$$

At integer X, p, n with $p, n > 0$ and $n = (XY + 1 - p)/2$, Binet's formula gives the exact identity

$$\begin{aligned} \log \frac{\psi_A(p)}{2\psi_P(n)} &= \log K + p\lambda_A - n\lambda_P + \mathcal{T}(X, p), \\ \mathcal{T}(X, p) &= \log(1 - e^{-2p\lambda_A}) - \log(1 - e^{-2n\lambda_P}). \end{aligned} \quad (39.21)$$

The non-tail expression has a smooth real extension; its center is

$$\mathcal{F}(X) = \frac{(XY + 1)\lambda_P(X)/2 + \log Y + \delta_0 - \log K(X)}{S(X)}. \quad (39.22)$$

Its non-tail log-ratio equals $\log Y + \delta_0$, the midpoint of the target log interval. At a different p , the non-tail part is exactly

$$\log Y + \delta_0 + S(X)(p - \mathcal{F}(X)). \quad (39.23)$$

39.4.2 An analytic decomposition with controlled derivatives

Let $a_0 = \log(2Y) > 0$, $z = 1/X$, and

$$v = \frac{1}{3 \log X + 4a_0}, \quad a_1 = \frac{Y+2}{Y}, \quad b_1 = \frac{1}{2Y^2}.$$

Write

$$\lambda_A = \log X + a_0 + \epsilon_a(z), \quad \lambda_P = \log X + 2a_0 + \epsilon_b(z), \quad \log K = \log Y + \epsilon_K(z).$$

The remainders have the following exact real-analytic descriptions near $z = 0$:

$$\begin{aligned} \epsilon_a(z) &= \log \frac{1 + a_1 z + \sqrt{(1 + a_1 z)^2 - z^2/Y^2}}{2}, \\ \epsilon_b(z) &= \log \frac{1 + b_1 z + \sqrt{(1 + b_1 z)^2 - z^2/(4Y^4)}}{2}, \\ \epsilon_K(z) &= \frac{1}{2} \log \frac{(1 + b_1 z)^2 - z^2/(4Y^4)}{(1 + a_1 z)^2 - z^2/Y^2}. \end{aligned} \quad (39.24)$$

Each logarithm has argument 1 at $z = 0$, and each square root uses the real branch equal to 1 there. Hence all three functions are analytic with zero constant terms.

Put $\epsilon_D = 2\epsilon_a + \epsilon_b$ and $\epsilon_{\text{diff}} = \epsilon_b - \epsilon_a$. Direct rearrangement of (39.22) gives

$$\mathcal{F}(X) = \frac{YX}{3} + \frac{2Ya_0}{3} \frac{X}{3 \log X + 4a_0} + \frac{1}{3} + vG(z, v), \quad (39.25)$$

where the exact two-variable analytic function is

$$G(z, v) = \frac{\frac{2Y}{3} \frac{\epsilon_{\text{diff}}(z)}{z} + \frac{2}{3}(a_0 + \epsilon_{\text{diff}}(z)) + 2\delta_0 - 2\epsilon_K(z) - \frac{2Ya_0}{3} v \frac{\epsilon_D(z)}{z}}{1 + v\epsilon_D(z)}. \quad (39.26)$$

The divided numerators vanish at zero, so their singularities are removable analytically. The denominator is 1 at $(z, v) = (0, 0)$. This establishes analyticity in two independent variables, not just a formal asymptotic series along a curve.

For completeness, the rearrangement may be verified without asymptotic manipulation. Substitute $\log X = (v^{-1} - 4a_0)/3$ in twice the numerator and denominator of (39.22). After clearing the common denominator and multiplying by $3vz$, the identity becomes

$$\begin{aligned} & (Y + z)(1 + 2a_0v + 3\epsilon_bv) + 6\delta_0vz - 6\epsilon_Kvz \\ &= (Y + 2Ya_0v + z)(1 + v\epsilon_D) \\ &+ v(2Y\epsilon_{\text{diff}} + 2z(a_0 + \epsilon_{\text{diff}}) + 6\delta_0z - 6\epsilon_Kz - 2Ya_0v\epsilon_D). \end{aligned}$$

This polynomial identity is checked by both independent algebra implementations in the packet.

Lemma 39.4.1 (Controlled second derivative). *As $X \rightarrow \infty$ with Y fixed,*

$$\mathcal{F}(X) = \frac{YX}{3} + \frac{2Ya_0}{3} \frac{X}{3\log X + 4a_0} + \frac{1}{3} + O(1/\log X), \quad (39.27)$$

$$\mathcal{F}''(X) = -\frac{2Ya_0}{9X(\log X)^2} (1 + O(1/\log X)). \quad (39.28)$$

The second assertion follows by controlled differentiation of an analytic remainder.

Proof. Analyticity of G yields bounded partial derivatives through order two on a sufficiently small closed neighborhood of $(0, 0)$. Set $R(X) = vG(z, v)$. Along the curve,

$$z' = -X^{-2}, \quad z'' = 2X^{-3}, \quad v' = -3v^2/X, \quad v'' = (3v^2 + 18v^3)/X^2.$$

As a function of two variables, R has derivative bounds

$$R_z = O(v), \quad R_v = O(1), \quad R_{zz} = O(v), \quad R_{zv} = O(1), \quad R_{vv} = O(1).$$

The multivariable chain rule gives

$$\begin{aligned} R'' &= R_{zz}(z')^2 + 2R_{zv}z'v' + R_{vv}(v')^2 + R_z z'' + R_v v'' \\ &= O\left(\frac{v}{X^3} + \frac{v^2}{X^2}\right) = O\left(\frac{1}{X^2(\log X)^2}\right), \end{aligned}$$

using $X^{-1} = O(v)$ eventually. The curved explicit term in (39.25) has exact second derivative

$$\frac{d^2}{dX^2} \left(\frac{2Ya_0}{3} Xv \right) = -\frac{2Ya_0v^2(1 - 6v)}{X}.$$

The linear term and the constant have second derivative zero. This proves (39.28); (39.27) follows directly from boundedness of G . In particular, the proof does not differentiate an uncontrolled $O(1/\log X)$ term. $\square$

39.4.3 Curvature along the actual lattice progression

Let

$$g(r) = \frac{\mathcal{F}(X_r) - p_0}{L_0}, \quad c_0 = q^3(q-1), \quad X_r = q^3 + c_0 r. \quad (39.29)$$

The chain rule and Lemma 39.4.1 yield

$$g''(r) = -\frac{K_*}{r(\log r)^2} (1 + O(1/\log r)), \quad K_* = \frac{2Y a_0 c_0}{9L_0} > 0. \quad (39.30)$$

The analytic constant K_* is unrelated to the compiler numeral K_0 . Because q was fixed, $\log X_r = \log r + O(1)$. There are therefore fixed $a_*, b_* > 0$ and an eventual threshold such that on each sufficiently large dyadic interval,

$$\frac{a_*}{N(\log N)^2} \leq |g''(r)| \leq \frac{b_*}{N(\log N)^2} \quad (N \leq r \leq 2N). \quad (39.31)$$

The constants do not depend on N or on the Fourier frequency introduced next.

39.5 A quantitative shrinking target argument

39.5.1 The two analytic estimates used

We invoke two classical estimates, with precisely the uniformity needed here.

Lemma 39.5.1 (Second derivative estimate). *If a real C^2 phase h on an interval containing N consecutive integers satisfies $\lambda \leq |h''| \leq C\lambda$, then*

$$\left| \sum e^{2\pi i h(r)} \right| \ll_C N\sqrt{\lambda} + \lambda^{-1/2}.$$

The constant depends only on the fixed comparability ratio C .

This is the second-derivative theorem stated by Robert, Theorem 1, Section 3.1, printed page 5 [8]. Translation of the interval does not change the bound.

Lemma 39.5.2 (Interval discrepancy). *For any real sequence $x_1, \dots, x_N$, any interval I in the unit circle, and any integer $H_f \geq 1$,*

$$|\#\{j : \{x_j\} \in I\} - N|I|| \ll \frac{N}{H_f + 1} + \sum_{h=1}^{H_f} \frac{1}{h} \left| \sum_{j=1}^N e^{2\pi i h x_j} \right|.$$

The implied constant is absolute and uniform in the interval I .

The normalized Erdős–Turán form in Mukhopadhyay, Ramaré and Viswanadham, Lemma 6, printed page 5 [9], implies this displayed unnormalized form. These primary sources were checked for the stated hypotheses and uniformity. Their estimates are analytic inputs; neither is inferred from finite numerical samples in this project.

39.5.2 Uniformity in every frequency being summed

For $1 \leq h \leq H_f$, apply Lemma 39.5.1 to $hg(r)$ on $N \leq r < 2N$, taking

$$\lambda = \frac{a_* h}{N(\log N)^2}, \quad C = b_*/a_*.$$

Multiplication by h leaves the comparability ratio fixed. Uniformly over all these frequencies,

$$\left| \sum_{N \leq r < 2N} e^{2\pi i h g(r)} \right| \ll \frac{\sqrt{hN}}{\log N} + \frac{\sqrt{N} \log N}{\sqrt{h}}. \quad (39.32)$$

There is no hidden dependence on h in the implied constant. The weighted sum in Lemma 39.5.2 is bounded by

$$\frac{\sqrt{N}}{\log N} \sum_{h \leq H_f} h^{-1/2} + \sqrt{N} \log N \sum_{h \leq H_f} h^{-3/2} \ll \frac{\sqrt{NH_f}}{\log N} + \sqrt{N} \log N.$$

Choose $H_f = \lfloor (\log N)^4 \rfloor$. Uniformly over every target interval, including intervals depending on N , the count error is

$$O\left(\frac{N}{(\log N)^4} + \sqrt{N} \log N\right) = o(N/\log N). \quad (39.33)$$

This quantitative estimate is the essential improvement over qualitative equidistribution. A fixed-interval density statement would not suffice for windows of width $1/\log r$.

39.5.3 The lower count with its explicit fixed constant

Let

$$\kappa_0 = \frac{\delta_0}{8L_0} > 0, \quad I_N = [0, \kappa_0/\log(2N)). \quad (39.34)$$

For sufficiently large N , this is a valid interval in $[0, 1)$. Its leading count is $\kappa_0 N/\log(2N)$. The error in (39.33) is eventually less than half that fixed positive leading term. Since $\log(2N) \leq 2\log N$ for $N \geq 2$, there are at least

$$\frac{\kappa_0 N}{2\log(2N)} \geq \frac{\delta_0}{32L_0} \frac{N}{\log N} \quad (39.35)$$

integers $r \in [N, 2N)$ such that

$$0 \leq \{g(r)\} < \frac{\kappa_0}{\log r}. \quad (39.36)$$

Indeed $r < 2N$ makes the interval I_N narrower than the pointwise interval in (39.36). Fractional part zero is allowed and causes no endpoint difficulty. The threshold may be enormous because κ_0 is a very small fixed constant. No numerical value or practical complexity estimate is asserted.

39.6 Exact Pell acceptance and every strict bound

For each sufficiently large r selected by (39.36), take

$$k_{\text{lat}} = \lfloor g(r) \rfloor, \quad p = p_0 + L_0 k_{\text{lat}}, \quad n = \frac{E_r + 1 - p}{2}. \quad (39.37)$$

The lattice makes p odd. Since E_r is even, n is an integer. The rounding error is exactly controlled:

$$0 \leq \mathcal{F}(X_r) - p = L_0 \{g(r)\} < \frac{\delta_0}{8 \log r}. \quad (39.38)$$

39.6.1 The exact ratio rather than a necessary real window

As $S(X_r) \sim \frac{3}{2} \log r$, eventually $S(X_r) \leq 2 \log r$. By (39.23), the non-tail logarithm at p lies between

$$\log Y + \frac{3}{4} \delta_0 \quad \text{and} \quad \log Y + \delta_0.$$

Equations (39.27) and (39.38) imply that p and n tend to infinity linearly with X_r . The two tail arguments in (39.21) tend to zero exponentially in $X_r \log X_r$. For example, $|\log(1 - z)| \leq 2z$ for $0 \leq z \leq 1/2$, so for all sufficiently large selected r the sum of the absolute tail terms is below $\delta_0/4$. Hence the full exact log-ratio is strictly between $\log Y$ and $\log Y + 2\delta_0 = \log(Y + 1)$. Exponentiating gives

$$Y < \frac{\psi_A(p)}{2\psi_P(n)} < Y + 1. \quad (39.39)$$

This proves exact acceptance by integer Pell values, not merely passage through a necessary approximate interval.

39.6.2 The positive deviation above one third

The center expansion yields the stronger asymptotic information

$$p = \frac{E_r}{3} + \frac{2Y a_0}{3} \frac{X_r}{3 \log X_r + 4a_0} + \frac{1}{3} + O(1/\log X_r). \quad (39.40)$$

In particular,

$$\frac{p}{E_r} \rightarrow \frac{1}{3}, \quad p - \frac{E_r}{3} \rightarrow +\infty, \quad \frac{n}{E_r} \rightarrow \frac{1}{3}, \quad d_{\text{def}} = 2n - p = E_r + 1 - 2p \sim \frac{E_r}{3}. \quad (39.41)$$

The fixed positive coefficient of $X_r/\log X_r$ matters: the limit $p/E_r \rightarrow 1/3$ alone would not justify the strict upper endpoint $E_r \leq 3p - 3$.

Proposition 39.6.1 (All eventual index conditions). *One finite threshold, enlarged if necessary, ensures that every selected r above it satisfies*

$$\begin{aligned} p \geq 13 \text{ odd}, \quad u < p, \quad \frac{p+1}{2} \leq n \leq p-1, \quad 2n+p-1 = E_r, \\ 2p+6 \leq E_r \leq 3p-3, \quad p < X_r q^4, \quad 4^{d_{\text{def}}} > X_r. \end{aligned} \quad (39.42)$$

Thus p lies in the precise interval (39.6) with $t = 1$, including its integer endpoints.

Proof. The first two assertions follow from $p \rightarrow \infty$ and fixed u . The quantity $E_r - 2p \sim E_r/3$ tends to infinity, yielding $2p+6 \leq E_r$. The deviation in (39.41) yields $3p - E_r \rightarrow +\infty$ and therefore $E_r \leq 3p - 3$. These give the stated bounds on n through (39.37). They also give $p \geq \lceil E_r/3 \rceil + 1$ and $p \leq \lfloor (E_r - 6)/2 \rfloor$ without losing an endpoint. The identity $2n + p - 1 = E_r$ is exact; its $+1$ cannot be discarded. Further,

$$\frac{p}{X_r q^4} \rightarrow \frac{1}{3q} < 1.$$

Since p and $X_r q^4$ are integers, eventual strict inequality also gives $p \leq X_r q^4 - 1$. Finally $d_{\text{def}} \sim E_r/3$ grows linearly in X_r , so $d_{\text{def}} \log 4 - \log X_r \rightarrow +\infty$. This proves the defect filter directly, independently of the omitted congruence. A finite list of eventual bounds has a common enlarged threshold. $\square$

39.6.3 Every outer and input sign

The exact recovery formula has asymptotic

$$\frac{F}{X_r} \longrightarrow \frac{Y}{3qQ} = \frac{q^2}{3(q^2 - 1)} < 1. \quad (39.43)$$

Indeed M, q, Q are fixed and $p/X_r \rightarrow Y/3$. Thus $F > q$ eventually. Since $C = W + 1 > 1$,

$$\frac{(K_0 + X_r)C - F}{X_r} \longrightarrow C - \frac{q^2}{3(q^2 - 1)} > 0.$$

The quotient z_{quot} in Lemma 39.3.1 is therefore a strictly positive integer. The loading inequalities and $\alpha_{\text{src}} > 0$ already hold by (39.11). Also $H > q > W = 2^u$ for every $r \geq 1$, so W is exactly the least input residue $2^u \bmod H$. With $e = u < p$, every input test is met. Finally $s = t = 1$ satisfies both finite ranges in Section 39.2.

39.7 The infinite family theorem

Theorem 39.7.1 (Infinite candidates with one congruence omitted). *Under the genuine fixed compiler and bootstrap premises inherited from Report 37, fix any valid complete75 modified compiler export and any $x \geq 1$. Choose q as in (39.11), for example $q = B^{2x+2}$, and fix the witnesses in (39.13). There is N_0 such that for every integer $N \geq N_0$, at least*

$$\frac{\delta_0 N}{32L_0 \log N}$$

distinct scales $w_r = 1 + (q - 1)r$ with $N \leq r < 2N$ admit odd indices p satisfying every condition of Report 37's exact negative-zero predicate except

$$2^p \equiv X_r \pmod{4Y(X_r + 1) + 3}. \quad (39.44)$$

Every compiler numeral is unchanged. The reduced predicate has infinitely many solutions on every such fixed compiler/input slice.

Proof. The discrepancy argument gives the lower count of r in (39.35). Lemma 39.3.1 places the selected odd index in the exact packing/transport lattice. Section 39.6 proves the exact ratio, all interval and defect conditions, and all outer/input tests with positive markers. Enlarge one threshold to cover every eventual claim. The progression is injective in r , so these are distinct scales. Taking arbitrarily large disjoint dyadic intervals supplies infinitely many solutions. None of these steps invokes (39.44). $\square$

What the count measures. Here N measures the progression index r , not w and not the bit length of the data. The counted actual scales lie in

$$[1 + (q - 1)N, 1 + 2(q - 1)N).$$

The constant and threshold may depend on the fixed compiler/input/ q . The theorem says at least the displayed real number of integer scales, equivalently the count is an integer no smaller than it. It asserts neither a density bounded away from zero nor a practical first accepted scale.

Why this is stronger than outer compatibility. The affine formulas alone do not force the narrow Pell ratio. The center has a curved $X/\log X$ correction to its linear term, and this curvature quantitatively distributes its lattice coordinate on the circle. The resulting $o(N/\log N)$ discrepancy error is small enough for the shrinking index windows. The old fixed- X bounded window cannot be used as though it were an unbounded density problem.

39.8 A fixed modulus Jacobi filter

The broad family theorem does not decide the main congruence. The new arithmetic input does decide it on a carefully chosen infinite subfamily. Write $\mathcal{J}(a, m)$ for the Jacobi symbol with positive odd denominator m , allowing the value zero when a, m are not coprime.

Lemma 39.8.1 (Fixed modulus character identity). *Let q be an even square with $3 \nmid q$. For every positive integer w , put*

$$h_0 = 4q^3 + 3, \quad X = q^3w, \quad H = 4q^6w + h_0. \quad (39.45)$$

Then

$$\mathcal{J}(X, H) = \mathcal{J}(w, h_0). \quad (39.46)$$

Consequently, for every positive odd p ,

$$2^p \equiv X \pmod{H} \implies \mathcal{J}(w, h_0) = -1. \quad (39.47)$$

No denominator is required to be prime or squarefree.

Proof. Both H and h_0 are 3 modulo 8. Since $H \equiv 3 \pmod{q}$ and $3 \nmid q$, we have $\gcd(q, H) = 1$. The integer q^3 is a square, so its Jacobi symbol modulo H is 1. Thus $\mathcal{J}(X, H) = \mathcal{J}(w, H)$. Also

$$\gcd(w, H) = \gcd(w, h_0).$$

If this gcd is greater than one, both sides of (39.46) are zero. Otherwise write $w = 2^\nu z$ with z positive and odd. The supplementary law gives

$$\mathcal{J}(2, H) = \mathcal{J}(2, h_0) = -1.$$

If $z = 1$, these equal factors finish the identity. If $z > 1$, apply reciprocity only to the coprime odd pairs. Since $H \equiv h_0 \pmod{z}$ and $H, h_0 \equiv 3 \pmod{4}$,

$$\begin{aligned} \mathcal{J}(z, H) &= (-1)^{(z-1)/2} \mathcal{J}(H, z) \\ &= (-1)^{(z-1)/2} \mathcal{J}(h_0, z) = \mathcal{J}(z, h_0). \end{aligned}$$

The two reciprocity signs cancel. Multiplying by the equal factors from 2^ν proves (39.46), including even w . Finally, for odd p ,

$$\mathcal{J}(2^p, H) = \mathcal{J}(2, H)^p = -1.$$

Jacobi symbols depend only on numerator residue classes, proving the necessary filter. $\square$

The laws used here are multiplicativity, the supplementary law for 2, and quadratic reciprocity for Jacobi symbols. NIST DLMF Section 27.9 explicitly extends the supplementary laws and reciprocity to relatively prime positive odd composite denominators [7]. The noncoprime case was handled before either reciprocity step. Symbol -1 is only a necessary test, not a sufficient criterion for a congruence hit.

39.8.1 Two fixed thinnings that force a miss

For a genuine compiler and input, choose the explicit scale

$$q = B^{2x+2} = (B^{x+1})^2.$$

It is an even square coprime to 3, as required by Lemma 39.8.1. Keep all markers and lattice constants from Sections 39.3–39.7. Choose either fixed thinning factor

$$T_{\text{thin}} = h_0 \quad \text{or} \quad T_{\text{thin}} = 4h_0, \quad (39.48)$$

and restrict

$$r = T_{\text{thin}}j, \quad w_j = 1 + (q-1)T_{\text{thin}}j \equiv 1 \pmod{h_0}, \quad j \geq 1. \quad (39.49)$$

Then $\mathcal{J}(w_j, h_0) = 1$, so (39.44) fails for *every odd* p at these scales. The choice $T_{\text{thin}} = h_0$ retains more progression indices and can give even w . The factor $4h_0$ from the original proposal also works; it has $w \equiv 1 \pmod{4h_0}$ and permits a shorter proof with individually positive reciprocity signs.

Arithmetic exclusion alone would not show that any of these scales meets the exact Pell ratio. That requires a fresh shrinking-target argument on the thinned sequence, proved next. One may not assume that the already-selected parent hits distribute among residue classes.

39.9 Infinitely many certified misses after fixed thinning

Define

$$g_T(j) = g(T_{\text{thin}}j).$$

All fixed data, including T_{thin} , precede the limit. Equation (39.30) and the exact chain rule give

$$\begin{aligned} g_T''(j) &= T_{\text{thin}}^2 g''(T_{\text{thin}}j) \\ &= -\frac{K_* T_{\text{thin}}}{j(\log j)^2} (1 + O(1/\log j)). \end{aligned} \quad (39.50)$$

Thus the curvature coefficient is multiplied by T_{thin} , not by its square after changing variables in the asymptotic denominator. The derivative-comparability ratio remains independent of the Fourier frequency.

Repeat Lemmas 39.5.1–39.5.2 on $N \leq j < 2N$. Uniformly for $1 \leq h \leq \lfloor (\log N)^4 \rfloor$,

$$\left| \sum_{N \leq j < 2N} e^{2\pi i h g_T(j)} \right| \ll \frac{\sqrt{hN}}{\log N} + \frac{\sqrt{N} \log N}{\sqrt{h}}.$$

The implied constant may now depend on the fixed thinning factor. The interval-count error is still

$$O\left(\frac{N}{(\log N)^4} + \sqrt{N} \log N\right) = o(N/\log N). \quad (39.51)$$

Use the narrower interval

$$I_{T,N} = \left[0, \frac{\kappa_0}{\log(2T_{\text{thin}}N)}\right).$$

Its leading count is $\kappa_0 N / \log(2T_{\text{thin}}N)$. Since T_{thin} is fixed, this dominates (39.51) eventually, leaving at least half the leading count. Enlarge the threshold so $N \geq 2T_{\text{thin}}$; then $\log(2T_{\text{thin}}N) \leq 2 \log N$, and the count is at least

$$\frac{\kappa_0 N}{2 \log(2T_{\text{thin}}N)} \geq \frac{\delta_0 N}{32 L_0 \log N}. \quad (39.52)$$

Every counted j satisfies

$$0 \leq \{g(T_{\text{thin}}j)\} < \frac{\kappa_0}{\log(T_{\text{thin}}j)}.$$

This is exactly the earlier pointwise hypothesis (39.36) at $r = T_{\text{thin}}j$. Thus all interior-margin, Binet-tail, strict-bound and positive-reconstruction arguments apply unchanged.

Theorem 39.9.1 (Infinite genuine comparison counterfamilies). *Under the inherited genuine compiler and bootstrap premises, fix any valid compiler export, any $x \geq 1$, and $q = B^{2x+2}$. Choose $T_{\text{thin}} = h_0$ or $4h_0$ as above. For all sufficiently large integers N , at least $\delta_0 N / (32L_0 \log N)$ indices $j \in [N, 2N)$ yield distinct scales (39.49) satisfying every condition of the reduced Report 37 predicate, with*

$$2^p \not\equiv X \pmod{H}.$$

Their positive reconstructions have restored index $R = -p$ and strictly positive remainder $1 \leq r_{\text{main}} < H$. In raw29 they satisfy every comparison except the main projection, which they violate. In positive21 they satisfy every retained comparison except the main norm, which they violate. Hence each indicated comparison is nonredundant relative to its own remaining conjunction on strictly positive integer tuples.

Proof. Equation (39.52) supplies infinitely many exact-ratio candidates after fixed thinning. The Jacobi identity excludes the main congruence for every selected odd p . Sections 39.10–39.11 provide all positive supplied coordinates and the literal residual values. Their nonzero remainder makes the indicated residual nonzero in each mode, while all others vanish. $\square$

Here N counts j . The corresponding original progression indices and actual scales lie in

$$[T_{\text{thin}}N, 2T_{\text{thin}}N), \quad [1 + (q-1)T_{\text{thin}}N, 1 + 2(q-1)T_{\text{thin}}N).$$

The same displayed constant does not mean that the parent r -count was silently reused. It follows from a fresh discrepancy proof on j , and its threshold now depends on T_{thin} . The result is not a positive natural density claim.

39.10 Complete strictly positive witness reconstruction

This section spells out the literal completion, so the result is not dependent on an unexamined sign assertion. The only potentially unsatisfied raw comparison will be the main projection comparison. The omitted congruence is not used anywhere in the completion.

[write] This reconstruction specializes Part V's (Section 37.5): (39.57) is the first and main block of (37.29), (39.61) the input block (37.34) on the branch $e = u$, and (39.62)–(39.63) the auxiliary block (37.35)–(37.36), which Part V in turn credits to the research tree's bootstrap note. The recurrence (39.58)–(39.59) is (37.33), that is, Lemma 33.3.2 of Report 33; the identities (39.64) are Lemma 33.3.3; and the divisibility $c^2 \mid \psi_A(m)$ is Lemma 33.3.4. The residual Table 11 specializes Part V's Table 9. What is new here is that every coordinate is positive on the family without the main congruence, and the two residual formulas of Section 39.11.

39.10.1 The literal retained system

The restored index is $R = k_{\text{first}} - hE - 1$. The outer system is

$$\begin{aligned} q &= (B - 1)J + 1, & C + \alpha_{\text{src}} + u - b &= q, \\ (K_0 + X)C &= F + z_{\text{quot}}(q - 1), & C &= Z + W, \\ R &= (q^2 - Z - qF)Q + M. \end{aligned} \quad (39.53)$$

The first and main comparisons are

$$\begin{aligned} \tau^2 - XY^2(XY^2 + 1)k_{\text{first}}^2 &= 1, & c &= k_{\text{first}}Y + \eta, \\ k_{\text{first}} &= \eta + \zeta, & a &= Y(X + 1), \\ D &= X + ac + g_a H, & D^2 - \Delta c^2 &= 1. \end{aligned} \quad (39.54)$$

The ordinary strong and auxiliary comparisons, with $T = ic^2$ and $U = jc - R$, are

$$T^2 = \Delta(f_{\text{aux}}^2 - 1), \quad T^2(U^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2, \quad U = of_{\text{aux}} - c. \quad (39.55)$$

The input block is

$$\kappa = u + \delta_{\text{in}}\Delta, \quad c = \kappa + \phi, \quad \mu_{\text{in}}^2 = 1 + \Delta\kappa^2, \quad \mu_{\text{in}} = W + a\kappa + \rho H. \quad (39.56)$$

In these two interfaces g_a and ρ are independent positive coordinates. The relation $g_a = \rho + \sigma$ belongs to a different signed interface and must not be inserted here.

39.10.2 First block and restored negative index

For a selected candidate set

$$\begin{aligned} c &= \psi_A(p), \quad D = \chi_A(p), \quad k_{\text{first}} = 2\psi_P(n), \quad \tau = \chi_P(n), \\ \eta &= c - k_{\text{first}}Y, \quad \zeta = k_{\text{first}} - \eta, \quad h = \frac{k_{\text{first}} + p - 1}{E}. \end{aligned} \quad (39.57)$$

The exact ratio gives $\eta, \zeta > 0$. Since $P \equiv 1 \pmod{E}$, the recurrence implies $\psi_P(n) \equiv n \pmod{E}$. Hence

$$k_{\text{first}} + p - 1 \equiv 2n + p - 1 = E \equiv 0 \pmod{E}.$$

Its numerator is positive, so h is a positive integer and $R = -p$ exactly. The two Pell identities give

$$D^2 - \Delta c^2 = 1, \quad \tau^2 - 1 = (P^2 - 1)\psi_P(n)^2 = XY^2(XY^2 + 1)k_{\text{first}}^2.$$

The factor 2 in k_{first} is essential for the first norm. Every outer comparison holds by the definitions and the exact lattice identities.

39.10.3 A recurrence for the exponent projections

Define $E_j^* = \chi_A(j) - a\psi_A(j)$. Since $a = A - 2$, the Pell recurrence gives

$$E_0^* = 1, \quad E_1^* = 2, \quad E_j^* = 2\psi_A(j) - \psi_A(j - 1) > \psi_A(j) \quad (j \geq 1). \quad (39.58)$$

Moreover

$$E_j^* \equiv 2^j \pmod{H}. \quad (39.59)$$

Indeed 2^j has the same initial values and satisfies the same recurrence modulo H , because $4-4A+1 = -H$. For $j \geq 3$, monotonicity of the Pell sequence gives

$$\psi_A(j) \geq \psi_A(3) = 4A^2 - 1 > H + X, \quad \psi_A(j) > H + q. \quad (39.60)$$

Here $a > X, q$, since $X, Y \geq q^3$. These inequalities will ensure positive quotients without assuming the main projection is divisible.

39.10.4 Every positive input coordinate

Use the fixed odd input index $e = u$ and set

$$\kappa = \psi_A(u), \quad \mu_{\text{in}} = \chi_A(u), \quad \delta_{\text{in}} = \frac{\kappa - u}{\Delta}, \quad \phi = c - \kappa, \quad \rho = \frac{\mu_{\text{in}} - a\kappa - W}{H}. \quad (39.61)$$

The recurrence modulo Δ , where $A^2 \equiv 1$, gives

$$\psi_A(j) \equiv \begin{cases} j, & j \text{ odd}, \\ jA, & j \text{ even} \end{cases} \pmod{\Delta}.$$

Thus δ_{in} is integral; $u \geq 3$ and strict Pell growth give $\kappa > u$, so it is positive. Since $u < p$, $\phi > 0$. Equation (39.59) and $W = 2^u \bmod H$ make ρ integral. Equation (39.60) and $W < q$ make it positive. The supplied root μ_{in} is the positive Pell root. All four input comparisons hold exactly.

39.10.5 Auxiliary completion for both odd index classes

The all-odd-index construction inherited from Report 37 may be verified directly. Define

$$m = \begin{cases} cp, & p \equiv 3 \pmod{4}, \\ 2cp, & p \equiv 1 \pmod{4}, \end{cases} \quad f_{\text{aux}} = \chi_A(m), \quad T_{\text{aux}} = \Delta\psi_A(m), \quad i = \frac{T_{\text{aux}}}{c^2}. \quad (39.62)$$

Let

$$\ell = p + 2m, \quad U = \frac{\chi_{T_{\text{aux}}}(\ell)}{T_{\text{aux}}}, \quad y_{\text{aux}} = \psi_{T_{\text{aux}}}(\ell), \quad j = \frac{U - p}{c}, \quad o = \frac{U + c}{f_{\text{aux}}}. \quad (39.63)$$

Here j denotes the supplied auxiliary coordinate, not the progression index of Section 39.9. All these auxiliary quantities are positive integers, as follows.

Write $m = pv_0$, with $v_0 = c$ or $2c$. The square-root coefficient of $(D + c\sqrt{\Delta})^{v_0}$ is $\psi_A(m)$. Its first odd binomial term contains v_0c , divisible by c^2 ; every later odd term contains c^3 . Therefore $c^2 \mid \psi_A(m)$ and $i > 0$ is integral. The norm gives

$$T_{\text{aux}}^2 = \Delta(f_{\text{aux}}^2 - 1).$$

Because A is even and p is odd, c is odd. The two choices in (39.62) both give $\ell \equiv 1 \pmod{4}$. In particular, the factor two for $p \equiv 1 \pmod{4}$ cannot be replaced by the other branch.

For every odd $\ell = 2h_{\text{aux}} + 1$, the quotient $\chi_T(\ell)/T$ is an integer polynomial $Q_{h_{\text{aux}}}(T^2)$ satisfying

$$Q_{h_{\text{aux}}}(0) = (-1)^{h_{\text{aux}}}(2h_{\text{aux}} + 1), \quad Q_{h_{\text{aux}}}(1 - A^2) = (-1)^{h_{\text{aux}}}\psi_A(2h_{\text{aux}} + 1). \quad (39.64)$$

These identities hold for all indices: $Q_0(t) = 1$, $Q_1(t) = 4t - 3$ and

$$Q_{h_{\text{aux}}+1}(t) = 2(2t - 1)Q_{h_{\text{aux}}}(t) - Q_{h_{\text{aux}}-1}(t).$$

The evaluations in (39.64) follow by the same recurrence and initial values, with the sign $(-1)^{h_{\text{aux}}}$ in the second identity. Thus the proof is not limited to the finitely many polynomial fixtures.

Here h_{aux} is even. Since $c \mid T_{\text{aux}}$ and $c \mid m$, the first identity gives $U \equiv \ell \equiv p \pmod{c}$. Modulo f_{aux} , the strong norm gives $T_{\text{aux}}^2 \equiv -\Delta = 1 - A^2$; the second identity gives $U \equiv \psi_A(\ell)$. Pell addition formulas give

$$\chi_A(2m) \equiv -1 \pmod{f_{\text{aux}}}, \quad \psi_A(2m) \equiv 0 \pmod{f_{\text{aux}}},$$

so $\psi_A(p + 2m) \equiv -c \pmod{f_{\text{aux}}}$. This proves integrality of j, o . Since $T_{\text{aux}} \geq c^2$, $\ell \geq 3$ and Pell growth gives $c > p$,

$$U \geq 4T_{\text{aux}}^2 - 3 > p.$$

Consequently $j, o > 0$. Finally the Pell norm at base T_{aux} is

$$T_{\text{aux}}^2(U^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2,$$

and $U = jc + p = jc - R = of_{\text{aux}} - c$. This verifies every comparison in (39.55) with positive $f_{\text{aux}}, i, j, o, y_{\text{aux}}$. No uniqueness or reasonable size of these auxiliary witnesses is claimed.

39.11 The literal one residual and the two different sums of squares

Set

$$g_a = \left\lfloor \frac{D - ac - X}{H} \right\rfloor, \quad r_{\text{main}} = D - ac - X - g_a H. \quad (39.65)$$

By (39.58)–(39.60), $D - ac > H + X$, so $g_a \geq 1$. By Euclidean division and (39.59),

$$0 \leq r_{\text{main}} < H, \quad r_{\text{main}} = (2^p - X) \bmod H. \quad (39.66)$$

Thus every supplied raw coordinate is positive even without knowing whether r_{main} is zero.

39.11.1 The exact supplied tuples

The raw29 tuple in saved order, using the aliases introduced above, is

$$(C, F, J, W, Z, a, \alpha_{\text{src}}, c, D, \delta_{\text{in}}, \eta, f_{\text{aux}}, g_a, h, i, j, k_{\text{first}}, \kappa, \mu_{\text{in}}, o, \phi, \\ q, \rho, s, \tau, w, y_{\text{aux}}, \zeta, z_{\text{quot}}).$$

The positive21 supplied tuple is

$$(J, F, \alpha_{\text{src}}, z_{\text{quot}}, f_{\text{aux}}, h, i, j, o, s, w, \tau, \eta, \zeta, g_a, y_{\text{aux}}, Z, W, \delta_{\text{in}}, \phi, \rho).$$

The eliminated positive coordinates in positive21 are recovered by triangular expressions. In particular its main root expression is

$$D' = X + ac + g_a H = D - r_{\text{main}} > 0, \quad (39.67)$$

which may differ from the raw supplied Pell root D .

39.11.2 Complete accounting of the raw comparisons

The following table records every raw residual. Positive21 retains the comparisons with raw indices 1, 2, 3, 5, 10, 11, 12, 13, 15, 16; the other eight are triangular definitions.

Table 11: Every raw comparison on the constructed tuple

Index	Comparison	Value or justification
0	$q = (B - 1)J + 1$	Exact repunit definition
1	$C + \alpha_{\text{src}} + u - b = q$	Positive loading slack
2	$(K_0 + X)C = F + z_{\text{quot}}(q - 1)$	Exact positive transport
3	$R = (q^2 - Z - qF)Q + M$	Exact packing with $R = -p$
4	$C = Z + W$	Fixed markers
5	$\tau^2 = 1 + XY^2(XY^2 + 1)k_{\text{first}}^2$	First Pell norm
6	$c = k_{\text{first}}Y + \eta$	Exact positive ratio gap
7	$k_{\text{first}} = \eta + \zeta$	Exact complementary gap
8	$a = Y(X + 1)$	Scale definition
9	$D = X + ac + g_a H$	Left minus right is r_{main}
10	$D^2 = 1 + \Delta c^2$	Exact raw main Pell norm
11	$T^2 = \Delta(f_{\text{aux}}^2 - 1)$	Ordinary strong norm
12	$T^2(U^2 - y_{\text{aux}}^2) = 1 - y_{\text{aux}}^2$	Auxiliary norm
13	$jc - R = of_{\text{aux}} - c$	Auxiliary congruences
14	$\kappa = u + \delta_{\text{in}}\Delta$	Input discriminant congruence
15	$c = \kappa + \phi$	Positive input gap
16	$\mu_{\text{in}}^2 = 1 + \Delta\kappa^2$	Positive input Pell norm
17	$\mu_{\text{in}} = W + a\kappa + \rho H$	Exact input projection

Static inspection of the saved source shows that g_a affects only raw comparison 9. Its finalizer contains exactly one square of each of the 18 residuals. Therefore its exact value on this tuple is

$$\text{SOS}_{\text{raw29}} = r_{\text{main}}^2. \quad (39.68)$$

This identity holds on the entire broad family. On the thinned family of Theorem 39.9.1, its value is strictly positive because $r_{\text{main}} \geq 1$.

39.11.3 The triangular main norm in positive21

The positive21 retained main norm uses D' from (39.67), not D . Hence its single possible residual is

$$(D')^2 - \Delta c^2 - 1 = (D - r_{\text{main}})^2 - D^2 = -r_{\text{main}}(2D - r_{\text{main}}). \quad (39.69)$$

The remaining nine retained comparisons vanish. Static dependency inspection confirms that g_a affects only positive21 comparison 4, which is raw comparison 10 after triangular elimination. Its full ten-term finalizer therefore has value

$$\text{SOS}_{\text{positive21}} = [r_{\text{main}}(2D - r_{\text{main}})]^2. \quad (39.70)$$

Since $D > H > r_{\text{main}}$, this vanishes if and only if $r_{\text{main}} = 0$. The two SOS formulas (39.68) and (39.70) are distinct and must not be conflated.

Corollary 39.11.1 (The remaining obstruction on this family). *Each selected candidate in the broad family becomes a full negative zero in both interfaces if and only if (39.44) holds. The thinned family of Theorem 39.9.1 is excluded: its two SOS values are respectively $r_{\text{main}}^2 > 0$ and $[r_{\text{main}}(2D - r_{\text{main}})]^2 > 0$. No congruence hit or full negative zero is supplied.*

The finite residual fixtures have nonzero residues, but they are explicitly main-block subsystem checks without the exact ratio and genuine compiler realization. They do not establish the genuine-family nonredundancy claim; that claim follows from the analytic existence proof together with the Jacobi obstruction.

39.12 A separate sharpened outer estimate

For every putative full negative zero in the unwrapped odd-input sector, the input inequalities imply

$$q > W = 2^b B^{2x} \geq 2B^2 > K_0.$$

Since $X \geq q^3$, it follows that

$$K_0(q^2 - 1) < q^3 \leq X, \quad q(q^2 - 1)(K_0 + X) < Xq^3.$$

Positive transport gives $F < (K_0 + X)C$. Since $Z < q < q^2$ and $M > 0$, negative packing gives the strict chain

$$\begin{aligned} p &< Q(Z + q(K_0 + X)C - q^2) - M \\ &< qQ(K_0 + X)C < CXq^3. \end{aligned} \tag{39.71}$$

Combining this with $tE \leq 3p - 3$ and $E = sXq^3$ yields

$$ts < 3C. \tag{39.72}$$

This strengthens the prior $ts < 3q$ and the looser estimate involving $C(1 + 1/q)$. It is separate from the family theorem and does not force the sign of R . The family constructed here satisfies (39.71) eventually and has $s = t = 1$.

[write] The prior bound $ts < 3q$ is Part V's, derived after (37.17); Part V's note there points here. The sharper (39.72) is proved only on the unwrapped odd-input sector, where $W = 2^u$ exactly.

39.13 Exact enumeration and unresolved questions

39.13.1 An integer enumeration independent of real endpoint decisions

The analytic selection proves existence but need not be the computational definition of the candidate set. Enumerate $r \geq 1$, set $w = w_r$, $s = t = 1$, and use Report 37's finite exact ratio search within (39.6). Increasing odd p by two increases $\psi_A(p)$ and decreases $\psi_P(n)$, so the ratio strictly increases. Its numerator grows by more than $(2A - 1)^2$, and $(2A - 1)^2 Y > Y + 1$; thus at most one odd p at each scale can pass (39.7). Exact integer binary search finds or rejects it. Retain it only if $p \equiv P_0 \pmod{L_0}$ and every fixed-marker test passes.

The broad theorem guarantees infinitely many retained scales. Restricting this enumeration to $r = T_{\text{thin}}j$ still leaves infinitely many, and each is a certified congruence miss by Theorem 39.9.1. This enumeration never has to decide whether a transcendental fractional part equals a shrinking-window endpoint. It uses enormous exact Pell integers if needed; termination at a fixed scale is not a practical complexity guarantee, and the theorem does not provide a feasible first-hit bound.

39.13.2 Questions that remain open

1. **The symbol minus one sector.** The fixed-character filter eliminates $\mathcal{J}(w, h_0) \in \{0, 1\}$. Does the broad exact-ratio family contain a main-congruence hit in the necessary sector $\mathcal{J}(w, h_0) = -1$? The filter is not sufficient, and no hit is proved.
2. **A numerically materialized genuine tuple.** Can one materialize a genuine compiler export and an accepted large- r tuple, then compute its exact positive residue in the miss subfamily? Existence and nonzeroness are already proved there, but no enormous example or residue numeral is supplied by the finite tests.
3. **Effective thresholds.** Can the analytic constants, Binet-tail estimates and all strict inequalities be made explicit enough for a useful numerical threshold? Finiteness of the threshold does not make it feasible.
4. **Additional modular obstructions.** The fixed Jacobi test can be computed without factoring h_0 or constructing Pell integers. Are there further fixed-modulus or character tests that constrain the unresolved sector? Any claimed joint distribution with the varying exponential modulus must be proved, not inferred from center discrepancy alone.
5. **Scope beyond this slice.** Other scales, markers, s, t , input branches and progression classes remain outside this family's parametrization. The present exclusions settle neither full raw29/positive21 negative-zero existence nor global restored-index positivity.

A hit in any passing full predicate would yield a full negative zero by Report 37 and the positive reconstruction. No such hit is claimed. The infinite certified misses prove nonredundancy of two specific comparisons in their respective systems; they do not prove emptiness of the full predicate.

[write] At this write (batch 88) all five questions are open, and so is Part V's yes-or-no question; the research tree records raw29 and positive21 as unresolved. A main-congruence hit with symbol -1 in this family would, by Theorem 37.5.1, be a full negative zero.

39.14 Reproducible evidence and its exact scope

[write] *Shipped names.* The packet's files are shipped with the prefix `16-nonredundant-`, the delivery directories flattened into the name (README): `ALL-BUT-MAIN-PROJECTION.md` is `16-nonredundant-evidence-ALL-BUT-MAIN-PROJECTION.md`, the independent review `16-nonredundant-evidence-independent-INDEPENDENT-REVIEW.md`, the Jacobi addendum and its audit `16-nonredundant-jacobi-JACOBI-ADDENDUM.md` and `16-nonredundant-jacobi-INDEPENDENT-AUDIT.md`, and the three checkers `code/16-nonredundant-evidence-check_unwrapped_family.py`, `code/16-nonredundant-evidence-independent-check_audit.py` and `code/16-nonredundant-jacobi-check_jacobi_addendum.py`. The four context snapshots are byte copies (three of research-tree files, one of the Report 37 manuscript) and are not shipped. At placement the three checkers, run directly on a copy of the delivery layout, passed in normal and optimized Python with receipts byte-identical to the frozen ones; the release verifier refuses to run on NTFS, whose file modes it cannot match. The README gives a rerun route. The programme's review executed none of these programs.

39.14.1 Frozen scientific evidence

The accompanying release preserves the complete follow-on packet and the separate later Jacobi addendum as immutable evidence. Its original proof is `ALL-BUT-MAIN-PROJECTION.md`; the independent

review explains the derivative control, strict bounds, auxiliary signs and literal residual dependencies. The scientific author checker uses the Python standard library. The independent checker uses exact symbolic algebra with SymPy, tested with version 1.14.0. They authenticate the four context snapshots as inert data. A third own checker corroborates the Jacobi addendum using exact integer arithmetic and a trial-factorization/Euler-criterion oracle. None of the three imports an upstream module or evaluates a saved arithmetic schedule.

The frozen scientific receipts are deterministic canonical JSON. The release replay compares stdout byte for byte with the expected files in both normal and optimized Python. Explicit exception checks remain active under `-O`. These replays corroborate the listed algebra and finite inventories, not the unbounded analytic theorem.

39.14.2 Finite checks and their limitations

The author receipt contains:

- Five formal multivariate polynomial identities: affine transport, negative packing, lattice progression, the positive21 shifted-root residual, and the analytic G rearrangement after clearing denominators
- 36 structural exponent/loading-bound fixtures, without materializing any actual fixed compiler export
- 864 outer algebra/parity fixtures, with no exact Pell-ratio or full-zero claim
- 48 exact main Pell/floor-residual fixtures, all with nonzero residues, explicitly limited to the main subsystem
- Authentication of the Report 37 TeX, projection JSON and two parity-source snapshots, plus static main-root checks

The independent portable receipt additionally contains exact symbolic curvature and remainder identities, both complete supplied-coordinate inventories and SOS suffixes, 432 abstract bound/lattice fixtures, 20 modular auxiliary fixtures with $p \geq 13$ and $p \equiv 1 \pmod{4}$, and 25 exact odd-quotient polynomial identities. The review also preserves historical evidence of five authenticated cached source identities and a 43-file unchanged Report 37 tree. That historical record is not a fresh five-source network check or a replay of all of Report 37.

The Jacobi receipt adds 21,109 binary-symbol/oracle comparisons, including 11,495 composite-denominator cases; 2,328 general-filter cases split into 994 symbol -1 , 990 symbol $+1$ and 344 zero cases; 11,640 odd-exponent checks; 384 thinning fixtures; and exact chain-factor, count-constant and residual identities. Its 1,192 even- w general-filter fixtures and 96 even- w fixtures in the $T_{\text{thin}} = h_0$ family guard the parity boundary. Boundary controls exercise nonsquare q , even p and $3 \mid q$. These are abstract arithmetic fixtures, not genuine compiler/Pell-ratio tuples.

The infinite count depends on the mathematical proof and the two stated analytic estimates. No numerical frequency experiment, floating-point sample, or symbolic check of finitely many indices establishes that count. No selected large- r genuine-compiler tuple has been materialized in this packet.

39.14.3 Authentication and execution boundaries

The release separates two questions. A manifest, exact source pins and archive checks establish which bytes are being reviewed or executed. The mathematical arguments and their assumptions establish

what those bytes claim and why the new theorem follows. A passing digest check is not a proof of mathematical truth; a valid theorem does not authenticate a substituted file.

The enclosing verifier authenticates the release and frozen source identities before executing the three own scientific checkers in an external temporary copy. It checks exact canonical output and strict receipt types rather than accepting a generic status string. Output paths are guarded so an enclosing replay cannot overwrite a packet, an expected receipt or an existing output. The frozen author checker itself allows replacement of an existing external output path; this limitation is left unchanged and is avoided by the enclosing safe workflow. The independent checker requires a fresh external output file. Default scientific runs use stdout only.

The release includes regression checks for tampered bytes/receipts and hostile ZIP members, plus deterministic PDF and ZIP build helpers. Archive validation rejects unsafe member paths and ambiguous inventories before extraction or execution. A release’s authenticity must ultimately be anchored by a digest obtained through a trusted channel; a manifest bundled with adversarial replacement content cannot authenticate itself. Detailed commands, exact replay gates and build dependencies are documented in the release’s `README.md` and `INTEGRITY.md`.

No mutation of inherited evidence. The original follow-on packet, later Jacobi packet and sealed Report 37 remain unchanged. The reproducible archive is external to the new release directory. Cached source scripts remain inert. PDF generation uses only the report’s own TeX and the permitted TeX toolchain, with no shell escape. Deterministic build checks and all-page visual inspection concern artifact quality, not an independent proof of the infinite theorem.

39.A Immutable source and proof identities

Two separately frozen source stages are preserved, both dated 3 October 2026. The first is the all-but-main follow-on. The SHA256 of its original `MANIFEST.sha256` is

b769c58935f5951b1ad4b23df0daa5a9c31e2a04ea734d7bb9916249e51e9c38

The proof `ALL-BUT-MAIN-PROJECTION.md` has SHA256

d67a98bd95a1a11a8b1dc24c508c601f92c6758f9c4d92d9d4d13acb85936575

The inherited Report 37 snapshot, 44,433 bytes, has SHA256

57d6598d60389b2fc283f89f47b29ef5905af259ebe0ea69433a8838f9749001

Its mathematical compiler/bootstrap premises and prior reduction are inherited unchanged.

The later Jacobi addendum’s `MANIFEST.sha256` has SHA256

27203cb0baa757bbbf05d166c379dd9da8ceeeed084ec131cc9558749dae5cf4

Its proof `JACOBI-ADDENDUM.md` has SHA256

16311c258666483ed3342e3129966018e33a17ff0ec71aceca50540b1a75c98a

The Jacobi own checker has SHA256

6e33f7d2452d510bed14c8097428b746c86b3b06719e4e2d5c987920feb05482

Its independent audit has SHA256

90307e9af8741b60e4a88bcb7c128e96def9a4f04396839ff71b16bce8acf15a

The addendum preserves the earlier family evidence unchanged. Its strengthened miss and nonredundancy conclusions are later results, not retroactive edits to that evidence stage.

The upstream source commit for the three code/data context snapshots is

d5bd4a67b41b89688a079997574a94bcd855bb83

The packet's source manifest records byte counts, immutable URLs, SHA256 and Git blob SHA1. The three identities are:

1. `compiler.snapshot.md`, 12,918 bytes. SHA256

68edf3bc40238ccf47b023d7358e4be62664a2d78a60b6fae19de14b67208117

Git blob SHA1 90aa8895f38e2f9cb3a13c7ecff9f3f2a8d1fbf3

2. `complete77.snapshot.py`, 14,539 bytes. SHA256

9222a13dc180bd2361877674c4bd957d18f2d9aef827852cf5d320a3d7ac7d7a

Git blob SHA1 35c38f27d3aa6e2c9a1048a2d340321d9aef0aa9

3. `projection.snapshot.json`, 74,001 bytes. SHA256

ea982f9585eb4e16d756dfc33c73041e73e5e18a38ea37c2302d57c1fe104c92

Git blob SHA1 60f726116071b587601dc2d7e4e318ae39b021a1

The independent review authenticated cached source bytes against the pinned records. Its attempted separate network retrieval of the `complete77` file returned a cache miss; it does not claim a second independent network authentication of that file. The present release preserves that limitation rather than rewriting historical evidence. An additional release review checked the full proof, review, both checker sources and the two primary analytic estimates, and confirmed exact isolated normal/optimized replay without modifying original bytes, modes or modification times.

39.B A compact map of the proof dependencies

The theorem uses the following logical chain:

1. Genuine compiler and immutable mask recipe imply the loading bounds and even M_C
2. Fixed q, W, C, Z and the scale progression give a common odd index lattice and exact affine outer coordinates
3. The exact Binet center has a controlled analytic remainder and nonzero curvature of order $1/(r \log^2 r)$
4. A frequency-uniform second-derivative estimate plus interval discrepancy yields a count of order $N/\log N$ in the shrinking target
5. Interior margin and exact tail control give the integer Pell ratio
6. The positive $X/\log X$ deviation and the lattice formulas give every strict index, outer and input condition
7. Pell and auxiliary identities reconstruct every positive coordinate, leaving one possible raw residual
8. Literal SOS dependencies give the two distinct residual-value formulas for the broad family

The Jacobi addendum then proves the fixed-character identity, selects a progression with symbol $+1$, and repeats the shrinking-target argument after fixed thinning. This certifies nonzero residuals and comparison nonredundancy. The symbol -1 sector and full negative-zero question remain open.

References of Report 39

- [1] *Research Report 37: Exact Full Negative Zero Obstruction for Positive Diophantine Interfaces*, 3 October 2026. Sealed TeX snapshot in the evidence packet; exact digest in Appendix 39.A. Compiler/bootstrap premises and the full-zero predicate are inherited from this report. [write] Printed as Part V. The snapshot `evidence/context/Report37.snapshot.tex` is a byte copy of that manuscript (arrival commit 3051d1446), not shipped.
- [2] *Infinite genuine-compiler candidates with the main exponential congruence omitted*, frozen follow-on proof and independent review, 3 October 2026. Complete packet and exact scientific checkers accompany this report; proof digest in Appendix 39.A. [write] Shipped as `16-nonredundant-evidence-ALL-BUT-MAIN-PROJECTION.md`, with the review `16-nonredundant-evidence-independent-INDEPENDENT-REVIEW.md`.
- [3] Vladimir Reshetnikov, ProveIt repository, *complete75 half-binomial compiler*, immutable commit d5bd4a67b41b89688a079997574a94bcd855bb83, equations (2)–(3), lines 34–46. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.md#L34-L46 [write] The packet copy `compiler.snapshot.md` is not shipped; it equals the current `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.md`.
- [4] Vladimir Reshetnikov, ProveIt repository, *explore fixed raw universal 77*, same immutable commit, mask formula lines 73–76 and position construction lines 83–107. Preserved as inert source text. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/verification/explore_fixed_raw_universal_77.py#L73-L76 [write] The packet copy `complete77.snapshot.py` is not shipped; it equals the current `Computability/HilbertTenthProblem/Papers/verification/explore_fixed_raw_universal_77.py`.
- [5] Vladimir Reshetnikov, ProveIt repository, *complete74 nonlinear index projection scout*, same immutable commit. Saved source inventories and finalizer data, not executed. https://github.com/VladimirReshetnikov/ProveIt/blob/d5bd4a67b41b89688a079997574a94bcd855bb83/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete74_nonlinear_index_projection_scout.json [write] The packet copy `projection.snapshot.json` is not shipped; it equals the current file (74,001 bytes).
- [6] *A fixed-modulus Jacobi obstruction and infinitely many certified misses*, frozen addendum and independent audit, 3 October 2026. Exact proof, checker and manifest identities are in Appendix 39.A; the unchanged packet accompanies this report. [write] Shipped as `16-nonredundant-jacobi-JACOBI-ADDENDUM.md`, `16-nonredundant-jacobi-INDEPENDENT-AUDIT.md` and `code/16-nonredundant-jacobi-check_jacobi_addendum.py`.
- [7] NIST Digital Library of Mathematical Functions, Section 27.9, *Quadratic Characters*, equations 27.9.2–27.9.3 and the final paragraph extending the laws to Jacobi symbols. Read 3 October 2026. <https://dlmf.nist.gov/27.9>
- [8] Olivier Robert, *On van der Corput’s k -th derivative test for exponential sums*, Indagationes Mathematicae (2016), DOI 10.1016/j.indag.2015.11.009. Theorem 1, Section 3.1, printed page 5. <https://perso.univ-st-etienne.fr/rool6510/robert-2015-indag.pdf>

- [9] Anirban Mukhopadhyay, Olivier Ramaré and G. K. Viswanadham, *Discrepancy estimates for generalized polynomials*, Lemma 6, printed page 5, primary manuscript. https://ramare-olivi.er.github.io/Maths/Discrepancy-V_5.pdf

Part VII

Failure of first index deletion: a full positive counterfamily for the 81- and 82-operation candidates

[write] Source 17 of this report, manuscript 02 of batch 88: *Research Report 41*, subtitled “A full positive counterfamily for the 81 and 82 operation candidates”, author line “Mathematical research report”, dated 3 October 2026; archive `Failure_of_First_Index_Deletion_Package.zip` (arrival commit `c5612efa1`, placed by `992aaafb3`); pins `3f4a974a5` (the two parent sources and the deletion scout) and `93c34e817` (the scaled-obstruction note, context only). Its sections, statements and equations are numbered 41.*k* here.

Abstract of the source. Deleting the first-index quotient from the validated normalized 85-operation and ordinary 86-operation sources produces syntactically valid 81-operation and 82-operation circuits, but destroys their intended ordinary-input soundness. For every unchanged genuine admissible six-numeral compiler instance and every positive ordinary input, we construct all seventeen supplied witnesses as strictly positive integers and make every one of the six surviving literal factors equal to $+1$. The construction uses a factorial exponent, a controlled radix and Chinese remaindering to fit an exact dyadic Pell family into the actual shifted-mask packing, sheared transport, input norm and shared quotient constraints. Both retained strong auxiliary forms are completed with the actual positive auxiliary quotient. The omitted coordinate would have to be $h = (k - R - 1)/(XY)$, whose least nonnegative numerator remainder is $25u - 1 > 0$. Thus no integer restoration is possible. This is a symbolic existence proof, not a materialized enormous numerical tuple. The 81/82 counts and degrees 168/124 describe rejected candidates; the established 85-operation minimum among the validated constructions is unchanged.

[write] *Which polynomials.* The parents are the research tree’s normalized85 universal polynomial (`complete85_auxiliary_bezout_projection.md`, 85 operations, exact degree 175) and ordinary86 polynomial (`complete86_ordinary_auxiliary_projection.md`, 86 operations, exact degree 131), at the pin `3f4a974a5`; the deletion candidates are those of the tree’s `first_index_quotient_deletion_scout.md`. The “82-operation candidate” of this Part, ordinary82 ($82 = 45M + 37A$, 17 witnesses, degree 124), is *not* the square/product 82-operation candidate of Part VIII ($82 = 45M + 37A$, 18 witnesses, degree 185), although the operation split is the same; Table 3 keeps the names apart.

A stale record figure (dated note, batch 88). The abstract’s closing clause, that “the established 85-operation minimum among the validated constructions is unchanged”, and the same sentence in the shipped `17-index-deletion-evidence-README.md` (line 6, “established85-operation upper bound”), were correct at the pin `3f4a974a5` (3 October 2026, 13:55 PDT). Since commit `20aafb9a5` (15:08 PDT the same day) the research tree’s record is *84 operations* ($84 = 47M + 37A$, 18 positive witnesses, uniform exact degree 187), with $F_{84} = \Delta \cdot F_{85}$ (Section 0.4). The substance stands: Report 41 changes no record, and the 85- and 86-operation parents remain valid. The delivered text is kept.

What the research tree did with it. The programme’s review `review_incoming_arithmetic_reports_39_46.md` (commit `5d8889017`) read the full counterfamily and the source correspondence and finds “the central arithmetic is consistent”: the factorial totients, the CRT choices, $55 \mid R$, the dyadic resonance, the strict ratio, the shared split ρ, σ , the transport quotient and the auxiliary index $2cp$; the remainder $25u - 1$ gives full positive zeros at every input, including a rejecting program; and the false strict endpoint at index 2 in the earlier note (the erratum in Section 41.13) is not used by the family. It called the result “an actionable transfer to the current84 source”. The same evening the tree carried it out: `complete80_first_index_deletion_collapse.md` (with `.py` and `.json`, commit `5d8889017`) deletes the same four first-index rows

from the 84-operation polynomial, obtaining a candidate with $80 = 45M + 35A$ operations, 17 positive witnesses and exact degree 180, proves the all-ring identities $F_{80} = \Delta \cdot F_{81}$ and $F_{84} + \Delta = (F_{80} + \Delta)N_k$ with $N_k = k - hE - R$, and transfers Report 41's normalized81 family to it. Its independent review `review_complete80_first_index_deletion_collapse.md` passes, with fresh normal and optimized receipts. So the complete80 candidate, too, is refuted on every positive input of every authentic compiler slice, the empty-set compiler included; the intake review records that no deletion is incorporated into the established universal polynomial. The tree's result is a transfer of Theorem 41.1.1, with Report 41 credited as its source, not an independent duplicate. The intake review does not discuss the smooth-radix refinement of Section 41.11, which the source itself says received only a coordinating review.

Relation to the other Parts. This is the same kind of failure as Part IV ("Failure of positive index restoration"), for a different interface: there the restored index of the signed19 child is negative; here $R = p > 0$ and the deleted quotient h is nonintegral. The source's "Report 33" is Part IV, and its "Reports 37 and 39" are Parts V and VI. Lemma 41.3.1 is Lemma 33.3.3 of Report 33 (also re-proved in Parts VI and VIII); Lemma 41.3.2 is the projection recurrence of Lemma 33.3.2, written for $g_j = (e_A(j) - 2^j)/H$; the scaled subsystem family of Section 41.12 is of the same kind as the tree's `first_index_scaled_obstruction.md` (pin 93c34e817, family $p = t(t + 2)$), and the theorem does not use it.

Notation in this Part. No symbol is renamed; the masks are written $M_{F,\text{src}}$ (shifted) and $M_{F,0}$ (native) as in Part V, and y_{aux} is the auxiliary coordinate. Letters that differ from Part V: $u = R/55$ is a family parameter, *not* Part V's input index (here the input index is $I = \ell x + b$, which is Part V's u , and $W = 2^I$); $t = L_*$ is the binary radix exponent with $q = 2^t$ (Part V's t is the first-representative quotient); m is the odd part of t ; e is a residue exponent, not an input index; $L_0 = 2XY$ and $M_0 = 4XY^2$ are dyadic bases (Part VI's L_0 is a lattice modulus); $a_0 = Y(X + 1)$ is Part V's a ; T_{tr} is the supplied transport quotient (Part V's z_{quot} ; Part VIII writes t_{tr}); T is the supplied auxiliary quotient and $S = \Delta\psi_A(m_{\text{aux}})$ a computed auxiliary coefficient, unlike the supplied coefficient S of Part VIII's free-coefficient candidate; $\mathcal{S}$ is the literal strong coefficient; $N_1, \dots, N_6$ are the six retained factors (Part VIII names them $N_{\text{first}}, \dots$).

The source's files are shipped with the prefix `17-index-deletion-`.

41.1 Conclusion and exact scope

The first-index equation is essential in these two exact sources. The counterfamily below preserves their compiler constants and the ordinary input. It satisfies the complete retained system, including the paid packing and final product subtraction, with positive supplied integers throughout.

Theorem 41.1.1 (Full collapse of both deletion candidates). *Fix any genuine admissible six-numeral compiler instance of the pinned normalized85 or ordinary86 source, and any ordinary input $x \in \mathbb{Z}_{>0}$. Its corresponding normalized81 or ordinary82 first-index-deleted candidate has seventeen strictly positive integer witnesses for which all six retained factors are +1 and the final polynomial is zero. For this same retained tuple the unique possible restored coordinate*

$$h = \frac{k - R - 1}{E}, \quad E = XY,$$

is positive rational and nonintegral. More precisely, for a positive integer $u \equiv 1 \pmod{4}$ constructed from the exact paid packing,

$$R = p = 55u, \quad n = 40u, \quad E = 2^{88u-1}, \quad (k - R - 1) \bmod E = 25u - 1.$$

The conclusion has two distinct levels. First, the positive zero-set projection obtained by dropping h has no inverse at the constructed tuple. Second, the deletion candidates accept *every* positive input for every genuine compiler instance. Taking a correct compiler for the empty computably enumerable set therefore refutes ordinary-input soundness of the proposed universal representation. For a compiler whose language is all positive integers, the acceptance statement is of course not itself an input-language contradiction; the failed coordinate restoration still holds.

What is unchanged. The normalized85 and ordinary86 parent results remain valid. Their retained first-index factor excludes this family. There is no new universal operation bound, no assertion that 85 is an absolute mathematical lower bound over all possible circuits, and no claim that the rejected candidates preserve the intended language merely because their arrays are live and topologically well formed.

Relationship to other interfaces. Report 33 refuted a restored-index positivity claim for signed19. Reports 37 and 39 concern raw29 and positive21, possible *negative* restored R , and their main comparisons; the global full-negative-zero question there remains open. Here $R = p > 0$ and the obstruction is the *nonintegrality* of the deleted first-index quotient h in different, validated 85/86 parents. This report neither settles the raw29/positive21 sign question nor invalidates those parents.

Evidence boundary. The theorem is proved by exact mathematical definitions and inequalities. No astronomical full tuple is numerically materialized, and no saved arithmetic schedule or upstream program is executed. Finite checks verify components and source correspondence; they are supplementary evidence rather than the proof of the universal quantifiers.

41.2 The exact source and fixed compiler contract

41.2.1 Authentic numerals and positive ports

The six fixed source ports, in order, are

`Bm1`, `Kconstant`, `twice_cell_bits`, `inner_bits`, `MC`, `MF`.

Their mathematical values are $B - 1, K, 2d, b, M_C, M_{F,\text{src}}$, respectively. Fix these values once and for all from an actual admissible compiler export. The proof uses the inherited properties

$$\begin{aligned} B = 2^d \geq 16, \quad \ell = 2d, \quad K > 0, \quad b > 0 \text{ odd}, \\ M_C \equiv 2 \pmod{4}, \quad M_{F,\text{src}} = M_{F,0} + B - 1, \quad 0 < M_C, M_{F,0} < B - 1. \end{aligned} \quad (41.1)$$

The literal `MF` is the *shifted* mask; the bound $M_{F,\text{src}} < B - 1$ would be false. The actual recipe also gives $M_{F,0} \equiv 4 \pmod{8}$ and further compiler restrictions. They remain satisfied because none of the fixed numerals is altered. Conditions (41.1) are properties of authentic exports, not a proposed replacement definition allowing arbitrary toy masks.

The provenance is explicit. The pinned coupled-index note states $M_C \equiv 2 \pmod{4}$ and the mask contract. The half-binomial compiler note defines the modified M_C , native $M_{F,0}$, and shifted export. The fixed raw universal proof and half-binomial note choose b, L, d as powers of five with $d = bL$, hence b is odd. These immutable primary notes are included in the source packet and cited in Appendix 41.B.

The ordinary positive input is x . The seventeen supplied witness ports are exactly

$$\begin{aligned} \text{Jrep}, F, \alpha, \text{transport_quotient}, f, i, \text{auxiliary_quotient}, s, w, \\ \text{tau_root}, \eta, \zeta, \text{y_aux}, Z, \delta, \rho, \sigma. \end{aligned} \quad (41.2)$$

We denote the two supplied quotients by T_{tr} and T , and the root and auxiliary coordinate by τ and y_{aux} . Divisions, factorials, exponentials and Pell recurrences used to *define* these witnesses are mathematical existence constructions. They are not additional counted circuit instructions or extra supplied ports.

41.2.2 Shared literal expressions

For the witnesses just listed, the retained source computes

$$\begin{aligned} q &= (B-1)J+1, & X &= wq, & Y &= sq^3, & E &= XY, \\ k &= \eta + \zeta, & c &= kY + \eta, & a_0 &= Y(X+1), & A &= a_0 + 2, \\ \Delta &= A^2 - 1, & H &= 4a_0 + 3, & D &= X + a_0c + (\rho + \sigma)H. \end{aligned} \quad (41.3)$$

The mathematical Pell base A must be distinguished from the literal register **A**, whose value is Δ . With

$$\begin{aligned} I &= \ell x + b, & C &= q - F - Z - \alpha - \ell x, & W &= C - Z, \\ \kappa &= I + \delta\Delta, & \mu &= W + a_0\kappa + \rho H, \\ Q &= q^2 - 1, & M &= (M_C + qM_{F,\text{src}})J, & R &= (q^2 - Z - qF)Q + M, \\ V &= c(Tf - 1) - Rf^2, \end{aligned} \quad (41.4)$$

the six surviving factors are

$$\begin{aligned} N_1 &= \tau^2 - XY^2(XY^2 + 1)k^2, \\ N_2 &= D^2 - \Delta c^2, \\ N_3 &= \mu^2 - \Delta \kappa^2, \\ N_4 &= \mathcal{S}(V^2 - y_{\text{aux}}^2) + y_{\text{aux}}^2, \\ N_5 &= (K + w)C + q - F - T_{\text{tr}}(q - 1), \\ N_6^{\text{N}} &= f^2 - \Delta(ic^2)^2, & \mathcal{S}^{\text{N}} &= \Delta^2(ic^2)^2, \\ N_6^{\text{O}} &= 1 + (ic^2)^2 - \Delta(f^2 - 1), & \mathcal{S}^{\text{O}} &= \Delta(f^2 - 1). \end{aligned} \quad (41.5)$$

The normalized mode uses the N superscript; the ordinary mode uses O. These are not alternative conventions for one register: they are the two actual strong source forms.

The source's first product is exact: **first_root_base** is XY^2k and **first_next** is $XY^2k + k$, so their product is $XY^2(XY^2 + 1)k^2$. There is no omitted factor of two. The product cone is

$$N_1N_2 \longrightarrow N_1N_2N_3 \longrightarrow N_1N_2N_3N_4 \longrightarrow N_1N_2N_3N_4N_5 \longrightarrow \prod_{j=1}^6 N_j.$$

The final subtraction is paid, and the candidate polynomial is

$$\mathcal{F}_{\text{cand}} = \prod_{j=1}^6 N_j - 1. \quad (41.6)$$

The historical register name **seven_units** contains six surviving factors.

41.2.3 The literal deletion and its cost

The original first-index factor is $N_h = k - hE - R$. Its unique consumer chain is

$$h \longrightarrow \text{hpm1} \longrightarrow \text{index_difference} \longrightarrow \text{norm_index} \longrightarrow \text{norm_product}.$$

Only the four named registers are deleted, and **all_units** is rewired from **norm_product** to **norm_four**. Every other definition and final subtraction remains. Thus, as a polynomial identity for arbitrary ring values,

$$\mathcal{F}_{\text{parent}} + 1 = (\mathcal{F}_{\text{cand}} + 1)(k - hE - R). \quad (41.7)$$

At a candidate zero, restoration would require $k - hE - R = 1$, so the unique possible inverse is $(k - R - 1)/E$.

Candidate	Multiplications	Additions	Total	Degree
Normalized81	46	35	81	168
Ordinary82	45	37	82	124

Both have seventeen positive witness ports. Degree assigns one to x and each supplied witness and zero to each fixed numeral. These are exact syntactic ledgers of *rejected candidates*, not sound universal bounds.

41.3 Pell identities used in the construction

For an integer $A \geq 2$ define

$$\chi_A(j) + \psi_A(j)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^j \quad (j \geq 0). \quad (41.8)$$

Both sequences are integral and satisfy $u_{j+2} = 2Au_{j+1} - u_j$, with initial values $(1, A)$ for χ and $(0, 1)$ for ψ . The norm identity is

$$\chi_A(j)^2 - (A^2 - 1)\psi_A(j)^2 = 1.$$

For $j \geq 3$, monotonicity and the recurrence give

$$(2A - 1)^{j-1} \leq \psi_A(j) < (2A)^{j-1}. \quad (41.9)$$

Indeed consecutive positive ψ terms increase, so each new term is greater than $(2A - 1)$ times its predecessor and smaller than $2A$ times it. The initial step is $\psi_A(1) = 1$ and $\psi_A(2) = 2A$.

Two further identities will be needed with their signs intact.

Lemma 41.3.1 (Odd auxiliary polynomials). *There are integer polynomials $Q_v(T)$, $v \geq 0$, such that*

$$\chi_S(2v + 1) = SQ_v(S^2), \quad Q_v(0) = (-1)^v(2v + 1),$$

and

$$Q_v(1 - A^2) = (-1)^v\psi_A(2v + 1).$$

Proof. Take $Q_0 = 1$, $Q_1 = 4T - 3$ and $Q_{v+2} = (4T - 2)Q_{v+1} - Q_v$. The doubled-step Pell recurrence proves the first identity. Evaluation at zero gives the second identity by induction. At $T = 1 - A^2$ the recurrence coefficient is $2 - 4A^2$; alternating the sign of the odd-index ψ_A sequence gives exactly this recurrence and the same two initial values. This proves the third identity. $\square$

Lemma 41.3.2 (Positive shared projection quotients). *For $a_0 = A - 2$ and $H = 4a_0 + 3 = 4A - 5$, put*

$$z_j = \chi_A(j) - a_0\psi_A(j), \quad g_j = \frac{z_j - 2^j}{H}.$$

Then every g_j is an integer, $g_0 = g_1 = 0$, $g_2 = 1$, and

$$g_{j+2} = 2Ag_{j+1} - g_j + 2^j. \quad (41.10)$$

The sequence is strictly increasing for $j \geq 1$.

Proof. The sequence z_j has initial values 1, 2 and the Pell recurrence. Subtracting the sequence 2^j yields the forcing term $(4A - 5)2^j = H2^j$, proving the recurrence and integrality. If $g_{j+1} > g_j \geq 0$, then

$$g_{j+2} - g_{j+1} = (2A - 1)g_{j+1} - g_j + 2^j > 0.$$

The base $g_2 = 1 > g_1 = 0$ completes the induction. $\square$

41.4 A factorial radix and controlled residue exponent

Now fix the genuine compiler and input from Theorem 41.1.1. Define

$$I = \ell x + b, \quad W = 2^I.$$

Choose any integer

$$L_* \geq \max(1100d, 16, K, W, \ell x), \quad (41.11)$$

and set

$$\begin{aligned} t &= L_*!, & N &= t/d, & q &= 2^t = B^N, & J &= \frac{q-1}{B-1}, \\ t &= 2^\nu m \quad (m \text{ odd}), & Q &= q^2 - 1, & M &= (M_C + qM_{F,\text{src}})J. \end{aligned} \quad (41.12)$$

Since $1100d \leq L_*$, the integer $1100d$ divides $L_*!$. Thus N is a positive multiple of 1100, $t \geq 64$, $\nu \geq 4$, and $K, W, \ell x \leq t$. In particular J is a positive integer.

Lemma 41.4.1 (The odd part of a factorial divides the radix minus one). *The choices in (41.12) satisfy $q \equiv 1 \pmod{m}$.*

Proof. For every odd prime power r^ν dividing $L_*!$, both $r^{\nu-1}$ and $r-1$ divide $L_*!$. The first divisibility follows from the r -adic valuation; the second from $r \leq L_*$. These two integers are coprime, so

$$\varphi(r^\nu) = r^{\nu-1}(r-1) \mid t.$$

Euler's theorem therefore gives $2^t \equiv 1 \pmod{r^\nu}$. Chinese remaindering over the odd prime powers gives $q \equiv 1 \pmod{m}$. No assertion that $\varphi(t)$ divides t is needed. $\square$

Let e_0 be the residue of M in $[0, m-1]$. Exactly one of $e_0, e_0 + m, e_0 + 2m, e_0 + 3m$ is 3 modulo 4, because m is odd. Choose that integer e . Then

$$e \equiv M \pmod{m}, \quad e \equiv 3 \pmod{4}, \quad 3 \leq e < 4m \leq t/4. \quad (41.13)$$

Let Z be the least strictly positive representative of $e - M$ modulo 2^ν . Thus

$$1 \leq Z \leq 2^\nu \leq t.$$

Because $q \equiv 1 \pmod{4}$ and $J = 1 + B + \dots + B^{N-1} \equiv 1 \pmod{4}$, the genuine mask residue gives $M \equiv M_C \equiv 2 \pmod{4}$. Hence $Z \equiv 1 \pmod{4}$.

Define the outer witnesses and exact packed value by

$$\begin{aligned} C &= W + Z, & F &= (K + 2^e)C, \\ \alpha &= q - F - 2Z - W - \ell x, \\ R &= (q^2 - Z - qF)Q + M. \end{aligned} \quad (41.14)$$

Substitution into the literal definitions (41.4) gives the intended C and W exactly. The positivity and arithmetic of these witnesses are established next, rather than assumed.

41.5 Outer positivity and exact packed arithmetic

41.5.1 All required outer margins

Since $C \leq 2t$, $K \leq t$ and $e < t/4$,

$$F \leq 2t(t + 2^{t/4}), \quad 2Z + W + \ell x \leq 4t.$$

For every $t \geq 64$,

$$2t^2 + 2t 2^{t/4} + 4t < 2^t = q. \quad (41.15)$$

Here is an elementary uniform verification. The function $2^{t/8}/t$ increases for $t \geq 64$ and is at least one there, so $t \leq 2^{t/8}$. Consequently

$$2t^2 + 4t \leq 4t^2 \leq 2^{t/2}, \quad 2t 2^{t/4} \leq 2^{t/2}.$$

Their sum is at most $2^{t/2+1} < 2^t$. Thus $\alpha > 0$, and F, Z, C are positive integers. In particular $F + Z < q$.

Set $G = q^2 - Z - qF$. Positivity and $F + Z < q$ give

$$2q - 1 \leq G \leq q^2 - q - 1.$$

The shifted mask bounds in (41.1) imply

$$0 < M < (B - 1)(1 + 2q)J = (q - 1)(1 + 2q).$$

These exact inequalities yield

$$(2q - 1)(q^2 - 1) < R < q^4 - q^3. \quad (41.16)$$

For the upper bound, expand $(q^2 - q - 1)(q^2 - 1)$ and add the strict upper bound $2q^2 - q - 1$ for M ; all lower-order terms cancel. The lower bound follows from $G \geq 2q - 1$ and $M > 0$.

In particular $R > q > t$. Also $2^t > 5t + 5$ for $t \geq 64$: it holds at 64 and is preserved under $t \mapsto t + 1$ since $2(5t + 5) > 5(t + 1) + 5$. Hence

$$R > 5t + 5. \quad (41.17)$$

These are the actual pretyping packed-index bounds; they are not a substitute for the packed equation, which already holds by definition.

41.5.2 The three simultaneous congruences

Modulo m , Lemma 41.4.1 gives $Q \equiv 0$, hence $R \equiv M \equiv e$. Modulo 2^ν , the radix $q = 2^t$ is zero, $Q \equiv -1$, and therefore

$$R \equiv Z + M \equiv e \pmod{2^\nu}.$$

Since m is odd, Chinese remaindering proves

$$R \equiv e \pmod{t}. \quad (41.18)$$

Moreover $2^{20} \equiv 1 \pmod{55}$, so $B^{20} \equiv 1 \pmod{55}$. The geometric sum J has N terms, where $1100 = 55 \cdot 20$ divides N . Grouping into blocks of 20, each block has the same residue modulo 55 and the number of blocks is divisible by 55. Thus

$$55 \mid J, \quad q = B^N \equiv 1 \pmod{55}, \quad 55 \mid M, Q, R.$$

This argument does not divide a congruence by $B - 1$, which need not be invertible modulo 55. Finally, (41.18), $4 \mid t$ and (41.13) give $R \equiv 3 \pmod{4}$.

Set

$$u = R/55. \quad (41.19)$$

Then u is a strictly positive integer and $u \equiv 1 \pmod{4}$, because $55u \equiv 3 \pmod{4}$. This parameter u is distinct from the fixed ordinary input index I .

41.6 An exact dyadic first and main family

Define

$$\begin{aligned} p &= R = 55u, & n &= 40u, & 2n - p &= 25u, \\ X &= 2^p, & Y &= 2^{33u-1}, & w &= 2^{p-t}, & s &= 2^{33u-1-3t}, \\ E &= XY, & a_0 &= Y(X+1), & A &= a_0 + 2, & \Delta &= A^2 - 1, & H &= 4a_0 + 3, & P &= 2XY^2 + 1, \\ D &= \chi_A(p), & c &= \psi_A(p), & \tau &= \chi_P(n), & k &= 2\psi_P(n). \end{aligned} \quad (41.20)$$

By (41.17), $p > t$ and

$$33u - 1 = 3p/5 - 1 > 3t.$$

Thus w, s are strictly positive integers, and the genuine asymmetric scales $X = wq$ and $Y = sq^3$ hold. The first and main Pell norms are exactly $+1$: indeed $P^2 - 1 = 4XY^2(XY^2 + 1)$, matching $k = 2\psi_P(n)$ in N_1 .

41.6.1 A strict ratio proved without asymptotics

Put $L_0 = 2XY = 2^{88u}$ and $M_0 = 4XY^2 = 2^{121u}$. The exponent identity

$$\frac{L_0^{p-1}}{2M_0^{n-1}} = Y \quad (41.21)$$

is exact: its exponent is $88u(55u - 1) - 1 - 121u(40u - 1) = 33u - 1$. Also

$$\begin{aligned} 2A - 1 &= L_0 \left(1 + \frac{1}{X} + \frac{3}{2XY} \right), & 2A &= L_0 \left(1 + \frac{1}{X} + \frac{2}{XY} \right), \\ 2P &= M_0 \left(1 + \frac{1}{2XY^2} \right), & 2P - 1 &> M_0. \end{aligned}$$

The Pell bounds (41.9) and (41.21) give

$$\frac{c}{kY} \geq \frac{(1 + 1/X + 3/(2XY))^{p-1}}{(1 + 1/(2XY^2))^{n-1}} > 1.$$

The last inequality follows because the numerator base exceeds the denominator base and $p > n$.

For the upper bound let $\varepsilon = 1/X + 2/(XY) < 2/X$. Since $p \geq 55$, we have $2p/2^p < 1/2$ and therefore $(p-1)\varepsilon < 1/2$. For any positive integer j with $j\varepsilon < 1/2$, the binomial coefficient estimate gives

$$(1 + \varepsilon)^j \leq \frac{1}{1 - j\varepsilon} < 1 + 2j\varepsilon.$$

Combining with the upper Pell bound and (41.21),

$$0 < \frac{c}{k} - Y < \frac{4pY}{X} = 110u2^{-22u} < 1. \quad (41.22)$$

The last bound holds at $u = 1$ and decreases: the ratio of successive values is $(u + 1)/(u2^{22}) < 1$. Set

$$\eta = c - kY, \quad \zeta = k - \eta. \quad (41.23)$$

Both are strictly positive integers, and they reproduce the exact literal source values $k = \eta + \zeta$, $c = kY + \eta$.

41.7 The actual input norm and the shared split

The input is still the prescribed x , so $I = \ell x + b$ is odd and at least 3. Define

$$\begin{aligned} \kappa &= \psi_A(I), & \mu &= \chi_A(I), & \delta &= \frac{\kappa - I}{\Delta}, \\ \rho &= g_I, & \sigma &= g_p - g_I, \end{aligned} \quad (41.24)$$

using Lemma 41.3.2. The coefficient of $\sqrt{\Delta}$ in the binomial expansion of $(A + \sqrt{\Delta})^I$ gives

$$\psi_A(I) \equiv IA^{I-1} \equiv I \pmod{\Delta},$$

because I is odd and $A^2 \equiv 1 \pmod{\Delta}$. Since $\psi_A(I) > I$ for $I \geq 3$, the supplied δ is a positive integer.

We have $2^I = W \leq t$, hence $I < t < p$. Lemma 41.3.2 therefore gives $\rho > 0$ and $\sigma > 0$. It also gives exactly

$$\begin{aligned} D &= X + a_0c + (\rho + \sigma)H, \\ \kappa &= I + \delta\Delta, \\ \mu &= W + a_0\kappa + \rho H. \end{aligned} \quad (41.25)$$

Indeed $X = 2^p$, $W = 2^I$, and $\rho + \sigma = g_p$. The Pell identity now proves $N_3 = \mu^2 - \Delta\kappa^2 = 1$.

This is the input norm with the *same* ρ required by the main root, and with both supplied parts of its split positive. It is not an independent substitute exponent witness, and no rescaling or change of the input has occurred.

41.8 The literal sheared transport quotient

By (41.18), $p \equiv e \pmod{t}$. Since $q = 2^t$ and $w = 2^{p-t}$,

$$w \equiv 2^e \pmod{q-1}.$$

Moreover $p > 5t + 5$ and $e < t/4$ imply $p - t > e$, so $w > 2^e$. The supplied transport quotient

$$T_{\text{tr}} = 1 + \frac{C(w - 2^e)}{q - 1} \quad (41.26)$$

is therefore a strictly positive integer. Using $F = (K + 2^e)C$, direct substitution gives

$$(K + w)C + q - F - T_{\text{tr}}(q - 1) = 1.$$

Thus $N_5 = 1$. The coefficient is the actual *sheared* $K + w$, not an older $K + X$ coefficient. No signed quotient or raw-mask substitution is used.

41.9 Both strong modes and the positive auxiliary quotient

At the constructed $p = R \equiv 3 \pmod{4}$, define

$$\begin{aligned} m_{\text{aux}} &= 2cp, & f &= \chi_A(m_{\text{aux}}), & i_N &= \frac{\psi_A(m_{\text{aux}})}{c^2}, & S &= \Delta\psi_A(m_{\text{aux}}), \\ y_{\text{aux}} &= \psi_S(p), & V &= \frac{\chi_S(p)}{S}, \\ o &= \frac{V+c}{f}, & j &= \frac{V+p}{c}, & T &= \frac{o+pf}{c}. \end{aligned} \quad (41.27)$$

Every quotient in this display is a positive integer, as we now prove.

41.9.1 Divisibility and the exact signs

Expand $(\chi_A(p) + c\sqrt{\Delta})^{2c}$. Its $\sqrt{\Delta}$ coefficient is $\psi_A(2cp)$. The term with one $c\sqrt{\Delta}$ factor is $2c^2\chi_A(p)^{2c-1}$; every higher odd term contains at least c^3 . Therefore $c^2 \mid \psi_A(m_{\text{aux}})$ and $i_N > 0$ is integral. The Pell norm gives

$$S^2 = \Delta(f^2 - 1). \quad (41.28)$$

In particular $S > c^2$ and $c \mid S$.

Set $v = (p-1)/2$, which is odd. Lemma 41.3.1 gives integral $V = Q_v(S^2)$. Modulo f , equation (41.28) says $S^2 \equiv 1 - A^2$. Modulo c , it says $S^2 \equiv 0$. Hence, retaining the minus signs,

$$V \equiv -c \pmod{f}, \quad V \equiv -p \pmod{c}. \quad (41.29)$$

Thus o and j are integers. Since $p \geq 3$, Pell growth gives

$$V \geq \frac{\chi_S(3)}{S} = 4S^2 - 3 > c > p,$$

so $o, j > 0$.

We also have $f^2 \equiv 1 \pmod{c}$. From $of = V + c$ and $V + p = jc$,

$$of + p = c(j + 1).$$

Multiplying this congruence modulo c by f gives $o + pf \equiv 0 \pmod{c}$. Thus the actual supplied T in (41.27) is a strictly positive integer. As $R = p$,

$$c(Tf - 1) - Rf^2 = of - c = V. \quad (41.30)$$

The Pell identity at base S is exactly

$$S^2(V^2 - y_{\text{aux}}^2) + y_{\text{aux}}^2 = 1. \quad (41.31)$$

41.9.2 Normalized source

Supply $i = i_N$. Then

$$f^2 - \Delta(ic^2)^2 = 1, \quad \Delta^2(ic^2)^2 = S^2.$$

The literal normalized strong factor is $N_6^N = 1$, and its literal auxiliary coefficient is $\mathcal{S}^N = S^2$. Equations (41.30)–(41.31) therefore give $N_4 = 1$.

41.9.3 Ordinary source

Instead supply $i = \Delta i_N$. Now

$$(ic^2)^2 = \Delta(f^2 - 1) = S^2,$$

so the ordinary strong factor $1 + (ic^2)^2 - \Delta(f^2 - 1)$ is $+1$, and the literal auxiliary coefficient $\mathcal{S}^O$ is again S^2 . The same positive f, T, V, y_{aux} satisfy the auxiliary equation.

The quantities o, j are mathematical intermediates proving positive integrality of T . They are *not* extra witness ports and are not supplied instead of T . Neither mode uses the missing h .

41.10 The full zero and the failed restoration

We have constructed all seventeen positive integers in the precise order

$$J, F, \alpha, T_{\text{tr}}, f, i, T, s, w, \tau, \eta, \zeta, y_{\text{aux}}, Z, \delta, \rho, \sigma.$$

Their literal register values match (41.3)–(41.4). The first/main norms, actual input norm, auxiliary norm, sheared transport factor and the appropriate strong factor all equal $+1$. The fully paid finalizer (41.6) is therefore exactly zero.

Since $P = 2XY^2 + 1 \equiv 1 \pmod{E}$, reduction of the ψ_P recurrence gives $\psi_P(n) \equiv n \pmod{E}$, hence $k \equiv 2n \pmod{E}$. Therefore

$$k - R - 1 \equiv 2n - p - 1 = 25u - 1 \pmod{E}.$$

Here $E = 2^{88u-1} > 25u$ and $25u - 1 > 0$, so this is the least nonnegative remainder. Also

$$k = 2\psi_P(40u) \geq 80u > 55u + 1 = R + 1.$$

Consequently $(k - R - 1)/E$ is positive rational but nonintegral. Equation (41.7) proves that no integer h can restore a parent zero at this retained tuple. This completes the proof of Theorem 41.1.1.

41.11 An optional smaller radix for the genuine recipe

This section is a later refinement of the factorial proof, preserved in its own frozen source packet. It received a separate coordinating review and author checks; it is not part of the earlier independent review. It changes neither the candidate arrays nor the conclusion. The original factorial construction remains intact in Sections 41.4–41.10; the new argument replaces only the choice of t and its outer positivity estimate.

The genuine recipe has the stronger property $d = 5^r$, with $r \geq 1$. Set

$$h_K = \lceil \log_2(K + 1) \rceil, \quad H_0 = \max(I, h_K).$$

Choose the least integer $\lambda \geq 0$ such that

$$t = 4400d2^\lambda \geq 4H_0. \tag{41.32}$$

The doubling count λ is only a mathematical selection parameter, not another supplied coordinate. Then

$$t = 2^{\lambda+4}5^{r+2}11, \quad \nu = \lambda + 4, \quad m = 275d, \quad N = t/d = 44002^\lambda.$$

Both Euler periods $\varphi(5^{r+2}) = 4 \cdot 5^{r+1}$ and $\varphi(11) = 10$ divide t , so $2^t \equiv 1 \pmod{m}$. Also $1100 \mid N$. Thus all radix and CRT arguments remain valid with

$$3 \leq e < 4m = 1100d \leq t/4.$$

Keep $q = 2^t$, J , Q , M , Z , C , F , α and the paid R defined exactly as before. The same two reductions give $R \equiv e \pmod{t}$, while the 1100-block argument gives $55 \mid R$ and the actual mask residues give $R \equiv 3 \pmod{4}$.

The former bounds $K, W \leq t$ are no longer asserted. Instead, (41.32) gives

$$I \leq t/4, \quad K < 2^{h_K} \leq 2^{t/4}, \quad W = 2^I \leq 2^{t/4}, \quad Z \leq 2^\nu \leq t.$$

Since $t \geq 22000 > 64$, we have $t \leq 2^{t/4}$ and $3t \leq 2^{t/4}$. Therefore

$$F = (K + 2^e)(W + Z) < 2^{t/2+2}, \\ 2Z + W + \ell x \leq 3t + 2^{t/4} \leq 2^{t/4+1}.$$

Both exponents $t/2 + 2$ and $t/4 + 1$ are smaller than $t - 1$. Their sum is less than 2^t , proving $\alpha > 0$. The same shifted-mask estimates yield the original packed bounds and $p = R > 5t + 5$. We also have $I \leq t/4 < t < p$, so the shared input split remains positive. Every subsequent dyadic, ratio, transport, auxiliary and restoration argument is unchanged.

Corollary 41.11.1 (The radix can be singly exponential at fixed compiler). *For each genuine compiler and positive input, Theorem 41.1.1 also holds with the smooth exponent (41.32). The least choice of λ satisfies*

$$t \leq \max(4400d, 8H_0).$$

Consequently the chosen radix $q = 2^t$ is at most singly exponential in x when the compiler is fixed.

Proof. If $\lambda = 0$, then $t = 4400d$. If $\lambda > 0$, minimality gives $t/2 < 4H_0$, so $t < 8H_0$. At fixed compiler, $I = 2dx + b$ is linear in x and h_K is constant. The already verified replacements prove all seventeen witness conditions and the same nonzero remainder $25u - 1$. $\square$

This is a bound on the radix selection only, not a small bound on the downstream Pell witnesses. It is not an operation-count improvement. The addendum has its own exact modular/CRT and finite-margin checks and separate review attribution; it does not retroactively change the independently reviewed factorial draft.

41.12 Earlier reductions and why the completion matters

The chronological stages are worth separating because a valid subsystem counterexample alone would not establish the theorem.

41.12.1 The noncircular bootstrap

The earlier reduction, preserved in `PROOF.md` with a separate audit, proves that *every* full positive zero of either deletion candidate satisfies

$$\begin{aligned} p = R, \quad R &\equiv 3 \pmod{4}, \quad R/2 < n < R, \\ d_{\text{def}} = 2n - R \text{ odd}, \quad 1 &\leq d_{\text{def}} \leq R - 2 < E, \\ (k - R - 1) \bmod E = d_{\text{def}} - 1, \quad \frac{k - R - 1}{E} &> 0. \end{aligned} \tag{41.33}$$

Its missing-coordinate question was therefore exactly whether the retained equations force $d_{\text{def}} = 1$. It did not assume the deleted equation or import a parent zero. Our full construction realizes $d_{\text{def}} = 25u > 1$ and settles that question negatively.

Here is the logical structure of that independent reduction. A product equal to one makes all six factors units. Congruences modulo 4 exclude negative units in the main, input, strong and auxiliary norms. A minimal-solution descent excludes the negative first norm. Thus transport is also +1, giving the genuine outer bounds before any dyadic decoding. Positive Pell classification and the strict ratio imply $p/2 < n < p$.

The projection $X \equiv 2^p \pmod{H}$, together with the actual scales, excludes $p \leq 5$ by elementary inequalities. Consequently $p \geq 6$ and

$$c = \psi_A(p) > A\Delta^2, \quad c > 2p, \quad c > 2R.$$

These are the noncircular bounds needed to apply the normalized rank argument or the ordinary relaxed-rank lemma. They yield a strong Pell index m with $p \mid m$, $c \mid m$ and $m > 2p$. The retained positive T then excludes the negative auxiliary branch and makes its derived o, j positive.

Odd auxiliary polynomials and the unsquared minus congruences recover $p = R$ and $p \equiv 3 \pmod{4}$. Finally $P \equiv 1 \pmod{E}$ gives (41.33). The full derivation, including the ordinary relaxed-rank proof and its smaller-fundamental-unit case, is retained in the accompanying source packet. The full counterfamily proof in Sections 41.4–41.10 does not depend on this classification theorem: it constructs its coordinates directly.

41.12.2 A genuinely scaled subsystem family

An intermediate family takes $r \geq 5$ with $r \equiv 1 \pmod{4}$ and

$$p = r(2r + 1), \quad n = (3r^2 + r)/2, \quad v = \frac{(2r + 1)(r + 1)}{2} - 1, \quad X = 2^p, \quad Y = 2^v.$$

It has defect $2n - p = r^2$ and exact dyadic balance

$$\frac{(2XY)^{p-1}}{2(4XY^2)^{n-1}} = Y.$$

The same Pell estimates yield $Y < c/k < Y + 1$ because

$$4p2^{v-p} < 1, \quad p - v = r^2 - (r - 1)/2.$$

The projection quotient is integral and positive, and the canonical auxiliary block can be completed. For a dyadic $q \geq 16$, choosing $r = q^2/2 + 1$ also gives the true scales $X = wq$, $Y = sq^3$ and the scalar bounds

$$(2q - 1)(q^2 - 1) < p < q^4 - q^3.$$

But scalar bounds are not the actual fixed-mask packed equation. That stage did not complete the same input, transport and packing simultaneously. The smaller materialized fixture $r = 5$, $p = 55$, $n = 40$, $X = 2^{55}$, $Y = 2^{32}$, $q = 16$ has remainder 24, but fails the scalar lower bound on R . It is only a subsystem check.

The upstream scaled-obstruction note at commit 93c34e817c7bf726becd47326d22ec5e511a4251 likewise leaves the input and auxiliary constraints unfinished. Its matching small fixture is prior subsystem context. The advance proved here is the factorial–CRT completion at the *actual paid* R , for every unchanged genuine compiler and input, with the shared ρ and both auxiliary source modes.

41.13 Reproducibility and the scope of computation

[write] *Shipped names.* The packet's files are shipped with the prefix `17-index-deletion-` (README): the frozen reduction note `PROOF.md` is `17-index-deletion-evidence-PROOF.md`, kept byte for byte with the erratum below; the final counterfamily proof is `17-index-deletion-evidence-FULL_COUNTERFAMILY.md`, the smooth-radix proof `17-index-deletion-smooth-SMOOTH_RADIX.md`, and the independently reviewed drafts `17-index-deletion-evidence-audit_bootstrap-original_sources-FULL_COUNTERFAMILY.reviewed.md` and `...-SCALED_FAMILY.reviewed.md`. Their SHA-256 digests were measured equal to the ones quoted below. The two reviewer snapshot ledgers `reviewed_snapshot.sha256` and `counterfamily_reviewed_snapshot.sha256` are not shipped; besides the files above they record the reviewer's snapshot of the preliminary `PROOF.md`, SHA-256 `8d61c1eb4b367e3d...2b5a87`, which differs from the shipped note and is historical. At placement the five checkers, run directly on a copy, passed in normal and optimized Python with receipts equal to the frozen ones after CRLF line ends were normalized; the release verifier refuses to run on NTFS (file modes) and compares stdout bytes that Windows writes with CRLF. The programme executed none of them.

The accompanying release contains the editable LaTeX source, deterministic PDF, the frozen factorial packet and separate smooth-radix addendum, a release manifest and hash inventory, and explicit replay/build/archive tools. The manifest authenticates the complete allowed regular-file inventory before a copied checker is invoked. The release verifier checks source pins and typed expected receipts before running newly authored active checkers in normal and optimized Python. It rejects extra or changed files rather than silently blessing them.

The active scientific checkers have canonical JSON stdout. Their `-expect` interface compares exact bytes; `-output` is optional and accepts only a fresh external path. The copied scientific packet and all predecessor sources remain immutable. A replay from a safely extracted and moved directory demonstrates independence from the original absolute workspace paths. Hostile archive and tamper tests are separate from mathematical evidence.

What the finite checks establish. The reduction checker performs static literal deletion/interface checks and exact mathematical recurrence, projection, rank, step-down, small-index, parity and scaled-subsystem checks. The full-construction checker tests factorial odd parts, literal CRT arithmetic, outer margins, shared input splits and both auxiliary forms. The separately authored smooth-radix checker adds 625 exact period cases, 27 height-selection and CRT cases, and 1009 finite margin checks. Its necessary-residue fixtures are not complete compiler instances. The independent auditor uses separately authored mathematical expressions and tests the full proof's vulnerable arithmetic and positivity steps. Normal and optimized runs must match their receipts exactly.

What they do not establish. The outer arithmetic fixtures use explicitly mock necessary-residue ports. They are not represented as complete compiler instances. The modest Pell fixtures do not simultaneously instantiate the entire factorial-radix family. No finite enumeration can replace the uniform theorem, and no recorded source arithmetic array is interpreted by a checker. Historical upstream check totals embedded in inert source documents describe those documents only; they are not new executions in this release.

A narrow erratum in the preserved preliminary note. The frozen reduction note `PROOF.md` states the strict bound $\psi_A(j) < (2A)^{j-1}$ for $j \geq 2$. At $j = 2$ equality holds. The correct strict range is $j \geq 3$, with the non-strict bound valid for $j \geq 1$. That note uses the estimate only at indices 4 and 5; the full counterfamily uses $p \geq 55$ and $n \geq 40$. Thus neither conclusion changes. Equation (41.9) states the correct range, and the frozen source bytes are preserved with this explicit erratum.

Independent mathematical review. The independent full-counterfamily audit reports PASS for the complete construction, including factorial totient divisibility, exact CRT, genuine shifted masks, packing margins, input/shared-split positivity, literal transport, both strong modes, positive T , and the final nonzero remainder. The independently reviewed draft had SHA256 789e0d876393c4cd74076c41b9d030d3e192939346505217db13cfd599ed48a0; those exact bytes, original audit and original receipts are preserved. The final text corrects the peripheral base-check phrase from $p = 4$ to $p = 5$ (the constructed range is $p \geq 55$) and spells out the already-used final inequalities $E = 2^{88u-1} > 25u$ and $k > R + 1$. The coordinating review checked these final bytes. Later active copies of the independent checkers change portability and output handling only, with before/after hashes and separately attributed author replay receipts. The independent reviewer did not review the subsequent packaging. The final proof SHA256 is

dc886e8991e32c331b9135b4ea6b3f73733656be3df6c2aa01576e1732fb83e1

The separately frozen smooth-radix proof SHA256 is

259ec165067555228eb97fd6ca37180feaba6861df4ce1cce94d73917b12d8ca

The manifests and replays establish byte identity; the mathematical arguments and their explicitly attributed reviews establish the theorem.

41.A Literal register correspondence

The following crosswalk fixes the notation used throughout the proof. In particular, literal A means Δ and literal R12 means a_0 .

Literal register	Mathematical value
repunit	$q - 1 = (B - 1)J$
q, Lbig, n2	q, q^2, q^3
wn2, sn2, UM	$X = wq, Y = sq^3, E = XY$
R10b, R10a	$k = \eta + \zeta, c = kY + \eta$
R12, a4m5, A	$a_0 = Y(X + 1), H = 4a_0 + 3, \Delta = A^2 - 1$
gamma_sum, R14	$\rho + \sigma, D = X + a_0c + (\rho + \sigma)H$
marked_rhs, W	$C = W + Z, W = 2^I$
odd_index, index_rhs	$I = 2dx + b, \kappa = I + \delta\Delta$
exponent_rhs	$\mu = W + a_0\kappa + \rho H$
gap, r_lhs	$q^2 - Z - qF, R = (q^2 - Z - qF)(q^2 - 1) + M$
kinner, local_rhs	$K + w, T_{\text{tr}}(q - 1)$
L16, aux_u_rhs	$f^2, V = c(Tf - 1) - Rf^2$
R16 in normalized81	$\Delta^2(ic^2)^2 = S^2$
R16 in ordinary82	$\Delta(f^2 - 1) = S^2$

The ordered product registers are `norm_pair`, `norm_triple`, `norm_four`, `all_units`, `seven_units`, and `polynomial`. They contain respectively two, three, four, five, six factors, and the six-factor product minus one.

41.B Source identity and primary references

The immutable source commit for the two parent arrays and deletion scout is

3f4a974a5ddf12c46302fc5d2edafc3730277923

The parent normalized85 JSON SHA256 is

e7ddc113f96cde37efc9d1973daffa5e4cef221db2c1d1c6ad1cf1772cd59edc

The parent ordinary86 JSON SHA256 is

f46dd919a038b6dd646489e64585331de30b8c91a380cf9611afbf288122a9a6

The deletion scout JSON SHA256 is

0004189e84fdc34e1ae7cbffa30a0e904ec6a0f15be18bcabebe242a303163f8

The provenance inventory records original paths, commit identifiers, Git blob identifiers and exact URLs. The enclosing release manifest adds SHA256 byte authentication for every included file.

References of Report 41

- [1] Vladimir Reshetnikov, *Complete85 auxiliary Bezout projection*, pinned primary proof at commit 3f4a974a5ddf12c46302fc5d2edafc3730277923. https://github.com/VladimirReshetnikov/ProveIt/blob/3f4a974a5ddf12c46302fc5d2edafc3730277923/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete85_auxiliary_bezout_projection.md.
- [2] Vladimir Reshetnikov, *Complete86 ordinary auxiliary projection*, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/3f4a974a5ddf12c46302fc5d2edafc3730277923/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete86_ordinary_auxiliary_projection.md.
- [3] Vladimir Reshetnikov, *First index quotient deletion scout*, same pinned commit. https://github.com/VladimirReshetnikov/ProveIt/blob/3f4a974a5ddf12c46302fc5d2edafc3730277923/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/first_index_quotient_deletion_scout.md.
- [4] Vladimir Reshetnikov, *Complete75 coupled index linear88*, same pinned commit, compiler contract and mask residues. https://github.com/VladimirReshetnikov/ProveIt/blob/3f4a974a5ddf12c46302fc5d2edafc3730277923/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_coupled_index_linear88.md.
- [5] Vladimir Reshetnikov, *Complete75 half binomial compiler*, same pinned commit, shifted mask export and powers-of-five parameters. https://github.com/VladimirReshetnikov/ProveIt/blob/3f4a974a5ddf12c46302fc5d2edafc3730277923/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/complete75_half_binomial_compiler.md.
- [6] Vladimir Reshetnikov, *Fixed raw universal 76 proof*, same pinned commit, odd b and $d = bL$. https://github.com/VladimirReshetnikov/ProveIt/blob/3f4a974a5ddf12c46302fc5d2edafc3730277923/Computability/HilbertTenthProblem/Papers/1980/FIXED_RAW_UNIVERSAL_76_PROOF.md.
- [7] Vladimir Reshetnikov, *First index scaled obstruction*, subsequent pinned upstream note at commit 93c34e817c7bf726becd47326d22ec5e511a4251; subsystem context only. https://github.com/VladimirReshetnikov/ProveIt/blob/93c34e817c7bf726becd47326d22ec5e511a4251/Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/first_index_scaled_obstruction.md.

[write] The seven research-tree files cited in these references are unchanged since their pins; the archive carries byte copies of all seven, of the JSON receipts of the two parents and of the scout, and of one further research-tree note, none of which is shipped (README). The parents are the 85/175 and 86/131 constructions of Section 0.4.

Part VIII

Candidates below the 84-operation polynomial: the free-coefficient 83 and square/product 82 charts

[write] This Part prints three manuscripts about two candidate reductions of the research tree’s 84-operation universal polynomial ($84 = 47M + 37A$, 18 positive witnesses, exact degree 187; commit 20aafb9a5, Section 0.4). Report 43 (Sections 43.k) studies the *free-coefficient 83* candidate of commit 0d9d1e017 ($83 = 46M + 37A$, 18 witnesses, source degree 111), which supplies the auxiliary coefficient S instead of paying $S = i\Delta c^2$. Report 45 (Sections 45.k) refutes the *square/product 82* candidate of commit 6ce2dcaaf ($82 = 45M + 37A$, 18 witnesses, exact degree 185), which supplies the products f^2 and Tf . Report 46 (Sections 46.k) measures the height of Report 45’s certificates and proves that none of Report 45’s outer tuples extends to the free-coefficient interface of Report 43. Report 46 imports both of the others and cannot be read without them, so the three form one Part, as Reports 33 and 34 form Part IV.

[write] *Status of both candidates at this write.* Both are refuted on their inherited compiler recipe, so neither lowers the record of 84 operations:

- The free-coefficient 83 candidate is refuted by the research tree’s `free_coefficient83_prime_outer_collapse.md` (commit 8cf6239b6, 3 October 2026, 18:33 PDT, with `.py`, `.json` and the independent review `review_free_coefficient83_prime_outer_collapse_math.md`): on every valid fixed-program slice it has infinitely many full positive zeros at every positive ordinary input, using prime main indices $p \equiv 1 \pmod{4}$. Report 43 is pinned to 0d9d1e017 and predates that commit; its statements that the ordinary-input language is open are superseded (dated note below). Report 46 cites the collapse (Section 46.1) and ships its independent audit.
- The square/product 82 candidate is refuted twice, by different constructions: by the tree’s `complete82_all_input_outer_collapse.md` (commit dc03992d9, 18:26 PDT, with `.py`, `.json` and two independent reviews that pass), and by Report 45, whose audit observation is dated 00:44 UTC on 4 October (17:44 PDT on 3 October), about forty minutes earlier, and which does not cite the tree’s note. Neither makes a priority claim; Report 45’s theorem is printed as an independent second route (note at Theorem 45.1.1).

“83” and “82” each name more than one polynomial in the programme; Table 3 lists them. In particular the *independent-gamma* 83-operation candidate ($83 = 47M + 36A$, 18 witnesses, degree 187) is a different polynomial, whose ordinary-input language is unresolved; none of these three reports treats it, and the programme’s review notes that Report 43’s lemmas “do not apply automatically” to it.

The programme’s review. `review_incoming_arithmetic_reports_39_46.md` (commit 5d8889017) reviewed all three archives within a stated scope; it executed no archived program. It did *not* review in full: Report 43’s inner-family addenda (Sections 43.5–43.8 and Appendix 43.A) and its other squarefree addenda (Lemmas 43.4.2 and 43.4.3 and Theorem 43.4.4); Report 45’s archived finite receipts, which it did not replay; and Report 46’s analytic expansions (Sections 46.7 and 46.8) and their claimed precision. Those parts rest on the sources’ own audits, which are shipped. Its findings are printed in the notes at the start of each source below.

Notation in this Part. The three sources share the macros for the Pell sequences $\chi_A(r)$, $\psi_A(r)$ and for the named factors, but not all their letters. Inside the sections of one source a letter has that source’s meaning; Table 13 lists the letters that change between the three sources or differ from Part V. One rendering was changed: Report 46 writes the free-coefficient auxiliary and strong factors N_a and N_s , and Reports 43 and 45 write the same factors N_{aux} and N_{strong} ; all three are printed as N_{aux} and N_{strong} . No other symbol was

renamed. The three sources write the shifted and native masks M_F and M_F^{native} , M_F^0 or $M_{F,0}$; these are Part V's $M_{F,\text{src}}$ and $M_{F,0}$.

Table 13: [write] Letters with different meanings in the three sources of Part VIII.

Letter	Report 43 / Report 45 / Report 46	Tempting false reading
S	43: the <i>supplied</i> free coefficient. 45: the computed coefficient $S = i\Delta c^2$. 46: $S = i\Delta c^2$ for square/product 82, and the supplied free coefficient in Theorem 46.1.3.	In 45 and in most of 46, S is not a witness.
y	43: the supplied auxiliary coordinate. 45: the exponent in $Y = 2^y$; the auxiliary coordinate is y_{aux} . 46: the exponent (yexp); y_{aux} is the square/product auxiliary witness and y_* the free-coefficient one.	43's y is 45's and 46's y_{aux} .
H, H_0	43: $H = 4a + 3$, and H_0 the parameter of Lemma 43.3.2. 45: $H = 4a_0 + 3$. 46: $H_0 = 4a_0 + 3$, H the parameter of Lemma 46.10.1, and $\mathcal{H}$ the height.	46's H_0 is 43's H , and 46's H is 43's H_0 .
a, a_0	43: $a = Y(X + 1)$. 45, 46: $a_0 = Y(X + 1)$ and a the exponent in $d = 5^a$; in Lemma 46.11.1, (a, b) is a fundamental unit and e its exponent.	43's a is 45's a_0 .
t	43: the transport quotient t_{tr} ; the radix exponent $t = dN$ (Section 43.8); an odd Pell index in Lemma 43.4.3. 45, 46: the radix exponent $t = 5^r$, with t_{tr} the transport quotient.	—
N	43: the second family parameter, $q = B^N$; N is also the norm value in Lemma 43.3.2. 45: $N = 5^{r-a}$, $q = B^N$. 46: $N = (R - 1)/T_0$, the bit length of the height minus one; N_* the norm value.	46's N is not a radix length.
C	43, 45: the marker $C = q - F - Z - \alpha - \ell x$. 46: the exponent $C = (p - 1)L$; the marker is C_{pack} .	—
L	43: an lcm in Appendix 43.A. 45: $L = K + 2^e$. 46: $L = p + y + 1$ (and L_1, L_0, M_0 dyadic exponents and bases).	None is Part V's compiler exponent.
u, v	43: $u = 17^{2j}$ (or 11^{12j+1}). 45, 46: $u = (R + \varepsilon)/6$ in branch A and $v = (R + 1)/28$ in branch B.	None is Part V's input index u ; the input index is $I = \ell x + b$.
T	43: the supplied auxiliary T ; the variable of $Q_v(Z)$ is Z . 45: the polynomial variable in $Q_j(T)$, and $T = t/5$ in branch B. 46: the polynomial variable, and $T_0 = 2pL - 1$.	45's and 46's square/product charts supply U_{aux} instead of T .
h	43, 45, 46: the retained first-index witness; also 45's choice index h_0 and 46's odd-polynomial index (Lemma 46.9.2) and Lagrange function $h(z)$.	—
Q, Q_s	$Q = q^2 - 1$ throughout; Q_v, Q_j, Q_h odd quotient polynomials; 45, 46: $Q_s = F_{\text{aux}} - \Delta i^2 c^4$.	—
P, P_5	$P = 2XY^2 + 1$ throughout; 45, 46: P_5 the product of the five retained factors and P_{82} the polynomial.	—
ϵ, ε	45: $\varepsilon = \pm 1$, the common sign of the index and transport factors; 46: the same sign written ϵ , and $\varepsilon_h, \varepsilon_0$ error terms.	—

Report 43. Structural obstructions and nonnative inner families for the free coefficient 83 candidate

[write] Source 18 of this report, manuscript 05 of batch 88: *Research Report 43*, author line “Research Report 43”, dated 4 October 2026; archive `Structural_Obstructions_for_the_Free_Coefficient`

`_83_Candidate_Package.zip` (arrival commit `c5612efa1`, placed by `992aaafb3`); pins `0d9d1e017` (the candidate) and `53c34b012` (the credited native-alias precursor). Its sections, statements and equations are numbered $43.k$ here.

Abstract of the source. The free-coefficient 83-operation arithmetic candidate is obtained from the verified scaled 84-operation construction by supplying its auxiliary coefficient directly. This report proves two complementary results. First, every full strictly positive candidate zero on a genuine fixed-compiler slice has auxiliary factor $N_{\text{aux}} = z^2 > 0$, where $z^2 \mid \Delta$ and $z = \gcd(|V|, y)$. On the additional squarefree-discriminant slice, the main, input and auxiliary factors are all 1. Second, an explicit infinite power-of-17 family satisfies the intended first, main, index, auxiliary and strong equations with main Pell rank $p \neq R$, and admits compatible positive input loading with the shared projection whenever $p > I$. Its completion to the genuine compiler is equivalent to a specified packing congruence, transport congruence and strict positivity test in two integer parameters. No solution of that interface at a rejected ordinary input is supplied. Ordinary-input language soundness remains **open**; the established universal construction remains $84 = 47M + 37A$, with 18 positive witnesses and exact degree 187. The results use classical Pell arithmetic. No priority claim or new universal-operation bound is made.

[write] *Superseded open status (dated note, batch 88).* The abstract’s “Ordinary-input language soundness remains **open**” and question 1 of Section 43.9 were open at the pin `0d9d1e017`. They are answered: the research tree’s `free_coefficient83_prime_outer_collapse.md` (commit `8cf6239b6`, 3 October 2026, 18:33 PDT, which is 4 October, 01:33 UTC, the date Report 43 carries) proves that on every valid fixed-program slice the candidate has infinitely many full positive zeros at every positive ordinary input, so the specialization to a compiler of the empty set gives a false acceptance; its independent review passes. The answer does not come through Report 43’s interface: the tree’s family uses prime main indices $p \equiv 1 \pmod{4}$ and a different outer tuple, not the power-of-17 family of Theorem 43.6.1, so questions 2–4 of Section 43.9 (the family’s mask condition and residual congruences, and the remaining divisor branches) stay open as mathematical questions, while question 5 presupposes a general soundness proof, which this candidate cannot have. The source’s remark that a prime-progression shortcut does not close the problem (end of Section 43.8) concerns its own interface and is not contradicted. No theorem of Report 43 is refuted. The open status is kept byte for byte in the delivered package: the shipped `data/18-free83-MANIFEST.json`, `data/18-free83-checks-MATH.normal.json`, `data/18-free83-checks-QA.json` and `data/18-free83-source-receipts-inner_replay.json` record “ordinary_input_language”: “OPEN”, and the unshipped delivery README says the same. The programme’s review: “The README’s open-language status is historical.”

The programme’s review (`review_incoming_arithmetic_reports_39_46.md`, commit `5d8889017`) read the three structural notes (shipped as `18-free83-source-structure-early_auxiliary_norm_lemma.md`, `-exceptional_negative_aux_exclusion.md` and `-even_parameter_divisor_classification.md`): the small-norm descent of Lemma 43.3.2, the reduction of a negative auxiliary factor to the single branch $f = 1$, $S = A$, $N_{\text{strong}} = -1$, $N_{\text{aux}} = -\Delta$ and its exclusion for both parities of the main index (Theorem 43.3.1), and the even- A divisor classification of Theorem 43.4.1, which “classifies values, not which values satisfy all other source constraints”. All consistent. The inner-family and other squarefree addenda were not reviewed in full (Part VIII’s opening note lists them).

Credits and overlaps. Section 43.5 extends the conditional auxiliary CRT completion of the tree’s `free_coefficient83_native_alias.md` (commit `53c34b012`), which the source credits in Section 43.1. Lemma 43.2.1 is Lemma 33.3.3 of Report 33 with the extra value $Q_v(1) = 1$; it is re-proved in Parts VI and VII and in Report 46. Lemma 43.3.2 is restated and proved again in Report 46 as Lemma 46.10.1 (a second proof, printed there). The source’s “Report 41” is Part VII. The tree’s `complete83_free_coefficient_scout.md` records degree 111 as source data; Report 43 does not use it. A nonunit zero that the tree materialized (commit `8d465637d`, factor values $(1, 1, 1, 1, 1, -675, -1)$ on numerals that are not a genuine compiler slice) is consistent with Report 43’s genuine-slice theorems; Report 43 does not cite it.

Layout. The source’s table of contents is omitted. Its release verifier is the only one of the five batch-88 verifiers that runs on Windows as delivered (README). The source’s files are shipped with the prefix `18-free83-`.

43.1 Scope and the exact arithmetic object

43.1.1 Three distinct questions

The following statements must be kept separate throughout.

1. *A full positive arithmetic zero* assigns positive integers to all 18 supplied witnesses and the ordinary input, retaining the six fixed compiler numerals, and makes the complete candidate polynomial zero.
2. *An intended inner completion* satisfies selected literal factors at their intended values and the indicated projection, while treating the packed index R as a target port. It need not realize the genuine outer packing and transport equations.
3. *A language counterexample* is a full positive arithmetic zero at an ordinary input known to be rejected by the unchanged genuine compiler. Neither extra witnesses at an already accepted input nor an inner completion alone is a language counterexample.

The structural theorems concern the first notion. The infinite families initially concern the second. The exact interface in Section 43.8 states what remains to pass from the second to the first. A separate rejected-input certificate would then be needed for the third.

The established 84-operation baseline is the previously independently audited construction, not an unrestricted lower bound. The present 83-operation candidate is also distinct from Report 41's first-index deletion and its 81/82-operation failures. Here the first-index factor is retained and is verified explicitly.

43.1.2 Relation to the existing native alias construction

The upstream note *Free-coefficient native index mismatch, and a four-bit mask obstruction*, at commit 53c34b0129bbde7b5d79e520ff3e004cf6f7c07a, already extends the family $p = t(t + 2)$, $n = t(t + 1)$, $R = 2n - 1$ to a positive retained index witness h and a conditional auxiliary CRT completion at $p \neq R$. Its condition $\gcd(c, p) \mid R - p$ is equivalent to $\gcd(c, p) \mid R$. That note also excludes its entire family at the $B = 16$ necessary-mask interface by a congruence modulo 17; it does not supply input or transport witnesses. Thus neither a wrong-index inner extension nor the conditional CRT idea is claimed here as a new invention.

The present synthesis adds the explicit infinitely many gcd-favorable 11- and 17-power parameters, the every-odd- p completion theorem, positive ordinary-input loading with its shared projection, the exact genuine outer interface, and the structural divisor obstructions. The predecessor's $B = 16$ argument does not automatically apply to these different families or to the genuine bases with d a power of 5. These comparisons are with the identified source, not a claim of priority over the literature.

43.1.3 Compiler slice and positive supplied domain

The six fixed ports are

$$B - 1, \quad K, \quad \ell, \quad b, \quad M_C, \quad M_F.$$

A *genuine slice* means these are produced by the complete inherited compiler recipe, not arbitrary numerals satisfying a few inequalities. In particular $B = 2^d$, $\ell = 2d$, d and b are powers of 5 with $d, b \geq 5$ and $b \leq d$, and $K > 0$. The source uses the shifted numeral

$$M_F = M_F^{\text{native}} + B - 1, \quad 0 < M_C, M_F^{\text{native}} < B - 1.$$

That shift is essential. Throughout, $x \geq 1$ is the ordinary input, and the 18 witnesses

$$J, F, \alpha, t_{\text{tr}}, f, h, S, T, s, w, \tau, \eta, \zeta, y, Z, \delta, \rho, \sigma$$

are strictly positive integers. Derived quantities may initially be signed. In particular we do not assume that V, C, W or the input norm's root is positive.

Define the paid expressions

$$q = (B - 1)J + 1, \quad X = wq, \quad Y = sq^3, \quad E = XY, \quad (43.1)$$

$$a = Y(X + 1) = E + Y, \quad A = a + 2, \quad \Delta = A^2 - 1, \quad H = 4a + 3, \quad (43.2)$$

$$k = \eta + \zeta, \quad c = kY + \eta, \quad D = X + ac + (\rho + \sigma)H. \quad (43.3)$$

Thus $A \geq 3$, $\Delta \geq 8$, $D > 0$ and $kY < c < k(Y + 1)$. The remaining definitions are

$$C = q - F - Z - \alpha - \ell x, \quad W = C - Z, \quad (43.4)$$

$$I = \ell x + b, \quad \kappa = I + \delta \Delta, \quad \mu = W + a\kappa + \rho H, \quad (43.5)$$

$$G = q^2 - qF - Z, \quad M = (M_C + qM_F)J, \quad R = G(q^2 - 1) + M, \quad (43.6)$$

$$V = c(Tf - 1) - Rf^2. \quad (43.7)$$

The seven literal factors are

$$N_{\text{first}} = \tau^2 - XY^2(XY^2 + 1)k^2, \quad (43.8)$$

$$N_{\text{main}} = D^2 - \Delta c^2, \quad N_{\text{input}} = \mu^2 - \Delta \kappa^2, \quad (43.9)$$

$$N_{\text{aux}} = S^2 V^2 - (S^2 - 1)y^2, \quad N_{\text{index}} = k - hE - R, \quad (43.10)$$

$$N_{\text{transport}} = (K + w)C + q - F - t_{\text{tr}}(q - 1), \quad (43.11)$$

$$N_{\text{strong}} = \Delta f^2 - S^2. \quad (43.12)$$

The complete polynomial is

$$F_{83} = N_{\text{first}} N_{\text{main}} N_{\text{input}} N_{\text{aux}} N_{\text{index}} N_{\text{transport}} N_{\text{strong}} - \Delta. \quad (43.13)$$

Consequently each factor at a zero is nonzero and divides Δ ; in particular its absolute value is at most Δ . No general unit normalization follows merely from this product equation.

43.1.4 Literal deletion and what the inverse says

In the verified 84-operation source, $S = i\Delta c^2$ is a paid multiplication from the supplied positive witness i . Replacing i by the supplied positive S and deleting that one row gives the exact all-ring identity

$$F_{83}(S = i\Delta c^2) = F_{84}. \quad (43.14)$$

The other rows and the seven-operation finalizer remain. The arithmetic ledger is therefore $83 = 46M + 37A$, with 18 supplied witnesses. The source's degree claim is 111; the independently audited 84 baseline has degree 187. Degree 111 is not needed by any theorem below and is not a substitute for soundness.

For $\Delta c^2 \neq 0$, the literal rational inverse is uniquely

$$i = \frac{S}{\Delta c^2}. \quad (43.15)$$

On the positive domain it is positive, but it need not be integral. Equation (43.14) proves the candidate accepts every input accepted by the 84 parent. Failure of integrality in (43.15) disproves this particular witness-set inverse, not equality of the ordinary-input languages.

43.2 Pell notation and elementary tools

For integer $A \geq 2$ define

$$\chi_A(r) + \psi_A(r)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^r \quad (r \geq 0).$$

Both sequences are integral, with initial values $(1, 0)$ and $(A, 1)$, and recurrence $u_{r+1} = 2Au_r - u_{r-1}$. The norm is 1. Addition and subtraction identities follow by multiplying the displayed powers. In particular

$$\psi_A(r + s) = \chi_A(r)\psi_A(s) + \psi_A(r)\chi_A(s), \quad (43.16)$$

$$\chi_A(r + s) = \chi_A(r)\chi_A(s) + (A^2 - 1)\psi_A(r)\psi_A(s), \quad (43.17)$$

$$\psi_A(2r) = 2\chi_A(r)\psi_A(r). \quad (43.18)$$

For $r \geq 1$, the positive sequences increase strictly, and for $r \geq 3$

$$(2A - 1)^{r-1} \leq \psi_A(r) < (2A)^{r-1}. \quad (43.19)$$

These inequalities follow by induction from the recurrence; the lower inequality uses $\psi_A(r - 1) < \psi_A(r)$. Pell classification for $u^2 - (A^2 - 1)v^2 = 1$ uses the fundamental positive pair $(A, 1)$: any pair with $v \geq 1$ has $u \geq A$, and multiplication by the inverse of that unit descends until $(1, 0)$. Thus all nonnegative solutions are the above pairs. We also use the usual integer Pell-unit classification for a general positive nonsquare discriminant in one conditional theorem.

Lemma 43.2.1 (Odd quotient identities). *There are polynomials $Q_v(Z) \in \mathbb{Z}[Z]$ such that*

$$\frac{\chi_S(2v + 1)}{S} = Q_v(S^2), \quad Q_v(0) = (-1)^v(2v + 1), \quad Q_v(1 - A^2) = (-1)^v\psi_A(2v + 1), \quad Q_v(1) = 1.$$

The quotient is an integer for every positive integer S .

Proof. Set $Q_0 = 1$, $Q_1 = 4Z - 3$ and $Q_{v+1} = (4Z - 2)Q_v - Q_{v-1}$. The odd subsequence of χ_S obeys this recurrence, proving the first identity. At $Z = 0$ the recurrence and first two values give $(-1)^v(2v + 1)$; at $Z = 1$ they give 1. At $Z = 1 - A^2$, multiplying the v th term by $(-1)^v$ gives the recurrence and initial values of $\psi_A(2v + 1)$, proving the remaining identity. These are polynomial identities, so reducing them modulo an integer does not require S to be invertible. $\square$

Lemma 43.2.2 (Residues of Pell second coordinates). *For $A \geq 2$, $m \geq 1$ and $t \geq 0$, modulo $\psi_A(m)$ the value $\psi_A(t)$ is either 0 or $\pm\psi_A(j)$ for some $1 \leq j < m$. If m is odd and t is even, j may be chosen even.*

Proof. Write $t = qm + j$ with $0 \leq j < m$. Addition and the norm identity give $\psi_A(t) \equiv \chi_A(m)^q\psi_A(j)$ and $\chi_A(m)^2 \equiv 1$. For even q use j itself; for odd q use $\chi_A(m)\psi_A(j) \equiv -\psi_A(m - j)$. If $j = 0$ the residue is zero. When m is odd and t even, j is even for even q , while $m - j$ is even for odd q . $\square$

43.3 An unconditional auxiliary square theorem

Theorem 43.3.1. *Every full strictly positive zero of (43.13) on a genuine fixed-compiler slice satisfies*

$$N_{\text{aux}} = z^2 > 0, \quad z^2 \mid \Delta, \quad z = \gcd(|V|, y).$$

No parity assumption on q or A , and no assumption $p = R$, is used.

43.3.1 Descent for a quadratic form near a square

Lemma 43.3.2. *Let $H_0 \geq 2$, $v \geq 0$ and $y > 0$ be integers, and put $N = H_0 v^2 - (H_0 - 1)y^2$.*

1. *If $N < 0$, then $N \leq -(H_0 - 1)$.*
2. *If $0 < N < H_0$, then $N = z^2$ and $\gcd(v, y) = z$ for a positive z . Every such solution descends to (z, z) by an integral unimodular map.*
3. *The solutions with $N = -(H_0 - 1)$ are exactly*

$$v = 2(H_0 - 1)\psi_{2H_0-1}(r), \quad y = \chi_{2H_0-1}(r), \quad r \geq 0.$$

Proof. When $y > v$, put $t = y - v \geq 1$. Direct expansion gives

$$-N = (H_0 - 1)t^2 + v(2(H_0 - 1)t - v). \quad (43.20)$$

If $N < 0$ and $v \leq 2(H_0 - 1)t$, the right side is at least $H_0 - 1$. Otherwise negativity gives

$$2(H_0 - 1)t < v < (H_0 - 1 + \sqrt{H_0(H_0 - 1)})t < (2H_0 - 1)t.$$

The transformation

$$v' = (2H_0 - 1)v - 2(H_0 - 1)y = v - 2(H_0 - 1)t, \quad y' = (2H_0 - 1)y - 2H_0 v = (2H_0 - 1)t - v \quad (43.21)$$

preserves N and has $0 < v' < v$, $y' > 0$. Induction proves the negative gap.

If $0 < N < H_0$ and $v \geq y + 1$, then $N = H_0(v^2 - y^2) + y^2 \geq H_0$, a contradiction. If $v = y$, then $N = v^2$. Otherwise (43.20) and $N > 0$ give $v > 2(H_0 - 1)t$; $N < H_0$ gives $v < (2H_0 - 1)t$, since the quadratic in v is increasing beyond its positive root and takes value $H_0 t^2$ at that endpoint. Thus (43.21) descends until a diagonal (z, z) . Its determinant is 1, so both it and its integral inverse preserve the gcd. This proves the second assertion, including the exact gcd.

At $N = -(H_0 - 1)$, descent ends where $v \leq 2(H_0 - 1)t$. Equation (43.20) forces $t = 1$ and $v = 0$ or $v = 2(H_0 - 1)$. The latter maps to $(0, 1)$. The inverse matrix is

$$\begin{pmatrix} 2H_0 - 1 & 2(H_0 - 1) \\ 2H_0 & 2H_0 - 1 \end{pmatrix}.$$

Its nonnegative iterates of $(0, 1)$ give exactly the formula in part 3. Conversely, every displayed pair is an integral positive- y solution by the Pell norm identity. This proves both inclusion and exhaustiveness. $\square$

43.3.2 Reduction to one negative branch

At a full zero write $d_s = N_{\text{strong}}$ and $N = N_{\text{aux}}$. The product gives $|d_s N| \leq \Delta$. The case $S = 1$ is impossible: then $d_s = \Delta f^2 - 1 \geq 7$, $d_s \mid \Delta$, and $\gcd(d_s, \Delta) = 1$. Hence $S \geq 2$ and $H_0 = S^2 \geq 4$.

Suppose $N < 0$. Lemma 43.3.2 gives

$$|d_s|(S^2 - 1) \leq \Delta. \quad (43.22)$$

If $f \geq 2$, then $S^2 = \Delta f^2 - d_s \geq 3\Delta$, contradicting the budget. Thus $f = 1$. If $d_s = -b_s < 0$, then $S^2 = \Delta + b_s$ and $b_s(\Delta + b_s - 1) \leq \Delta$, forcing $b_s = 1$. If $d_s > 0$, then $\Delta = S^2 + d_s$, and the budget becomes $(d_s - 1)(S^2 - 2) \leq 2$. For $d_s = 1$ this would require $A^2 - S^2 = 2$, impossible. For $d_s \geq 2$

the only possible square $S^2 \geq 4$ is $S^2 = 4$, $d_s = 2$, giving $A^2 = 7$, also impossible. Therefore the only negative possibility is

$$f = 1, \quad S = A, \quad N_{\text{strong}} = -1, \quad N_{\text{aux}} = -\Delta. \quad (43.23)$$

For $N > 0$, one always has $N < S^2$. Indeed $N \leq \Delta/|d_s|$. For $f \geq 2$ use $S^2 \geq 3\Delta$; for $f = 1, d_s < 0$ use $S^2 > \Delta$. For $f = 1, d_s > 0$ the value $d_s = 1$ was excluded, and $d_s S^2 > S^2 + d_s = \Delta$ for $d_s \geq 2, S^2 \geq 4$. Lemma 43.3.2 already proves the theorem on the positive branch. It remains to exclude (43.23) on the genuine source.

43.3.3 What can legitimately be recovered in that branch

In (43.23), $N_{\text{strong}}N_{\text{aux}} = \Delta$. Thus the other five integer factors have product 1 and are individually units. Only at this point are unit arguments available.

The first norm cannot be -1 . Put $u_0 = XY^2 > 1$. A solution $\tau^2 - u_0(u_0 + 1)k^2 = -1$ has $u_0k < \tau < (u_0 + \frac{1}{2})k$ for every $k \geq 1$, since $u_0 > 1$. The inverse unit gives $0 < (2u_0 + 1)k - 2\tau < k$ and an integral companion root preserving the norm. Taking its nonzero absolute value yields an impossible infinite descent. Main and input norms cannot be -1 modulo 4, since $\Delta \equiv 0$ or 3. Hence

$$N_{\text{first}} = N_{\text{main}} = N_{\text{input}} = 1, \quad N_{\text{index}} = N_{\text{transport}} = \epsilon \in \{1, -1\}. \quad (43.24)$$

Rewrite transport as

$$N_{\text{transport}} = (K + w)C + 1 - F - (t_{\text{tr}} - 1)(q - 1).$$

Since $K + w \geq 2$ and the final terms are nonpositive, $C \leq -1$ would give $N_{\text{transport}} \leq -2$. Thus $C \geq 0$ and $F + Z < q$. The shifted mask bounds give

$$2q - 1 \leq G \leq q^2 - q - 1, \quad 0 < M < (q - 1)(1 + 2q).$$

It follows directly that

$$(2q - 1)(q^2 - 1) < R < q^4 - q^3, \quad 0 < R < a < \Delta, \quad R + 2 < E. \quad (43.25)$$

These are untyped integer bounds: q has not been assumed even or a power of B .

Put $P = 2XY^2 + 1$. The fundamental unit for $\tau^2 - XY^2(XY^2 + 1)k^2 = 1$ has pair $(P, 2)$: coefficient 1 places a square strictly between consecutive squares, coefficient 2 gives P , and the inverse unit descends any larger solution. Thus $\tau = \chi_P(n)$, $k = 2\psi_P(n)$, $n \geq 1$. Since $P \equiv 1 \pmod{E}$, the recurrence gives $\psi_P(n) \equiv n \pmod{E}$. From the index unit and (43.25),

$$2n = R + \epsilon + vE, \quad v \geq 0, \quad n \geq (R - 1)/2 \geq 24.$$

The main positive root gives $D = \chi_A(p)$, $c = \psi_A(p)$. Since $P > A$ and $c > k$, growth forces $p > n$, hence $p \geq 25$. Also $c > 2R$, and independently

$$c > A\Delta^2, \quad \psi_A(6) - A\Delta^2 = A(31A^4 - 30A^2 + 5) > 0. \quad (43.26)$$

No strong-rank theorem or equality $p = R$ has been used.

43.3.4 Excluding both parities of the main rank

Let $B_0 = 2A^2 - 1$. Lemma 43.3.2 and the literal V give

$$V = \pm 2\Delta\psi_{B_0}(r) \quad (r \geq 0), \quad V \equiv -R \pmod{c}. \quad (43.27)$$

For odd p , $\gcd(A, c) = 1$, because $\psi_A(p) \equiv \pm 1 \pmod{A}$. Duplication gives $2A\psi_{B_0}(r) = \psi_A(2r)$. Lemma 43.2.2 with its parity refinement therefore makes V congruent modulo c to zero or

$$\pm \frac{\Delta\psi_A(j)}{A}, \quad 2 \leq j \leq p-1, \quad j \text{ even.}$$

Every nonzero displayed magnitude is at least 2Δ and is less than $c/2$. For the upper bound it suffices to use $j = p-1$ and

$$Ac = A^2\psi_A(p-1) + A\chi_A(p-1) > 2\Delta\psi_A(p-1).$$

Thus none is the least-absolute residue $-R$, since $0 < R < \Delta < c/2$. Zero is excluded as well.

For even $p = 2m$, let $b_0 = \psi_{B_0}(m)$, so $c = 2Ab_0$. Equation (43.27) forces R even. Divide by 2 and reduce modulo b_0 :

$$\pm \Delta\psi_{B_0}(r) \equiv -R/2 \pmod{b_0}.$$

The left side is zero or a signed $\Delta\psi_{B_0}(j)$, $1 \leq j < m$. Every nonzero magnitude is at least Δ and less than $b_0/2$, since

$$b_0 > (2B_0 - 1)\psi_{B_0}(m-1) > 2\Delta\psi_{B_0}(m-1).$$

By (43.26), $b_0 > \Delta^2/2 > R$. Again $0 < R/2 < \min(\Delta, b_0/2)$ is impossible. Both signs of V and $r = 0$ are covered. This excludes the only negative branch and completes Theorem 43.3.1; divisibility $z^2 \mid \Delta$ follows from the full product.

43.4 Conditional divisor classification and squarefree obstructions

Theorem 43.4.1 (Even parameter norm values). *Let $A \geq 2$ be even, $\Delta = A^2 - 1$, and suppose the nonzero represented integer $N = x_0^2 - \Delta y_0^2$ divides Δ . Then either N is a positive square, or $N < 0$ and $\Delta/|N|$ is a square. This is a necessary classification of represented norm values, not a sufficiency assertion for arbitrary listed divisors or for full compiler witnesses.*

Proof. Write $\Delta = d_0 m^2$ with d_0 squarefree, and $|N| = rz^2$ with r squarefree. Since $\Delta \equiv 3 \pmod{4}$, m, d_0 are odd and $d_0 \equiv 3 \pmod{4}$. Divisibility gives $z \mid m$, $z \mid x_0$. Set $X_0 = |x_0|/z$, $Y_0 = m|y_0|/z$, and $\varepsilon = \operatorname{sgn} N$. Then $X_0^2 - d_0 Y_0^2 = \varepsilon r$. Each prime of r divides X_0 because $|N| \mid x_0^2$. If such a prime did not divide d_0 , it would divide Y_0 too, contradicting the squarefree valuation on the right. Hence $r \mid d_0$ and $r \mid X_0$.

The integers

$$U = \frac{X_0^2 + d_0 Y_0^2}{r}, \quad W_0 = \frac{2X_0 Y_0}{r}$$

satisfy $U^2 - d_0 W_0^2 = 1$. The norm is odd, so X_0, Y_0 have opposite parity; U is odd. Let $\alpha + \beta\sqrt{d_0}$ be the fundamental positive integer Pell unit. Since $A + m\sqrt{d_0}$ has even first coordinate, α is even: if α were odd, β would be even, and all powers would have odd first coordinate. With α even, β odd and first-coordinate parity alternates with the exponent. Thus $U = \chi_\alpha(2h)$ for some $h \geq 0$.

For $\varepsilon = 1$, $X_0^2/r = (U+1)/2 = \chi_\alpha(h)^2$, forcing $r = 1$. For $\varepsilon = -1$, $(d_0/r)Y_0^2 = (U+1)/2 = \chi_\alpha(h)^2$, forcing $r = d_0$. This yields $N = z^2$ or $N = -d_0 z^2$ and $\Delta/|N| = (m/z)^2$, respectively. Zero-coordinate cases obey the same identities, or follow immediately from $N \mid \Delta$. $\square$

If Δ is squarefree then A must be even: odd $A \geq 3$ would make $8 \mid \Delta$. Theorems 43.3.1 and 43.4.1 give

$$N_{\text{main}}, N_{\text{input}} \in \{1, -\Delta\}, \quad N_{\text{strong}} \in \{-1, \Delta\}, \quad N_{\text{aux}} = 1. \quad (43.28)$$

Lemma 43.4.2 (A first-coordinate divisibility obstruction). *For $A \geq 2$, $m \geq 3$ and $p \geq 0$,*

$$\psi_A(m) \mid \chi_A(p) + \epsilon, \quad \epsilon \in \{1, -1\} \implies 2m \mid p, \quad \chi_A(p) \equiv 1 \pmod{\psi_A(m)}.$$

Conversely $2m \mid p$ gives the displayed residue 1, hence divisibility of $\chi_A(p) - 1$. For $m = 2$, divisibility by $\psi_A(2) = 2A$ of $\chi_A(p) \pm 1$ at least forces p even.

Proof. Put $f = \psi_A(m)$, $D_m = \chi_A(m)$ and $p = qm + j$, $0 \leq j < m$. Modulo f , $\chi_A(p)$ is $\chi_A(j)$ for even q and $\chi_A(m - j)$ for odd q . For $1 \leq j < m$, the interior values lie strictly between 1 and $f - 1$, since $f - \chi_A(m - 1) = A\psi_A(m - 1) \geq 2$. At $j = 0$, q odd, the residue of $D_m = Af - \psi_A(m - 1)$ is $f - \psi_A(m - 1)$, again strictly between 1 and $f - 1$ for $m \geq 3$. Only $j = 0$, q even gives ± 1 , and it gives $+1$. The reverse implication follows from addition and $D_m^2 \equiv 1$. For $m = 2$, an odd p has $\chi_A(p) \equiv 0 \pmod{A}$, excluding ± 1 . $\square$

Lemma 43.4.3 (Small target quotient obstruction). *Let $S \geq 2$, $k_0 \geq 6$, $c = \chi_S(k_0)$, and $1 < L < S^2 - 1$. There is no odd positive t and no sign ϵ such that*

$$\epsilon \frac{\chi_S(t)}{S} \equiv -L \pmod{c}.$$

Proof. The Pell pair at $2k_0$ is $(-1, 0)$ modulo c . Period and reflection therefore reduce $\chi_S(t)$ to a signed $\chi_S(j)$ with j odd and $1 \leq j \leq k_0$. If k_0 is even, $\gcd(S, c) = 1$; division modulo c gives representatives $\pm \chi_S(j)/S$ with $j < k_0$, each of magnitude less than $c/2$. If k_0 is odd, divide modulo c/S ; $j = k_0$ becomes zero and the remaining magnitudes are less than $(c/S)/2$. In either case the smallest nonzero magnitude is 1, and the next is

$$\chi_S(3)/S = 4(S^2 - 1) + 1 > L.$$

Finally $c \geq \chi_S(6) > S(2S - 1)^5 > S(S^2 - 1)^2 > 2SL$. Thus L itself is below half the applicable modulus and lies in the forbidden gap between the representatives. It is neither zero nor 1. $\square$

Theorem 43.4.4 (Squarefree main and input normalization). *Every full strictly positive candidate zero on a genuine slice with squarefree Δ satisfies $N_{\text{main}} = N_{\text{input}} = N_{\text{aux}} = 1$.*

Proof. By (43.28), if either main or input equals $-\Delta$, the other is 1, $N_{\text{strong}} = -1$, and the first, index and transport factors are units. The first is $+1$ by descent, and index and transport have the same sign. Thus (43.25) applies, in particular $1 < R < a < \Delta$ and $C \geq 0$.

Suppose first $N_{\text{input}} = -\Delta$. Squarefreeness gives $\Delta \mid \mu$; write $\mu = \Delta b_0$. Then $\kappa^2 - \Delta b_0^2 = 1$, so $\kappa = \chi_A(t)$ for $t \geq 0$. Modulo Δ this is A^t , hence 1 or A . But $\kappa \equiv I \pmod{\Delta}$, whereas genuine input loading and $C \geq 0$ give

$$1 < I = 2dx + b < q + d < 2q < A < \Delta.$$

Here $2dx < q$, $b \leq d$, and $q \geq B = 2^d > d$. The residue I is neither 1 nor A , a contradiction.

Now suppose $N_{\text{main}} = -\Delta$. Squarefreeness yields $c = \chi_A(p)$ and $D = \Delta\psi_A(p)$. The retained first/index units give $k = 2\psi_P(n)$ and $n \geq 24$. Since $c > kY > \psi_A(n) > \chi_A(n - 1)$, one has $p \geq n \geq 24$. The strong equation gives $S = \chi_A(m)$, $f = \psi_A(m)$, $m \geq 1$. The auxiliary equation gives

$$V = \epsilon \chi_S(t)/S, \quad t > 0 \text{ odd}, \quad \epsilon = \text{sgn } V.$$

Lemma 43.2.1 and $S^2 = 1 + \Delta f^2$ imply $V \equiv \epsilon \pmod{f}$. The literal V implies $V \equiv -c \pmod{f}$, so $f \mid c + \epsilon$.

For $m \geq 3$, Lemma 43.4.2 gives $2m \mid p$ and $c \equiv 1 \pmod{f}$. Also $c \geq \chi_A(2m) = 1 + 2\Delta f^2$. Since $R < \Delta$ and $T \geq 1$,

$$V \geq c(f-1) - Rf^2 > c(f-1) - c/2 > 0.$$

Thus $\epsilon = 1$, forcing $c \equiv -1 \pmod{f}$ as well, impossible because $f > 2$. For $m = 2$, Lemma 43.4.2 forces $p = 2k_0$ with $k_0 \geq 12$; for $m = 1$ put $k_0 = p \geq 24$. In either case $c = \chi_S(k_0)$. Put $L = Rf^2$. Then $1 < L < \Delta f^2 = S^2 - 1$, while the literal quotient gives $V \equiv -L \pmod{c}$. Lemma 43.4.3 excludes both cases. $\square$

Two alternatives remain on this slice. If $N_{\text{strong}} = \Delta$, then $N_{\text{first}} = 1$ and $N_{\text{index}} = N_{\text{transport}} \in \{1, -1\}$. If $N_{\text{strong}} = -1$, then

$$N_{\text{first}} N_{\text{index}} N_{\text{transport}} = -\Delta. \quad (43.29)$$

The latter divisor/sign equation is unresolved. The former still has the free-coefficient rank problem. Neither is automatically a parent zero.

43.5 Auxiliary completion with a weaker rank condition

Theorem 43.5.1. *Let $A \geq 2$, $p \geq 3$ odd, $c = \psi_A(p)$, $D = \chi_A(p)$ and $R > 0$. If $\gcd(c, p) \mid R$, there exist positive integers f, S, T, y satisfying the literal auxiliary and strong equations $N_{\text{aux}} = 1$, $N_{\text{strong}} = \Delta$. The construction does not require $p = R$.*

Proof. Set

$$f = \chi_A(2p), \quad b_0 = \psi_A(2p) = 2Dc, \quad S = \Delta b_0.$$

Then $f^2 - \Delta b_0^2 = 1$, so $N_{\text{strong}} = \Delta$, $c \mid S$, $f^2 \equiv 1 \pmod{c}$, and $\gcd(c, f) = 1$. Odd p gives odd c . Let $j_0 = 1$ if $p \equiv 3 \pmod{4}$ and $j_0 = 3$ if $p \equiv 1 \pmod{4}$. Choose a positive solution of

$$z \equiv j_0 p \pmod{8p}, \quad z \equiv R \pmod{c}. \quad (43.30)$$

The gcd of the moduli is $\gcd(c, p)$, and it divides $j_0 p$. The CRT therefore says that (43.30) is compatible if and only if $\gcd(c, p) \mid R$. A positive representative exists whenever it is compatible. It satisfies $z \equiv 3 \pmod{4}$.

Write $z = 2v + 1$, with v odd, and set

$$V = \chi_S(z)/S = Q_v(S^2), \quad y = \psi_S(z).$$

By Lemma 43.2.1, $V \equiv -z \equiv -R \pmod{c}$. Coefficientwise in the quadratic ring modulo f , the Pell pair at $4p$ is $(-1, 0)$ and that at $8p$ is $(1, 0)$. Hence $\psi_A(z) \equiv \psi_A(j_0 p) \equiv c \pmod{f}$; for $j_0 = 3$ use $\psi_A(3p) = (2f + 1)c$. Since $S^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f}$, the odd quotient identity gives $V \equiv -c \pmod{f}$. It follows that

$$T = \frac{V + c + Rf^2}{cf} \quad (43.31)$$

is a positive integer: its numerator is divisible by each of the coprime numbers c, f . Equation (43.31) is exactly $V = c(Tf - 1) - Rf^2$. Finally the Pell norm at S gives $S^2 V^2 - (S^2 - 1)y^2 = 1$. $\square$

The inverse (43.15) in this construction is $2D/c$. It is nonintegral because $c > 1$ is odd and $\gcd(c, D) = 1$. The gcd hypothesis is necessary and sufficient for this particular CRT pair; the theorem does *not* claim it is necessary for every possible auxiliary completion. For example $A = 2, p = 3$ gives $c = 15$; $R = 1$ fails that recipe's congruence compatibility modulo 3, without ruling out other recipes.

When $p \equiv 3 \pmod{4}$ and $\gcd(c, p) = 1$, a simpler choice is

$$f = D, \quad S = \Delta c, \quad z \equiv p \pmod{4p}, \quad z \equiv R \pmod{c}. \quad (43.32)$$

The pair at $2p$ is $(-1, 0)$ modulo f , and that at $4p$ is $(1, 0)$; the same proof gives (43.31), $N_{\text{aux}} = 1$, $N_{\text{strong}} = \Delta$. Here the inverse is simply $1/c$. These divisions define existential integer witnesses; they do not add instructions to the saved arithmetic circuit.

43.6 The power of 17 inner family

Theorem 43.6.1. *For every $j \geq 0$, set*

$$u = 17^{2j}, \quad p = 35u, \quad n = 30u, \quad R = 60u - 1, \quad (43.33)$$

$$X = 2^p, \quad Y = 2^{7u-1}, \quad E = XY, \quad P = 2XY^2 + 1, \quad (43.34)$$

$$a = Y(X + 1), \quad A = a + 2, \quad \Delta = A^2 - 1, \quad H = 4a + 3, \quad (43.35)$$

$$D = \chi_A(p), \quad c = \psi_A(p), \quad \tau = \chi_P(n), \quad k = 2\psi_P(n). \quad (43.36)$$

Positive $\eta, \zeta, h, \gamma, f, S, T, y$ exist such that

$$k = \eta + \zeta, \quad c = kY + \eta, \quad D = X + ac + \gamma H, \quad N_{\text{first}} = N_{\text{main}} = N_{\text{index}} = N_{\text{aux}} = 1, \quad N_{\text{strong}} = \Delta.$$

Here $p \neq R$, and the literal inverse $S/(\Delta c^2) = 1/c$ is nonintegral. The parameters are not yet a genuine outer completion.

43.6.1 Exact resonance and strict witness inequalities

The Pell norms give $N_{\text{main}} = 1$ and, because $P^2 - 1 = 4XY^2(XY^2 + 1)$, $N_{\text{first}} = 1$. Put $L_0 = 2XY$ and $M_0 = 4XY^2$. Exponent arithmetic gives

$$L_0^{p-1} = 2YM_0^{n-1}. \quad (43.37)$$

Indeed the exponent equality is equivalent to $p(p - n) = 7u(2n - p)$, and $35 \cdot 5 = 7 \cdot 25$. Moreover

$$\begin{aligned} 2A - 1 &= L_0 \left(1 + \frac{1}{X} + \frac{3}{2XY} \right), & 2A &= L_0 \left(1 + \frac{1}{X} + \frac{2}{XY} \right), \\ 2P &= M_0 \left(1 + \frac{1}{2XY^2} \right), & 2P - 1 &> M_0. \end{aligned}$$

Use (43.19) and (43.37). The numerator perturbation in $2A - 1$ exceeds the denominator perturbation in $2P$, and $p > n$, giving $c > kY$. For the upper bound let $\varepsilon = 1/X + 2/(XY) < 2/X$. Since $(p - 1)\varepsilon < 1/2$,

$$(1 + \varepsilon)^{p-1} \leq \frac{1}{1 - (p - 1)\varepsilon} < 1 + 2(p - 1)\varepsilon.$$

Here the first inequality follows by bounding binomial coefficients by powers of $p - 1$. Thus

$$0 < \frac{c}{k} - Y < \frac{4pY}{X} = 70u 2^{-28u} < 1. \quad (43.38)$$

The last inequality holds for all $u \geq 1$, for example from $u \leq 2^{u-1}$ and $70 < 2^{27u+1}$. Therefore $\eta = c - kY$ and $\zeta = k - \eta$ are positive integers.

As $P \equiv 1 \pmod{E}$, $k \equiv 2n \pmod{E}$. Since $2n = R + 1$ and $\psi_P(n) > n$,

$$h = (k - R - 1)/E = (k - 2n)/E \in \mathbb{Z}_{>0}, \quad N_{\text{index}} = 1. \quad (43.39)$$

This verifies the retained index equation; no index deletion is involved.

43.6.2 The shared main projection

Define $z_r = \chi_A(r) - a\psi_A(r)$. Then $z_0 = 1$, $z_1 = 2$, and $z_{r+1} = 2Az_r - z_{r-1}$. The sequence 2^r obeys this recurrence modulo $H = 4A - 5$, since its discrepancy is a multiple of H . Hence $z_r \equiv 2^r \pmod{H}$. For $r \geq 1$, positivity and strict increase give

$$z_{r+1} > (2A - 1)z_r. \quad (43.40)$$

It follows that $z_p > 2^p = X$ and

$$\gamma = (z_p - X)/H \in \mathbb{Z}_{>0}. \quad (43.41)$$

The literal main projection is therefore satisfied.

43.6.3 Coprimality at every parameter value

Since $17^2 \equiv 1 \pmod{24}$, $u \equiv 1 \pmod{24}$. The exact expression

$$A = 2^{42u-1} + 2^{7u-1} + 2$$

gives $A \equiv 3 \pmod{5}$, $A \equiv 0 \pmod{7}$ and $A \equiv 0 \pmod{17}$. Modulo 5, the recurrence coefficient $2A$ is 1 and the ψ sequence has period six $0, 1, 1, 0, -1, -1$. As $p \equiv 5 \pmod{6}$, $c \equiv -1 \pmod{5}$. Modulo 7 and 17 the coefficient is 0; $p \equiv 3 \pmod{4}$ gives $c \equiv -1$ in both. The only possible prime divisors of $p = 35 \cdot 17^{2j}$ are 5, 7 and 17. Therefore

$$\gcd(c, p) = 1 \quad \text{for every } j \geq 0. \quad (43.42)$$

The simplified auxiliary extension (43.32) now completes the theorem. Since $R - p = 25u - 1 > 0$, the ranks differ for every parameter.

43.7 Ordinary input loading within the inner family

Fix a genuine compiler and a positive ordinary input x . Its input rank $I = \ell x + b$ is odd and at least 3. For any family member with $p > I$, put $W = 2^I$ and define

$$\kappa = \psi_A(I), \quad \mu = \chi_A(I), \quad \delta = \frac{\kappa - I}{\Delta}, \quad (43.43)$$

$$\rho = \frac{z_I - W}{H}, \quad \sigma = \gamma - \rho. \quad (43.44)$$

For odd I , $\psi_A(I) \equiv I \pmod{\Delta}$: the recurrence has $A^2 \equiv 1 \pmod{\Delta}$, or this follows from the power expansion. Also $\psi_A(I) > I$, so δ is a positive integer. The projection congruence and (43.40) make ρ a positive integer. For $p > I$, monotonicity gives

$$z_p - z_I \geq z_p - z_{p-1} > (2A - 2)z_{p-1} > 2^p = X.$$

Therefore $\sigma = (z_p - z_I - X + 2^I)/H > 0$. These definitions simultaneously satisfy

$$\kappa = I + \delta\Delta, \quad \mu = W + a\kappa + \rho H, \quad D = X + ac + (\rho + \sigma)H, \quad N_{\text{input}} = 1.$$

This is a shared positive projection, not two unrelated Pell equations. It uses no acceptance fact, native computation history or mask decoding. It does not yet realize the register $W = C - Z$ with actual outer witnesses.

43.8 The exact remaining compiler interface

Fix all six genuine numerals and x , and choose $j \geq 0$ in Theorem 43.6.1. Choose a second integer $N \geq 1$ and put

$$t = dN, \quad q = 2^t = B^N, \quad 3t \leq 7u - 1, \quad p > I. \quad (43.45)$$

Then

$$w = 2^{p-t}, \quad s = 2^{7u-1-3t}, \quad J = \frac{q-1}{B-1}, \quad Q = q^2 - 1, \quad M = (M_C + qM_F)J, \quad e = p \bmod t.$$

Both w, s are positive integers, $X = wq$, $Y = sq^3$, and $0 \leq e < t$.

Theorem 43.8.1 (Intended-factor interface equivalence). *With the choices above and $W = 2^I$, the family realizes its prescribed $R = 60u - 1$ and $N_{\text{transport}} = 1$ with all remaining outer witnesses positive if and only if the following tests hold:*

$$R \equiv M \pmod{Q}, \quad (43.46)$$

$$G = (R - M)/Q, \quad Z = (-G) \bmod q \in \{1, \dots, q-1\}, \quad F = (q^2 - Z - G)/q \geq 1, \quad (43.47)$$

$$F \equiv (K + 2^e)(W + Z) \pmod{q-1}, \quad (43.48)$$

$$F + 2Z + W + \ell x < q. \quad (43.49)$$

The residue in (43.47) is the least nonnegative residue and is required to be nonzero. In particular F, Z are uniquely forced by (j, N) whenever the tests succeed.

Proof. For necessity, the literal packing equation implies (43.46) and $G = q^2 - qF - Z$. Positivity with $W = C - Z$ gives $\alpha = q - F - 2Z - W - \ell x > 0$, hence $1 \leq Z < q$ and (43.49). Reducing the packing identity modulo q forces exactly the residue and F in (43.47). The equation $N_{\text{transport}} = 1$ is equivalent to

$$(K + w)(W + Z) - F = (t_{\text{tr}} - 1)(q - 1).$$

Since $w = 2^{p-t} \equiv 2^e \pmod{q-1}$, this implies (43.48).

Conversely suppose all four tests hold. Set

$$\alpha = q - F - 2Z - W - \ell x, \quad t_{\text{tr}} = 1 + \frac{(K + w)(W + Z) - F}{q - 1}. \quad (43.50)$$

The first is positive integral by (43.49); the second is integral by (43.48). The scale restriction implies $p > 2t$, so $w > q$. Since $0 < F < q$ and $W + Z \geq 1$, its numerator is positive, proving $t_{\text{tr}} > 0$. The definitions give $C = W + Z$, recover the exact packing R , and make $N_{\text{transport}} = 1$. Sections 43.6–43.7 supply all other positive witnesses. Thus the six factors other than strong are 1, strong is Δ , and (43.13) vanishes. $\square$

The iff is deliberately limited to this family with its prescribed R and intended transport factor 1. It does not classify accidental product zeros obtained by changing R , abandoning these family coordinates, or using different divisor values. In a generic zero, $q = B^N$ has not been proved. Here it is a construction choice.

43.8.1 Necessary restrictions that cannot be discarded

Because $J \mid Q$ and $J \mid M$, packing requires $J \mid R$. For the power-of-17 family:

1. N must be odd. If N were even, odd d gives $q \equiv 1$, $J \equiv 0$ and $Q \equiv 0 \pmod{3}$, forcing $R \equiv 0$; but $R = 60u - 1 \equiv 2 \pmod{3}$.
2. For odd N , one has $q \equiv 2$, $J \equiv 1 \pmod{3}$ and $Q \equiv 0$. Thus the genuine shifted masks must satisfy

$$M_C + 2M_F \equiv 2 \pmod{3}. \quad (43.51)$$

This has not been shown automatic from the compiler recipe.

3. If $3 \mid N$, then B has order 3 modulo 7 and $7 \mid J$. Since $R \equiv 4 \cdot 2^j - 1 \pmod{7}$, this requires $j \equiv 1 \pmod{3}$.

These restrict this family, not every positive candidate zero.

Strict positivity sharpens the generic packing lower bound. From $F = q - 2Z - W - \ell x - \alpha$ one gets

$$G = (2q - 1)Z + q(W + \ell x + \alpha) \geq q(W + \ell x + 3) - 1.$$

Therefore any intended family completion must satisfy

$$R \geq [q(W + \ell x + 3) - 1](q^2 - 1) + M. \quad (43.52)$$

The algebraic lower bound is attained at $Z = \alpha = 1$ when the resulting F is positive; this does not assert transport or packing congruence solvability there.

For $j = 0$, the 17-family has $u = 1$, $R = 59$ and $Y = 2^6$. The scale bound demands $t \leq 2$, whereas a genuine compiler requires $t = dN \geq 5$. This smallest exact inner fixture cannot complete on any genuine positive-input slice. Large exact Pell integers in a fixture are therefore not evidence of a full compiled witness.

43.8.2 Why a prime progression shortcut does not close the problem

Replacing a power parameter by primes might simplify a local gcd test, but packing uses the full modulus $Q = q^2 - 1$, not merely its divisor J . For the 17-family shape the congruence is $60u \equiv M + 1 \pmod{Q}$; after removing $\gcd(60, Q)$, the modulus still has size comparable to q^2 . For the original 11-family it is $416u \equiv M + 1 \pmod{Q}$. On an unwrapped transport branch $F = L(W + Z)$ with $L = K + 2^e$ fixed,

$$R = Q\{q^2 - qLW - (1 + qL)Z\} + M.$$

The progression step as Z varies is $Q(1 + qL)$, at least of order q^3 , within a packing range below q^4 . No prime-existence argument at these actual moduli is supplied. This identifies a gap in a shortcut; it proves neither solvability nor insolvability of the interface.

43.9 Conclusions and sharply posed next questions

The structural theorem excludes all negative auxiliary factors on every genuine slice, and on a squarefree discriminant excludes both swapped main/input values $-\Delta$. It does not normalize all factors on a general zero. In particular a nonunit auxiliary square $z^2 \mid \Delta$ remains possible when Δ is not squarefree, and (43.29) remains unsettled even on the squarefree slice.

The inner family proves that the intended first/main/index/auxiliary/ strong block, with the exact main projection and compatible positive ordinary-input loading, does not force $p = R$. The old strong-rank proof relied on the deleted divisibility $\Delta c^2 \mid S$; importing that proof after freeing S would assume what is missing.

The principal remaining questions are therefore concrete.

1. For an actual genuine compiler and a known rejected x , do there exist j, N satisfying (43.45) and all four tests of Theorem 43.8.1? An affirmative result plus the rejection certificate would give a language counterexample for this candidate.
2. Does the exact compiler recipe force or forbid the shifted-mask condition (43.51)? This can eliminate genuine slices for this family without changing any numeral.
3. Can the residual congruences be excluded for every j, N in the 17-family or the original 11-family? Such an exclusion would be useful, but it would not establish general soundness.
4. Can (43.29) be resolved, and can nonsquarefree nonunit auxiliary squares and other remaining divisor branches be controlled without presuming units or $p = R$?
5. If a general soundness proof exists, can it recover a native ordinary-input witness by a new decoding argument even though the literal positive integer witness inverse fails?

Theorems about the entire positive zero set, intended-family constructions, finite checks and inherited compiler facts have different scopes. Keeping those boundaries explicit is essential. No 83-operation universal representation and no new false-accepted ordinary input is established.

43.A The original power of 11 family

The earlier independently reviewed family remains useful as a distinct construction. For $j \geq 0$ let

$$u = 11^{12j+1}, \quad p = 247u, \quad n = 208u, \quad R = 416u - 1, \quad X = 2^p, \quad Y = 2^{57u-1}.$$

Define $E, a, A, \Delta, H, P, D, c, \tau, k$ as in (43.36). The same proof gives all inner equations and the shared projection. Here the resonance is

$$p(p - n) = 57u(2n - p), \quad 247 \cdot 39 = 57 \cdot 169 = 9633,$$

and the strict ratio estimate is

$$0 < c/k - Y < 4pY/X = 494u 2^{-190u} < 1.$$

Thus $\eta, \zeta > 0$, $h = (k - R - 1)/E > 0$, and $N_{\text{index}} = 1$. Projection and input loading are exactly those of Sections 43.6 and 43.7.

To prove $\gcd(c, p) = 1$ for every j , the only primes to exclude are 11, 13 and 19. Put

$$L = \text{lcm}(12, 18, 13(13^2 - 1), 19(19^2 - 1)) = 622440.$$

Integer arithmetic gives $11^{12} \equiv 1 \pmod{L}$, so $u \equiv 11 \pmod{L}$. For primes 13 and 19, this fixes A modulo the prime through the binary exponents and fixes p modulo the order bound $|\mathrm{SL}_2(\mathbb{F}_\ell)| = \ell(\ell^2 - 1)$ for the determinant-one Pell companion matrix. At $u = 11$ the exact residues are

$$(A, c) \equiv (4, 1) \pmod{13}, \quad (A, c) \equiv (11, 18) \pmod{19}.$$

They persist at every j . Modulo 11, $u \equiv 1 \pmod{10}$ gives $A \equiv 8$ and $\Delta \equiv 8$. In the quadratic ring in characteristic 11, Frobenius gives

$$\psi_A(11r) \equiv \Delta^5 \psi_A(r) = -\psi_A(r) \pmod{11}.$$

Since $\psi_8(247) \equiv 10$ and $12j + 1$ is odd, $c \equiv 1 \pmod{11}$. This proves the universal gcd claim. Theorem 43.5.1 supplies the auxiliary extension at $R = 416u - 1$ with inverse $2D/c$ nonintegral.

The exact interface is Theorem 43.8.1 with $3dN \leq 57u - 1$ and these new p, R . Its unavoidable restrictions include $\gcd(N, 6) = 1$. Indeed d is an odd power of 5 with $5 \mid d$, so $B \equiv -1 \pmod{11}$; even N would give $11 \mid J$ but $R \equiv -1 \pmod{11}$. Also B has order 3 modulo 7, so $3 \mid N$ would give $7 \mid J$, while $u \equiv 4$ and $R \equiv 4 \pmod{7}$.

The smallest materialized fixture is $u = 11$, $p = 2717$, $n = 2288$, $R = 4575$, with Y exponent 626. Its main coefficient has 9,082,305 bits, but it still cannot complete on any genuine positive-input slice: $I \geq 15$ gives $W \geq 32768$, and positivity forces $q > W$. Even the relaxed values $q \geq 32769$, $\ell x \geq 10$, $M \geq 1$ in (43.52) force $R > 10^{18}$. The proof and independent review of this original family are preserved in the reproducibility packet. No extra offset family is used here.

43.B Exact source correspondence and evidence boundaries

43.B.1 Immutable candidate identification

The target is `complete83_free_coefficient_scout.json`. Its repository commit is

0d9d1e0174df30d82d80d758e9ba8e793203126e.

The repository directory is

Computability/HilbertTenthProblem/Papers/
research-wip/native-stream-queue.

The preserved read-only retrieval receipt records Git blob SHA-1

71edcd445eb45561613d40fbf8908d586f5a782c

The exact JSON SHA-256 is

682d37d4cedcd3c4a086ed3a2e55f272892670edb0424fc0fd8242337f1bf016

The compact canonical `packet.source` SHA-256 is

7834fa4ab4baa9faba720f572301d7f2ef5782a077c5e5adee931c89f46d240c

The independent auxiliary audit freshly fetched and matched these bytes. This report preserves that receipt and rechecks the cryptographic identities locally. It does not claim a new independent network retrieval.

43.B.2 Register correspondence

The mathematical Pell parameter $A = a + 2$ is *not* the source register named **A**; that register is Δ . The following correspondence is literal, with no evaluation of the stored arithmetic schedule.

Source register or path	Mathematical expression
q, wn2, sn2, UM	q, X, Y, E
R10b, R10a, R12, R14	k, c, a, D
a4m5, A, Ac2	$H, \Delta, \Delta c^2$
marked_rhs, W	$C, W = C - Z$
odd_index, index_rhs, exponent_rhs	I, κ, μ
gap_product, gap	$(q - 1)(q - F)$ and $q^2 - qF - Z$
mask, r_lhs	$(M_C + qM_F)J = M$ and R
index_difference, norm_index	$k - hE$ and $k - hE - R$
aux_u_rhs, R16, L16	V, S^2, f^2
norm_aux, norm_strong	$S^2(V^2 - y^2) + y^2$ and $\Delta f^2 - S^2$
polynomial	Product of all seven factors minus Δ

All 25 supplied ports are live: 18 witnesses, x , and six fixed numerals. The one-row deletion leaves the five retained factor dependency cones unchanged. The local report checker verifies the whole source fingerprint, the literal 84-to-83 deletion, topological well-formedness, liveness, operation counts and selected mathematical paths. It treats every saved row as inert data.

43.B.3 Which proof or audit supports which claim

The packet contains the full author proofs and independent review texts. The source catalogue and SHA-256 inventory distinguish original bytes from this report's new code and outputs.

1. *Auxiliary square theorem.* The two structural author notes on small-norm descent and exceptional negative exclusion are independently reviewed in `source/audits/auxiliary_square.md`. The review strengthens the divisibility conclusion to $z = \gcd(|V|, y)$ and explicitly avoids q/A parity and $p = R$ assumptions.
2. *Conditional structural results.* The even-parameter norm-value classification and all- f squarefree swapped-norm exclusions are separate proof components. Their separate PASS review is preserved at `source/audits/squarefree_addendum.md`; the earlier core audit did not cover them.
3. *Original inner theorem and interface.* The general completion, 11-family, input loading, and intended-factor iff are independently reviewed in `source/audits/inner_family.md`. Its two scope repairs are retained: projection growth begins at $r \geq 1$, and the iff fixes the prescribed R and $N_{\text{transport}} = 1$.
4. *17-family.* The simplified construction and its outer restrictions have a distinct PASS review in `source/audits/simple_family.md`. The original 11-family review is preserved independently.
5. *84 baseline.* The earlier PASS review is preserved in `source/audits/baseline84.md`. It authenticates the scaled source, whole-ring identity with the 85 parent, $47M + 37A$, 18 positive witnesses and degree 187. Its inherited full compiler chain is not freshly re-audited here.

43.C Reproduction and finite check scope

[write] The source packet is shipped with the prefix `18-free83-`: the structural notes as `18-free83-source-structure-*.md`, the inner-family proofs as `18-free83-source-inner-*.md`, the five independent reviews as `18-free83-source-audits-*.md`, the checker as `code/18-free83-check_math.py`. Its eleven research-tree byte copies under `source/immutable/` are not shipped (README), nor are four historical freeze manifests that only list hashes, most of them of files outside the curated selection. The release verifier `code/18-free83-verify_release.py` is the only one of the five batch-88 verifiers that runs on Windows as delivered; the package supplies no independent digest, so its trust anchor must be the archive’s SHA-256 (README).

The compact ZIP contains this complete LaTeX source and PDF, a frozen selection of the mathematical sources and independent reviews, an authenticated source inventory, a new standard-library checker, deterministic build scripts and receipts. Original research packets are not modified. No upload, publication or repository write is part of this work.

Only newly written report code is executed. In particular no upstream Python, compiler builder, historical verifier or saved arithmetic schedule is executed or imported. Upstream JSON is parsed only for hashes, row structure and symbolic correspondence. The numerical probes evaluate new explicit Pell recurrences and mathematical formulas.

The local checker corroborates the descent, represented norm values, residue identities, first-coordinate obstruction, 17-family modular coprimality, the exact $j = 0$ inner equations, shared input projections, auxiliary quotient congruences, and algebraic interface rewrites. It also preserves the original 11-family proof and independent exact-fixtured receipt without unnecessarily rematerializing its multi-million-bit integers. Huge auxiliary witnesses are certified by the unbounded proof and modular identities, not printed or claimed materialized.

Finite checks supplement the proofs. Their success does not prove any unbounded theorem, genuine packing solution, rejection certificate or language statement. Normal and optimized Python runs use explicit errors and must produce identical receipts. The replay script works after extraction to a separate directory and verifies source hashes before checking formulas. The PDF build uses fixed metadata and the archive uses fixed timestamps and sorted paths. Exact byte reproduction requires the recorded toolchain. All PDF pages are rendered and visually reviewed before final sealing; render images are omitted from the compact archive.

References of Report 43

- [1] V. Reshetnikov repository, *A free auxiliary coefficient: 83 paid gates, with a failed full-witness inverse*, pinned candidate note and source, commit `0d9d1e0174df30d82d80d758e9ba8e793203126e`. [Immutable candidate note](#). [write] Unchanged since the pin; the packet’s byte copies of the note and its JSON are not shipped. Refuted since by the tree’s commit `8cf6239b6` (note at the start of Part VIII).
- [2] *Modified compiler masks for the half-binomial75*, preserved inherited compiler recipe, especially Sections 1–2 for fixed widths and shifted masks; included in the source packet. [write] A byte copy of the research tree’s `complete75_half_binomial_compiler.md`; not shipped.
- [3] *Independent positive-zero proof for the auxiliary quotient85*, pinned mathematical review, especially Sections 2 and 4; included in the source packet. [write] A byte copy of the research tree’s `review_complete85_auxiliary_bezout_math.md`; not shipped.

- [4] *Free-coefficient native index mismatch, and a four-bit mask obstruction*, upstream precursor, commit 53c34b0129bbde7b5d79e520ff3e004cf6f7c07a. [Immutable precursor note](#). [write] Unchanged since the pin; the packet's byte copy is not shipped.
- [5] *Independent source audit: scaled 84-operation polynomial*, 3 October 2026, preserved in the source packet. [write] Shipped as 18-free83-source-audits-baseline84.md.
- [6] *Free-coefficient83: rigorous divisor-branch obstructions*, structural proof notes and independent audits, 3–4 October 2026; precise components and hashes are in the source inventory. [write] Shipped as 18-free83-source-structure-*.md, with the reviews 18-free83-source-audits-auxiliary_square.md and 18-free83-source-audits-squarefree_addendum.md.
- [7] *Free-coefficient83: a weakened rank condition and an infinite nonnative inner family*, with the separately audited power-of-17 addendum, 3 October 2026; included with both independent reviews. [write] Shipped as 18-free83-source-inner-PROOF.md and 18-free83-source-inner-SIMPLE_FAMILY_ADDENDUM.md, with the reviews 18-free83-source-audits-inner_family.md and 18-free83-source-audits-simple_family.md.

Report 45. Full positive collapse of the auxiliary square product 82 candidate

[write] Source 19 of this report, manuscript 06 of batch 88: *Research Report 45*, author line “Research Report 45”, dated 4 October 2026; archive `Full_Positive_Collapse_of_the_Square_Product_82_Candidate_Package.zip` (arrival commit `c5612efa1`, placed by `992aaafb3`); pin `6ce2dcaaf` (the candidate and its review); its audit’s source observation was made at repository HEAD `6bf7f30d0`. Its sections, statements and equations are numbered $45.k$ here.

Abstract of the source. For every fixed genuine numeral tuple emitted by the inherited compiler of the auxiliary square/product 82-operation candidate, and every positive ordinary input, we construct infinitely many strictly positive 18-witness tuples that zero its complete polynomial. The actual packed index is controlled by a power-of-five Chinese remainder construction. The shared input projections and retained first-index factor are restored. When the fixed mask residue is one modulo three, the index and transport factors are both -1 , so their signs cancel. A direct Pell extension at the actual packed index completes the auxiliary block with even main coordinate c , and its supplied auxiliary coordinate is nonsquare. Specialization to the genuine empty-set compiler proves failure of the inherited ordinary-input representation. A different numeral compiler is not ruled out. The validated universal baseline remains $84 = 47M + 37A$, with 18 positive witnesses and degree 187. The full proof, explicit source pins, research and independent audit packets, and bounded path-preserving replay are supplied. This is neither a smaller universal bound nor a priority claim.

[write] *The same theorem twice.* Theorem 45.1.1 and Corollary 45.1.2 state the result of the research tree’s `complete82_all_input_outer_collapse.md` (commit `dc03992d9`, 3 October 2026, 18:26 PDT): the square/product 82 candidate has infinitely many full positive zeros at every positive input on every inherited compiler slice, the empty-set compiler included. The two proofs are different constructions. The tree’s family keeps the original positive outer slack and makes all five retained factors $+1$. Report 45 uses a power-of-five radix and Chinese remaindering with the two branches $(p, n) = (4u, 3u)$ and $(20v, 14v)$; when the mask residue m_3 is 1 its index and transport factors are *both* -1 ; its main coordinate c is even, so it needs a new auxiliary completion with $i = 1$, $F_{\text{aux}} = \Delta c^4 + 1$ and the actual packed index R as the auxiliary Pell index (Lemma 45.10.1). Report 45 is therefore printed as a second, independent route to the tree’s theorem. Its audit observation (Appendix 45.A) is dated 00:44 UTC on 4 October, 17:44 PDT on 3 October, about forty minutes before `dc03992d9`; Report 45 does not cite the tree’s note, and neither text claims priority. The programme’s review (`review_incoming_arithmetic_reports_39_46.md`, commit `5d8889017`) read the complete counterfamily and calls it “a different construction of full false-input zeros for the exact 82-row source already refuted by the WIP all-input proof”; its even main rank and the matching negative signs are “useful new structure”, not a normalization theorem for every zero and not a saving in a valid universal polynomial. The review cross-read the branch CRT bounds and positivity estimates algebraically and *did not replay the archived finite receipts*.

A superseded sentence (dated note, batch 88). Section 45.1 and question 3 of Section 45.12 say that Report 43’s ordinary-input language question remains open. It was answered by the tree’s `free_coefficient83_prime_outer_collapse.md` (commit `8cf6239b6`): the free-coefficient candidate is refuted on every input (note at the start of Part VIII). The source’s statement that its own theorem settles neither direction for that candidate remains true.

A reconstructed proof. A workspace replacement interrupted the preparation of this report (Appendix 45.A). The research proof (shipped as `19-square-product82-counterfamily-COUNTERFAMILY.md`) and the audit packet were recovered from retained records and checked again; `data/19-square-product82-RECOVER.Y.json` and `data/19-square-product82-counterfamily-HISTORICAL_LOST_RELEASE.json` separate byte-identical recoveries from reconstructed files. The reconstructed proof is authenticated by fresh hashes and by the fresh independent audit (`19-square-product82-audit-audit.md`), not by the historical hashes of the lost release, which no longer match by design.

Credits and overlaps. The candidate, its gate ledger, its degree 185, the forward identity (45.16), the conditional odd- c auxiliary extension and the nonsquare zeros at accepted inputs are the tree’s (commit `6ce2dcaaf`), as

the source’s Attribution section says; Lemma 45.10.1 extends that extension to even c . Lemma 45.6.1 is the Pell estimate (41.9) of Report 41 and (43.19) of Report 43; Lemma 45.8.1 is Lemma 41.3.2 of Report 41, and through it Lemma 33.3.2 of Report 33; the odd polynomials of (45.66) are Lemma 33.3.3; the strict ratio window of Lemma 45.7.1 uses the method of Section 41.6. Report 45 cites Report 41 (“first-index-deleted circuit”, Part VII) but none of these lemmas. Report 46 (below) imports Report 45’s whole construction.

Layout. The source’s table of contents is omitted. Its Appendix 45.A read the pin table from the file `source_pins.tex` and Appendix 45.B the row list from `source_rows.txt`; both are shipped unchanged as `data/19-square-product82-source_pins.tex` and `data/19-square-product82-source_rows.txt`, and their contents are printed in place here, so that this article builds on its own. The source’s files are shipped with the prefix `19-square-product82-`; Report 46 shipped a byte copy of the whole Report 45 release, which is not shipped again.

45.1 The theorem and its scope

45.1.1 The exact candidate

The object is `complete82_auxiliary_square_product_chart` in Vladimir Reshetnikov’s *ProveIt* repository, frozen at commit

6ce2dcaaf49d28de43ce9396a094643308fb0595.

Its source and independent repository review establish a complete $82 = 45M + 37A$ arithmetic circuit, 18 strictly positive supplied witnesses, and uniform exact degree 185 [1, 2]. The two freed coordinates replace the parent products f^2 and Tf . The first-index equation remains in the circuit.

A *genuine inherited compiler tuple* is a fixed numeral tuple produced by the unchanged fixed-program compiler inherited through the validated 84-operation parent, including its half-binomial mask convention. It is not an arbitrary numerical tuple satisfying some necessary mask tests. Write $P_{82,C}$ for the polynomial specialized to such a fixed compiler tuple C .

[write] *Second route.* The research tree proved the same statement in `complete82_all_input_outer_col_lapse.md` (commit dc03992d9) by a different construction; the note at the start of this source compares the two.

Theorem 45.1.1 (Full positive counterfamily). *For every genuine inherited compiler tuple C and every $x \in \mathbb{Z}_{>0}$, there are infinitely many distinct $\xi \in \mathbb{Z}_{>0}^{18}$ such that*

$$P_{82,C}(x, \xi) = 0.$$

All six fixed numeral ports and the ordinary input are unchanged. The constructed tuples have unbounded repunit coordinate J . Each has even main Pell coordinate c , main Pell index $p \neq R$ for the actual packed index R , and nonsquare supplied auxiliary coordinate F_{aux} .

Corollary 45.1.2 (Inherited language failure). *Every genuine inherited compiler slice of this candidate accepts all positive inputs. In particular, the genuine compiler of the empty computably enumerable set gives a false positive at every positive input. Thus the candidate fails to preserve the inherited universal representation.*

Proof. The theorem gives acceptance of every $x \in \mathbb{Z}_{>0}$. The validated parent represents the empty set with its genuine compiler tuple, so it has no positive witnesses at any positive input. The candidate uses those same fixed numerals and has positive witnesses at each such input. $\square$

The empty-set specialization does not require materializing its astronomically large fixed constants. It follows from the theorem's quantifiers. This proves more than failure of a particular inverse on witnesses at an already accepted input. It does not prove that the same polynomial could never be universal under some different, unprovided numeral compiler or another allowed numeral domain. Indeed the arithmetic construction works for every fixed tuple satisfying the explicit consequences in Section 45.3; it does not classify tuples outside those hypotheses.

45.1.2 The neighboring reports remain distinct

Report 41 concerns a first-index-deleted circuit and its failures. Here the index factor, the strictly positive quotient h , and the coefficient $S = i\Delta c^2$ are retained and verified. Report 43 concerns the separate free-coefficient 83-operation candidate; its ordinary-input language question remains open. Neither its soundness nor its failure follows from the present theorem.

The validated universal baseline remains $84 = 47M + 37A$, with 18 positive witnesses and uniform exact degree 187 [3]. These are achieved counts, not lower bounds on all arithmetic circuits. The failed candidate still has formal degree 185 and 82 paid operations; those source facts do not make it a universal representation.

45.1.3 What was previously established and what is added

The repository introduced the candidate and proved its exact forward identity, a conditional auxiliary extension for odd c , and nonsquare candidate zeros at already accepted parent inputs. Its independent review explicitly left ordinary-input soundness unresolved. The present counterfamily completes the genuine outer packing, input and transport conditions for every positive input and every inherited compiler, including all three fixed-mask residues modulo three. The repository's earlier claims remain credited within their stated scope.

This report contains the entire quantified proof. Exponentiation, division by proved divisors, and Pell powering are definitions of existential witnesses, not additional free circuit operations. Full enormous witnesses need not be printed. Source Python and saved arithmetic rows are treated only as inert data; neither is run or interpreted as a schedule. The reproducible checks are bounded corroboration of the proof, not a finite-search substitute for it.

45.2 The complete source and its retained factors

45.2.1 Fixed ports and all witness coordinates

The six fixed ports, in source order, are

`Bm1, Kconstant, twice_cell_bits, inner_bits, MC, MF,`

with respective values $B - 1, K, \ell, b, M_C, M_F$. The ordinary input is x . The 18 supplied witness values, in source order, will be

$J, F, \alpha, t_{tr}, F_{aux}, h, i, U_{aux}, s, w,$
 $\tau, \eta, \zeta, y_{aux}, Z, \delta, \rho, \sigma.$

Their literal source names are

Jrep	F	alpha	transport_quotient	L16	h
i	auxiliary_Tf	s	w	tau_root	eta
zeta	y_aux	Z	delta	rho	sigma

The packing witness F is distinct from $F_{\text{aux}} = \text{L16}$. The source register **A** is Δ , not the Pell parameter A below. The source register **R14** is the main root D ; **R10a** is c , and **r_lhs** is the actual packed R .

The exact paid definitions are

$$q = (B - 1)J + 1, \quad X = wq, \quad Y = sq^3, \quad E = XY, \quad (45.1)$$

$$a_0 = Y(X + 1), \quad A = a_0 + 2, \quad H = 4a_0 + 3, \quad \Delta = A^2 - 1, \quad (45.2)$$

$$k = \eta + \zeta, \quad c = kY + \eta, \quad D = X + a_0c + (\rho + \sigma)H. \quad (45.3)$$

The source computes $\Delta = a_0^2 + H$, equal to $(a_0 + 2)^2 - 1$. The packing and input relations are

$$C = q - F - Z - \alpha - \ell x, \quad W = C - Z, \quad (45.4)$$

$$I = \ell x + b, \quad \kappa = I + \delta\Delta, \quad \mu = W + a_0\kappa + \rho H, \quad (45.5)$$

$$G = q^2 - qF - Z, \quad Q = q^2 - 1, \quad M = (M_C + qM_F)J, \quad (45.6)$$

$$R = GQ + M. \quad (45.7)$$

In particular R is not a freely supplied target. It must be restored by its actual paid expression.

45.2.2 The full output identity

The five retained factors are

$$N_{\text{first}} = \tau^2 - XY^2(XY^2 + 1)k^2, \quad (45.8)$$

$$N_{\text{main}} = D^2 - \Delta c^2, \quad (45.9)$$

$$N_{\text{input}} = \mu^2 - \Delta\kappa^2, \quad (45.10)$$

$$N_{\text{index}} = k - hE - R, \quad (45.11)$$

$$N_{\text{transport}} = (K + w)C + q - F - t_{\text{tr}}(q - 1). \quad (45.12)$$

Write $P_5 = N_{\text{first}}N_{\text{main}}N_{\text{input}}N_{\text{index}}N_{\text{transport}}$. The actual auxiliary expressions are

$$S = i\Delta c^2, \quad V = c(U_{\text{aux}} - 1) - RF_{\text{aux}}, \quad (45.13)$$

$$N_{\text{aux}} = S^2V^2 - (S^2 - 1)y_{\text{aux}}^2, \quad N_{\text{strong}} = \Delta F_{\text{aux}} - S^2. \quad (45.14)$$

The complete output is $P_5N_{\text{aux}}N_{\text{strong}} - \Delta$. With the proof expression $Q_s = F_{\text{aux}} - \Delta i^2 c^4$, this gives the all-value identity

$$\boxed{P_{82} = \Delta(P_5N_{\text{aux}}Q_s - 1)}. \quad (45.15)$$

Indeed $N_{\text{strong}} = \Delta Q_s$ identically. No zero-only substitution or division is used. A product equation is never silently replaced by the assertion that each of its factors equals +1.

The core costs $39M + 36A = 75$ operations; the full finalizer costs $6M + 1A = 7$. All 82 rows and all 25 supplied ports are live in the repository review. The complete inert row contract is reproduced in Appendix 45.B. The gate ledger, formal degree, and original forward identity are source/review results, not conclusions inferred from numerical tests in this report.

45.2.3 The forward map and its inverse restriction

The removed parent rows are $L16 = f \cdot f$ and $\text{auxiliary_Tf} = T \cdot f$. All other rows are retained. Thus

$$P_{82}(F_{\text{aux}} = f^2, U_{\text{aux}} = Tf) = P_{84} \quad (45.16)$$

over every commutative ring $[1, 2]$. On positive integer coordinates, the literal inverse requires F_{aux} to be a square and its positive square root to divide U_{aux} . The candidate supplies these coordinates without enforcing those conditions. The later nonsquare interval shows that every tuple in our family falls outside that literal image, while the all-input theorem separately establishes the language failure.

45.3 The inherited recipe consequences

Fix a genuine compiler once and for all. Its recipe gives

$$B = 2^d, \quad d = 5^a \ (a \geq 1), \quad \ell = 2d, \quad b \in \{5, 25, 125, \dots\}, \quad K > 0. \quad (45.17)$$

The inherited radix bound gives $2^b \geq 16$; together with the power-of-five choice it forces $b \geq 5$. Only odd $b \geq 3$ is needed below. The symbol a denotes the exponent in $d = 5^a$; a_0 denotes the paid Pell expression in (45.2). For the native second mask M_F^0 after the half-binomial modification,

$$M_F = M_F^0 + B - 1, \quad 0 < M_C, M_F^0 < B - 1, \quad M_C \equiv 2 \pmod{4}. \quad (45.18)$$

Here M_F^0 is not a supplied port. The source uses the shifted M_F , which can exceed B .

The residue follows directly from the authenticated recipes. The baseline native first mask is

$$M_C^{\text{old}} = B - 1 - \sum_{e \in \mathcal{E}, e \neq 1} (2^b)^e.$$

The Start exponent $e = 0$ occurs and contributes 1. All other subtracted powers are zero modulo four, while $B - 1 \equiv 3 \pmod{4}$. Thus $M_C^{\text{old}} \equiv 2 \pmod{4}$. The half-binomial compiler subtracts $2(2^b)^{e^*}$ at a nonzero ignored dummy exponent, which leaves that residue unchanged. It also modifies the native second mask and then supplies its shifted value as in (45.18) [4, 5]. The baseline chooses b and the cell-block length to be powers of five, so their product d is a power of five, and the inherited input width is $\ell = 2d$.

Every other genuine compiler restriction stays in force because none of the six fixed numerals is changed. Only (45.17)–(45.18) are used in the construction. This is why the proof applies to every genuine compiler while the bounded diagnostic masks need not themselves encode genuine programs.

45.4 A power of five radix and the two CRT constructions

Fix any $x \in \mathbb{Z}_{>0}$, and set $I = \ell x + b$ and $W = 2^I$. The number I is odd and at least three. Choose any integer $r \geq a$ large enough that

$$t = 5^r \geq 4 \max\{I, \text{bitlength}(K), 61, 16\}. \quad (45.19)$$

Here $\text{bitlength}(K) = \lfloor \log_2 K \rfloor + 1$. In particular $t \geq 625$. Put

$$q = 2^t, \quad N = 5^{r-a}, \quad J = \frac{q-1}{B-1}, \quad Q = q^2 - 1, \quad M = (M_C + qM_F)J. \quad (45.20)$$

Then $q = B^N$ and $J = 1 + B + \dots + B^{N-1} \in \mathbb{Z}_{>0}$. The length N is odd.

Lemma 45.4.1 (Radix and mask residues). *Let $m_3 \in \{0, 1, 2\}$ be the residue of $M_C + 2M_F$ modulo three. Then*

$$q \equiv 2 \pmod{5}, \quad Q \equiv 3 \pmod{5}, \quad q \in \{2, 4\} \pmod{7}, \quad Q \in \{3, 1\} \pmod{7}, \quad (45.21)$$

$$q \equiv 2 \pmod{3}, \quad J \equiv 1 \pmod{3}, \quad M \equiv m_3 \pmod{3}, \quad (45.22)$$

$$q \equiv 0 \pmod{4}, \quad J \equiv 1 \pmod{4}, \quad M \equiv 2 \pmod{4}. \quad (45.23)$$

Thus Q is a unit modulo five and seven.

Proof. Since $t = 5^r \equiv 1 \pmod{4}$, $2^t \equiv 2 \pmod{5}$. Since t is one or two modulo three, 2^t is two or four modulo seven. The numbers t, d, N are odd, so $q, B \equiv -1 \pmod{3}$; the odd-length alternating repunit sum is one modulo three. Since B, q are divisible by four, the repunit is one modulo four, and the last congruence uses $M_C \equiv 2 \pmod{4}$. $\square$

The variable Z will control the actual R through an affine expression. The CRT conditions simultaneously arrange $R \equiv 3 \pmod{4}$, a transport residue $p \equiv e \pmod{t}$, and exact matching $2n = R + \varepsilon$.

45.4.1 The nonzero residues modulo three

If $m_3 = 2$ or $m_3 = 1$, put

$$\varepsilon = \begin{cases} 1, & m_3 = 2, \\ -1, & m_3 = 1, \end{cases} \quad e = \begin{cases} 8, & K \not\equiv 1 \pmod{5}, \\ 9, & K \equiv 1 \pmod{5}. \end{cases}$$

Define

$$L = K + 2^e, \quad \beta = Q(1 + qL), \quad A_* = Q(q^2 - q(LW + 1 - \varepsilon)) + M. \quad (45.24)$$

For $e = 8$, $1 + qL \equiv 3 + 2K \pmod{5}$, which vanishes only at $K \equiv 1$. In that exceptional case $e = 9$ gives $1 + qL \equiv 2 \pmod{5}$. Therefore β is a unit modulo $t = 5^r$.

Choose the unique positive representative $Z \leq 4t$ satisfying

$$Z \equiv 1 \pmod{4}, \quad A_* - \beta Z \equiv 3e2^{-1} - \varepsilon \pmod{t}. \quad (45.25)$$

The notation 2^{-1} denotes the modular inverse of two, not a nonintegral coordinate. The second condition fixes one class modulo t , and $\gcd(4, t) = 1$ permits CRT. Define

$$F = L(W + Z) + 1 - \varepsilon, \quad (45.26)$$

$$\alpha = q - F - 2Z - W - \ell x, \quad (45.27)$$

$$R = (q^2 - qF - Z)Q + M = A_* - \beta Z. \quad (45.28)$$

Uniform positivity will be proved below. These definitions restore the source's actual $C = W + Z$ and actual packed R .

Since $Z \equiv 1 \pmod{4}$, Lemma 45.4.1 gives $R \equiv 3 \pmod{4}$. Also $Q \equiv 0 \pmod{3}$, so $R \equiv m_3 \pmod{3}$. Consequently

$$u = \frac{R + \varepsilon}{6}, \quad p = 4u, \quad n = 3u, \quad y = 2u - 1 \quad (45.29)$$

are integers. For $\varepsilon = 1$, $R \equiv 11 \pmod{12}$ and u is even; for $\varepsilon = -1$, $R \equiv 7 \pmod{12}$ and u is odd. Multiplying the second congruence in (45.25) by $2/3$ in $\mathbb{Z}/t\mathbb{Z}$ gives

$$p \equiv e \pmod{t}, \quad 2n = R + \varepsilon. \quad (45.30)$$

Both two and three are units modulo t . The actual integer p is defined by (45.29); modular division introduces no nonintegral witness.

45.4.2 The zero residue modulo three

If $m_3 = 0$, choose $h_0 \in \{1, \dots, 12\}$ such that, with $e = 5h_0$,

$$1 + q(K + 2^e) \not\equiv 0 \pmod{5}, \quad 1 + q(K + 2^e) \not\equiv 0 \pmod{7}. \quad (45.31)$$

Modulo five, $2^{5h_0} \equiv 2^{h_0}$ takes four distinct values as h_0 runs over its classes modulo four; at most one is forbidden. Modulo seven, $2^{5h_0} \equiv 2^{2h_0}$ takes three distinct values over classes modulo three; at most one is forbidden. CRT modulo twelve therefore leaves at least $(4 - 1)(3 - 1) = 6$ valid choices. In particular $5 \leq e \leq 60$.

Put $\varepsilon = 1$ and

$$L = K + 2^e, \quad \beta = Q(1 + qL), \quad A_* = Q(q^2 - qLW) + M, \quad T = t/5. \quad (45.32)$$

The coefficient β is invertible modulo seven and T . The moduli $4, 7, T$ are pairwise coprime, so choose the unique positive representative $Z \leq 28T$ with

$$\begin{aligned} Z &\equiv 1 \pmod{4}, \\ A_* - \beta Z &\equiv -1 \pmod{7}, \\ A_* - \beta Z &\equiv 7h_0 - 1 \pmod{T}. \end{aligned} \quad (45.33)$$

Define F, α, R by (45.26)–(45.28), with $\varepsilon = 1$. Then $R \equiv -1 \pmod{28}$, so

$$v = \frac{R + 1}{28}, \quad p = 20v, \quad n = 14v, \quad y = 15v - 1 \quad (45.34)$$

are integers. The last congruence yields $28v \equiv 7h_0 \pmod{T}$. Because T is a power of five, division by seven is legitimate and gives $4v \equiv h_0 \pmod{T}$. Multiplication by five now gives

$$p \equiv 5h_0 = e \pmod{t}, \quad 2n = R + 1 = R + \varepsilon. \quad (45.35)$$

The residue $R \equiv 0 \pmod{3}$ only implies $v \equiv 1 \pmod{3}$. There is no attempted inversion modulo three in this branch. In both branches y denotes the exponent for $Y = 2^y$ and is distinct from the auxiliary witness y_{aux} .

45.5 Uniform positivity and the real packed index interval

Lemma 45.5.1 (Outer positivity and size margins). *Both CRT constructions give positive integers J, F, Z, α , with*

$$1 \leq Z \leq 6t, \quad F + Z < q, \quad (2q - 1)(q^2 - 1) < R < q^4 - q^3. \quad (45.36)$$

Also $R > q^3 > 100t$, and

$$p > I, \quad p > t + e, \quad y \geq 3t. \quad (45.37)$$

Proof. The CRT bounds are $Z \leq 4t$ or $Z \leq 28t/5$, both at most $6t$. Put $U = 2^{t/4}$, a positive real used only for estimates. The sizing condition and $e \leq 60$ give $W, K, 2^e \leq U$. For $t \geq 625$, $13t < U$: the starting value is verified by $(13 \cdot 625)^4 < 2^{625}$, and $2^{t/4}/t$ increases thereafter because its logarithmic derivative is $(\log 2)/4 - 1/t > 0$. Thus $W + Z < 2U$, $L \leq 2U$, and

$$0 < F \leq 4U^2 + 2, \quad 2Z + W + \ell x \leq 13t + U < 2U.$$

As $U > 4$, $4U^2 + 2 + 2U < U^4 = q$. Equation (45.27) proves $\alpha > 0$ and also $F + Z < q$. Integrality gives $F + Z \leq q - 1$.

The actual $G = q^2 - qF - Z$ obeys

$$2q - 1 \leq G \leq q^2 - q - 1. \quad (45.38)$$

For the lower bound, $F \leq q - 1 - Z$ implies $G \geq q + (q - 1)Z \geq 2q - 1$; the upper bound uses $F, Z \geq 1$. The shifted masks satisfy

$$0 < M < (q - 1)(1 + 2q), \quad (45.39)$$

because $M_C < B - 1$, $M_F < 2(B - 1)$, and $J = (q - 1)/(B - 1)$. Thus the strict lower R bound follows, and the upper endpoint is

$$(q^2 - q - 1)(q^2 - 1) + (q - 1)(1 + 2q) = q^4 - q^3.$$

For $q \geq 3$, $(2q - 1)(q^2 - 1) > q^3$ since the difference is $q^3 - q^2 - 2q + 1 > 0$. Hence $R > 2^{3t} > 100t$.

In the first branch $u = (R + \varepsilon)/6 > R/7 > 14t$; in the second $v = (R + 1)/28 > R/28 > 3t$. Thus $u \geq 3$ or $v \geq 2$. The first branch gives $p > 56t$ and $y > 28t - 1$; the second gives $p > 60t$ and $y > 45t - 1$. Using $I \leq t/4$ and $e \leq 60 < t$ proves (45.37). $\square$

Define the supplied scales

$$w = 2^{p-t}, \quad s = 2^{y-3t}. \quad (45.40)$$

They are positive integers, and the actual source computes

$$X = wq = 2^p, \quad Y = sq^3 = 2^y. \quad (45.41)$$

There is no circular choice: first fix the compiler and input, then t, q, J , then CRT data Z, F, α, R , then p, n, y , and finally the scales and Pell coordinates.

45.6 Pell coordinates and exact resonance

For any integer $A \geq 2$, define $\chi_A(j), \psi_A(j)$ for $j \geq 0$ by

$$(A + \sqrt{A^2 - 1})^j = \chi_A(j) + \psi_A(j)\sqrt{A^2 - 1}. \quad (45.42)$$

Equivalently these are coordinates in the quadratic ring. They are integers, positive for $j \geq 1$, and satisfy

$$\chi_A(j)^2 - (A^2 - 1)\psi_A(j)^2 = 1, \quad (45.43)$$

$$\psi_A(0) = 0, \quad \psi_A(1) = 1, \quad \psi_A(j + 2) = 2A\psi_A(j + 1) - \psi_A(j). \quad (45.44)$$

Lemma 45.6.1 (Pell growth bounds). *The sequence $\psi_A(j)$ is strictly increasing for $j \geq 0$. For $j \geq 3$,*

$$(2A - 1)^{j-1} \leq \psi_A(j) < (2A)^{j-1}. \quad (45.45)$$

Also $\psi_A(j) > j$ for $j \geq 2$.

Proof. The initial values $0, 1, 2A$ increase. If the two latest positive terms increase, the recurrence gives $\psi_A(j + 1) > (2A - 1)\psi_A(j) > \psi_A(j)$ for $j \geq 2$. Iteration from $\psi_A(1) = 1$ proves the lower bound. The upper inequality $\psi_A(j + 1) < 2A\psi_A(j)$ becomes strict when the subtracted term is positive, giving the stated strict bound for $j \geq 3$. The lower estimate and the initial case $\psi_A(2) = 2A > 2$ give $\psi_A(j) > j$. $\square$

Using the actual X, Y from (45.41), define

$$a_0 = Y(X + 1), \quad A = a_0 + 2, \quad \Delta = A^2 - 1, \quad H = 4a_0 + 3, \quad (45.46)$$

$$P = 2XY^2 + 1, \quad D = \chi_A(p), \quad c = \psi_A(p), \quad (45.47)$$

$$\tau = \chi_P(n), \quad k = 2\psi_P(n), \quad E = XY. \quad (45.48)$$

Here P is a Pell parameter, distinct from P_{82} . The norm identities give exactly

$$D^2 - \Delta c^2 = 1, \quad \tau^2 - XY^2(XY^2 + 1)k^2 = 1, \quad (45.49)$$

since $P^2 - 1 = 4XY^2(XY^2 + 1)$ and $k = 2\psi_P(n)$.

Put $L_0 = 2XY$ and $M_0 = 4XY^2$. Both families satisfy the exact resonance

$$L_0^{p-1} = 2YM_0^{n-1}. \quad (45.50)$$

Taking exponents of two shows that this is equivalent to

$$p(p - n) = (y + 1)(2n - p).$$

For (45.29), both sides are $4u^2$; for (45.34), both sides are $120v^2$. Thus this is an exact integer identity rather than an asymptotic estimate. In both families $p > n$.

45.7 The strict positive ratio and the retained index factor

Lemma 45.7.1 (A strict one-unit ratio window). *The coordinates in (45.48) satisfy*

$$0 < \frac{c}{k} - Y < 1. \quad (45.51)$$

Thus $\eta = c - kY$ and $\zeta = k - \eta$ are positive integers and restore $k = \eta + \zeta$, $c = kY + \eta$.

Proof. The exact identities

$$\begin{aligned} 2A - 1 &= L_0 \left(1 + \frac{1}{X} + \frac{3}{2XY} \right), \\ 2A &= L_0 \left(1 + \frac{1}{X} + \frac{2}{XY} \right), \\ 2P &= M_0 \left(1 + \frac{1}{2XY^2} \right), \quad 2P - 1 > M_0 \end{aligned}$$

and Lemma 45.6.1, together with (45.50), give

$$\frac{c}{kY} > \frac{(1 + 1/X + 3/(2XY))^{p-1}}{(1 + 1/(2XY^2))^{n-1}} > 1.$$

The numerator base is larger, both bases exceed one, and $p > n$.

For the upper bound put $z = 1/X + 2/(XY) < 2/X$, using $Y > 2$. The upper Pell estimate for c and lower estimate for $\psi_P(n)$ give $c/(kY) < (1 + z)^{p-1}$. Since $p \geq 12$, $(p - 1)z < 2p/2^p < 1/2$. For an integer $j \geq 1$ and $0 < jz < 1/2$, the bound $\binom{j}{k} \leq j^k$ gives

$$(1 + z)^j \leq \sum_{k \geq 0} (jz)^k = \frac{1}{1 - jz} < 1 + 2jz.$$

Consequently

$$0 < \frac{c}{k} - Y < \frac{4pY}{X}. \quad (45.52)$$

The right side is $8u2^{-2u} < 1$ in the first branch for $u \geq 3$, and $40v2^{-5v} < 1$ in the second for $v \geq 2$. The starting comparisons are $24 < 64$ and $80 < 1024$; successive-term ratios are $(u+1)/(4u) < 1$ and $(v+1)/(32v) < 1$. This proves (45.51). Multiplying by $k > 0$ gives $0 < \eta < k$, and hence $\zeta = k - \eta > 0$. $\square$

As $P = 2XY^2 + 1 \equiv 1 \pmod{E}$, the recurrence for $\psi_P(n)$ reduces modulo E to the recurrence for the sequence n . Thus $\psi_P(n) \equiv n \pmod{E}$. Define

$$h = \frac{k - 2n}{E}. \quad (45.53)$$

This is an integer and is strictly positive because $n \geq 2$ and $\psi_P(n) > n$. The actual retained source factor becomes

$$N_{\text{index}} = k - hE - R = 2n - R = \varepsilon. \quad (45.54)$$

Its possible value -1 is a factor sign in the complete polynomial, not a nonpositive supplied witness. No first-index equation or domain condition has been deleted.

45.8 The exact shared main and input projections

The input root uses ρ , while the main root uses $\rho + \sigma$. Solving two Pell equations separately would not suffice; these shared projections must also be restored with positive coordinates.

Lemma 45.8.1 (Increasing integral projection quotients). *With $a_0 = A - 2$ and $H = 4A - 5$, set*

$$z_j = \chi_A(j) - a_0\psi_A(j), \quad g_j = \frac{z_j - 2^j}{H}.$$

Then every g_j is integral, $g_0 = g_1 = 0$, $g_2 = 1$, and

$$g_{j+2} = 2Ag_{j+1} - g_j + 2^j. \quad (45.55)$$

For every $j \geq 1$, $g_{j+1} > g_j \geq 0$.

Proof. The sequence z_j has initial values 1, 2 and satisfies the Pell recurrence. The sequence 2^j satisfies it modulo H , since

$$2^{j+2} - 2A2^{j+1} + 2^j = -(4A - 5)2^j = -H2^j.$$

Thus $z_j \equiv 2^j \pmod{H}$, proving integrality. Subtracting the recurrences gives (45.55). The value $g_2 = 1$ follows from $z_2 = 4A - 1$. If $g_{j+1} > g_j \geq 0$, then

$$g_{j+2} - g_{j+1} = (2A - 1)g_{j+1} - g_j + 2^j > 0.$$

Induction begins with $g_2 > g_1 = 0$. $\square$

Because $p > I \geq 3$, put

$$\rho = g_I, \quad \sigma = g_p - g_I, \quad \kappa = \psi_A(I), \quad \mu = \chi_A(I), \quad \delta = \frac{\kappa - I}{\Delta}. \quad (45.56)$$

The first two are positive integers. As I is odd, the binomial expansion of the Pell power gives

$$\psi_A(I) = \sum_{j=0}^{(I-1)/2} \binom{I}{2j+1} A^{I-2j-1} \Delta^j \equiv IA^{I-1} \equiv I \pmod{\Delta},$$

using $A^2 \equiv 1 \pmod{\Delta}$. Since $\psi_A(I) > I$, δ is a positive integer. These choices restore

$$\kappa = I + \delta\Delta, \quad (45.57)$$

$$\mu = 2^I + a_0\kappa + \rho H = W + a_0\kappa + \rho H, \quad (45.58)$$

$$D = 2^p + a_0c + g_p H = X + a_0c + (\rho + \sigma)H, \quad (45.59)$$

$$N_{\text{input}} = \mu^2 - \Delta\kappa^2 = 1. \quad (45.60)$$

The W here is the actual $C - Z$ because (45.27) fixes $C = W + Z$. The prescribed ordinary input is never changed or replaced by an independent exponent port.

45.9 Positive transport and its matching sign

The two CRT branches gave $p \equiv e \pmod{t}$ and $p > t + e$. Thus $p - t - e$ is a strictly positive multiple of t . Since $q = 2^t$,

$$w = 2^{p-t} \equiv 2^e \pmod{q-1}, \quad w > 2^e.$$

Set

$$t_{\text{tr}} = 1 + \frac{(W + Z)(w - 2^e)}{q - 1}. \quad (45.61)$$

This is a strictly positive integer. Using the literal $C = W + Z$ and $F = (K + 2^e)C + 1 - \varepsilon$,

$$\begin{aligned} N_{\text{transport}} &= (K + w)C + q - F - t_{\text{tr}}(q - 1) \\ &= (K + w)C + q - ((K + 2^e)C + 1 - \varepsilon) - (q - 1) - C(w - 2^e) \\ &= \varepsilon. \end{aligned} \quad (45.62)$$

Combining the first, main, input, index and transport factors yields

$$P_5 = 1 \cdot 1 \cdot 1 \cdot \varepsilon \cdot \varepsilon = 1. \quad (45.63)$$

For $m_3 = 1$, both the index and transport factors are -1 . This matched sign is essential to covering that entire fixed-mask residue class. Both quotients h and t_{tr} are strictly positive there. Requiring all five individual factors to be $+1$ would miss this branch of the all-input proof.

45.10 Direct auxiliary completion including even c

The repository's conditional extension assumed odd c to combine conditions modulo c and four. Every tuple here has even c . The actual $R \equiv 3 \pmod{4}$ lets us use R itself as the auxiliary Pell index, with no coprimality assumption on c .

Lemma 45.10.1 (All-parity auxiliary completion). *Let Δ, c, R be positive integers with $\Delta c^2 > 1$ and $R \equiv 3 \pmod{4}$. Define*

$$i = 1, \quad S = \Delta c^2, \quad F_{\text{aux}} = \Delta c^4 + 1, \quad V = \frac{\chi_S(R)}{S}, \quad y_{\text{aux}} = \psi_S(R). \quad (45.64)$$

Then $V \in \mathbb{Z}_{>0}$, $V \equiv -R \pmod{c}$, and

$$U_{\text{aux}} = \frac{V + RF_{\text{aux}}}{c} + 1 \quad (45.65)$$

is a positive integer. The literal argument $V = c(U_{\text{aux}} - 1) - RF_{\text{aux}}$ is restored, with $N_{\text{aux}} = 1$ and $Q_s = 1$.

Proof. Write $R = 2j + 1$. There are integer polynomials $Q_j(T)$ satisfying

$$\chi_S(2j + 1) = SQ_j(S^2), \quad (45.66)$$

$$\begin{aligned} Q_0(T) &= 1, & Q_1(T) &= 4T - 3, \\ Q_{j+2}(T) &= (4T - 2)Q_{j+1}(T) - Q_j(T), \\ Q_j(0) &= (-1)^j(2j + 1). \end{aligned} \quad (45.67)$$

The first identity and recurrence follow from the first-coordinate Pell recurrence at increments of two, whose coefficient is $2\chi_S(2) = 4S^2 - 2$. The powers one and three give the initial values. Evaluation at zero and induction prove (45.67).

Therefore the division defining V is integer-polynomial division. It is positive because the Pell coordinate is positive. Since $R \equiv 3 \pmod{4}$, j is odd. Because $c \mid S$,

$$V = Q_j(S^2) \equiv Q_j(0) = -R \pmod{c}.$$

Also $F_{\text{aux}} \equiv 1 \pmod{c}$, so $V + RF_{\text{aux}}$ is divisible by c . This numerator is positive, proving (45.65) integral and positive. Rearrangement restores the literal V . Finally the Pell norm at S gives

$$S^2V^2 - (S^2 - 1)y_{\text{aux}}^2 = 1,$$

and $Q_s = F_{\text{aux}} - \Delta i^2 c^4 = 1$ by definition. □

Apply the lemma to the already constructed actual Δ, c, R . No oddness condition on c is invoked. In fact p is divisible by four in both families, and the recurrence modulo two gives $\psi_A(p) \equiv p \pmod{2}$. Hence c is even. The main index p is even whereas the actual R is odd, so $p \neq R$.

The positive main norm also proves that F_{aux} is nonsquare. Here $c > 1$, $\Delta > 1$ and $D^2 - \Delta c^2 = 1$, so $D > c$. Directly,

$$(cD)^2 - F_{\text{aux}} = c^2 - 1 > 0, \quad (45.68)$$

$$F_{\text{aux}} - (cD - 1)^2 = c(2D - c) > 0. \quad (45.69)$$

Thus

$$(cD - 1)^2 < F_{\text{aux}} < (cD)^2. \quad (45.70)$$

These are consecutive squares. This describes how the witnesses fail the literal parent inverse; the language failure itself already follows from the all-input theorem.

45.11 All eighteen witnesses and completion of the proof

The full tuple is listed below in exact source order. No new port is introduced by these mathematical definitions.

Witness	Definition	Integrality and positivity
J	$(q - 1)/(B - 1)$ with $q = B^{5^r - a}$	Repunit identity
F	$(K + 2^e)(W + Z) + 1 - \varepsilon$	Lemma 45.5.1
α	$q - F - 2Z - W - \ell x$	Lemma 45.5.1
t_{tr}	$1 + (W + Z)(w - 2^e)/(q - 1)$	Section 45.9
F_{aux}	$\Delta c^4 + 1$	Lemma 45.10.1
h	$(k - 2n)/E$	Section 45.7
i	1	Immediate
U_{aux}	$(V + RF_{\text{aux}})/c + 1$	Lemma 45.10.1
s	2^{y-3t}	Lemma 45.5.1
w	2^{p-t}	Lemma 45.5.1
τ	$\chi_P(n)$	Positive Pell coordinate
η	$c - kY$	Lemma 45.7.1
ζ	$k - \eta$	Lemma 45.7.1
y_{aux}	$\psi_S(R)$	Positive Pell coordinate
Z	Positive CRT representative	Section 45.4
δ	$(\psi_A(I) - I)/\Delta$	Section 45.8
ρ	g_I	Lemma 45.8.1
σ	$g_p - g_I$	$p > I$ and Lemma 45.8.1

Proof of Theorem 45.1.1. Choose any sufficiently large r satisfying (45.19). The applicable CRT branch and Lemma 45.5.1 supply positive outer witnesses and scaling margins. The exact Pell resonance and bounds give positive η, ζ, h and first/main norms equal to one. The shared projection lemma and odd input index give positive δ, ρ, σ , restoring both root definitions and the input norm. The positive transport quotient gives the same sign as the retained index factor, so $P_5 = 1$. Lemma 45.10.1 supplies the remaining positive coordinates with $N_{\text{aux}} = Q_s = 1$. The complete paid identity (45.15) therefore gives $P_{82} = \Delta(1 - 1) = 0$.

Every supplied coordinate is in the list above. All fixed numerals and the ordinary input x were held fixed. Infinitely many r satisfy the sizing condition, and $J = (2^{5^r} - 1)/(B - 1)$ strictly increases with r . Thus the witness tuples are distinct and infinite. Section 45.10 establishes even c , $p \neq R$, and nonsquare F_{aux} for every member. $\square$

No numerical mask fixture was used as a genuine compiler in this argument. No full compiler execution or explicit rejection search is needed to specialize it to the genuine empty-set compiler.

45.12 Consequences and further research questions

The earlier repository projection and nonsquare extension were correct within their stated scope. The all-input construction shows that their auxiliary freedom can be coupled to the actual packing, input and transport constraints for every inherited compiler. Both the matched negative branch and the direct even- c extension are necessary to cover the full quantifier proved here.

Several questions remain open.

1. **A repair below the validated baseline.** Can a different paid arithmetic restriction replace the two deleted products with enough information to recover soundness at a lower total count? Restoring exactly f^2 and Tf returns to 84 operations, but no optimality theorem follows. A repair needs a complete source and a proof on its entire positive existential zero set.

2. **A different numeral compiler.** The theorem covers every tuple satisfying the displayed arithmetic consequences, and hence the full inherited compiler family. A proposal outside that family must specify a different effective compiler and prove both language directions. Merely changing a convenient residue does not provide such a compiler.
3. **The free-coefficient 83 candidate.** Its auxiliary structure is different. Does its outer interface admit a full false positive at a genuinely rejected input, or does another constraint restore soundness? Report 43's language question is still open and is not settled by transferring an operation count.
4. **Sign control.** Requiring both index and transport factors to equal +1 would block this family's $m_3 = 1$ branch, but not its $m_3 = 0, 2$ branches. A sign-based repair therefore needs more than an observation about the matched negative factors.
5. **Smaller examples and formal verification.** A smaller explicit counterexample on an independently certified genuine compiler slice, or a formalized proof of the quantified construction, would be useful additional evidence. Neither is needed for the empty-set specialization. Finite subsystem checks should continue to be labeled separately from full compiler examples.

These questions are not completed repairs, improved universal bounds, or novelty claims. The pins identify the object and evidence studied; they do not constitute a global priority survey.

45.A Source provenance and explicit pins

The immutable repository source is commit [6ce2dcaaf49d28de43ce9396a094643308fb0595](#). The main references are the [candidate proof and source chart](#) and the [independent repository review](#). All eight source files below were fetched read-only at that commit and matched their previously recorded SHA-256 values byte for byte. Long digests are split into two visual halves; concatenate the halves to recover the complete digest.

File	SHA-256
FIXED_RAW_UNIVERSAL_76_PROOF.md	75a13b5fd0c44c3ec91366306fb5876a 3ea2f867e464b84cebd46ca08d060f87
HALF_PARAMETER_PELL_92_PROOF.md	c1132ffe3610ff4132ca0a82312e2477 4328f4e9b3ed3b6f641055a7f29bfa7b
complete75_half_binomial_compiler.md	68edf3bc40238ccf47b023d7358e4be6 2664a2d78a60b6fae19de14b67208117
complete82_auxiliary_square_product_ch art.json	7c029ad047c9db4652621334cb4c57d1 781b5833dfac19de734ef0cb4c05fc7a
complete82_auxiliary_square_product_ch art.md	10b83a1800550674298c797e97f4e3cc adb45e7f4c532045e869e03b55caf1c9
complete82_auxiliary_square_product_ch art.py	5ff4a91d10551eafedd1d44673f64f10 aa2ed283ab071d2c969d38410999f6dc
complete84_scaled_strong_output.md	01eb7df6688c08c6788a3c7a1cc272ca 5ffc1a87734fd622dbec8a64ae974ade
review_complete82_auxiliary_square_pro duct_chart.md	02a12fd0426ad57b7c2bc822bc93266e d6511334e98cfb712cdd9f893d75722c

The source/research distinction matters. The repository review proves the candidate's formal source claims but does not contain the all-input counterfamily. That later proof and its separate mathematical

audit are preserved in the accompanying research packets, with their own file inventories. The top-level release manifest also covers the article, build utilities, evidence and receipts.

A workspace replacement interrupted report preparation. The repository snapshots were recovered exactly by their immutable commit and hashes; the research material was recovered from retained records and checked again. The release’s recovery record distinguishes any byte-identical recoveries from reconstructed files and fresh receipts. No reconstructed file is authenticated merely by assigning an old hash to it. This packaging event does not alter the theorem or its source polynomial.

The independent audit’s historical read-only source observation was recorded at 2026-10-04 00:44:40 UTC. It reported the target review path’s last commit as the frozen commit, dated 2026-10-03 23:59:58 UTC. Repository HEAD then was 6bf7f30d0352f7596e70928b3d4f304914075907, an unrelated Report 32 publication audit dated 2026-10-04 00:30:07 UTC. This is a dated identification, not a statement about later repository state or a global novelty search.

45.B The complete literal row contract

Each line below has an output name, operation, and two operands. The 82 rows are transcribed in exact order from the pinned JSON. They are inert data, never evaluated as a saved source schedule. Section 45.2 provides the exact mathematical meanings of all factor cuts and the final output.

```

1 tau_square * tau_root tau_root
2 repunit * Bm1 Jrep
3 q + repunit 1
4 Lbig * q q
5 n2 * Lbig q
6 wn2 * w q
7 sn2 * s n2
8 UM * wn2 sn2
9 R10b + eta zeta
10 ksn2 * R10b sn2
11 first_root_base * UM ksn2
12 first_next + first_root_base R10b
13 first_product * first_root_base first_next
14 norm_first - tau_square first_product
15 R10a + ksn2 eta
16 R12 + UM sn2
17 cam2 * R10a R12
18 D1 + wn2 cam2
19 gamma_sum + rho sigma
20 a4 * 4 R12
21 a4m5 + a4 3
22 gam * gamma_sum a4m5
23 R14 + D1 gam
24 L15 * R14 R14
25 a_square * R12 R12
26 A + a_square a4m5
27 c2 * R10a R10a
28 Ac2 * A c2
29 norm_main - L15 Ac2
30 norm_pair * norm_first norm_main
31 q_minus_F - q F
32 q_minus_FZ - q_minus_F Z
33 C_after_alpha - q_minus_FZ alpha
34 scaled_t * twice_cell_bits x
35 marked_rhs - C_after_alpha scaled_t
36 W - marked_rhs Z
37 odd_index + scaled_t inner_bits
38 index_product * delta A
39 index_rhs + odd_index index_product
40 difference_multiple * index_rhs R12
41 exponent_partial + W difference_multiple

```

```

42 modulus_multiple * rho a4m5
43 exponent_rhs + exponent_partial modulus_multiple
44 mu2 * exponent_rhs exponent_rhs
45 kappa2 * index_rhs index_rhs
46 scaled_kappa2 * A kappa2
47 norm_input - mu2 scaled_kappa2
48 norm_triple * norm_pair norm_input
49 aux_y2 * y_aux y_aux
50 hpm1 * h UM
51 index_difference - R10b hpm1
52 gap_product * repunit q_minus_F
53 gap + gap_product q_minus_FZ
54 Lm1 - Lbig 1
55 rproduct * gap Lm1
56 qMF * q MF
57 mask_factor + MC qMF
58 mask * mask_factor Jrep
59 r_lhs + rproduct mask
60 norm_index - index_difference r_lhs
61 kinner + Kconstant w
62 innerC * kinner marked_rhs
63 transport_partial + innerC q_minus_F
64 local_rhs * transport_quotient repunit
65 norm_transport - transport_partial local_rhs
66 auxiliary_Tf_minus_one - auxiliary_Tf 1
67 auxiliary_c_Tf * R10a auxiliary_Tf_minus_one
68 auxiliary_R_f2 * r_lhs L16
69 aux_u_rhs - auxiliary_c_Tf auxiliary_R_f2
70 H2 * aux_u_rhs aux_u_rhs
71 aux_square_gap - H2 aux_y2
72 aux_coefficient_root * i Ac2
73 R16 * aux_coefficient_root aux_coefficient_root
74 scaled_f_square * A L16
75 norm_strong - scaled_f_square R16
76 L17 * R16 aux_square_gap
77 norm_aux + L17 aux_y2
78 norm_four * norm_triple norm_aux
79 norm_product * norm_four norm_index
80 all_units * norm_product norm_transport
81 seven_units * all_units norm_strong
82 polynomial - seven_units A

```

45.C Replay and the limits of executable evidence

45.C.1 Proof and bounded corroboration

Sections 45.3–45.11 prove the unbounded quantifiers. The independent mathematical audit returned PASS after checking all CRT branches, strict positivity, source-register restoration, shared projections, the direct even- c auxiliary extension, and the paid finalizer. Its independent checker provides corroboration, not a finite proof of the theorem.

The counterfamily checker records all eight source pins, all 82 literal row definitions in order, the 18-witness interface, and the 45M + 37A ledger. Its handwritten sparse polynomial cuts include the complete 17-term output identity. Its bounded evidence comprises:

- 70 exponent-choice residue cases and all five nonzero-branch $K \bmod 5$ cases;
- 1,683 outer diagnostic cases, 561 per m_3 branch, reaching $t = 15,625$ and 62,500-bit packed R ;
- 21 exact small Pell families, 63 shared input cases, 180 auxiliary extensions including 90 with even c , and 150 projection cases.

The separately authored independent checker records 70 exponent-control cases, 2,520 outer diagnostic cases (840 per branch), 18 exact Pell subsystem fixtures, five exact even- c auxiliary completions, 288 all-parity auxiliary congruences, three source pins, and 19 critical literal rows. Its outer packed R values reach 12,500 bits.

[write] The two packets are shipped with the shorter prefixes `19-square-product82-counterfamily-` and `19-square-product82-audit-` (README). The path-preserving replay below therefore runs only in an extraction of the arrival archive on a POSIX system: on Windows the delivered verifier and `replay.py` build inventory keys with backslashes, and the audit checker writes its receipt with CRLF line ends. At placement both passed once these two portability defects were bypassed in a scratch copy, and the checkers run directly passed in normal and optimized Python with receipts equal to the frozen ones up to CRLF. The programme did not replay these receipts.

The large outer fixtures stop at manageable integers such as q, R, p, n, y . They never construct the associated 2^p or enormous Pell coordinates. Their numerical masks satisfy necessary arithmetic conditions but are not certified genuine compiler outputs. The small exact Pell fixtures test subsystem lemmas and are not presented as full genuine compiler counterexamples. The language conclusion comes from the general proof specialized to the genuine empty-set compiler.

45.C.2 Path-preserving replay

The two research packets are sibling directories under `packets/`:

```
square-product82-counterfamily-recovered-20261004
square-product82-independent-audit-recovered-20261004.
```

The sibling layout matters: the audit checker locates the counterfamily source relative to its own directory. Renaming a packet or extracting only a checker breaks that contract. The report wrapper preserves both names and copies them into a temporary replay directory before running the checkers. The audit checker's intentional receipt write is thereby isolated from the frozen release copy; the reproduced receipt must match the saved receipt.

From any working directory, replace `REPORT45` below with the absolute extracted release path:

```
python3 REPORT45/replay.py
python3 -O REPORT45/replay.py
python3 REPORT45/build_pdf.py --output /tmp/Report45.pdf
python3 REPORT45/verify_release.py
python3 REPORT45/archive_release.py --output /tmp/Report45.zip
```

Choose new PDF and ZIP output paths if those files already exist. The utilities preserve existing outputs rather than silently overwriting them. Standard-library Python is sufficient for replay, inventory verification and archive construction. Rebuilding the PDF additionally requires the recorded `pdflatex` toolchain and packages.

The wrapper checks packet hashes, runs each inspected independent checker normally and with `-O`, verifies the saved receipts, and checks all packet bytes again afterward. Upstream Python is authenticated as bytes only; the saved 82-row arithmetic schedule is never imported or interpreted. No compiler recipe, physical experiment, repository write, or public publication is part of replay.

The LaTeX build uses a fixed source epoch and suppresses volatile PDF metadata. The archive uses sorted paths, fixed timestamps, fixed permissions, and fixed compression settings. Byte-identical PDF

and ZIP reproduction is checked with the recorded toolchain. Different TeX or compression versions may change rendering bytes without changing the mathematics or source pins. The final release records the independent manuscript review, all-page visual inspection and isolated replay; temporary page images and TeX intermediates are omitted from the compact ZIP.

45.D Attribution

The candidate, source ledger, forward identity, odd- c conditional projection and earlier nonsquare observation are credited to the cited *ProveIt* source and repository review. This report presents the separately preserved counterfamily proof and independent audit as a standalone article, with expanded details and reproducible packaging. No claim is made to having introduced the classical Pell recurrences, CRT, the original chart, or a new universal-operation bound.

References of Report 45

- [1] Vladimir Reshetnikov's *ProveIt* repository, *A two-coordinate 82-gate chart that erases the auxiliary restriction*, source, receipt and proof at commit `6ce2dcaaf49d28de43ce9396a094643308fb0595`. Immutable reference in Appendix 45.A. [write] Unchanged since the pin; the packet's byte copies of the note, JSON and Python are not shipped. Refuted since (Part VIII).
- [2] *Independent review of the auxiliary square/product 82 chart*, same repository and commit. Its source count, exact degree, forward identity and conditional results are distinguished from its then-open ordinary-input question. [write] The research tree's `review_complete82_auxiliary_square_product_chart.md`, unchanged; its byte copy is not shipped.
- [3] *A complete universal polynomial in 84 operations*, `complete84_scaled_strong_output.md`, frozen source snapshot. The validated baseline and inherited compiler chain remain in force; their entire universality proof is not repeated here. [write] The research tree's file of that name, unchanged; byte copy not shipped.
- [4] *A 76-operation fixed-index universal certificate*, `FIXED_RAW_UNIVERSAL_76_PROOF.md`, frozen source snapshot, especially the fixed machine, power-of-five widths and mask recipe. [write] `Computability/HilbertTenthProblem/Papers/1980/FIXED_RAW_UNIVERSAL_76_PROOF.md`; byte copy not shipped.
- [5] *Modified compiler masks for the half-binomial75*, `complete75_half_binomial_compiler.md`, frozen source snapshot, especially the dummy-bit modification and shifted second mask. [write] The research tree's file of that name; byte copy not shipped.
- [6] *Full positive collapse of the auxiliary square/product 82 chart*, `COUNTERFAMILY.md`, 4 October 2026; research proof packet with checker and receipts. [write] Shipped as `19-square-product82-countfamily-COUNTERFAMILY.md` (a reconstruction; SHA-256 `690c5a1d...770b`, measured).
- [7] *Independent audit of the full inherited compiler collapse of square/product82*, `audit.md`, 4 October 2026; PASS mathematical review and independent checker packet. [write] Shipped as `19-square-product82-audit-audit.md`, with `19-square-product82-audit-manuscript_review.md`.

Report 46. Exact certificate height and an even-rank obstruction

[write] Source 20 of this report, manuscript 08 of batch 88: *Research Report 46*, author line “Research Report 46”, dated 4 October 2026; archive `Exact_Certificate_Height_and_an_Even_Rank_Obstruction_Package.zip` (arrival commit `c5612efa1`, placed by `992aaafb3`); pins `6ce2dcaaf` (square/product 82), `0d9d1e017` (free-coefficient 83) and `8cf6239b6` (the tree’s prime-index collapse). Its sections, statements and equations are numbered $46.k$ here.

Abstract of the source. We quantify the square/product82 counterfamily constructed in Report45. For every fixed outer tuple in that family, the displayed auxiliary completion uniquely minimizes the maximum of all 18 positive supplied witnesses, even when every positive auxiliary multiplier is allowed. Its largest coordinate is $\psi_{\Delta c^2}(R)$, with exact bit length $(R-1)(2p(p+y+1)-1)+1$. This is an exact cubic in the actual packed index R , with leading coefficient $4/3$ or $25/14$. An 18-coordinate ledger, uniform error bounds, and compiler/input bounds retain the power-of-five radix jumps. The numerical witness height is doubly exponential in the radix exponent, not triply exponential. A separately audited extension gives a convergent all-orders forward log-height formula, a convergent algebraic inverse, an explicit first exponential inverse correction, and a convergent all-orders Lagrange inverse algorithm with a geometric tail bound.

Separately, no even-main-rank outer tuple with $A \geq 4$ even, $c = \psi_A(p)$ for even $p \geq 2$, odd R , and retained factor product $P_5 = 1$ extends to a zero of the literal free-coefficient83 interface. The proof covers all positive auxiliary variables, both norm signs, and nonsquarefree $\Delta = A^2 - 1$. We prove the required small-norm descent and the correct underlying Pell-lattice reduction. These results do not minimize over different outer tuples. The even-rank obstruction is compatible with a separately audited odd-prime-index collapse of the unchanged inherited free83 recipe. Frozen source packets, independent audits, inspected bounded replay, and reproducible LaTeX accompany the proofs.

[write] *What it imports.* The source’s “Report45” is Report 45 above (Sections 45. k) and its “Report43” is Report 43 (Sections 43. k). Section 46.2 restates the construction of Sections 45.3–45.11 as a recap, and imports Theorem 45.1.1 as a proved result; Theorem 46.1.3 is about the free-coefficient interface (43.13) of Report 43. The “Report41” it distinguishes is Part VII. The certificates whose height is measured belong to a chart that is already refuted; the programme’s review calls the result “controlled witness-size information for one already-refuted chart”.

The programme’s review (`review_incoming_arithmetic_reports_39_46.md`, commit `5d8889017`) read the full even-rank proof, its source notes, the height proof and the auxiliary-minimum proof. It finds that the proof of Theorem 46.1.3 correctly passes to the fundamental Pell unit (Lemma 46.11.1) “rather than incorrectly assuming every strong solution is a power of $A + \sqrt{\Delta}$ ”, that the theorem is scoped to a fixed outer tuple and does not contradict the tree’s odd-prime counterfamily, “which changes the outer tuple”, and it summarizes Theorems 46.1.1 and 46.1.2 (11 exact lengths and 7 upper bounds; $8 \mid c$ and $0 < R < c/2$ in the minimality argument). On growth it agrees with the source: the height is doubly exponential in t and, eventually for a fixed compiler, in the numerical input x , not in the binary length of x . *The analytic logarithm and inverse expansions of Sections 46.7 and 46.8 and their claimed precision were not reviewed*; they rest on the source’s own independent audits, shipped as `20-cert-height-expansion-audit-AUDIT.md` and `20-cert-height-inverse-audit-AUDIT.md`.

Same lemma twice. Lemma 46.10.1 is Lemma 43.3.2 of Report 43, with H_0 written H ; Report 46 restates and proves it so that its theorem is self-contained, and credits Report 43. Its proof is printed as a second proof. Lemma 46.9.2 is Lemma 43.2.1 of Report 43 and Lemma 33.3.3 of Report 33; Lemma 46.9.1 is the classical classification of Pell solutions.

Notation. The source’s N_a and N_s are printed N_{aux} and N_{strong} , the renderings of Reports 43 and 45 for the same factors; this is the one rendering change of this Part. Its macro `\bl` (bit length, $\text{bl}(z) = \lfloor \log_2 z \rfloor + 1$) is kept; it is the function written bitlength in Parts I and II. Table 13 lists its other letters.

Layout and files. The source’s table of contents is omitted. Its packet directories were given shorter names when its files were shipped with the prefix `20-cert-height-` (the README maps them; for example

square-product82-height-release-20261004/ is 20-cert-height-release-). The complete Report 45 release that Report 46 carried is identical, file for file (47 files), to the contents of Report 45's own archive, shipped once under 19-square-product82- (its manuscript and PDF are not shipped; the manuscript is printed above); of the selected Report 43 sources, the manuscript and PDF are copies of Report 43's (not shipped), the early small-norm note is shipped once under 18-free83-, and the candidate JSON is a research-tree file. One independent checker hard-codes an absolute path of the author's workspace; the delivered `replay.py` patches that one line in a temporary copy (Section 46.13; README).

46.1 Results, quantifiers, and source boundary

The two circuits in this report are distinct. The *auxiliary square/product82* candidate frees the supplied products f^2 and Tf while retaining the paid auxiliary coefficient $S = i\Delta c^2$. The *free-coefficient83* candidate instead retains f and T and supplies an independent positive coefficient root S . Neither is Report41's first-index-deleted82 candidate.

Report45 proves that every genuine inherited compiler slice of the square/product82 candidate accepts every positive ordinary input by an explicit strictly positive 18-witness counterfamily [1]. We use exactly that construction. The fixed six numerals and ordinary input are never changed. Its all-input existence theorem is an imported proved result; the quantitative and nonextension results below are proved here in full. The recovered Report45 proof and all original source bytes are supplied as inert data, with exact identities in Appendix 46.A.

For a positive integer z , let $\text{bl}(z) = \lfloor \log_2 z \rfloor + 1$. The *height* of a supplied witness tuple is its numerical maximum; intermediate circuit values such as S, V, c are not extra supplied witnesses. Write y for the exponent called `yexp` in the construction, and reserve y_{aux} for its auxiliary Pell witness.

Theorem 46.1.1 (Exact height of the displayed completion). *Fix a genuine compiler, a positive ordinary input, and one Report45 outer tuple. Set*

$$L = p + y + 1, \quad C = (p - 1)L, \quad T_0 = 2pL - 1, \quad N = (R - 1)T_0.$$

For the displayed completion $i = 1$, the largest of all 18 supplied coordinates is strictly $y_{\text{aux}} = \psi_{\Delta c^2}(R)$. In particular, if $\mathcal{H}$ denotes its height, then

$$2^N < \mathcal{H} = y_{\text{aux}} < 2^{N+1}, \quad \text{bl}(\mathcal{H}) = N + 1. \quad (46.1)$$

The exact branch polynomials for $\text{bl}(\mathcal{H})$ are

$$\begin{aligned} \text{A, } \epsilon = +1 : & \quad (4R^3 + 4R^2 - 7R + 2)/3, \\ \text{A, } \epsilon = -1 : & \quad (4R^3 - 12R^2 + 9R + 2)/3, \\ \text{B :} & \quad (25R^3 + 25R^2 - 39R + 3)/14. \end{aligned} \quad (46.2)$$

The 18-row ledger in Section 46.5 has 11 exact bit lengths and seven certified upper bounds; it does not assert 18 exact lengths.

Theorem 46.1.2 (Unique minimum with the outer tuple fixed). *Fix the same 14 non-auxiliary supplied coordinates. Among every positive integer choice of $(i, F_{\text{aux}}, U_{\text{aux}}, y_{\text{aux}})$ making the complete square/product82 polynomial zero, the displayed completion is the unique minimizer of the maximum of all 18 supplied coordinates. Its minimum is exactly $\psi_{\Delta c^2}(R)$ and has (46.1)–(46.2). It also simultaneously minimizes the four auxiliary coordinates; y_{aux} and U_{aux} are uniquely minimized at this completion, while F_{aux} alone is minimal for every admissible Pell index when $i = 1$.*

Theorem 46.1.3 (Even-rank nonextension to free83). *Let $A \geq 4$ be even, $\Delta = A^2 - 1$, $p \geq 2$ even, $c = \psi_A(p)$, and R odd. Retain five outer factors with product $P_5 = 1$. There are no positive integers f, S, T, y_* satisfying*

$$V = c(Tf - 1) - Rf^2, \quad (46.3)$$

$$N_{\text{aux}} = S^2V^2 - (S^2 - 1)y_*^2, \quad N_{\text{strong}} = \Delta f^2 - S^2, \quad (46.4)$$

$$N_{\text{aux}}N_{\text{strong}} = \Delta. \quad (46.5)$$

There is no sign restriction on the intermediate integer V and no squarefreeness assumption on Δ . Thus no Report45 outer tuple can be transferred to free83 merely by changing these four auxiliaries.

The first two results are exact within each fixed outer tuple. They give no lower bound for certificates from a different outer construction. The third excludes one broad invariant sector of free83. During final preparation, a separate commit-pinned construction was independently audited: the unchanged inherited free83 compiler recipe has infinitely many full positive zeros at every positive ordinary input on every valid fixed-program slice, using prime main indices $p \equiv 1 \pmod{4}$ [12]. That all-input collapse is compatible with our nonextension theorem for fixed even main indices. It excludes neither other 83-operation circuits nor different coefficient recipes. No new universal operation bound, generic soundness theorem, or priority claim is made here.

46.2 The fixed counterfamily and the literal interfaces

We record the construction contract sufficiently explicitly to identify every coordinate used below. The full Chinese-remainder existence and positivity proof is in the supplied Report45 packet [1, 2]. Our notation deliberately separates the height exponent C from the packing expression C_{pack} .

46.2.1 Compiler and packing data

The six genuine fixed ports are $B - 1, K, \ell, b, M_C, M_F$, where

$$B = 2^d, \quad d = 5^a, \quad a \geq 1, \quad \ell = 2d, \quad K > 0,$$

$b \geq 5$ is an odd power of five, and the inherited masks satisfy $M_F = M_{F,0} + B - 1$, $0 < M_C, M_{F,0} < B - 1$, $M_C \equiv 2 \pmod{4}$. These are consequences of a genuine compiler, not a replacement definition by necessary numerical conditions.

For $x \in \mathbb{Z}_{>0}$, set $I = \ell x + b$, $W = 2^I$, and choose

$$M_* = \max\{I, \text{bl}(K), 61\}, \quad r = \max\{a, \lceil \log_5(4M_*) \rceil\}, \quad t = 5^r, \quad q = 2^t. \quad (46.6)$$

Larger r also give family members; (46.6) is the least admissible choice for the global input bounds. All height and minimality statements hold for every allowed r . Define

$$J = \frac{q-1}{B-1}, \quad Q = q^2 - 1, \quad M_{\text{mask}} = (M_C + qM_F)J, \quad m_3 = (M_C + 2M_F) \bmod 3.$$

The packing parameters e, Z, ϵ are those of Report45, with $1 \leq e \leq 60$, $1 \leq Z \leq 6t$, and

$$F = (K + 2^e)(W + Z) + 1 - \epsilon, \quad (46.7)$$

$$\alpha = q - F - 2Z - W - \ell x,$$

$$R = (q^2 - qF - Z)(q^2 - 1) + M_{\text{mask}}.$$

The residue-controlled branch parameters are

branch	R	p	n	y
A ($m_3 \in \{1, 2\}$)	$6u - \epsilon$	$4u$	$3u$	$2u - 1$
B ($m_3 = 0$)	$28v - 1$	$20v$	$14v$	$15v - 1$

(46.8)

Here $\epsilon = +1$ if $m_3 = 2$, $\epsilon = -1$ if $m_3 = 1$, and $\epsilon = +1$ in Branch B. The CRT choice ensures $p \equiv e \pmod{t}$; it uses the actual packed R above. The inherited proof supplies

$$t \geq 625, \quad 0 < J, \alpha < q, \quad F \leq 2^{t/2+2} + 2, \quad W + Z < 2^{t/4+1}, \quad (46.9)$$

$$2^{3t} < R < q^4 - q^3, \quad R \equiv 3 \pmod{4}, \quad p \equiv 0 \pmod{4}, \quad p > I, \quad p > 56t, \quad y \geq 3t. \quad (46.10)$$

In particular I is odd, $p \geq 16$, $y \geq 3$, $n \geq 3$, $R \geq 8$, and $R < 2p$ directly from (46.8).

46.2.2 All witness definitions and retained factors

For $A \geq 2$, our Pell notation is

$$(A + \sqrt{A^2 - 1})^j = \chi_A(j) + \psi_A(j)\sqrt{A^2 - 1}. \quad (46.11)$$

Both sequences obey $z_{j+2} = 2Az_{j+1} - z_j$ with initial pairs $(\chi_0, \chi_1) = (1, A)$, $(\psi_0, \psi_1) = (0, 1)$. Set

$$\begin{aligned} w &= 2^{p-t}, \quad s = 2^{y-3t}, \quad X = wq = 2^p, \quad Y = sq^3 = 2^y, \quad E = XY, \\ a_0 &= Y(X + 1), \quad A = a_0 + 2, \quad \Delta = A^2 - 1, \quad H_0 = 4a_0 + 3, \quad P = 2XY^2 + 1, \\ D &= \chi_A(p), \quad c = \psi_A(p), \quad \tau = \chi_P(n), \quad k = 2\psi_P(n), \\ \eta &= c - kY, \quad \zeta = k - \eta, \quad h = (k - 2n)/E, \\ g_j &= \frac{\chi_A(j) - a_0\psi_A(j) - 2^j}{H_0}, \quad \rho = g_I, \quad \sigma = g_p - g_I, \quad \delta = \frac{\psi_A(I) - I}{\Delta}, \\ t_{\text{tr}} &= 1 + \frac{(W + Z)(w - 2^e)}{q - 1}. \end{aligned}$$

These expressions define integers by the proved construction. The ratio estimate needed later is

$$0 < \frac{c}{k} - Y < \frac{4pY}{X} < 1. \quad (46.12)$$

Let $\kappa = \psi_A(I)$, $\mu = \chi_A(I)$, $C_{\text{pack}} = q - F - Z - \alpha - \ell x = W + Z$. The five retained factors are

$$\begin{aligned} N_{\text{first}} &= \tau^2 - XY^2(XY^2 + 1)k^2 = 1, \\ N_{\text{main}} &= D^2 - \Delta c^2 = 1, \quad N_{\text{input}} = \mu^2 - \Delta \kappa^2 = 1, \\ N_{\text{index}} &= k - hE - R = \epsilon, \\ N_{\text{transport}} &= (K + w)C_{\text{pack}} + q - F - t_{\text{tr}}(q - 1) = \epsilon. \end{aligned}$$

Their product is $P_5 = 1$, including the branch with two negative factors. The shared projections remain literally $\kappa = I + \delta\Delta$, $\mu = W + a_0\kappa + \rho H_0$, and $D = X + a_0c + (\rho + \sigma)H_0$.

46.2.3 The square/product82 auxiliary block

The exact polynomial identity is

$$S = i\Delta c^2, \quad V = c(U_{\text{aux}} - 1) - RF_{\text{aux}}, \quad (46.13)$$

$$\begin{aligned} N_{\text{aux}} &= S^2V^2 - (S^2 - 1)y_{\text{aux}}^2, \quad Q_s = F_{\text{aux}} - \Delta i^2c^4, \\ P_{82} &= \Delta(P_5N_{\text{aux}}Q_s - 1). \end{aligned} \quad (46.14)$$

In particular the strong factor is $\Delta F_{\text{aux}} - S^2 = \Delta Q_s$; we have not dropped the paid finalizer or substituted separate equations for the polynomial. The displayed completion is

$$i = 1, \quad S_0 = \Delta c^2, \quad F_{\text{aux}} = \Delta c^4 + 1, \quad V = \frac{\chi_{S_0}(R)}{S_0}, \quad y_{\text{aux}} = \psi_{S_0}(R), \quad U_{\text{aux}} = \frac{V + RF_{\text{aux}}}{c} + 1. \quad (46.15)$$

Its source-ordered 18 supplied coordinates are

$$J, F, \alpha, t_{\text{tr}}, F_{\text{aux}}, h, i, U_{\text{aux}}, s, w, \tau, \eta, \zeta, y_{\text{aux}}, Z, \delta, \rho, \sigma. \quad (46.16)$$

All are strictly positive. The four varying auxiliaries in Theorem 46.1.2 are precisely positions 5, 7, 8, 14.

46.3 Pell estimates with an explicit error budget

For $a \geq 2$ and $j \geq 3$, recurrence and strict increase give

$$(2a - 1)^{j-1} \leq \psi_a(j) < (2a)^{j-1}. \quad (46.17)$$

Indeed $\psi_2 = 2a > 2a - 1$, and $\psi_{j+1} > (2a - 1)\psi_j$ for $j \geq 2$ because $\psi_j > \psi_{j-1} > 0$. The upper recurrence inequality is strict once the subtracted term is positive. Likewise

$$a(2a - 1)^{j-1} < \chi_a(j) < a(2a)^{j-1} \quad (j \geq 2), \quad (46.18)$$

since $\chi_2/\chi_1 = 2a - 1/a > 2a - 1$ and all subsequent ratios exceed $2a - 1$ while remaining below $2a$.

Put

$$L = p + y + 1, \quad C = (p - 1)L, \quad F_0 = 4C + 2L - 2, \quad T_0 = 2pL - 1, \quad N = (R - 1)T_0. \quad (46.19)$$

The exact branch resonance is equivalently

$$L_1 = p + 2y + 2, \quad (n - 1)L_1 = C - y - 1, \quad nL_1 = pL. \quad (46.20)$$

These follow by substitution in (46.8); no approximate exponent matching is used.

Lemma 46.3.1 (Uniform perturbation bound). *If $p \geq 16$, $0 \leq z < 2^{1-p}$, and $0 \leq j \leq 4p^2$ is integral, then $1 \leq (1 + z)^j < 32/31$, except that the harmless value 1 occurs when $jz = 0$.*

Proof. The ratio of consecutive values of $p^2/2^p$ is $(p+1)^2/(2p^2) < 1$ for $p \geq 3$. Thus $jz < 8p^2/2^p \leq 1/32$ at and above $p = 16$. Using $\binom{j}{k} \leq j^k$, the binomial sum is at most the geometric series $(1 - jz)^{-1} < 32/31$ when $jz > 0$. $\square$

Set $d_0 = X^{-1} + 2/(XY)$, so

$$2A = 2^L(1 + d_0), \quad 0 < d_0 < 2^{1-p}, \quad 2A - 1 > 2^L. \quad (46.21)$$

Every perturbation exponent below is at most $4p^2$; in particular $2p(R - 1) < 4p^2$, $4p - 2 \leq 4p^2$, and $n < p$. The norm parameter satisfies

$$2^{2L-2} < \Delta < 2^{2L-2}(1 + d_0)^2. \quad (46.22)$$

The lower endpoint follows from $(XY + Y + 2)^2 - 1 > (XY)^2$.

46.4 Exact dominant bit lengths

For the displayed completion we henceforth write $S = S_0$. Equations (46.17) and (46.21) yield

$$2^C < c < 2^C(1 + d_0)^{p-1} < (32/31)2^C. \quad (46.23)$$

Combining with (46.22),

$$2^{T_0-1} < S < 2^{T_0-1}(1 + d_0)^{2p}. \quad (46.24)$$

Since S is integral, $2S - 1 > 2^{T_0}$. Applying the Pell bounds at rank R gives the strict dyadic enclosure

$$2^N < (2S - 1)^{R-1} \leq y_{\text{aux}} < (2S)^{R-1} < 2^N(1 + d_0)^{2p(R-1)} < (32/31)2^N < 2^{N+1}. \quad (46.25)$$

Thus $\text{bl}(y_{\text{aux}}) = N + 1$ exactly. Also $V = \chi_S(R)/S > (2S - 1)^{R-1} > 2^N$, while its norm gives

$$V^2 = (1 - S^{-2})y_{\text{aux}}^2 + S^{-2} < y_{\text{aux}}^2$$

because $y_{\text{aux}} > 1$. Hence $\text{bl}(V) = N + 1$ and $V < y_{\text{aux}}$.

The strong witness obeys

$$2^{F_0} < F_{\text{aux}} = \Delta c^4 + 1 < (32/31)2^{F_0} + 1 < 2^{F_0+1},$$

where the combined perturbation exponent is $4p - 2$ and $F_0 \geq 1$. Consequently $\text{bl}(F_{\text{aux}}) = F_0 + 1$.

For U_{aux} , division by c must be controlled before adding the remaining positive terms. Since $2\Delta - 1 > 2^{2L-1}$ and $2S - 1 > (2\Delta - 1)c^2$,

$$\begin{aligned} V/c &> (2\Delta - 1)^{R-1}c^{2R-3} > 2^{N-C}, \\ V/c &< (2\Delta)^{R-1}c^{2R-3} < 2^{N-C}(1 + d_0)^{2p(R-1)-p+1} < (32/31)2^{N-C}. \end{aligned}$$

The exponent identity

$$N - F_0 = (2R - 6)C + (2R - 4)L - R + 3 \geq R \quad (46.26)$$

holds for $R \geq 8$ and $C \geq 15$ (already the C term exceeds $2R - 3$). Because $F_0 \geq C$, $c < 2^{C+1}$, and $R + 1 \leq 2^{R-2}$ for $R \geq 8$, we obtain

$$RF_{\text{aux}} + c < (R + 1)2^{F_0+1} \leq 2^{N-1}.$$

Dividing by $c > 2^C$ and adding to V/c proves

$$2^{N-C} < U_{\text{aux}} < (32/31 + 1/2)2^{N-C} < 2^{N-C+1}. \quad (46.27)$$

Thus $\text{bl}(U_{\text{aux}}) = N - C + 1$. In particular the additive terms do not create a binary carry beyond this exact length.

46.5 The complete 18-coordinate ledger

Let $\lambda = \lceil \log_2(W + Z) \rceil \leq \lceil t/4 \rceil + 1$. Here and below a real logarithm merely describes an exact integer bound; a replay may compute $\lceil \log_2 z \rceil$ as $\text{bl}(z - 1)$.

Port	Coordinate	Relation	Bit length or upper bound
1	J	$\leq$	t
2	F	$\leq$	$\lceil t/2 \rceil + 3$
3	α	$\leq$	t
4	t_{tr}	$\leq$	$p - 2t + \lambda + 2$
5	F_{aux}	$=$	$4C + 2L - 1$
6	h	$=$	$C - p - 2y + 1$
7	i	$=$	1
8	U_{aux}	$=$	$N - C + 1$
9	s	$=$	$y - 3t + 1$
10	w	$=$	$p - t + 1$
11	τ	$=$	pL
12	η	$\leq$	$C - p + \lceil \log_2 p \rceil + 3$
13	ζ	$\leq$	$C - y + 1$
14	y_{aux}	$=$	$N + 1$
15	Z	$\leq$	$\lfloor \log_2(6t) \rfloor + 1$
16	δ	$=$	$(I - 3)L + 3$
17	ρ	$=$	$(I - 2)L + 1$
18	σ	$=$	$(p - 2)L + 1$

46.5.1 Packing, transport, and pure powers

The J, F, α, Z bounds are immediate from (46.9); s, w, i have their exact lengths by definition. For transport, use $q - 1 > 2^{t-1}$ and $w - 2^e < w$:

$$t_{\text{tr}} < 1 + 2^{p-2t+\lambda+1} < 2^{p-2t+\lambda+2}.$$

The exponent is positive because $p > 56t$. The table gives a convenient integer upper bound, without claiming it is sharp.

46.5.2 The first Pell block and ratio witnesses

Set $M_0 = 2^{L_1} = 4XY^2$ and $d_1 = 1/(2XY^2)$, so $2P = M_0(1 + d_1)$, $2P - 1 = M_0 + 1$, and $d_1 < 2^{1-p}$. The resonance and Lemma 46.3.1 give

$$2^{C-y} < k < (32/31)2^{C-y} < 2^{C-y+1}. \quad (46.28)$$

To justify subtracting $2n$ in h , a strict lower margin is needed:

$$\psi_P(n) \geq (M_0 + 1)^{n-1} \geq M_0^{n-1} + (n-1)M_0^{n-2} > M_0^{n-1} + n,$$

since $n \geq 3$ and $M_0 \geq 4$. Hence $k - 2n > 2M_0^{n-1} = 2^{C-y}$. Dividing by $E = 2^{L-1}$, and using (46.28) for the upper endpoint, proves the exact length of h .

From (46.18),

$$2^{nL_1-1} < \tau < 2^{nL_1-1}(1 + d_1)^n < 2^{nL_1} = 2^{pL}.$$

Thus $\text{bl}(\tau) = pL$ by (46.20). Finally (46.12) yields

$$0 < \eta < 4pkY/X < 2^{C-p+3}p, \quad 0 < \zeta < k,$$

which proves their two one-sided rows.

46.5.3 The input-loader coordinates

For odd j , put $d_j = (\psi_A(j) - j)/\Delta$. The step-two recurrence gives

$$d_1 = 0, \quad d_3 = 4, \quad d_{j+2} = (4A^2 - 2)d_j - d_{j-2} + 4j.$$

Induction gives $d_j > d_{j-2} \geq 0$ for odd $j \geq 3$ and

$$d_{j+2} > (4A^2 - 3)d_j > (2A - 1)^2 d_j.$$

Consequently

$$2^{(I-3)L+2} \leq d_I < \frac{\psi_A(I)}{\Delta} < 2^{(I-3)L+2}(1 + d_0)^{I-1} < 2^{(I-3)L+3}.$$

The lower equality is allowed at $I = 3$, where $\delta = d_3 = 4$. This proves the exact δ row in all cases.

The sequence g_j from Section 46.2 has $g_0 = g_1 = 0$, $g_2 = 1$, and

$$g_{j+1} = 2Ag_j - g_{j-1} + 2^{j-1}.$$

Its strict increase implies $g_j > (2A - 1)^{j-2}$ for $j \geq 3$. For the upper bound, $g_j \geq 2^{j-2}$ for $j \geq 2$, so $2^{j-1} \leq 2g_j$ and $g_{j+1} \leq (2A + 2)g_j$. Therefore, for $3 \leq j \leq p$,

$$(2A - 1)^{j-2} < g_j \leq (2A + 2)^{j-2}.$$

Since $2A + 2 = 2^L(1 + X^{-1} + 3/(XY))$ and the new perturbation is less than 2^{1-p} when $Y \geq 8$, the uniform budget yields

$$2^{(j-2)L} < g_j < (32/31)2^{(j-2)L}.$$

This proves the $\rho = g_I$ row. Since $I \leq p - 1$,

$$\sigma \geq g_p - g_{p-1} > (2A - 2)g_{p-1} > 2^{(p-2)L}, \quad \sigma < g_p < 2^{(p-2)L+1},$$

where $2A - 2 > 2^L$. This establishes its exact row without losing the required strict lower endpoint to subtraction.

46.5.4 Dominance and the exact cubic

Every non-auxiliary row is at most $F_0 + 1$. For the two largest inner bounds use $pL < 4C + 2L - 1$ and $\lceil \log_2 p \rceil + 3 \leq p$ for $p \geq 16$; the other comparisons follow from $I < p$, $y \geq 3t$, $p > 56t$, and $\lambda \leq \lceil t/4 \rceil + 1$. Together with (46.26), the exact lengths of F_{aux} , U_{aux} , i , and (46.25), these comparisons show that y_{aux} strictly exceeds all other supplied coordinates.

In Branch A put $z = R + \epsilon$. Then $p = 2z/3$, $L = z$, and

$$\text{bl}(\mathcal{H}) = (R - 1)\left(\frac{4}{3}(R + \epsilon)^2 - 1\right) + 1.$$

In Branch B put $z = R + 1$. Then $p = 5z/7$, $L = 5z/4$, and

$$\text{bl}(\mathcal{H}) = (R - 1)\left(\frac{25}{14}(R + 1)^2 - 1\right) + 1.$$

Expansion proves (46.2), whose integrality is guaranteed by the branch congruences. This completes Theorem 46.1.1.

46.6 Radix dependence, global bounds, and genuine jumps

The least-radix formula (46.6) implies

$$\max\{d, 4M_*\} \leq t \leq \max\{d, 20M_*\}. \quad (46.29)$$

If $t > d$, minimality gives $t/5 < 4M_*$, so $t < 20M_*$ strictly. An integer-only implementation starts at $d = 5^a$ and repeatedly multiplies by five until reaching $4M_*$; no floating-point logarithm is needed.

The packed index is much closer to q^4 than the coarse lower bound q^3 suggests. Expanding (46.7),

$$R = q^4 - q^3F - q^2(Z + 1) + qF + Z + M_{\text{mask}}.$$

All three final summands are positive, and $R < q^4$. Hence

$$0 < 1 - R/q^4 < e_t := 4 \cdot 2^{-t/2} + 2 \cdot 2^{-t} + (6t + 1)2^{-2t}. \quad (46.30)$$

For $t \geq 625$, $6t + 1 < 2^{t/2}$ and $2 \cdot 2^{-t} + 2^{-3t/2} < 2^{-t/2}$. Thus $e_t < 5 \cdot 2^{-t/2} < 1/100$.

It follows from the exact cubics, uniformly along the actual construction as $t \rightarrow \infty$, that

$$\text{bl}(\mathcal{H}) = \gamma 2^{12t} (1 + O(2^{-t/2})), \quad \log_2 \log_2 \mathcal{H} = 12t + \log_2 \gamma + o(1), \quad (46.31)$$

where $\gamma = 4/3$ in Branch A and $\gamma = 25/14$ in Branch B. The branch is fixed by the compiler's mask residue. These asymptotics supplement the exact formulas rather than supply their floors.

For a fully explicit envelope, direct subtraction in the three cubics shows for $R \geq 100$ that

$$\frac{5}{4}R^3 < \text{bl}(\mathcal{H}) < 2R^3.$$

For example the weakest lower polynomial, Branch A with $\epsilon = -1$, differs from $\frac{5}{4}R^3$ by $R^3/12 - 4R^2 + 3R + 2/3 > 0$ for $R \geq 100$; all other lower comparisons and upper comparisons are larger-margin coefficient inequalities. From $R > (99/100)q^4$ and $(5/4)(99/100)^3 > 1$, we conclude

$$2^{12t} < \text{bl}(\mathcal{H}) < 2^{12t+1}, \quad 2^{2^{12t}} < \mathcal{H} < 2^{2^{12t+1}}. \quad (46.32)$$

For the strict lower height endpoint, use integrality: $\text{bl}(\mathcal{H}) > 2^{12t}$ implies $N \geq 2^{12t}$, then $\mathcal{H} > 2^N$. There is no omitted third exponential level.

With the least allowed radix, (46.29) gives the all-input, compiler-dependent bounds

$$2^{2^{12 \max\{d, 4M_*\}}} < \mathcal{H} < 2^{2^{12 \max\{d, 20M_*\} + 1}}. \quad (46.33)$$

For every fixed compiler, eventually I dominates $\text{bl}(K)$ and 61, and $4I > d$. Thus

$$2^{2^{48I}} < \mathcal{H} < 2^{2^{240I+1}}, \quad I = \ell x + b. \quad (46.34)$$

The variable x here is the numerical input value, not its binary encoding length. Thus the double-exponential description refers to t and, eventually for a fixed compiler, the input value x ; it is not a double-exponential bound in $\text{bl}(x)$. These are bounds, not a smooth leading asymptotic in x . For example with $a = 1$, $M_* = 156$ allows $t = 625$ while $M_* = 157$ forces $t = 3125$. On each radix plateau the actual R can still change through packing data. Replacing t everywhere by one constant multiple of x would erase a real feature of the construction.

46.7 Convergent log-height expansion and a controlled inverse

This section incorporates a separately completed and independently audited extension [8, 9]. The forward result is an absolutely convergent identity in five dependent small arguments. The inverse result combines a convergent algebraic cubic inverse with the first exponentially small correction. It is not a complete all-orders exponential inverse expansion.

46.7.1 An exact identity and its convergent expansion

Keep the displayed completion, and write

$$\lambda_A = A + \sqrt{A^2 - 1}, \quad c = \frac{\lambda_A^p - \lambda_A^{-p}}{2\sqrt{A^2 - 1}}, \quad S = \frac{(\lambda_A^p - \lambda_A^{-p})^2}{4},$$

$$\lambda_S = S + \sqrt{S^2 - 1}, \quad B_* = 2p(R - 1), \quad a_* = A^{-2}, \quad b_* = S^{-2}, \quad u_* = \lambda_A^{-2p}, \quad v_* = \lambda_S^{-2R}.$$

The symbols B_*, a_*, b_* are local analytic variables, not the compiler radix or fixed compiler ports. Let

$$g(z) = \ln \frac{1 + \sqrt{1 - z}}{2} \quad (0 \leq z < 1), \quad c_j = \frac{\binom{2j}{j}}{2^j 4^j}.$$

All of d_0, a_*, b_*, u_*, v_* belong to $(0, 1)$.

Theorem 46.7.1 (Exact forward log-height formula). *The exact identity*

$$(\ln 2)(\log_2 \mathcal{H} - N) = B_*[\ln(1 + d_0) + g(a_*)] + 2(R - 1) \ln(1 - u_*)$$

$$+ Rg(b_*) - \frac{1}{2} \ln(1 - b_*) + \ln(1 - v_*) \quad (46.35)$$

has the absolutely convergent all-orders expansion

$$(\ln 2)(\log_2 \mathcal{H} - N) = \sum_{j \geq 1} \left[B_* \frac{(-1)^{j+1} d_0^j}{j} - B_* c_j a_*^j - \frac{2(R - 1) u_*^j}{j} \right.$$

$$\left. + \left(\frac{1}{2j} - R c_j \right) b_*^j - \frac{v_*^j}{j} \right]. \quad (46.36)$$

After truncation at $j = k \geq 1$, the absolute error in natural-log units is at most

$$\mathcal{R}_k = \frac{B_* d_0^{k+1}}{k + 1} + \frac{B_* a_*^{k+1}}{2(k + 1)(1 - a_*)} + \frac{2(R - 1) u_*^{k+1}}{(k + 1)(1 - u_*)}$$

$$+ \frac{(R + 1) b_*^{k+1}}{2(k + 1)(1 - b_*)} + \frac{v_*^{k+1}}{(k + 1)(1 - v_*)}. \quad (46.37)$$

Divide this bound by $\ln 2$ for the error in $\log_2 \mathcal{H}$.

Proof. The identities

$$\lambda_A = 2^L(1 + d_0)e^{g(a_*)}, \quad 2S = \lambda_A^{2p}(1 - u_*)^2/2,$$

and the auxiliary Pell formula give

$$\ln \mathcal{H} = (R - 1) \ln(2S) + Rg(b_*) - \frac{1}{2} \ln(1 - b_*) + \ln(1 - v_*).$$

Substitution proves (46.35), including the coefficient R on $g(b_*)$ and the denominator correction. Differentiating gives

$$g'(z) = -\frac{(1-z)^{-1/2} - 1}{2z},$$

so integration of the convergent binomial series with $g(0) = 0$ yields $g(z) = -\sum_{j \geq 1} c_j z^j$. The logarithm series then prove (46.36); each component is absolutely convergent. For (46.37), use the alternating remainder for $\ln(1+d_0)$, $0 < c_j \leq 1/(2j)$, and geometric tail bounds for the other arguments. For the b_* coefficient the triangle inequality gives $|1/(2j) - Rc_j| \leq (R+1)/(2j)$, requiring no guessed sign. $\square$

This formula separates main and auxiliary radical corrections from their conjugate-power corrections. Grouping them by j does not assert a single global asymptotic ordering in the ordinary input, nor remove the radix jumps. The arguments are dependent; that dependence does not affect convergence of the finite sum of convergent series.

Proposition 46.7.2 (Uniform leading correction). *For the family, and for the real interpolation below,*

$$\left| \log_2 \mathcal{H} - N - \frac{B_* d_0}{\ln 2} \right| < \frac{B_* d_0^2}{\ln 2}. \quad (46.38)$$

In particular $\log_2 \mathcal{H} - N > 0$, and

$$\log_2 \mathcal{H} - N = \frac{2p(R-1)}{\ln 2} 2^{-p} (1 + 2^{1-y} + O(d_0)).$$

The constant-one absolute remainder is the explicit statement (46.38), before this rearrangement.

Proof. Under $p \geq 16$, $y \geq 3$, $R \geq 8$, $R < 2p$, one has

$$a_*, u_*, b_*, v_* \leq d_0^2/16. \quad (46.39)$$

Indeed $a_* < 2^{-2p-2y}$ and $d_0 \geq 2^{-p}$. Also $\lambda_A > 2XY + 1 = 2^L + 1$, so $u_* < 2^{-2pL}$. For real as well as integral p , convexity gives

$$\lambda_A^{2p} > 2^{2pL} + 2p 2^{(2p-1)L} > 2^{2pL} + 2,$$

whence $S > 2^{2pL-2}$, $b_* < 2^{-4pL+4}$, and $v_* < 2^{-2R(2pL-2)}$ since $\lambda_S > S$. Each of the latter three is at most 2^{-2p-4} by the stated hypotheses, proving (46.39).

Use $|\ln(1+d_0) - d_0| \leq d_0^2/2$, $|g(z)| \leq z/[2(1-z)]$, and $|\ln(1-z)| \leq z/(1-z)$. All four small arguments are below $1/2$. In (46.35), the five error contributions after division by $B_* d_0^2$ are bounded respectively by

$$\frac{1}{2}, \quad \frac{1}{16}, \quad \frac{1}{8p}, \quad \frac{1}{16p}, \quad \frac{1}{8}.$$

Their sum is at most $179/256 < 1$ for $p \geq 16$. This proves (46.38); positivity follows from $d_0 < 1$. $\square$

46.7.2 Real analytic interpolation and the cubic branch

For each branch separately, extend p, y to affine real functions on $R \geq 100$:

	$p(R)$	$y(R)$
A+	$2(R+1)/3$	$(R+1)/3 - 1$
A−	$2(R-1)/3$	$(R-1)/3 - 1$
B	$5(R+1)/7$	$15(R+1)/28 - 1$

Use the positive real formulas above to define $S(R)$ and

$$\mathcal{H}(R) = \frac{\sinh(R\theta(R))}{\sinh \theta(R)}, \quad \theta(R) = \operatorname{arcosh} S(R) > 0.$$

At the integer family parameters this agrees exactly with the Pell witness. It is an analytic interpolation, not a claim that noninteger parameters are chart witnesses.

$A, \lambda_A, \lambda_A^p$, and then S strictly increase with R . For fixed $\theta > 0$, the hyperbolic quotient strictly increases with $R > 1$. For fixed $R > 1$, its logarithmic derivative in θ is $R \coth(R\theta) - \coth \theta > 0$, since

$$\frac{d}{dz}(z \coth z) = \frac{\sinh z \cosh z - z}{\sinh^2 z} > 0 \quad (z > 0).$$

Thus $\mathcal{H}(R)$ and $\log_2 \mathcal{H}(R)$ are strictly increasing on each branch, independently of integer bit-length results.

Let $N(R)$ be the cubic bit length minus one:

$$\begin{aligned} A+ : & (4R^3 + 4R^2 - 7R - 1)/3, \\ A- : & (4R^3 - 12R^2 + 9R - 1)/3, \\ B : & (25R^3 + 25R^2 - 39R - 11)/14. \end{aligned} \tag{46.40}$$

Direct differentiation gives, uniformly for $R \geq 100$,

$$N'(R) > 3R^2, \quad 0 < N''(R) < 12R, \quad B_*(R) < 2R^2, \quad 0 < B'_*(R)/B_*(R) \leq 3/R. \tag{46.41}$$

For example B_* is respectively $\frac{4}{3}(R^2 - 1)$, $\frac{4}{3}(R - 1)^2$, and $\frac{10}{7}(R^2 - 1)$. The cubic is increasing and unbounded on this interval.

Theorem 46.7.3 (One exponential inverse correction). *Fix an actual or interpolated $R \geq 100$, write $\ell_h = \log_2 \mathcal{H}(R)$, and let $r_0 \geq 100$ be the unique root $N(r_0) = \ell_h$. Put $p_0 = p(r_0)$, $y_0 = y(r_0)$, and*

$$\mathcal{T}(r) = \frac{2p(r)(r-1)}{(\ln 2)N'(r)} 2^{-p(r)}.$$

Then

$$R = r_0 - \mathcal{T}(r_0) + \varepsilon_h, \quad |\varepsilon_h| \leq 64(2^{-p_0-y_0} + 2^{-2p_0}). \tag{46.42}$$

Proof. Write $\ell_h = N(R) + E(R)$, where (46.38) makes $E(R) > 0$. Thus $r_0 > R$. For $\delta_r = r_0 - R$ the mean-value theorem gives $\delta_r = E(R)/N'(\xi)$, $R < \xi < r_0$. Using (46.41), $\ln 2 > 2/3$, and (46.38) gives $0 < \delta_r < 2d_0 < 1$. Put $q_* = 2^{-p(R)}$. Since $d_0 = q_*(1 + 2/Y) < 2q_*$,

$$\left| E(R) - \frac{B_*(R)q_*}{\ln 2} \right| \leq \frac{B_*(R)}{\ln 2} (2q_*/Y + 4q_*^2).$$

Also $B_*/((\ln 2)N') < 1$. Division by $N'(\xi)$ therefore contributes error at most $2q_*/Y + 4q_*^2$. Replacing its principal denominator $N'(\xi)$ by $N'(R)$ contributes at most

$$\frac{B_*q_*}{\ln 2} \frac{N'(\xi) - N'(R)}{N'(\xi)N'(R)} < \frac{4q_*\delta_r}{R} < \frac{20q_*^2}{R} \leq q_*^2/5.$$

Here $N'(\xi) - N'(R) < 12\xi(\xi - R)$ and $\delta_r < 4q_*$ suffice; the displayed 20 is deliberately loose. Furthermore

$$0 < \mathcal{T}(r) < 2^{-p(r)}, \quad |\mathcal{T}'/\mathcal{T}| \leq B'_*/B_* + N''/N' + p' \ln 2 < 7/r + 1 < 2.$$

Thus changing $\mathcal{T}(R)$ to $\mathcal{T}(r_0)$ costs less than $2q_*\delta_r < 8q_*^2$. Summing the errors gives less than $2q_*/Y + 13q_*^2$. Finally $p', y' < 1$ and $\delta_r < 1$ imply $q_* < 2 \cdot 2^{-p_0}$ and $Y^{-1} < 2 \cdot 2^{-y_0}$. The total is bounded by $8 \cdot 2^{-p_0-y_0} + 52 \cdot 2^{-2p_0}$, proving the stated looser constant 64. $\square$

The full rational prefactor in $\mathcal{T}(r)$ tends to $1/(3\ln 2)$ in either A branch and $4/(15\ln 2)$ in B. Replacing it by this limit incurs an extra $O(2^{-p_0}/r_0)$ term, generally too large to preserve (46.42). Its evaluation point and full coefficient matter.

46.7.3 Convergent algebraic inverse to all orders

Let $\gamma = 4/3$ in A and $\gamma = 25/14$ in B, and put $s_h = (\ell_h/\gamma)^{1/3}$. The large cubic root has

$$\begin{aligned} \text{A+ : } r_0 &= s_h - \frac{1}{3} + \frac{25}{36}s_h^{-1} - \frac{11}{81}s_h^{-2} + O(s_h^{-3}), \\ \text{A- : } r_0 &= s_h + 1 + \frac{1}{4}s_h^{-1} + 0s_h^{-2} + O(s_h^{-3}), \\ \text{B : } r_0 &= s_h - \frac{1}{3} + \frac{142}{225}s_h^{-1} - \frac{104}{2025}s_h^{-2} + O(s_h^{-3}). \end{aligned}$$

Direct substitution into (46.40) proves the displayed coefficients. For convergence at all algebraic orders write $N(r) = \gamma r^3 + b_2 r^2 + b_1 r + b_0$, $r = s_h h$, $w_h = 1/s_h$. The equation becomes

$$h^3 + (b_2/\gamma)w_h h^2 + (b_1/\gamma)w_h^2 h + (b_0/\gamma)w_h^3 = 1.$$

At $(w_h, h) = (0, 1)$ its derivative in h is three. The analytic implicit-function theorem gives a unique convergent power series $h(w_h)$ near zero; its recursively determined coefficients supply all algebraic orders of the large real cubic root. No explicit convergence radius is asserted. Theorem 46.7.3 adds the first controlled exponential correction beyond this algebraic branch.

46.8 A convergent all-orders analytic inverse algorithm

A further separately audited extension [10, 11] supplies an all-orders analytic inverse with an explicit geometric tail. It is a Lagrange derivative algorithm in an auxiliary parameter; it does not claim that every inverse exponential sector has been rearranged into a canonical ordered transseries.

46.8.1 Holomorphic formulas without logarithmic winding

Fix one branch and a real center $r_0 \geq 100$. On the closed complex disk $|r - r_0| \leq 1/4$, write

$$p(r) = \alpha_*(r + \epsilon), \quad L(r) = \lambda_*(r + \epsilon), \quad y(r) = L(r) - p(r) - 1,$$

where $(\alpha_*, \lambda_*, \epsilon)$ is $(2/3, 1, 1)$, $(2/3, 1, -1)$, or $(5/7, 5/4, 1)$. Put $p_0 = p(r_0)$, $y_0 = y(r_0)$, $q_0 = 2^{-p_0}$. Here 2^z means $\exp(z \ln 2)$. Define, in this order,

$$\begin{aligned} d_c &= 2^{-p} + 2 \cdot 2^{-p-y}, & a_c &= 2^{-2p-2y}(1 + d_c)^{-2}, \\ \ell_A &= L \ln 2 + \ln(1 + d_c) + g(a_c), & u_c &= \exp(-2p\ell_A), \\ b_c &= 16u_c^2/(1 - u_c)^4, & \ell_S &= 2p\ell_A - \ln 2 + 2 \ln(1 - u_c) + g(b_c), \\ v_c &= \exp(-2r\ell_S). \end{aligned}$$

The logarithms $\ln(1 + z)$ and $\ln(1 - z)$ are their Taylor branches at zero, and g is the convergent series from Theorem 46.7.1. These formulas specify holomorphic logarithms ℓ_A, ℓ_S ; they never take a principal log S or principal arcosh S on the complex disk, where the values of S may wind around zero. Let $N(r)$ be the branch cubic (46.40), and define

$$\begin{aligned} E_c(r) &= \frac{1}{\ln 2} \{ 2p(r)(r - 1)[\ln(1 + d_c) + g(a_c)] + 2(r - 1)\ln(1 - u_c) \\ &\quad + rg(b_c) - \frac{1}{2} \ln(1 - b_c) + \ln(1 - v_c) \}. \end{aligned} \tag{46.43}$$

On the real axis this is exactly $\log_2 \mathcal{H}(r) - N(r)$. We prove all required branch and nonvanishing assertions, rather than assuming that these formulas are well defined.

Lemma 46.8.1 (Uniform complex-disk bounds). *On a neighborhood of $|r - r_0| \leq 1/4$, E_c and*

$$D_c(r; r_0) = \frac{N(r) - N(r_0)}{r - r_0}, \quad D_c(r_0; r_0) = N'(r_0), \quad f(r) = E_c(r)/D_c(r; r_0)$$

are holomorphic, and on the closed disk

$$|E_c(r)| < 18r_0^2 q_0, \quad |D_c(r; r_0)| > 2r_0^2, \quad |f(r)| < \varepsilon_0 := 16q_0 < 1/8. \quad (46.44)$$

Proof. Uniformly, $p_0 \geq 66$, $y_0 \geq 32$, and $|p(r)| < r_0$. The slopes of p, y are positive and less than one, so

$$|2^{-p}| \leq 2^{1/4} q_0 < (5/4)q_0, \quad |2^{-y}| < (5/4)2^{-y_0}, \quad |d_c| < 2q_0 \leq 2^{-65}.$$

Also $|2^{-2p-2y}| < 4q_0^2 2^{-2y_0}$ and $|1 + d_c|^{-2} < 2$, whence $|a_c| < q_0^2$. The convergent series give, with $\Phi = \ln(1 + d_c) + g(a_c)$,

$$|\ln(1 + d_c)| < 4q_0, \quad |g(a_c)| < q_0^2, \quad |\Phi| < 5q_0. \quad (46.45)$$

Write $h_c = r - r_0$ and $z_0 = r_0 + \epsilon$. The inequality $\operatorname{Re}((z_0 + h_c)^2) \geq z_0^2 - 2z_0|h_c| - |h_c|^2$ and $2/3 \leq \alpha_* \lambda_* \leq 25/28$ imply

$$\operatorname{Re}(pL) \geq \frac{2}{3} \left(r_0^2 - \frac{5}{2}r_0 + \frac{7}{16} \right) \geq \frac{3}{5}r_0^2.$$

The last comparison follows by direct subtraction for $r_0 \geq 100$. As $5r_0 q_0 < 1$, (46.45) gives

$$\operatorname{Re}(p\ell_A) \geq \frac{3}{5}r_0^2 \ln 2 - 1 > \frac{1}{2}r_0^2 \ln 2.$$

Therefore

$$|u_c| < 2^{-r_0^2} \leq q_0^4, \quad |b_c| \leq \frac{16q_0^8}{(1 - q_0^4)^4} < 32q_0^8 < q_0^4. \quad (46.46)$$

Here $p_0 < r_0$ and $r_0^2 \geq 4r_0$ justify the first comparison. This validates the Taylor branches involving u_c, b_c . In particular

$$\ell_S = 2pL \ln 2 - \ln 2 + Q_c, \quad Q_c = 2p\Phi + 2\ln(1 - u_c) + g(b_c), \quad |Q_c| < 11r_0 q_0.$$

The cubic $r(r + \epsilon)^2$ differs from its center value by at most

$$\frac{1}{4}(3r_0^2 + 4r_0 + 1) + \frac{1}{16}(3r_0 + 2) + \frac{1}{64} < r_0^2.$$

Since $\alpha_* \lambda_* < 1$, it follows that

$$\operatorname{Re}(rpL) \geq \frac{2}{3}r_0(r_0 - 1)^2 - r_0^2 \geq \frac{3}{5}r_0^3.$$

Using $22r_0^2 q_0 < 1$ and $|r| \leq r_0 + 1/4$ gives

$$\operatorname{Re}(r\ell_S) \geq \frac{6}{5}r_0^3 \ln 2 - (r_0 + 1/4) \ln 2 - 1 > r_0^3 \ln 2, \quad |v_c| < 2^{-2r_0^3} \leq q_0^4.$$

The elementary inequalities $5r_0 q_0 < 1$ and $22r_0^2 q_0 < 1$ hold at $r_0 = 100$ using $q_0 \leq 2^{-66}$; their left sides decrease thereafter because $p'_0 \geq 2/3$.

Now $|2p(r-1)| < 2r_0^2$ and $|r-1| < r_0$. The first numerator term in (46.43) is less than $10r_0^2q_0$. By (46.46), the remaining terms have total modulus below

$$4r_0q_0^4 + 2r_0q_0^4 + q_0^4 + 2q_0^4 = (6r_0 + 3)q_0^4 < r_0^2q_0.$$

Since $1/\ln 2 < 3/2$, this gives $|E_c| < (33/2)r_0^2q_0 < 18r_0^2q_0$.

If $\gamma < 2$ is the cubic's leading coefficient, then exactly

$$D_c(r_0 + h_c; r_0) = N'(r_0) + \frac{1}{2}N''(r_0)h_c + \gamma h_c^2.$$

The real derivative bounds imply $|D_c| > 3r_0^2 - (3/2)r_0 - 1/8 > 2r_0^2$. Thus D_c never vanishes and $|f| < 9q_0 < 16q_0$. All series arguments have modulus strictly below one, with strict uniform bounds on the compact disk. Those bounds persist on a slightly larger disk, proving holomorphy on a neighborhood of its closure. $\square$

Theorem 46.8.2 (Lagrange inverse with an explicit tail). *Put $\rho_0 = 1/(8\varepsilon_0) > 1$. The unique small solution of*

$$h(z) = -z f(r_0 + h(z)), \quad h(0) = 0, \quad (46.47)$$

is holomorphic on a neighborhood of $|z| \leq \rho_0$ and has $|h(z)| < 1/4$ there. Its convergent Lagrange series is

$$h(z) = \sum_{n \geq 1} \frac{(-z)^n}{n!} \left. \frac{d^{n-1}}{dr^{n-1}} [f(r)^n] \right|_{r=r_0}. \quad (46.48)$$

At $z = 1$, $r = r_0 + h(1)$ is the exact inverse of $\log_2 \mathcal{H}(r) = N(r)$ in the disk. For every $K \geq 0$,

$$\left| h(1) - \sum_{n=1}^K \frac{(-1)^n}{n!} \left. \frac{d^{n-1}}{dr^{n-1}} [f(r)^n] \right|_{r=r_0} \right| \leq \frac{1}{4} \frac{\rho_0^{-K-1}}{1 - \rho_0^{-1}}. \quad (46.49)$$

Proof. On $|h| = 1/4$ and $|z| \leq \rho_0$, $|zf(r_0 + h)| < \rho_0\varepsilon_0 = 1/8 < |h|$. Rouché's theorem gives exactly one zero, counting multiplicity, of $h + zf(r_0 + h)$ in the disk. It is therefore simple. The analytic implicit-function theorem, together with uniqueness, patches the local solution into one holomorphic function throughout the parameter disk and its neighborhood. Lagrange inversion at zero gives (46.48). At $z = 1$, multiplying (46.47) by $D_c(r_0 + h; r_0)$ gives $N(r_0 + h) + E_c(r_0 + h) = N(r_0)$, the desired inverse equation. Finally Cauchy's coefficient estimate on $|z| = \rho_0$ is $|[z^n]h(z)| \leq (1/4)\rho_0^{-n}$. Summing its geometric tail proves (46.49) and absolute convergence at $z = 1$. $\square$

For an actual $R \geq 100$, take $N(r_0) = \log_2 \mathcal{H}(R)$ as in Theorem 46.7.3. Its localization gives $0 < r_0 - R < 2d_0 < 1/4$, so this real R is the unique root above; conjugation symmetry also makes $h(1)$ real. At an arbitrary center $r_0 = 100$, the inverse of the target $N(100)$ may instead lie slightly below 100. The disk theorem then concerns the local analytic continuation, not an inverse constrained to the half-line $R \geq 100$. This endpoint caveat does not affect the application starting from an actual $R \geq 100$, where $r_0 > R \geq 100$.

The first Lagrange term is $-E_c(r_0)/N'(r_0)$ and retains all forward scales. Its leading 2^{-p_0} part agrees with Theorem 46.7.3, while (46.49) makes the $n \geq 2$ terms uniformly $O(q_0^2)$. Every finite derivative in (46.48) is computable by power-series arithmetic on the explicitly specified holomorphic functions. This is a genuinely convergent all-orders algorithm with a computable error bound. Its ordering is by the artificial parameter z , and it does not smooth the input's radix jumps.

46.9 All positive auxiliary completions and unique minimality

We now hold the entire outer tuple fixed and allow every positive auxiliary choice in (46.13). First, $A \equiv 2 \pmod{4}$ and $p \equiv 0 \pmod{4}$. Modulo eight, the ψ_A recurrence repeats 0, 1, 4, 7, so $8 \mid c$. Moreover

$$c \geq (2A - 1)^{p-1} \geq 3^{p-1} > 4p > 2R$$

for $p \geq 12$. In particular $0 < R < c/2$.

Lemma 46.9.1 (Integer-coordinate Pell classification). *For integral $S \geq 2$, every solution of $x^2 - (S^2 - 1)y_*^2 = 1$ with $x, y_* > 0$ is $(x, y_*) = (\chi_S(m), \psi_S(m))$ for a unique $m \geq 1$.*

Proof. The discriminant is strictly between $(S - 1)^2$ and S^2 . For $y_* = 1$ necessarily $x = S$. For $y_* > 1$, $\sqrt{S^2 - 1}y_* < x < Sy_*$ and $x > (S - 1)y_*$. Multiplication by $S - \sqrt{S^2 - 1}$ gives integral coordinates

$$x' = Sx - (S^2 - 1)y_* > 0, \quad 0 < y'_* = Sy_* - x < y_*.$$

They have the same norm. Descent ends at $(S, 1)$ and then $(1, 0)$; reversal proves the classification, and strict growth gives uniqueness. Only the integer-coordinate Pell problem is used, not a claim about all algebraic integers of the quadratic field. $\square$

Lemma 46.9.2 (Odd quotient polynomial). *There are integer polynomials Q_h such that*

$$\chi_S(2h + 1) = SQ_h(S^2), \quad Q_0(Z) = 1, \quad Q_1(Z) = 4Z - 3, \quad Q_{h+2} = (4Z - 2)Q_{h+1} - Q_h.$$

They satisfy

$$Q_h(0) = (-1)^h(2h + 1), \quad Q_h(1 - A^2) = (-1)^h\psi_A(2h + 1). \quad (46.50)$$

Proof. The step-two recurrence for χ proves the first assertion and the two initial values. The constant term follows by induction. For the second identity, the sequence $(-1)^h\psi_A(2h + 1)$ has initial values 1, $1 - 4A^2$ and recurrence coefficient $2 - 4A^2$, exactly the specialized polynomial recurrence. $\square$

Proposition 46.9.3 (Complete auxiliary classification). *For the fixed Report45 outer tuple, all positive completions are parametrized by an arbitrary $i \in \mathbb{Z}_{>0}$ and a positive rank $m \equiv R$ or $-R \pmod{c}$. They are exactly*

$$S = i\Delta c^2, \quad F_{\text{aux}} = \Delta i^2 c^4 + 1, \quad y_{\text{aux}} = \psi_S(m), \quad V = \chi_S(m)/S, \\ U_{\text{aux}} = (V + RF_{\text{aux}})/c + 1.$$

Their least rank is R and their next rank is $c - R$.

Proof. Because $P_5 = 1$, the zero condition is $N_{\text{aux}}Q_s = 1$. Both factors are integers. Since $4 \mid S$, $N_{\text{aux}} \equiv y_{\text{aux}}^2 \pmod{4}$, so $N_{\text{aux}} = -1$ is impossible. Necessarily $N_{\text{aux}} = Q_s = 1$, forcing the stated F_{aux} . Apply Lemma 46.9.1 to $x = |SV|$ and $y_{\text{aux}} > 0$; this yields $SV = \pm\chi_S(m)$ and $y_{\text{aux}} = \psi_S(m)$.

The recurrence modulo S gives $\chi_S(2j) \equiv (-1)^j$ and $\chi_S(2j + 1) \equiv 0$. Therefore m must be odd. Writing $m = 2h + 1$, let $v_m = \chi_S(m)/S > 0$. Since $c \mid S$, Lemma 46.9.2 implies $v_m \equiv (-1)^h m \pmod{c}$, and hence $v_m \equiv 1 \pmod{4}$. On the other hand the literal V relation requires $V \equiv -R \equiv 1 \pmod{4}$, since $F_{\text{aux}} \equiv 1 \pmod{c}$ and $R \equiv 3 \pmod{4}$. Thus $V = -v_m$ is impossible; all signs have been treated.

For positive V , the congruence holds precisely when $m \equiv R \pmod{c}$ with $m \equiv 3 \pmod{4}$, or $m \equiv -R \pmod{c}$ with $m \equiv 1 \pmod{4}$. As $4 \mid c$, these are simply the stated two residue classes. They make U_{aux} integral and positive, and the resulting norm and strong factor are one, proving sufficiency as well as necessity. Finally $0 < R < c/2$ identifies the first two positive ranks as $R, c - R$. $\square$

Proof of Theorem 46.1.2. At fixed S , both Pell coordinates strictly increase with rank. For $m \geq 2$,

$$\psi_S(m) = \sum_{\substack{1 \leq k \leq m \\ k \text{ odd}}} \binom{m}{k} S^{m-k} (S^2 - 1)^{(k-1)/2}$$

is strictly increasing in real $S \geq 2$: all terms are nondecreasing and the $k = 1$ term is strictly increasing. For odd $m \geq 3$,

$$\chi_S(m)/S = \sum_{\substack{0 \leq k \leq m-1 \\ k \text{ even}}} \binom{m}{k} S^{m-k-1} (S^2 - 1)^{k/2}$$

has the same property. Every admissible rank is at least $R \geq 3$, and $S = i\Delta c^2$ increases strictly with i . Thus y_{aux} and V are strictly larger unless $i = 1, m = R$. The strong witness strictly increases with i , and $U_{\text{aux}} = (V + RF_{\text{aux}})/c + 1$ has the same unique minimum. i itself is minimal at one and F_{aux} at every admissible rank with $i = 1$, establishing the simultaneous-coordinate assertion.

Let $\mathcal{H}_* = \psi_{\Delta c^2}(R)$. Every completion has $y_{\text{aux}} \geq \mathcal{H}_*$, strictly unless it is the displayed one. Theorem 46.1.1 shows that this displayed y_{aux} strictly exceeds all fixed outer coordinates and the other three auxiliaries. Hence it attains the minimum of the maximum of all 18 coordinates, and every other completion has strictly larger maximum. Proposition 46.9.3 fixes all remaining coordinates uniquely. $\square$

46.10 Small-norm descent for the free83 interface

To make the nonextension theorem self-contained, we prove the exact arithmetic lemma imported by its original research packet from Report43 [7, 6]. The variable y_* below is the positive auxiliary coordinate, not the construction exponent y .

[write] *Second proof (source 20).* The next lemma is Lemma 43.3.2 of Report 43, with that lemma's H_0 , N and y written H , N_* and y_* ; Report 46 restates and proves it to keep Theorem 46.1.3 self-contained, and credits Report 43 [7]. The two descents are the same transformation (43.21). The programme's review read Report 43's version directly.

Lemma 46.10.1 (Small norms and the negative equality orbit). *Let $H \geq 2$, $v \geq 0$, $y_* > 0$ be integers and $N_* = Hv^2 - (H - 1)y_*^2$.*

1. *If $N_* < 0$, then $N_* \leq -(H - 1)$.*
2. *If $0 < N_* < H$, then N_* is a positive perfect square.*
3. *If $N_* = -(H - 1)$, every nonnegative- v solution is*

$$v = 2(H - 1)\psi_{2H-1}(r), \quad y_* = \chi_{2H-1}(r), \quad r \geq 0.$$

Proof. If $N_* < 0$, then $y_* > v$. Put $t_* = y_* - v \geq 1$; then

$$-N_* = (H - 1)t_*^2 + v(2(H - 1)t_* - v). \quad (46.51)$$

If $v \leq 2(H - 1)t_*$, this is at least $H - 1$. Otherwise negativity implies

$$v < (H - 1 + \sqrt{H(H - 1)})t_* < (2H - 1)t_*.$$

The integral transformation

$$\begin{aligned} v' &= (2H - 1)v - 2(H - 1)y_* = v - 2(H - 1)t_*, \\ y'_* &= (2H - 1)y_* - 2Hv = (2H - 1)t_* - v \end{aligned}$$

has $0 < v' < v$, $y'_* > 0$, and preserves N_* by expansion. Induction on v proves the first assertion.

If $0 < N_* < H$, then $v \geq y_* + 1$ would give $N_* = H(v^2 - y_*^2) + y_*^2 \geq H$, impossible. Thus $y_* \geq v$. When $y_* = v$ the norm is v^2 . Otherwise $t_* \geq 1$ and positivity in (46.51) requires $v > 2(H - 1)t_*$. Also $v \geq (2H - 1)t_*$ would give $N_* \geq Ht_*^2 \geq H$, since the quadratic in v is increasing beyond its positive root. The same descent has positive coordinates and smaller v , so eventually $y_* = v$; the norm is a square.

For $N_* = -(H - 1)$, descend while $v > 2(H - 1)t_*$. At a terminal point (46.51) forces $t_* = 1$ and $v = 0$ or $2(H - 1)$. The latter maps to $(v', y'_*) = (0, 1)$. The inverse transformation has matrix

$$\begin{pmatrix} 2H - 1 & 2(H - 1) \\ 2H & 2H - 1 \end{pmatrix}.$$

Its iterates on $(0, 1)$ are exactly the displayed Pell formula, as verified by the initial value and recurrence. Conversely every such iterate has the required norm. This includes $v = 0$ at $r = 0$. $\square$

Proposition 46.10.2 (Exhaustion of norm signs). *Under (46.4)–(46.5) with $A \geq 4$, $f, S, y_* > 0$, one has $S \geq 2$. If $N_{\text{aux}} > 0$, then $N_{\text{aux}} = z^2$ for some positive integer z with $z^2 \mid \Delta$. If $N_{\text{aux}} < 0$, then necessarily*

$$f = 1, \quad S = A, \quad N_{\text{strong}} = -1, \quad N_{\text{aux}} = -\Delta, \quad V = \pm 2\Delta\psi_{2A^2-1}(r) \quad (r \geq 0). \quad (46.52)$$

Proof. Both norm factors are nonzero divisors of Δ and have the same sign. If $S = 1$, then $N_{\text{strong}} = \Delta f^2 - 1 > 1$ is coprime to Δ , contradicting divisibility. Set $H = S^2 \geq 4$.

Suppose $N_{\text{aux}} < 0$. The small-norm lemma gives $|N_{\text{strong}}|(H - 1) \leq \Delta$, in particular $H - 1 \leq \Delta$. If $f \geq 2$, the strong equation and $|N_{\text{strong}}| \leq \Delta$ instead give $H = \Delta f^2 - N_{\text{strong}} \geq 3\Delta$, a contradiction. Thus $f = 1$. Write $b_* = -N_{\text{strong}} \geq 1$; then $H = \Delta + b_*$ and $b_*(\Delta + b_* - 1) \leq \Delta$, forcing $b_* = 1$. Consequently $S = A$, $N_{\text{strong}} = -1$, $N_{\text{aux}} = -\Delta$. The equality orbit of Lemma 46.10.1, with $v = |V|$, gives the last formula.

Suppose $N_{\text{aux}} > 0$; then $N_{\text{strong}} > 0$. We show $N_{\text{aux}} < H$. If $f \geq 2$, $H \geq 3\Delta > \Delta/N_{\text{strong}} = N_{\text{aux}}$. If $f = 1$, then $\Delta = H + N_{\text{strong}}$. The value $N_{\text{strong}} = 1$ would imply $A^2 - S^2 = 2$, impossible for positive integers, so $N_{\text{strong}} \geq 2$. Now $N_{\text{strong}}H > H + N_{\text{strong}} = \Delta$, because $H \geq 4$, giving $H > \Delta/N_{\text{strong}} = N_{\text{aux}}$ again. Lemma 46.10.1 makes N_{aux} a positive square, and its divisibility gives $z^2 \mid \Delta$. $\square$

46.11 The nonsquarefree Pell lattice and residue separation

A tempting but incorrect shortcut is to assume every strong- Δ solution has $f = \chi_A(m)$ and $S = \Delta\psi_A(m)$. When Δ is nonsquarefree this can miss solutions. For example, $A = 26$, $\Delta = 675$ has $f = 2, S = 45$ satisfying $675 \cdot 2^2 - 45^2 = 675$; the least positive $\chi_{26}(m)$ is 26, not 2. The correct lattice is the following.

Lemma 46.11.1 (Underlying integer Pell unit). *Let $A \geq 4$ be even and $\Delta = db_0^2$ with d squarefree. There exist even $a \geq 2$, odd $b \geq 1$, and odd $e \geq 1$ with*

$$a^2 - db^2 = 1, \quad A + b_0\sqrt{d} = (a + b\sqrt{d})^e.$$

Put $C_* = \psi_a(e)$ and $\delta_* = a^2 - 1 = db^2$. Then

$$b_0 = bC_*, \quad \Delta = \delta_* C_*^2, \quad C_* \psi_A(t) = \psi_a(et) \quad (t \geq 0). \quad (46.53)$$

Every positive strong- Δ solution is exactly

$$\Delta f^2 - S^2 = \Delta \iff f = \chi_a(m), \quad S = \delta_* C_* \psi_a(m), \quad m \geq 1. \quad (46.54)$$

Moreover $\gcd(\chi_a(m), b_0) = 1$ for every $m \geq 0$.

Proof. Since $\Delta \equiv 3 \pmod{4}$, both b_0 and d are odd, and $d \equiv 3 \pmod{4}$. Also $d > 1$ because Δ is not a square. Among positive integer-coordinate norm-one units for d , choose the least one $u_0 = a + b\sqrt{d} > 1$; existence is supplied by $A + b_0\sqrt{d}$. Every positive unit is a power of u_0 : choose k with $u_0^k \leq u < u_0^{k+1}$; the quotient is an integer-coordinate norm-one unit in $[1, u_0)$, hence is one by minimality. Its coordinates have the claimed signs because for a positive real norm-one unit $w \geq 1$, the conjugate is w^{-1} , so its two coordinates are nonnegative. This proves generation without requiring a classification of the full ring of integers.

If a were odd, the norm equation modulo four would force b even, and all powers would have odd first coordinate, contrary to even A . Thus a is even, b is odd, and modulo two the coefficients of u_0^k alternate between $(1, 0)$ and $(0, 1)$. Therefore the exponent e representing $A + b_0\sqrt{d}$ is odd. Comparing coordinates and then powers proves (46.53).

If $\Delta f^2 - S^2 = \Delta$, then $b_0^2 \mid S^2$, so $b_0 \mid S$. Write $S = b_0 s$; then $s^2 = d(f^2 - 1)$, and squarefreeness gives $s = dv$. Thus $f^2 - dv^2 = 1$, and generation gives $f = \chi_a(m)$, $v = b\psi_a(m)$ for $m \geq 1$ because $S > 0$. This proves (46.54); its converse follows from the norm.

The norm also implies $\gcd(\chi_a(m), b) = 1$. Suppose an odd prime ℓ_0 divides both $\chi_a(m)$ and C_* . In the commutative ring $\mathbb{F}_{\ell_0}[s]/(s^2 - \delta_*)$, the element $u = a + s$ is a unit of norm one. The two divisibilities imply $u^{2m} = -1$ and $u^{2e} = 1$. Raising these identities to e and m , respectively, gives $-1 = 1$ because e is odd, impossible. This ring argument remains valid if its quadratic is reducible or has a repeated root. The prime two does not divide $b_0 = bC_*$. Thus $\gcd(\chi_a(m), b_0) = 1$ as claimed. $\square$

Lemma 46.11.2 (Opposite-parity Pell residues). *For $a \geq 2$, $m \geq 1$, and $F_* = \chi_a(m)$, the sets*

$$\{\pm\psi_a(t) \bmod F_* : t \geq 0 \text{ even}\}, \quad \{\pm\psi_a(t) \bmod F_* : t \geq 0 \text{ odd}\}$$

are disjoint.

Proof. For the formal unit $u = a + \sqrt{a^2 - 1}$, its coefficientwise identity modulo F_* is $u^{2m} = -1$; this follows from the norm and $\chi_a(m) \equiv 0$. Reduction modulo $2m$ therefore changes only a sign. Reflection of a reduced rank t to $2m - t$ preserves the ψ residue, because $u^{2m-t} = -u^{-t}$ has second coefficient $\psi_a(t)$. Both operations preserve parity. Hence every residue has a signed representative $\pm\psi_a(j)$ with $0 \leq j \leq m$ and $j \equiv t \pmod{2}$.

For $m = 1$ the representatives are zero and one modulo $a \geq 2$. For $m \geq 2$ write $b_1 = \psi_a(m)$, $b_2 = \psi_a(m-1)$. Strict increase and recurrence give $b_1 > (2a-1)b_2$, while $F_* = ab_1 - b_2$. Consequently $b_1 + b_2 < F_*$: indeed $F_* - (b_1 + b_2) = (a-1)b_1 - 2b_2 > 0$. Distinct-parity representatives have distinct indices, so a nonzero difference has absolute value less than F_* and a positive sum is at most $b_1 + b_2 < F_*$. Neither can be zero modulo F_* . Zero and nonzero representatives are included in this argument. $\square$

46.12 Proof of the free83 nonextension theorem

Assume for contradiction that (46.3)–(46.5) hold under Theorem 46.1.3's hypotheses.

46.12.1 Positive auxiliary norms reduce to one

By Proposition 46.10.2, $N_{\text{aux}} = z^2 > 0$ with $z^2 \mid \Delta = db_0^2$. As d is squarefree, $z \mid b_0$. Multiplying the strong equation by z^2 yields

$$\Delta(zf)^2 - (zS)^2 = \Delta.$$

Lemma 46.11.1 therefore gives $zf = \chi_a(m)$ for some $m \geq 1$. Its coprimality conclusion forces $z = 1$. Thus every positive branch has

$$N_{\text{aux}} = 1, \quad N_{\text{strong}} = \Delta, \quad f = \chi_a(m), \quad S = \delta_* C_* \psi_a(m), \quad m \geq 1. \quad (46.55)$$

This is a proof of normalization for this interface, rather than an assumed generic unit theorem for the full free83 polynomial.

46.12.2 The unit branch contradicts rank parity

By Lemma 46.9.1, $N_{\text{aux}} = 1$ implies

$$SV = \varepsilon \chi_S(t_*), \quad y_* = \psi_S(t_*), \quad t_* \geq 1, \quad \varepsilon \in \{+1, -1\}.$$

Divisibility by S forces t_* odd. Write $t_* = 2h + 1$. The strong identity makes $S^2 \equiv -\Delta = 1 - A^2 \pmod{f}$, so Lemma 46.9.2 gives

$$V \equiv \varepsilon Q_h(1 - A^2) = \varepsilon(-1)^h \psi_A(t_*) \pmod{f}.$$

The literal expression (46.3) instead gives $V \equiv -c = -\psi_A(p) \pmod{f}$. Multiplying the resulting signed congruence by C_* , and using (46.53), yields

$$\psi_a(et_*) \equiv \pm \psi_a(ep) \pmod{\chi_a(m)}.$$

The first rank is odd because e, t_* are odd; the second is even because p is even. This contradicts Lemma 46.11.2. We multiplied by C_* ; no unproved division by C_* modulo f is involved. The value of R disappears in this step, so this positive-branch obstruction does not require R to be odd.

46.12.3 The negative branch contradicts the literal parity

Proposition 46.10.2 forces the entire negative branch (46.52), and in particular makes V even. But $f = 1$ in (46.3) gives

$$V = c(T - 1) - R.$$

The recurrence modulo two gives $\psi_A(p) \equiv p \pmod{2}$, so c is even, whereas R is odd. The displayed expression for V is therefore odd, a contradiction. The equality orbit's $V = 0$ case is also excluded. Positive and negative branches exhaust the nonzero product $N_{\text{aux}}N_{\text{strong}} = \Delta > 0$, proving Theorem 46.1.3.

46.12.4 Application to the authenticated Report45 tuples

Those tuples have $A = Y(X + 1) + 2 \equiv 2 \pmod{4}$, $p \equiv 0 \pmod{4}$, $c = \psi_A(p)$, $R \equiv 3 \pmod{4}$, and $P_5 = 1$, exactly as recorded in Section 46.2. Hence all satisfy the theorem's hypotheses. The independent free83 audit initially certified the arithmetic theorem conditionally on these outer invariants because the historical construction directory was unavailable. The recovered authenticated Report45 construction supplied here verifies the invariants and closes that source-level application. It does not extend the conclusion to other outer tuples.

46.13 Verification, reproducibility, and limitations

46.13.1 Proofs and finite checks have different roles

The unbounded conclusions are proved by Sections 46.3–46.12. Numerical tests corroborate them and expose implementation or transcription mistakes; they do not prove the quantifiers over genuine compilers, inputs, or auxiliary ranks. No astronomical full tuple is materialized. Inner Pell component fixtures and diagnostic outer CRT fixtures remain disjoint, and diagnostic masks are never represented as genuine compiled programs.

The preserved author height packet [3] reports 3,458 active checks, six inner fixtures, 74 odd input-loader ranks, 27 separate outer fixtures, 241 rational-error cases, and 401 cubic-envelope ranks. Its largest materialized integer is 307,945 bits under a 500,000-bit cap. Seven tamper scenarios, each under normal and optimized Python, reject altered source pins, wrong exact floors, or a reduced hard cap. The separate auxiliary checker treats sign/index classes, Pell descent, modulo-four norm exclusion, and positive- i monotonicity.

The independent height audit [4] gives a mathematical PASS and independently checks nine inner fixtures, 45 input-loader ranks, 37 transport/pure-power cases, 29 outer fixtures, 11,640 minimal-radix cases, 5,703 cubic inequalities, 985 rational budgets, 69 exhaustively located Pell solutions, and 45 auxiliary-sector cases. Its largest checked Pell coordinate is 389,069 bits under a 450,000-bit cap. It also proves the fixed-outer minimum-height corollary used in Theorem 46.1.2.

The free83 research packet [5] supplies separate bounded parity, odd-index coprimality, nonsquarefree-lattice, and strong-divisor tests. Its independent proof audit [6] is read-only mathematical review; no nonexistent numerical audit script is implied. The additional report-level checker corroborates the newly restated small-norm descent and odd-polynomial identities. All active checks use explicit failures, so Python optimization does not disable them.

The separately preserved analytic author checker makes 355 active checks at 450 decimal digits, over nine real parameter cases. The independent analytic checker makes 8,719 active checks, including 18 real cases through $R = 512$ with fractional ranks, up to 983 decimal digits, and 12 exact rational inverse-series coefficients. Its checks of (46.42) have largest sampled error-to-bound ratio below 0.0151. These high-precision calculations are corroboration, not interval-certified proofs. The uniform bounds are proved analytically in Section 46.7 and separately audited [9].

The all-orders inverse author checker supplies 82,094 active checks, including 3,601 exact rational grid cases and 2,880 complex samples over the three branches. It controls the tiny conjugates through logarithmic bounds rather than treating floating underflow as exact arithmetic. Its complex sampling is not interval certification; Section 46.8 proves the uniform disk bounds and convergent inverse directly. Its independent checker contributes 26,373 active checks, including 2,409 exact rational branch/center cases, 864 complex samples, nine real inverse cases through five Lagrange coefficients, and 36 roots on the large artificial-parameter circle. Its largest sampled geometric-tail error-to-bound ratio is below 0.01503.

46.13.2 Portable inspected replay

The ZIP preserves the scientific packets byte for byte and inventories every file. Its report-level `replay.py` runs only an explicit allowlist of inspected, independently authored bounded checkers. Upstream repository Python, compiler recipes, and saved arithmetic source rows are never imported, evaluated, or executed. Their `.py` and JSON files are hashed or parsed strictly as data.

One independent audit checker contains a historical absolute source root. The wrapper verifies that exact line and its frozen file hash, copies the checker to a temporary tree, and substitutes only that root with the copied sibling Report45 packet path. It records both frozen and executed-copy hashes. No scientific packet is modified. All checker outputs are compared with their frozen receipts in normal and `python -O` modes, and packet hashes are checked before and after.

The report can be replayed with `python3 replay.py`, rebuilt with `python3 build_pdf.py -output /tmp/Report46-rebuilt.pdf`, and checked with `python3 verify_release.py`. The build uses ordinary local LaTeX with shell escape disabled; it never runs research code. The bounded checkers require Python 3; the two independent analytic checkers additionally use `mpmath 1.3.0`, as recorded in their receipts. All other replay checkers use the standard library. The supplied build receipts, replay receipt, and PDF-layout QA record the observed outcomes. The release contains editable LaTeX as well as the PDF. The ZIP manifest excludes itself and the ZIP to avoid a self-hash cycle.

46.13.3 What remains outside the conclusion

1. The height is the exact height of this family and the exact minimum among auxiliary completions with its outer tuple fixed. It is not the globally smallest certificate of square/product82.
2. The input bounds preserve the genuine radix jumps. They do not assert a single smooth leading constant in the ordinary input.
3. The free83 theorem requires the specified even-rank sector and $P_5 = 1$. Its nonextension result is compatible with the separately audited all-input collapse through odd prime ranks [12]; it cannot be promoted into generic soundness.
4. The author packet’s old “pending audit” wording is preserved as source history. The separate completed audit supplies the final core verdict; no frozen source bytes were edited to update prose.
5. The forward series, algebraic cubic inverse, and Lagrange analytic inverse algorithm are genuinely convergent in their stated domains. The explicitly isolated exponential sector is the first one; the Lagrange algorithm is not a fully sorted canonical inverse transseries. No numerical interval certification is claimed.

46.14 Further research questions

1. Can a different genuine outer construction reduce the height below this doubly exponential family, perhaps to singly exponential height in a suitable input parameter? The fixed-outer minimum prevents an auxiliary-only improvement but says nothing about that broader search.
2. Can sharper packing analysis quantify the height oscillations inside each radix plateau and across the factor-five jumps without discarding the compiler dependence?
3. Can the convergent Lagrange inverse be reorganized into a canonical ordered expansion of all exponential sectors, with uniform tail bounds after that regrouping? The analytic algorithm and its geometric tail are established here; such a sector-by-sector reorganization requires additional justification.
4. Beyond the known odd-prime-index construction [12], which outer invariants characterize free83 completions? The positive branch’s opposite-parity obstruction suggests distinguishing main-rank parity from the independent negative-branch condition on R .

5. Can the nonsquarefree-lattice argument extend beyond $P_5 = 1$ to classify the actual divisor budgets contributed by nonunit retained factors? A proof must control those factors rather than import a unit normalization from a different circuit.
6. Can the seven one-sided ledger entries be sharpened to exact lengths while preserving all genuine compiler branches? This would refine coordinate accounting, although it cannot change the exact maximum already proved here.

46.A Source identities and attribution

The candidate source is from Vladimir Reshetnikov’s *ProveIt* repository. The square/product82 snapshots are frozen at commit

6ce2dcaaf49d28de43ce9396a094643308fb0595.

The free83 source snapshot belongs to commit

0d9d1e0174df30d82d80d758e9ba8e793203126e.

The delivered Report45 PDF has SHA256

8c635d4b4b32e574923b299c93627ca1cd21817d8402e2725970967bd2b72897.

The recovered construction proof is newly authenticated, not claimed byte-identical to a historical lost proof. Its exact SHA256 is

690c5a1dc237bd53a9582bfbe01fcd8a176174e6b116089a1842532f83c1770b.

The selected Report43 small-norm proof and literal free83 JSON are preserved unchanged. Their SHA256 values are respectively

84c5c68411514717e6f2643c9d225bf529675c7123bcbbef65fd1a8b2f44ea1
682d37d4cedcd3c4a086ed3a2e55f272892670edb0424fc0fd8242337f1bf016.

Appendix 46.B and `PACKET_INVENTORY.json` locate and pin every frozen file. Research and audit authors are attributed by their preserved packet roles rather than invented personal names.

46.B Packet map and execution boundary

All paths below are relative to the ZIP root’s `packets/` directory.

square-product82-height-release-20261004/

Author proofs `HEIGHT.md`, `AUX_MINIMUM.md`, recovered `BASE_COUNTERFAMILY.md`, original inert source snapshots, three bounded checkers, normal/optimized receipts, and author manifest.

square-product82-height-independent-audit-20261004/

Independent mathematical audit, checker, source bindings, normal/optimized receipts, and hash manifest. Only its temporary checker copy receives the documented source-root substitution.

free83-even-rank-obstruction-20261004/

Nonextension proof, source notes, independent bounded checker, normal/optimized receipts, and original manifest.

free83-even-rank-independent-audit-20261004/

The independent mathematical PASS audit, preserved unchanged.

square-product82-height-expansion-20261004/

Separately sealed analytic proof, independent bounded Decimal checker, normal/optimized receipts, and author manifest.

square-product82-height-expansion-audit-20261004/

Independent analytic PASS audit, source bindings, fresh checker, normal/optimized receipts, and hash manifest.

square-product82-inverse-all-orders-20261004/

Separately sealed all-orders analytic inverse proof, complex-disk checker, normal/optimized receipts, and author manifest.

square-product82-inverse-all-orders-audit-20261004/

Independent all-orders analytic inverse proof audit, fresh checker, normal/optimized receipts, source bindings, and hash manifest.

Manuscript-specific review packets**report46-even-rank-manuscript-audit-20261004/**

The self-contained free83 proof and its recovered Report45 application.

square-product82-report46-manuscript-audit-20261004/

The forward/first-inverse section.

square-product82-report46-allinverse-manuscript-audit-20261004/

The all-orders inverse section. Exact reviewed excerpts and hashes accompany all three review statements.

square-product82-report45-release-20261004/

The complete delivered Report45 release, including its PDF, LaTeX, recovered author/audit packets, source pins, and historical receipts. No executable file in this nested release is run by Report46 replay.

report43-selected-sources/

Delivered Report43 PDF and LaTeX, the exact early small-norm proof, and the inert literal free83 source JSON. This is a selected-source collection, not a claim to contain the whole Report43 release.

free83-prime-collapse-independent-audit-20261004/

Late source-status reconciliation: commit-pinned odd-prime-index collapse proof and independent audit. This separate construction is cited for the current free83 status; no upstream executable is run.

References of Report 46

- [1] *Research Report45: Full positive collapse of the auxiliary square product 82 candidate*. Delivered 4 October 2026. Complete authenticated release preserved in this archive. [write] Printed above as Report 45 (Sections 45.k); the nested release copy is not shipped again.
- [2] Recovered Report45 construction proof, *Full positive collapse of the auxiliary square/product82*, BASE_COUNTERFAMILY.md, 4 October 2026. New recovered identity is explicitly distinguished from historical lost identities. [write] Byte copy of 19-square-product82-counterfamily-COUNTERFAMILY.md; not shipped again.
- [3] Height research packet, *Quantitative height of the full square/product82 counterfamily* and *Exact auxiliary minimum for the fixed outer counterfamily*, HEIGHT.md and AUX_MINIMUM.md, 4 October 2026. [write] Shipped as 20-cert-height-release-HEIGHT.md and 20-cert-height-release-AUX_MINIMUM.md.
- [4] Independent height audit, *Exact height and minimum positive auxiliary completion*, AUDIT.md, 4 October 2026. Core verdict: PASS. [write] Shipped as 20-cert-height-independent-audit-AUDIT.md.
- [5] Even-rank nonextension research packet, *No free83 auxiliary extension of the new even-main-rank square/product82 family*, PROOF.md, 4 October 2026. [write] Shipped as 20-cert-height-er-PROOF.md.
- [6] Independent proof audit, *Even-main-rank free83 nonextension*, AUDIT.md, 4 October 2026. Arithmetic theorem verdict: PASS, including nonsquarefree Δ . [write] Shipped as 20-cert-height-er-audit-AUDIT.md; the manuscript audits are 20-cert-height-er-ms-audit-AUDIT.md and its earlier round 20-cert-height-er-ms-audit-history-initial-review-0154-AUDIT.md.
- [7] Report43 structural packet, *Auxiliary sign and small-norm descent for the free-coefficient83 candidate*, early_auxiliary_norm_lemma.md, recovered from the delivered Report43 release. Required lemma restated and proved in Section 46.10. [write] Shipped as 18-free83-source-structure-early_auxiliary_norm_lemma.md; printed as Lemma 43.3.2.
- [8] Separate analytic research packet, *Convergent log-height expansion and cubic inversion*, LOG_HEIGHT.md, 4 October 2026. Exact forward expansion and first exponential inverse correction, not a complete inverse transseries. [write] Shipped as 20-cert-height-expansion-LOG_HEIGHT.md.
- [9] Independent analytic audit, *Separate analytic height follow-on*, AUDIT.md, 4 October 2026. Verdict: PASS for the exact forward formula, tail, monotone real interpolation, first inverse correction, and convergent algebraic inverse. [write] Shipped as 20-cert-height-expansion-audit-AUDIT.md; the manuscript review of Section 46.7 is 20-cert-height-ms-audit-REVIEW.md.
- [10] Separate analytic inverse research packet, *A convergent all-orders analytic inverse algorithm*, INVERSE_ALL_ORDERS.md, 4 October 2026. Lagrange inversion with a uniform disk argument and explicit Cauchy tail. [write] Shipped as 20-cert-height-inverse-INVERSE_ALL_ORDERS.md.
- [11] Independent all-orders inverse audit, 4 October 2026. Verdict: PASS for the winding-free holomorphic definitions, disk bounds, simple-root continuation, Lagrange formula, geometric tail, and endpoint scope. Preserved in the release packet. [write] Shipped as 20-cert-height-inverse-audit-AUDIT.md; the manuscript review of Section 46.8 is 20-cert-height-allinverse-ms-audit-REVIEW.md.
- [12] Vladimir Reshetnikov, *Free-coefficient83 prime outer collapse*, ProveIt repository, commit 8cf6239b6d805b08f106c7fe31ea4b5b46601722. [Commit-pinned source proof](#). Independently audited 4 October 2026; source bytes, hash binding, and PASS review are preserved with this report. Its odd-prime-index construction is distinct from the fixed even-index family studied here. [write] Unchanged since that commit. The packet's byte copies of the tree's sources are not shipped; its audit is shipped as 20-cert-height-prime-audit-AUDIT.md.